

ReaScratch Manual

Pitch-locked audio and video scratching in plain REAPER

Version 0.7.3 (October 21, 2026)

Turn the jog wheel of a DJ controller (or a rotary encoder or plain knob) to **scratch the audio without changing its pitch**, while **REAPER's video window shows the frame that matches the play position**.

Runs on REAPER 7.x (Windows / macOS) alone. The basic features need no extensions and no separate FFmpeg install (for what some features such as the LOAD button use, see “What you need” under “Setup”).

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Features

- **Pitch-locked scratching**: small slices of the audio (grains) always play at normal speed, so the pitch stays the same even when you play backwards or freeze.
- **Video follows the audio**: the edit cursor follows the scratch position, and the video window shows that frame.
- **Built-in all-intra (every frame a keyframe) conversion**: H.264 video, which is slow to seek and play backwards, can be converted to fast-seeking MJPEG with REAPER alone (if FFmpeg is on the PC, it is used automatically so that no frames shift).
- **Works with many MIDI controllers**: the Reloop Buddy and the APC40 mkII (tested on the real hardware), other DJ controllers such as the DDJ-FLX4 (presets not yet tested on the hardware are marked ⚠ in the preset list), encoders, and even ordinary MIDI knobs.
- **Crossfader cuts**: open and close the sound instantly with the crossfader (chirps, transformers and so on). You can set where the sound starts (the cut-in gap) and how much fader travel is required to open fully. While the sound is cut, the video disappears too.
- **8 hot cues**: store positions on the pads (keys `1`–`8`) and jump to them. A short press jumps and keeps playing; a long press returns to where you were when you let go (press PLAY while holding to keep playing after release). Positions are saved in the project.
- **Memory cues (up to 32)**: like rekordbox / CDJ memory cues, store many positions and step back and forth through them with `CALL`. The recalled position becomes the CUE point (while playing, it jumps and keeps playing; while stopped, it moves there and stays stopped). Positions are saved in the project.
- **SLIP mode**: when on, playback keeps running in the background while you scratch or hold a hot cue. When you let go, playback continues from "where it would have been if you had not touched anything" (like SLIP on rekordbox / CDJs). Off by default.
- **Beat-quantized jumps (quantize)**: when on, hot cues, memory cue CALL, the return when you release a held hot cue, and CUE all jump on the next beat instead of the moment you press, while playing (same as QUANTIZE on rekordbox / CDJs). Presses up to 0.03 s after a beat jump at once. BPM and beat positions are estimated automatically when a video is imported. If the estimate fails, tap the `T` key or the `bpm` box on the screen on the beat while playing, or type the BPM into the `bpm` box (while stopped, or with Shift+click). You can also set them in setup. Off by default.
- **Sample switching (up to 8)**: import up to 8 videos and switch between them instantly with the `1`–`8` buttons at the bottom of the screen, the `F1`–`F8` keys, or the DDJ-FLX4 pads (`SAMPLER` mode). Each sample remembers its last play position, CUE, hot cues, memory cues and BPM, and resumes where it left off when you come back. The play/stop state is kept too. With SLIP on, samples you are not using keep running in the background while playing, and when you come back they play from "where they would be if they had kept playing".
- **Two decks plus crossfader to layer video (deck B)**: create a second deck (deck B) during setup and the crossfader moves smoothly between the two decks' sound and video. Video is composited with REAPER's built-in Video processor. You can choose the blend around the middle (Mix, Add, Subtract, Multiply, Overlay, Soft light, Hard light, Luma key or Chroma key) and a transition, where the crossfader's movement replaces one picture with the other (29 types such as wipes, zooms and spins). On the DDJ-FLX4, deck B is played from the right deck (Deck 2).

- **Tempo and SYNC**: change the playback speed without changing the pitch (range ±8%, ±16% or ±50%, or ±5 to ±60 BPM; default ±20 BPM). The video stretches with it. The `sync` button (`BEAT SYNC` on the DDJ-FLX4) matches the other deck's BPM and lines up the beats (same as BEAT SYNC in rekordbox; when the BPMs are close to double or half, it matches at double or half). For example, set two samples to the same BPM and switch between them on the beat.
- **Songs whose tempo changes**: for live recordings and other songs whose tempo wavers or changes partway, the beat positions can be kept as a map (the same idea as rekordbox's dynamic beat grid). A map is made in three ways: when importing a video or with LOAD, the beats are followed one by one and the tempo turns out to move (songs with a steady tempo keep one BPM as before); when rekordbox's XML has tempo changes; and when you press `Shift` + `T` where the grid is off while playing. Everything that uses beats (beat-synced jumps, loops, SYNC, the beat lines, the LFOs, and so on) follows the map. The BPM readout shows the BPM where the playback is, and SYNC keeps matching the other deck's tempo where it is playing. Typing the BPM again or tapping it with `T` goes back to one tempo
- **Musical key and KEY SYNC**: when a video is imported or loaded with LOAD, its key is detected and shown in the `key` window above the word `grain`, like `Am` (white for the sample's own key; with the key knob turned, the key you hear, in the lamp colour, and the semitones in small text at the window's top right). A `?` after it means the detection was a close call. Click the window or press `U` to turn the key knob so this deck matches the other deck's key (KEY SYNC: the same key or its relative, such as Am and C, the nearer way, up to 6 semitones). `Shift` +click or `Shift` + `U` goes back to the sample's own key. The detection is libKeyFinder's method (the one Mixxx uses) ported to Lua: on 120 EDM test tracks it found the same key for 57%, and most of the others were close keys (a fifth away or the relative). Check by ear before relying on it.
- **Metronome**: a click on the beats of a deck (the first beat of each bar higher). REAPER's own metronome only sounds while REAPER plays, and the decks play with REAPER stopped, so ReaScratch makes its own. It sounds in the headphones only (or in the main output too), and follows the playing deck, the SYNC base deck, or each deck. Turn it on/off with the `N` key, a click on the window in the record's corner, or a MIDI button (see "Listening to a metronome while you play" (p. 203)).
- **Video effects (15 types)**: colour (Tint, Hue rotate, Invert, Threshold), shape (Mirror, Kaleidoscope, LoRez, Shift RGB, Disc, Zoom blur), redrawn pictures (Line art, Halftone), beat-synced (Strobe, Trails) and an old videotape look (VHS), on each deck's picture and on the mixed picture (in projects that have deck B). Their amounts can also rise and fall with the beat (LFO). Each video effect also comes with a matching sound effect (echo, bitcrusher, chorus and so on).
- **Pad modes, loops and filter**: besides hot cues, the pads can play auto loops, effects while held (TOUCH FX), a slicer, short loops (BOUNCE LOOP) and pitch (PITCH PLAY). There is also CUE LOOP, which starts a loop from a hot cue's position, and a DJ filter (8 types; turning it also changes the colour tone of the layered picture) (on the Reloop Buddy, the APC40 mkII and many other presets).
- **Brake and spinback**: a brake that slows down gradually when you stop with PLAY (its length in seconds or in beats), PAUSE quantize that makes the stop wait for the beat, a motor start-up when playback starts (all off by default), and a spinback on the `B` key that throws the record backwards and slows to a stop.
- **Scratching from envelopes (patterns)**: write the record movement and the crossfader into REAPER envelopes and the deck scratches exactly that when it plays the section. Place the included patterns (baby, chirp, tear, flare, transformer and more) as automation items, rearrange them by copying, moving and editing points, and save them as your own patterns. Play while recording in REAPER and your jog and crossfader movements are written to the envelopes, so you can replay them later.

- **Try it without a controller**: in the JSFX window, grab the record with the mouse and turn it, or drag the zoomed waveform, to scratch; click or drag the whole-track waveform to search; use the PC keyboard for play/stop, CUE and freeze.
- **Loading cues from rekordbox**: load the hot cues, memory cues, BPM and beat grid, and key from an XML exported from rekordbox. Traktor's and VirtualDJ's track data can be loaded too ("Loading cues from rekordbox" (p. 26), "Traktor and VirtualDJ track data"). You can also write a sample's cues and BPM out to an XML for rekordbox or to ReaScratch track data ("Writing out track data").
- **Loading with the LOAD button**: put a video or audio file chosen in the Media Explorer into a deck while you play (Windows). An audio-only file shows a spinning speaker picture instead of video ("Loading with the LOAD button (from the Media Explorer)" (p. 23)). You can also drag a file from the Media Explorer or the Windows Explorer onto a deck's JSFX window ("Loading by dragging a file" (p. 25)).
- **Previewing a Media Explorer file**: the `P` key (on many controllers, a button too) previews the file chosen in the Media Explorer. The sound goes to the headphones only and the picture to the centre of the deck's record only, never to the video window ("Previewing a Media Explorer file (P key)" (p. 25)).
- **Headphones, capture and your own presets**: you can also listen ahead on headphones to a deck the crossfader has cut ("Listening to a deck on headphones (Cue)" (p. 202)), turn a performance into each deck's sound and video plus the finished video ("Capturing a performance" (p. 214)), and make the settings for a controller without a preset on setup's `Learn` tab ("Check by touch / make a preset" (p. 194)).

How it differs from other DJ software

ReaScratch has the same controls as rekordbox and other DJ software, such as hot cues, SLIP, quantize and SYNC. On top of that, it has the following points for making otoMAD videos, which other DJ software mostly does not have.

- **Scratching without a pitch change**: grains always play at normal speed, so the pitch stays the same when you scratch or stop. The `pitch — tape` bar under the record changes it smoothly all the way to a record-like pitch that follows the speed ("Reading the screen" (p. 38)).
- **A performance captured as material to edit**: each deck's sound, and video cut and joined to match the exact frames the deck showed, are laid out as a REAPER project. With ffmpeg, it also makes mp4 files with no shifted frames ("Capturing a performance" (p. 214)).
- **Scratches you program**: place techniques such as the baby and the chirp as patterns on the beat, fit them to the track's swing, rearrange and save them, and play them from the pads ("Scratching from envelopes (patterns)" (p. 207), "Fitting the swing" (p. 209)).
- **Fix a performance afterwards**: your hand movements and the crossfader are recorded into envelopes (automation items), and you can fix the details by moving points ("Recording a performance into envelopes" (p. 211)).
- **Decks inside a DAW**: a deck is a REAPER track, so you can add other effects (VST and so on) to that track. Decks can play side by side with song tracks, in time with REAPER's tempo (`REAPER tempo` in "9. Beat-quantized jumps" (p. 48)). A recorded performance is saved as a project and can be rendered in REAPER.
- **Video effects for otoMAD**: each of the 15 video effects comes with a matching sound effect. The crossfader controls 9 blends and 29 transitions between the two decks ("Sound and video effects" (p. 31), "Two decks with layered video" (p. 200)).

- **Many controllers**: there are 122 presets. Even old models and the APC40, which is not a DJ controller, can play two decks from one unit. For a controller without a preset, you can make your own on setup's `Learn` tab ("Preset list" (p. 65), "Check by touch / make a preset" (p. 194)).
- **Working with other DJ software and hardware**: hot cues, memory cues, BPM and beat grid, and key can be loaded from a collection XML exported from rekordbox ("Loading cues from rekordbox" (p. 26)). NI Traktor Kontrol controllers that talk HID (S4 MK3, S3, S2 MK3 and others) connect through the included bridge program ("HID controllers" (p. 197)). There are also presets for the CDJ-2000NXS2 and CDJ-3000 as MIDI controllers (not yet tested on the hardware).
- **Free**: ReaScratch is free, and under GPL-3.0-or-later you may change it and share it ("License & credits" (p. 228)). REAPER keeps working after its 60-day evaluation, but if you keep using it, you are expected to buy a license (60 US dollars for personal use).

Files in the kit

Folder / file	What it is
`Effects/ReaScratch/reascratch.jsfx`	The scratch audio engine (JSFX plug-in)
`Effects/ReaScratch/reascratch_cue.jsfx`	The headphone mix (on the `Cue` track; the `Cue mix` knob blends cue and main. The metronome is added wherever the `Cue mix` knob is)
`Effects/ReaScratch/reascratch_perf.jsfx`	The performance lanes (`DJ performance lanes`). Setup puts it on each deck's engine track; it records what you do on the deck onto REAPER's timeline while REAPER records, and plays it back in performance playback mode (see "Recording and playing a performance on REAPER's timeline" (p. 212))
`Effects/ReaScratch/reascratch_logo.png`	Title logo of the plug-in screen (keep it in the same folder as the JSFX; without it, plain text is shown)
The other images in `Effects/ReaScratch/` (`tex_front.png`, `knob_big.png`, `platter.png` and the other `*.png`)	Images for the plug-in screen (frame, panels, knobs, lamps, switches, TBKgao logo mark, record, help watermark). Keep them in the same folder as the JSFX. It works without them, but with a simpler look
`Effects/ReaScratch/LICENSE_Saike_filters.txt`	License (MIT) of the DJ filter models in the JSFX, ported from Saike's JSFX (Joep Vanlier)
`Effects/ReaScratch/LICENSE_ntsc_zhuker.txt`	License (Apache 2.0) of the VHS video effect, ported from zhuker's ntsc
`Scripts/ReaScratch/reascratch_setup.lua`	Initial setup: creates the tracks, loads the JSFX and registers the other scripts
`Scripts/ReaScratch/reascratch_import_video.lua`	Imports a video. Places it, extracts a WAV (with FFmpeg if available, otherwise with REAPER itself) and loads it into the JSFX. Warns you if the extracted audio goes silent from some point to the end
`Scripts/ReaScratch/reascratch_reencode_mjpeg.lua`	Re-encodes a video to all-intra (AVI/MJPEG) and swaps it in (FFmpeg if available, otherwise REAPER itself)
`Scripts/ReaScratch/reascratch_video_sync.lua`	Syncs the video window to the scratch position. While it runs, it immediately stops REAPER's own play and pause (to keep the picture and sound from drifting apart; recording is not stopped). Run it again to stop it. If you answer Yes in setup, it starts automatically with REAPER
`Scripts/ReaScratch/reascratch_video_reopen.lua`	Reopens the video window (use it when the video window has frozen after a monitor or projector was switched off/on or unplugged; fullscreen is kept)

Folder / file	What it is
`Scripts/ReaScratch/reascratch_load_test.lua`	Load test (about 40 s). Each deck scratches by itself while the script measures dropouts, JSFX load and screen stalls, and saves the result to a text file. See "Testing on a low-spec PC (load test)" (p. 218)
`Scripts/ReaScratch/reascratch_pattern.lua`	Places a scratch pattern on a deck (as envelope automation items). See "Scratching from envelopes (patterns)" (p. 207)
`Scripts/ReaScratch/reascratch_pattern_save.lua`	Saves a pattern you rearranged under a name and adds it to the pattern menu
`Scripts/ReaScratch/reascratch_record_lanes.lua`	Turns writing recorded performance movements into envelopes on or off (on by default). See "Recording a performance into envelopes" (p. 211)
`Scripts/ReaScratch/reascratch_lanes.lua`	Shows or hides the envelope lanes by group (hand-made scratch, recorded effects, performance lanes). See "Showing only the lanes you use" (p. 208)
`Scripts/ReaScratch/reascratch_take.lua`	Turns performance capture on/off (each deck's audio and video, and the finished mix as seen on screen). See "Capturing a performance" (p. 214)
`Scripts/ReaScratch/reascratch_perf_play.lua`	Turns performance playback mode (`PERF PLAY` on the toolbar) on/off. While it is on, REAPER's playback plays the performance lanes back. See "Recording and playing a performance on REAPER's timeline" (p. 212)
`Scripts/ReaScratch/reascratch_perf_play_decks.lua`	Chooses which decks performance playback mode plays (both, deck A only or deck B only)
`Scripts/ReaScratch/reascratch_export.lua`	Write the hot cues, memory cues, CUE, BPM, beat grid and key of the decks' samples to an XML for rekordbox or to ReaScratch track data. See "Writing out track data"
`Scripts/ReaScratch/reascratch_rekordbox.lua`	Choose the XML exported from rekordbox, Traktor's `collection.nml` or VirtualDJ's `database.xml` and load its hot cues, memory cues, BPM, beat grid and key into the samples. See "Loading cues from rekordbox" (p. 26)
`Scripts/ReaScratch/reascratch_patterns.lua`	The included scratch patterns (baby, chirp, tear, flare, crab, transformer, stutter, freeze and more; not run directly)
`Scripts/ReaScratch/reascratch_deck_mix.eel`	Video processor code that layers deck B's picture over deck A's. Setup installs it in the Video processor on the `video` track (no need to open it)
`Scripts/ReaScratch/reascratch_lib.lua`	Shared library for the scripts (not run directly)

Folder / file	What it is
`Scripts/ReaScratch/reascratch_browser.lua`	Loads the file chosen in the Media Explorer into a deck with the LOAD button (on controllers that have one), loads files dragged onto a deck's JSFX window, and previews with the `P` key. Called by the sync script, so not run directly. The LOAD button needs the js_ReaScriptAPI extension and the preview needs js_ReaScriptAPI and SWS; loading by dragging needs no extension
`Scripts/ReaScratch/reascratch_black.png`	A black picture placed where a sample slot is empty (emptied, or nothing loaded yet), so the video window does not keep showing the last picture. Used by the scripts
`Scripts/ReaScratch/reascratch_audio_only.avi`	The spinning speaker video placed where a sample has sound only (it turns with the record. Used by the scripts)
`Scripts/ReaScratch/reascratch_lights_off.ps1`	A small program that turns every controller light off after REAPER closes (Windows). It starts with the video sync script and waits for REAPER to close
`Scripts/ReaScratch/reascratch_ctl_presets.lua`	Writes and reads controller presets made on the setup screen's `Learn` tab (called by the sync script; not run directly). Saved to `Data/ReaScratch_controllers/` in the REAPER resource folder (`<controller name>.reascratch-preset`)
`ProjectTemplates/ReaScratch.RPP`	A project template that is already set up
`Data/ReaScratch/reascratch_screenset.ini`	The ReaScratch window layout (a REAPER window set). Setup adds it to REAPER's window sets when you answer Yes to "Use the ReaScratch window layout?"
`Data/ReaScratch_samples/`	Where the WAVs read by the JSFX are stored (the import script writes them). The `(empty).wav` marks an empty sample slot (a silent file; if you delete it, setup or import recreates it). `rs_load_a1.wav` to `rs_load_b8.wav` hold the sound loaded with the LOAD button (setup makes them), and files such as `rs_load_a1_src.txt` record which source each sound was made from
`tools/allintra_ffmpeg.cmd` / `.command`	Alternative standalone conversion scripts for users with FFmpeg installed
`HID bridge/`	A set of programs that turns NI Traktor Kontrol S2 / S3 / S4 (HID) into MIDI (contents below the table). See "HID controllers (NI Traktor Kontrol S4, S3, S2)" (p. 197)
`ReaScratch_manual_en.pdf`	This manual (the full text plus screen diagrams and the preset table; searchable, copyable text, and the contents link to each chapter). `ReaScratch_manual.pdf` is the Japanese edition

Folder / file	What it is
`README_en.md`	A short guide with just the setup steps and troubleshooting. `README.md` is the Japanese edition
`ReaScratch_screenshot_en.png` / `ReaScratch_screenshot.png`	An example of REAPER after setup (toolbar 3, the video window, the JSFX windows of decks A and B), with REAPER in English / Japanese
`LICENSE` / `CREDITS.md`	The license (GNU GPL v3.0 or later) and the credits

Contents of `HID bridge/`: `hid_bridge.py`, `ReaScratch-HID-Bridge.exe` (Windows), `build_mac.command` / `run_mac.command` (macOS), `report_windows.cmd` / `report_mac.command` (a report for testers) and `THIRD_PARTY_LICENSES.txt` (licenses of the parts bundled in the exe).

Setup

What you need

- **REAPER 7.x** (Windows / macOS; tested with 7.80)
- **A MIDI controller** (you can also try it with the mouse and keyboard). Some controllers need the manufacturer's driver.
  - Reloop Buddy (Windows): needs Reloop's “ASIO Driver”. Without it, the jog and buttons do not respond (see “Troubleshooting” (p. 219)).
  - Some Hercules models, CDJs and others: noted under “Preparation” in the table and in “Notes by model” of the “Preset list” (p. 65).
  - NI Traktor Kontrol (HID): uses the included `HID bridge` and loopMIDI (see “HID controllers” (p. 197)).
- **Used when present** (the basic features work without them)
  - FFmpeg: used for the following. For where ReaScratch looks for it, see “Heavy video” (p. 216).
    - Importing a video and converting it to all-intra (so that the frames match the source exactly)
    - Making videos when capturing a performance
    - Loading a file by dragging it onto a JSFX window (Windows only)
    - Loading with the LOAD button (from the Media Explorer; Windows only)
  - The js_ReaScriptAPI extension (install it with ReaPack): needed for the LOAD button and for previewing a Media Explorer file (the `P` key).
  - The SWS extension (install it with ReaPack): needed for previewing a Media Explorer file (the `P` key).

1. Copy the files

You can install with ReaPack or copy the kit's folders by hand. Use one of the two.

With ReaPack

(you can also get updates via ReaPack)

1. In REAPER, choose `Extensions` > `ReaPack` > `Import repositories...`, paste this URL and click `OK`: `https://raw.githubusercontent.com/tobokegao/ReaScratch/main/index.xml`
2. In `Extensions` > `ReaPack` > `Browse packages...`, find `ReaScratch`, right-click it, choose `Install`, then click `Apply`.
3. The installed files and their locations are the same as when copied by hand. The manual PDFs and the HID bridge are not included; get them from the GitHub Releases if you need them.
4. Restart REAPER and proceed to "2. Initial setup".

By hand

1. Connect your MIDI controller to the PC and start REAPER.
2. In REAPER, choose `Options` > `Show REAPER resource path in explorer/finder`. This opens the **REAPER resource folder** where plug-ins and scripts live (it contains folders such as `Effects` and `Scripts`).
3. Copy the kit's **`Effects` / `Scripts` / `Data` / `ProjectTemplates`** folders into that folder as they are (merging with the folders of the same name).
   - **macOS note**: dragging onto a folder of the same name in Finder "replaces" it by default, deleting what was already in `Scripts` and the others. **Hold Option while dragging and choose "Merge"** (if "Merge" does not appear, open each of the 4 folders and copy their contents into the matching folders).
4. Restart REAPER.

2. Initial setup

1. Open `Options` > `Preferences` > `Audio` > `MIDI Inputs` and check that your controller's MIDI input is enabled (Enabled).
   - To light the controller's buttons, also enable the same controller under `MIDI Outputs`. Setup points the deck tracks' MIDI output there (if you enable it later, run setup again; see "Button lights" (p. 194)).
2. Open `Actions` > `Show action list...`.
3. Click `New action...` > `Load ReaScript...`, load `Scripts/ReaScratch/reascratch_setup.lua`, select it in the list and click `Run`.
   - If you installed with ReaPack, `reascratch_setup.lua` is already in the list: type `reascratch_setup` in the filter box, select it and click `Run`.
   - First (after checking that the JSFX is installed), a menu "Language / 言語 (ReaScratch messages)" appears with three choices: `Auto (follows REAPER's language pack: …)` (the end shows the language Auto gives right now), `English` and `日本語`. The current setting is checked. `Auto` gives Japanese when the filename of the language pack chosen in REAPER (`langpack=` under `[REAPER]` in reaper.ini) is a Japanese one (it contains `Japanese`, `jpn` or `日本`, or starts with `ja_`, `jp_` and the like), and English otherwise (also when no language pack is used). Closing the menu without choosing keeps the current setting. The setting is stored in REAPER on this PC (for all projects); the rest of this setup run and the messages of the other scripts use it. The menu appears every time you run setup.
   - Three tracks, `video`, `audio` and `scratch engine`, are created, and the JSFX is inserted on `scratch engine`.
   - The MIDI input of `scratch engine` is set to "All MIDI inputs". You choose your controller with the JSFX's `Preset` (see below).
   - The other scripts are registered in the action list too (search for "`reascratch`" to find them).
   - You are asked "Make a second deck (deck B)?". Yes creates 4 tracks, `video B`, `audio B` (both muted), `scratch engine B` (deck B's JSFX) and `video B view` (shows deck B's picture; just above or just below `video`, to suit the project's video settings), plus a Video processor on the `video` track that layers the two pictures. If you answer No, the answer is stored in the project and you are not asked again for that project (to add deck B later, create an empty track named `scratch engine B` and run setup). See "Two decks with layered video (deck B)" (p. 200).

- A `Cue` track for headphones is created. It is not sent to the master but straight to the headphone output. If an output has "Phones" or similar in its name, that one is used; otherwise a menu “Which output goes to your headphones (Cue)? (to change it later: ROUTE on the Cue track)” lists the outputs and you pick the one your headphones are on (for example `Analog 7 / Analog 8` on an RME Fireface UCX II). The choice is saved in REAPER's settings on this PC and you are not asked again. See “Listening to a deck on headphones (Cue)” (p. 202).
- A box titled “ReaScratch setup: video conversion height (smaller = lighter, coarser)” asks for `Height (720 / 540 / 360)` (default 720; another number becomes the nearest of the three). This is the height used when converting videos to all-intra; smaller is lighter on the PC and coarser. Choose 540 or 360 on a slower PC. The value is kept in REAPER's settings on this PC and applies to all projects (Cancel keeps the current value).
- If REAPER is set to draw the video ahead and cache it (REAPER's default), a notice “REAPER is set to draw the video ahead and cache it.” appears. With this, the crossfader and video effects show up about 0.2–0.5 s late. Right-click the video window (`View` > `Video`) and choose `Performance tweaks (experimental)` > `Video cache` > `All media and video FX are not cached` . It works at once; no need to restart REAPER.
- Next you are asked “Start the video sync (reascratch_video_sync.lua) automatically when REAPER starts?”. **We recommend Yes.** A few lines are added to REAPER's `Scripts/__startup.lua` (a file REAPER runs every time it starts), and from then on the picture follows automatically as soon as you open a project that contains ReaScratch, without starting the sync script by hand. Projects without ReaScratch are left alone. To stop auto-sync, run setup again and answer No; only the added lines are removed (the rest of `__startup.lua` stays as it was).
- Then you are asked “Use the ReaScratch window layout?”. Yes puts ReaScratch's 11 buttons on toolbar 3 (see “ReaScratch buttons (toolbar 3)” (p. 17)) and adds a window set “ReaScratch” (this toolbar at the top left, the video window at the top right at 16:9) in a free slot. If toolbar 3 already has other buttons, or window sets #01-#10 are all in use, you are asked before they are replaced (what was there is kept; see “Things that depend on your setup” (p. 224)). REAPER reads window sets when it starts, so **restart REAPER**, then switch with the action `Screenset: Load window set #NN` (NN = the number in the message). The same kind of action switches back to another layout.

4. Open the video window with `View` > `Video` .
5. Choose your controller. Open the JSFX window with the [FX] button of the `scratch engine` track, press `setup` in a bottom corner (usually the bottom right; when you made deck B, deck A's window is mirrored and it is at the bottom left) and pick your model from the `Preset` menu (manufacturer → model → `deck 1` / `deck 2` ; see “Preset list” (p. 65)). If you made deck B, pick `deck 2` in the JSFX window of `scratch engine B` . For a controller without a preset, make one in setup's `Learn` tab (see “Check by touch / make a preset” (p. 194)). Without a controller, `Custom` is fine.

> **Tip: new projects from now on** Once setup has been run, you can start straight from `File > Project templates > ReaScratch` (the template only contains deck A's `scratch engine` ; deck B, the `Cue` track and the controller's button lights (the MIDI output) are created and set when you run setup after opening the template).

ReaScratch buttons (toolbar 3)

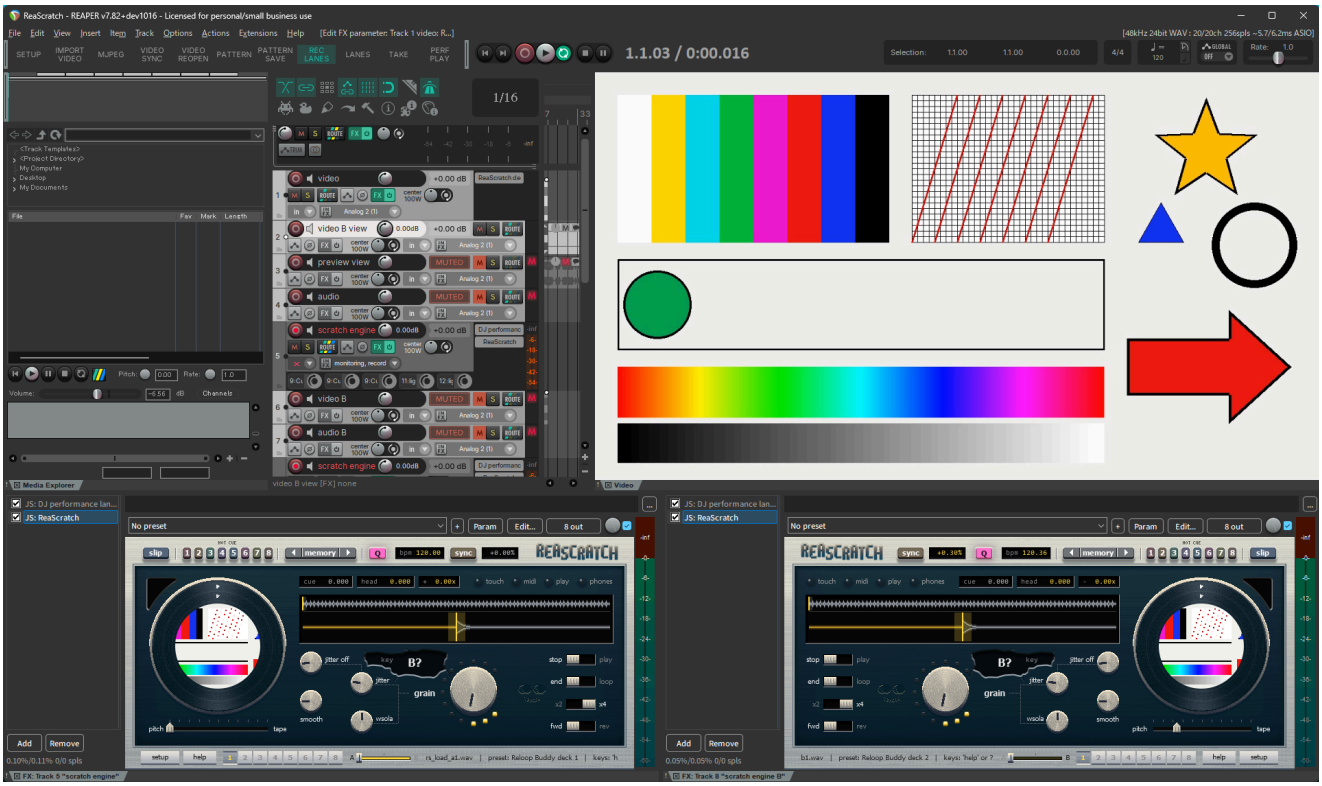


Figure 1 An example of the ReaScratch window layout (toolbar 3 at the top left, the video window at the top right, the JSFX screens of decks A and B below)

If you answer Yes to the window layout prompt, the following 11 buttons appear at the top left of the window set "ReaScratch" (it is REAPER's toolbar 3; with another layout, open it with the action `Toolbar: Open/close toolbar 3`). Each button does the same as running the script of the same name from the action list.

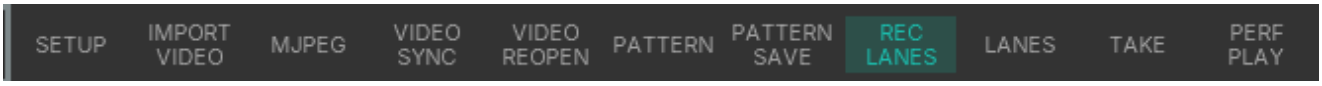


Figure 2 ReaScratch buttons (toolbar 3)

Button	Script it runs	What it does
`SETUP`	`reascratch_setup.lua`	The initial setup (the steps on this page). You can run it any number of times (to add deck B or the `Cue` track, re-point the MIDI output, change the message language, and so on)
`IMPORT VIDEO`	`reascratch_import_video.lua`	Imports a video as a sample ("Step 1: Import a video" (p. 19))
`MJPEG`	`reascratch_reencode_mjpeg.lua`	Converts a video to all-intra (AVI/MJPEG) and swaps it in. It converts the selected video items, or the video of deck A's current sample when nothing is selected ("Heavy video" (p. 216))
`VIDEO SYNC`	`reascratch_video_sync.lua`	Starts / stops the video sync (each press toggles it; "Step 2: Start the sync" (p. 20))

Button	Script it runs	What it does
`VIDEO REOPEN`	`reascratch_video_reopen.lua`	Reopens a video window that has stopped ("Projector output" (p. 201))
`PATTERN`	`reascratch_pattern.lua`	Places a scratch pattern on a deck ("Placing a pattern" (p. 207))
`PATTERN SAVE`	`reascratch_pattern_save.lua`	Saves a rearranged pattern under a name ("Saving a rearranged pattern" (p. 208))
`REC LANES`	`reascratch_record_lanes.lua`	Switches whether the performance being recorded is written to envelopes. The button is lit while on (on by default; "Recording a performance into envelopes" (p. 211))
`LANES`	`reascratch_lanes.lua`	Shows or hides the envelope lanes by group ("Showing only the lanes you use" (p. 208))
`TAKE`	`reascratch_take.lua`	Turns performance capturing on / off. The button is lit while on ("Capturing a performance" (p. 214))
`PERF PLAY`	`reascratch_perf_play.lua`	Turns "play the performance" on / off. The button is lit while on ("Recording and playing a performance on REAPER's timeline" (p. 212))

Basic use

All the scripts can be run by typing `reascratch` into the search box of `Actions` > `Show action list...`. With the window layout, the buttons at the top left ("ReaScratch buttons (toolbar 3)" (p. 17)) run the same scripts.

Step 1: Import a video

Run `reascratch_import_video.lua` and choose a video file (MP4 / MOV / AVI and so on).

- The video is placed on the `video` track, and its audio is extracted to `Data/ReaScratch_samples/<name>.wav` and loaded into the JSFX automatically.
- An item for viewing the waveform is placed on the `audio` track (muted automatically).
- With deck B, the video goes into the deck whose window (the JSFX window of `scratch engine` or `scratch engine B`) was opened last. When a video item on the `video` or `video B` track is selected, it goes into that track's deck. When neither tells the deck, you are asked which deck to use (see "Two decks with layered video (deck B)" (p. 200)).
- Up to 8 sample slots are available. If the current sample (the lit number among `1 – 8` at the bottom of the screen) is empty, the video goes there. If it is in use, you are asked "The current slot N (name) is in use." with "Yes: put it in the empty slot …" and "No: replace slot …" (when all 8 are in use: "Slots 1-8 are all in use. Replace the current slot …?"). A replaced sample's CUE, hot cues, memory cues and BPM are reset (since it is a new video).
- Each sample's video sits at its own place on the timeline (sample 1 at 0 s, sample 2 at 1 hour (3600 s), sample 3 at 2 hours, …). The video sync script moves the edit cursor to the place of the sample in use.
- If you select a video item on the `video` track before running the script, that video is re-imported as the sample for that place.
- There is a length limit (the total of all 8 samples; with REAPER 7.80, about 23 minutes of 48 kHz stereo, about 25 minutes at 44.1 kHz, or about 11.5 minutes at 96 kHz; the imported audio keeps the video's sample rate; older REAPER versions may allow less). Beyond this limit, a notice informs you that only the first portion of the audio was loaded, and the JSFX's waveform view shows it in red. To use the rest, cut the video shorter before importing.
- The imported audio is loaded in pieces (so that the other deck, if it is playing, does not drop out while loading). With long videos or on slower PCs, it can take a few seconds before the deck you imported into starts to sound. The same goes for opening a project; the currently selected sample is loaded first.
- Importing a video with the same name replaces `Data/ReaScratch_samples/<name>.wav` with the new audio (and the converted `<name>_mjpeg.avi` too). If a different video has the same filename, the audio of older projects gets overwritten as well, so rename the video before importing.
- The script checks whether the video is already all-intra. If it is, it says "This video is already all keyframes (…). No conversion needed." and finishes. If not, it asks "Convert it to all keyframes (AVI/MJPEG) and replace it?" (after Yes you choose the height: 720, 540 or 360); **as a rule, answer Yes** (this prevents the picture from stuttering while scratching). A video that is already all-intra but taller than the height chosen in setup gets the question whether to convert it smaller.

Step 2: Start the sync

If you answered Yes to auto-sync during setup, skip this step (it is already syncing).

If you answered No, run `reascratch_video_sync.lua`.

- This keeps the JSFX's read position and REAPER's edit cursor in sync.
- To stop syncing, run this script again (it toggles on and off).

Step 3: Scratch

1. Check that **record arm (the red circle)** and **input monitoring (the speaker icon)** are on for the `scratch engine` track.
2. Turn the controller's jog or knob to scratch (set `Preset` to your model first, in step 5 of the setup).
    - **PLAY button**: plays automatically at normal speed (1x) while your hand is off (can also be switched with `Untouched transport` in the JSFX).
    - **CUE button**: works like a CDJ. While playing, it returns to the CUE point and stops. While stopped, the current position becomes the CUE point and it plays only while you hold the button (release to return to the CUE point and stop; press PLAY while holding to keep playing after release).
    - **Hot cues (pads 1–8)**: on the DDJ-FLX4, switch the pad mode to `HOT CUE` and use the 8 pads (on the Reloop Buddy, hold `PERFORMANCE MODES` and press the `HOT CUE` pad to switch, then use the 8 pads).
        - Press an empty pad: stores the current read position (with quantize `Q` on, it is set to the nearest beat)
        - Short-press a stored pad: jumps there and keeps playing (playback starts even when stopped)
        - Hold a stored pad: plays from there only while held; on release it returns to the previous position and the previous play/stop state
        - Press PLAY while holding: keeps playing after you let go of the pad
        - SHIFT + pad: clears that pad's position
        - The boundary between short and long presses is `Hot cue hold time` (default 300 ms). Stored positions are saved in the project file.
    - **Sample switching (pads 1–8)**: on the DDJ-FLX4, set the pad mode to `SAMPLER`; the 8 pads choose samples 1–8 (set by the `DDJ-FLX4` preset; not yet tested on the real hardware). Pressing the pad of an empty sample does nothing.
    - **Crossfader**: Deck 1 is the left deck and Deck 2 the right one. Moving the fader to the opposite end silences the deck and hides its picture (set by the preset on controllers with a crossfader). The picture is hidden only while the video sync (Step 2) is running.
3. When you finish, save the project in REAPER (`File` > `Save project`). The imported samples, CUE, hot cues, memory cues, BPM, `Preset` and the other settings are saved in the project and come back when you open it.

Without a controller (mouse and keyboard)

Open the JSFX window (the [FX] button of `scratch engine`) and play there.

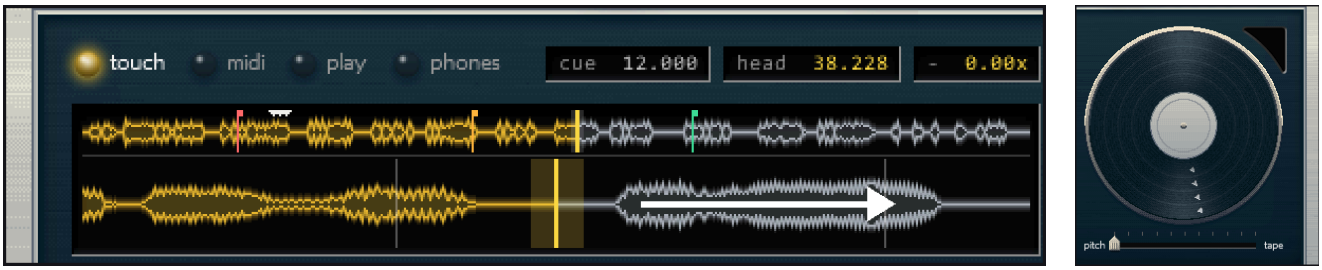


Figure 3 Scratching with the mouse (left: dragging the lower waveform to the right, i.e. backwards; right: grabbing the record)

- **Grab the record and turn it** (usually on the right; on the left in deck A's window when you made deck B): scratch. Clockwise moves forward, counter-clockwise moves back. One turn moves 1.8 seconds of the sample (same as a record at 33 rpm). You can grab anywhere on the record and keep turning without letting go. Holding the mouse button counts as "touching the jog"; releasing counts as taking your hand off. Hold Shift for 1/4 sensitivity. The thin line on the record shows the angle of the current read position (it turns at 33 rpm while playing with `play`). Near the centre the angle cannot be read reliably, so movement there is not counted.
- **Grab the zoomed waveform at the bottom and drag left/right**: scratch. The point you dragged follows the cursor, so dragging the waveform left moves forward and dragging it right moves back (while playing, the waveform flows from right to left). While dragging counts as "touching the jog"; releasing counts as taking your hand off (with `play` it runs on at 1x, with `stop` it stops). When you get near the edge of the screen, let go and grab again (like repositioning your hand on a record). Hold Shift while dragging for 1/4 sensitivity (the cursor and the waveform then no longer move together).
- **Click or drag the upper (whole-track) waveform**: search (like needle search on a CDJ or in rekordbox). It moves to the clicked position at once and plays from there at 1x. If you keep the button down and move left or right, short bits of sound at the cursor position play in a stutter (it does not fast-forward through the parts in between, so there is no fast-forward "zzzt"). When you release, it stops there with `stop` or keeps playing from there with `play`. During search the picture is updated only about 10 times a second, and the exact frame appears the moment you release
- **Right-click the waveform**: makes that spot the CUE point (upper or lower waveform). Right-clicking right next to (within 3 px of) a coloured hot cue line clears that hot cue
- The coloured lines on the waveform are hot cues (1 = red, 2 = orange, 3 = green, 4 = purple, 5 = sky blue, 6 = pink, 7 = lime, 8 = copper; the lower waveform also shows the numbers)
- The small white triangles hanging from the top edge of the waveform are memory cues (the lower waveform also shows faint lines). Right-clicking right next to (within 3 px of) a triangle clears that memory cue.
- `1` – `8` **at the bottom of the screen**: sample switching. The current sample's number is pressed in and lit. Empty samples have faint numbers and do nothing when clicked. Hover over a button to see the sample's filename in the strip at the bottom of the screen

- **The 8 small pads at the top of the screen (on the frame)**: hot cues 1–8. Click to jump (an empty one stores the current position; hold it to return when you let go; with CUE LOOP on, it loops from that cue). Right-click or Shift+click clears it. Clicking the `HOT CUE` label above the pads turns CUE LOOP on/off
- `◂` `memory` `▸` **at the top of the screen (on the frame)**: click `memory` to store the current position; right-click or Shift+click clears the memory cue at the CUE point. `◂` `▸` are CALL (previous / next) and light up while there is a memory cue to call in that direction
- Keyboard (works only after you click the JSFX window once, while it has focus; if REAPER's main window has focus, Space starts REAPER's playback)
  - If Space still starts REAPER's playback while the JSFX window has focus, click the `+` button at the top right of the JSFX window and tick `Send all keyboard input to plug-in` (running `reascratch_setup.lua` ticks it for you; the template has it ticked already)
  - `Space`: same as the PLAY button (toggles play/stop when your hand is off)
  - `C`: same as the CUE button (while playing: return to the CUE point and stop. While stopped: the current position becomes the CUE point and it plays while held; release to return and stop, or press PLAY while holding to keep playing)
  - `Shift` + `C`: make the current read position the CUE point
  - `1` – `8`: same as hot cue pads 1–8 (press `Space` while holding to keep playing after release). `Shift` + `1` – `8` clears them (the keys are read by their Japanese-keyboard symbols, so on an English keyboard `Shift` + `7` clears pad 6, and pad 7 cannot be cleared from the keyboard)
  - Hold `X`: counts as touching the jog (stops on the spot and keeps sounding the same grain: a freeze). Release to go back
  - `M`: store the current read position as a memory cue. `Shift` + `M`: clear the memory cue at the CUE point (the one you just called; if there is none at the CUE point, the one closest to the read position)
  - `←` / `→`: CALL (previous / next memory cue; hold to keep stepping). While playing, `←` skips memory cues you passed less than 0.5 s ago and goes to the one before (press again to go further back)
  - `L`: CUE LOOP on/off (pressing a hot cue loops from there). `Shift` + `L`: enter / exit a loop of the current length at the current position. `J` / `K`: halve / double the loop length (default 4 beats)
  - `S`: SLIP mode on/off (same as the `slip` button at the top right)
  - `B`: spinback (throws the record backwards, slows down and stops; good for endings)
  - `Q`: quantize (beat-quantized jumps) on/off (same as the `Q` button at the top)
  - `?`: show / close the key list (`Esc` closes it too)
  - `T`: tap along with the beat while playing; from the 4th tap the BPM is calculated and entered, and the position of the last tap becomes the beat position. `Shift` + `T`: make the current read position a beat (most accurate if you stop on the start of a beat first). With a BPM, the nearest beat moves there. The first time the whole grid moves; pressing it again somewhere else stretches only the beats between there and the place you pressed before (how to fix a song that drifts; see “Songs whose tempo changes” below)
  - `↑` / `↓`: tempo up/down by 0.1%. `PageUp` / `PageDown`: by 1% (all of these turn SYNC off)
  - `U`: KEY SYNC (match the other deck's key; same as clicking the `key` window above `grain`). `Shift` + `U`: the key knob back to 0

- `Y`: SYNC on/off (same as the `sync` button at the top). `Shift` + `Y`: cycle the tempo range (within %: ±8% → ±16% → ±50%; within BPM: ±5 → ±10 → ±15 → ±20 → ±30 → ±40 → ±60 BPM)
- `F1` – `F8`: switch to samples 1–8 (empty samples are ignored). `F2` and up are taken by REAPER unless `Send all keyboard input to plug-in` is ticked
- Hold `Z`: crossfader closed (no sound, no picture). Drag the waveform while tapping `Z` to practise cuts
- Hold `,` / `.`: nudge (the job of the jog's outer ring). While playing, slows down / speeds up by 4% and returns to normal speed on release (for matching beats by ear). While stopped, moves the position slowly
- `[` / `]`: crossfader 5% left / right. With `Shift` (`{` / `}`), all the way. Whichever deck's window you press it in, both decks take the same position (moving the controller's crossfader takes over from there)
- `9` / `0`: this deck's volume fader down / up by 10% (volume and picture strength; same job as a channel fader)
- `H`: this deck on headphones (Cue) on/off (same as clicking the `phones` lamp)
- `N`: the metronome on/off (same as clicking the window in the record's corner; see "Listening to a metronome while you play" (p. 203))
- `P`: preview the file chosen in the Media Explorer (sound in the headphones only, picture only in this deck's record). Press again to stop ("Previewing a Media Explorer file (P key)" (p. 25))
- `V`: this deck's video effect on/off (same as FX ON on the Buddy; returns to the amount before it was turned off). `Shift` + `V`: cycle the amount 25% → 50% → 75% → 100%. Choose the effect type on the setup `FX` tab

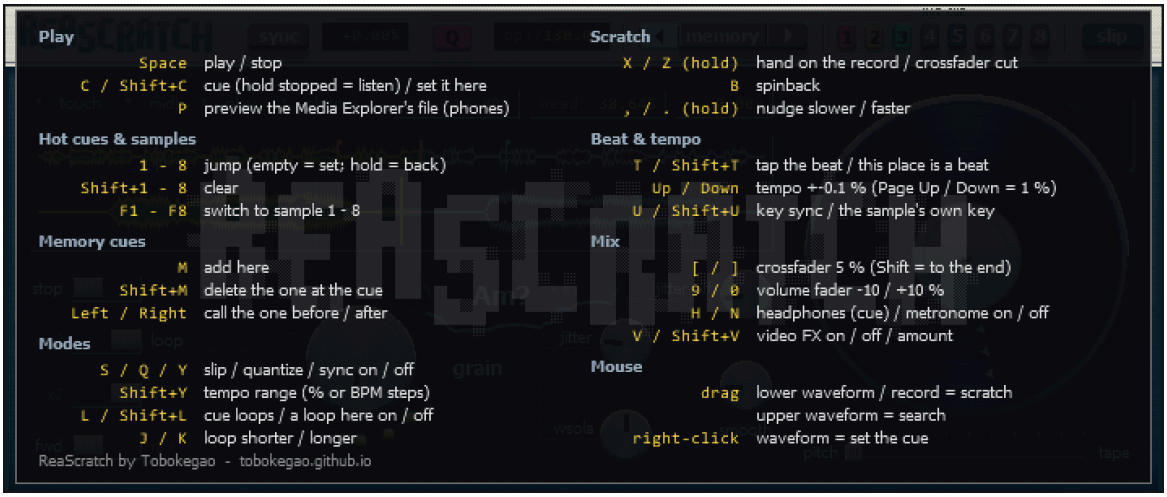


Figure 4 The key list (the screen shown by the help button or the ? key)

Loading with the LOAD button (from the Media Explorer)

The controller's LOAD button loads the file chosen in REAPER's Media Explorer into a deck's sample slot, so you can bring in the next video or track while you play. The LOAD buttons are mapped in the `Reloop Buddy` preset (`LOAD A` = deck A, `LOAD B` = deck B), the APC40 (`REC ARM` 1 / 5; `SHIFT` + a pad in the grid's 3rd row loads into that sample) and many controllers that have LOAD buttons, such as the DDJ-FLX4 (see "Preset list" (p. 65)).

1. Open the Media Explorer from REAPER's `View` > `Media Explorer` and click the file to load (if it is closed, pressing LOAD opens it).
2. Press `LOAD A` or `LOAD B`.
   - `LOAD` alone: into an empty slot. While the deck is playing, into an empty slot after the current one (the slot playing is not read again).
   - `SHIFT` + `LOAD`: replaces the slot in use.
3. The file is loaded when its audio has been converted (the video keeps following meanwhile). A stopped deck switches to that slot; a playing deck keeps playing its current slot (switch with the `1`–`8` buttons at the bottom of the screen or the pads). The BPM and beat position are estimated automatically.
4. Requirements: the video sync script (`reascratch_video_sync.lua`) running, the js_ReaScriptAPI extension (install it with ReaPack) and FFmpeg (used to convert the audio to WAV; looked for in the places listed in "Heavy video", method A). Windows only (FFmpeg is run through a batch file; on macOS, pressing it says so).
5. Video files (mp4, mov, avi, mkv, webm, wmv, m4v, mpg, mpeg, flv, ts, mts) and audio files (wav, mp3, flac, ogg, aif, aiff, m4a, aac, opus, wma) can be loaded. An audio-only file becomes a slot with no video, shown as a spinning speaker picture instead (one turn per platter turn of about 1.8 s, adjusted so the looped length is a whole number of turns and the picture carries on when it loops — the looped length is the sample's length, or, with a BPM and `At end of sample` set to loop, the loop's length in whole bars; it turns back and forth as you scratch); a video with no sound gets silence as long as the video.
6. If the video is not all keyframes (H.264 and so on), or is taller than the height chosen in setup, FFmpeg converts it to all keyframes (MJPEG) in the background after the audio is loaded, and swaps in only the picture when it is done (`<name>_mjpeg.avi` appears next to the original video; playing is not interrupted). A message appears next to the mouse during the conversion and when it finishes. If a `_mjpeg.avi` of the same video already exists with the same number of frames, it is used at once. Videos added by methods other than LOAD are not converted automatically (see "Heavy video" (p. 216)).
7. The audio is written to a fixed file per slot (`Data/ReaScratch_samples/rs_load_a1.wav` to `rs_load_b8.wav`); loading into the same slot again overwrites it. When you reopen the project without saving it, or undo a LOAD, the sound file is written again from its source (a message appears next to the mouse; if the source file is missing, the message says so).
8. If the JSFX does not know these files yet (for example a JSFX added before running the setup), the first LOAD reloads the JSFX, so stop the deck before pressing it. While the deck is playing, it shows `stop the deck for the first LOAD (it reloads the JSFX to prepare)` and loads nothing.
9. Messages appear briefly next to the mouse (for example `select a file in the Media Explorer` when no file is chosen, `no empty slot after the current one (stop the deck to use the ones before it)` while playing with no empty slot after the current one, `file not found (a database view of the Media Explorer cannot be read: …)` when the file was chosen in a database view of the Media Explorer: open its folder and choose it there).
10. To use another controller's buttons, set the numbers in the `Browser` group of the setup screen's `More` tab: `Browser section MIDI channel` (the buttons' MIDI channel) and the note numbers `LOAD note` (into an empty slot) and `LOAD replace note` (replace). On some controllers the LOAD button sends on a different MIDI channel from the BROWSE knob (common on units that use a separate channel for each deck). For those, enter numbers in `LOAD button (any ch)` (into an empty slot) and `LOAD replace (any ch)` (replace) instead of

`LOAD note` and `LOAD replace note`. The number is `(channel − 1) × 128 + note`; look up the channel and note in the unit's MIDI list. The presets set them for the units they cover.

Loading by dragging a file

You can also drag a file from the Media Explorer or the Windows Explorer onto a deck's JSFX window. It loads the same way as `LOAD` (the same requirements and messages; js_ReaScriptAPI is not needed).

- Drop it on one of the `1`–`8` buttons at the bottom: it goes into that slot (replacing what was there).
- Drop it anywhere else: it goes into an empty slot, as with `LOAD` (while playing, an empty slot after the current one).
- Drop several files at once and they go into empty slots one by one (dropped on a button: the first into that slot, the rest into empty slots). Folders cannot be loaded.

Previewing a Media Explorer file (P key)

When you preview a video with the play button of REAPER's Media Explorer, its picture shows in the video window (the screen the audience sees). With ReaScratch's preview, the sound goes only to the headphones and the picture only to the record in the JSFX window.

1. In the Media Explorer, click a video or audio file to select it.
2. Select a deck's JSFX window and press `P`. The preview's picture shows in the centre (label) of that deck's record, with its position below it, such as `PREVIEW 0:12`. The sound goes to the same headphone output as the `Cue` track (not through the crossfader or `Cue mix`).
3. Press `P` again to stop. Select another file and press `P` to switch to it. It also stops when you press `P` on the other deck, press `LOAD`, or the file plays to its end.

You can also preview with a MIDI controller's button (the same as `P`). On the Reloop Buddy and Reloop Ready, hold the mode button (`PERFORMANCE MODES`) and press `LOAD`. On the APC40 mkII and mkI, press `REC ARM` (`RECORD ARM` on the mkI) of column 2 (column 6 for deck B); it lights while previewing. Most other controllers get a preview button from their preset too (on those with pad mode buttons and `LOAD`: `LOAD` pressed while holding the mode button; each controller's button is in "Metronome and preview buttons by controller"). For a controller without a preset, or to use another button, enter `(channel − 1) × 128 + note` in `PREVIEW button` in the `Browser` group of the setup screen's `More` tab. With `Mode button + LOAD` on, `LOAD` pressed while holding the mode button previews instead. Then, on controllers that switch their pad modes themselves and send other notes in each mode (those using `Pad modes by notes`), ReaScratch's pad mode changes when you let go of the mode button, not when you press it (and not at all if you previewed with `LOAD` while holding it).

- What you need: the video sync script (`reascratch_video_sync.lua`) running, the js_ReaScriptAPI and SWS extensions (both from ReaPack), and a `Cue` track with a headphone output chosen. Without that output it does not play. The picture needs deck B (with deck A only, you get the sound only).
- The first preview makes a `preview view` track right below `video B view`. It holds the preview's picture and never shows in the video window. If you delete or move it, the next preview makes it again.

- When `Auto-play selected media` in the Media Explorer's `Options` menu is on, just selecting a file starts the Media Explorer's own preview, and its picture shows in the video window. Turn it off.
- REAPER has no setting that keeps the Media Explorer's own preview out of the video window (checked with REAPER 7.80).

Loading cues from rekordbox

You can load the hot cues, memory cues, BPM and beat grid, and key you set in rekordbox into the samples of the same tracks.

1. In rekordbox, choose `File` → `Export Collection in xml format` and save the XML
2. Run `reascratch_rekordbox.lua` from REAPER's action list and choose that XML. The samples now in the decks that are in rekordbox get its values (a list shows which samples were found and which were not)
3. From then on, importing a video and LOAD also read from this XML by themselves (the XML is remembered: choose it again only when you export it to another place after editing cues in rekordbox; saving over the same file needs nothing)
4. How a track is found: first the track with the same file, else the one with the same filename (without the extension). What ReaScratch adds to the name when importing (such as `_mjpeg`) is ignored
5. What a sample gets: hot cues A–H → 1–8 (a loop hot cue: its start), memory cues (up to 32), CUE = the first memory cue (where rekordbox loads a track), BPM and beat grid (the first tempo, lined up so bars start on the grid), key. Values rekordbox does not have (such as the CUE of a track without memory cues) stay as they are. For a track whose tempo changes, the beats follow its tempo changes (“Songs whose tempo changes”)
6. rekordbox's values take priority over ReaScratch's own estimates
7. Note: when rekordbox analysed a different file than the video you imported (an MP3 of the song and its music video, for example), the song may start at a different time in each, and the cues will be off. Loading the video file into rekordbox too and setting the cues there avoids this

Traktor and VirtualDJ track data

The same `reascratch_rekordbox.lua` also reads Traktor's and VirtualDJ's track data. It tells which program a file is from by its contents.

- **Traktor**: `collection.nml` (on Windows, `Documents¥Native Instruments¥Traktor <version>¥collection.nml`). Hot cues 1–8 (a loop: its start), cues that are not hot cues and load cues → memory cues, BPM and the beat grid marker (a bar's start), key
- **VirtualDJ**: `database.xml` (on Windows, `Documents¥VirtualDJ¥database.xml`; tracks on an external drive are in that drive's `VirtualDJ¥database.xml`). Cues numbered 1–8 → hot cues, other cues and saved loops → memory cues, BPM and the beat grid (a bar's start), key
- ReaScratch only reads these files and never changes them. To be sure the cues you set in Traktor or VirtualDJ are included, close that program first (it saves on closing)
- Note: Traktor and VirtualDJ reading has been tested with files built from public documentation and converter tools, not yet with files written by Traktor or VirtualDJ themselves. Please report values that come

in wrong

Writing out track data

`reascratch_export.lua` writes the hot cues, memory cues, CUE, BPM and beat grid, and key of the samples now in the decks' slots to a file. When run, it asks which kind to write.

- **Yes = an XML for rekordbox**: read it through rekordbox's "rekordbox xml". In rekordbox, turn on `rekordbox xml` under `Preferences` → `View` → `Layout`, and set this file as `rekordbox xml` under `Advanced` → `Database`. Select the tracks that appear under `rekordbox xml` on the left, right-click, and choose `Import To Collection`. ReaScratch only writes a new file and never touches rekordbox's database. The CUE is added as a memory cue (rekordbox loads a track at its first memory cue). A sample whose tempo changes gets one tempo per stretch between its tempo changes
- **No = ReaScratch track data**: a file for reading back into ReaScratch. It has the same form as the rekordbox XML, plus the CUE as it is (not added as a memory cue), the auto gain loudness, and the beat map as it is. To read it back, choose this file in `reascratch_rekordbox.lua`
- The file written for each track: the video the sample came from (LOAD's record, else the video at the slot's place on the video track; for a video converted to MJPEG, the original file beside it if there is one), else the sample's WAV. The WAV holds the whole sound of the video from its start, so the times match the original video
- When the same file is in two slots, it is written once (the first one found)
- Where: with the js_ReaScriptAPI extension, a save dialog lets you choose. Without it, the file goes to `Data¥ReaScratch_export` in REAPER's resource folder, named by the date and time
- Note: whether importing into tracks already in rekordbox's collection overwrites or adds to their cues has not been tested yet. With a collection that matters, back up rekordbox first (`File` → `Library` → `Backup Library`). Writing to Traktor and VirtualDJ is not available yet, because it would mean changing their own data

Pad modes, loops, filter and effects

Besides hot cues, there are modes that play loops and effects from the pads, a DJ filter, and sound and video effects. Which buttons do what differs by controller (see "Preset list" (p. 65) below). This chapter uses the Reloop Buddy and the APC40 mkII as examples.

Pad modes

Each deck has one pad mode. How you choose it depends on the controller.

- **Reloop Buddy**: hold `PERFORMANCE MODES` (the mode button) and press the pad printed with the mode
- **APC40 mkII**: the 8 `DEVICE CONTROL` buttons of the selected deck (`TRACK SELECT`) (on the APC40 mkI, the buttons in the same place)
- **Other controllers**: the controller's own pad mode buttons (`HOT CUE`, `PAD FX`, `BEAT LOOP` and so on) are mapped to the mode closest to their label

The chosen mode is shown in the strip at the bottom of the screen, for example `pads: AUTO LOOP`. After choosing a new preset, and right after opening, it is `HOT CUE`.

#	Mode	Buddy label	APC40 mkII label
1	HOT CUE	HOT CUE	◂ DEVICE
2	AUTO LOOP	AUTO LOOP	DEVICE ▸
3	TOUCH FX	TOUCH FX	◂ BANK
4	FILTER TYPE	SAMPLER	BANK ▸
5	SLICER	SLICER	DEV. ON/OFF
6	BOUNCE LOOP	BOUNCE LOOP	DEV. LOCK
7	PITCH PLAY	NEURAL MIX	CLIP/DEV. VIEW
8	NEURAL MIX	LOOPER	DETAIL VIEW

Pads 1–8 in each mode:

- **HOT CUE**: hot cues 1–8 (see "Step 3: Scratch" (p. 20) above). SHIFT + pad: clear
- **AUTO LOOP**: loops of 1/8, 1/4, 1/2, 1, 2, 4, 8 and 16 beats. Press the same pad again to exit. Pressing another pad during a loop keeps the start and changes the length. SHIFT + pad: loop only while held (exits on release)
- **TOUCH FX**: effects while held (1 Echo, 2 Bitcrusher, 3 Vibrato, 4 Invert, 5 Threshold, 6 Mirror, 7 Kaleidoscope, 8 Shift RGB). Any number can be layered, and they act on both sound and picture. SHIFT + pad: the second row of effects while held (1 Tint, 2 Hue rotate, 3 Strobe, 4 Disc, 5 Line art, 6 Halftone, 7 Zoom blur, 8 VHS;

covering all video effect types together with the first row). Press a pad twice quickly to latch it on (twice again to turn it off; both rows)

- **FILTER TYPE**: choose the filter type (the 8 types under "Filter" (p. 30) below). The chosen pad lights brightly.
- **PATTERN**: choose FILTER TYPE again while in the FILTER TYPE mode to switch to it (again to go back). Pads 1–8 play the scratch patterns assigned to them (at first Baby, Baby x2, Chirp, Tear, Flare (1 click), Crab, Transformer, Stutter 16th). With quantize (`Q`) on it starts on the next beat, and immediately when off; when the pattern ends it counts as letting go (with slip on, back to where the song would be). Touching the record while it plays gives your hand priority. The pad playing a pattern lights brightly. While recording it is recorded as `Rec A-1` ... like hand movements. Change the assignments in the `reascratch_pattern.lua` menu ("Put on a PATTERN pad", see "Placing a pattern" (p. 207)). It works while the video sync script runs. If the deck's `Swing` is above 50, the pad patterns swing too (see "Fitting the swing" (p. 209))
- **SLICER**: divides one bar (4 beats) into 8 and jumps to the same place within the beat in the pressed pad's slice (the beat does not shift). SHIFT + pad: jump to that slice and loop half a beat while held
- **BOUNCE LOOP**: loops only while held. Lengths 1/8, 1/6, 1/4, 1/3, 1/2, 2/3, 1 and 2 beats (including triplets).
- **PITCH PLAY**: plays hot cue 1 at a pitch set by the pad (pads 1–4 = +1 to +4 semitones, pads 5–8 = −3, −2, −1, 0 semitones; the tempo does not change). It is pressed like a hot cue, and if hot cue 1 is empty, the current position is stored as hot cue 1. The pitch stays until you change mode. SHIFT + pad: choose the filter type
- **NEURAL MIX**: pads 1–4 = the memory cue buttons MEMORY, CALL ◂, CALL ▸ and DELETE. Pads 5–8 = while held, a fast spinback, reverse play at 1x, half speed, fast forward (on release, a playing deck goes back to its speed and a stopped deck stops). `SHIFT` + pads 5–8: stop after the move
- When **SLIP** (the `S` key, `slip` on screen) is on, releasing a loop that runs while held (AUTO LOOP with SHIFT, BOUNCE LOOP, SLICER, NEURAL MIX 5–8) goes back to where the track would have been if it had kept running. When it is off, it continues from where you let go
- With SLIP off, SLICER remembers the bar of the first pad you pressed and keeps jumping within that bar while you keep pressing pads (it forgets one bar after your last release, or when you touch the jog, stop, or change mode). On samples without a BPM it works as one bar at 120 BPM, and `SLICER: no BPM` is shown in the strip at the bottom of the screen
- The strength of TOUCH FX is remembered for each effect (100% at first). On the Buddy, turn the FX knob while holding the pad to change it smoothly, or use `FX PARAM` ◂ ▸ for 10% steps (also while it is latched)
- On the APC40, memory cues and the moves while held are always on the grid's 4th and 5th rows, so in mode 8 the top two rows' 8 pads choose the **video blend** (Add, Subtract, Multiply, Overlay, Soft light, Hard light, Luma key, Chroma key; pressing the lit one again gives Mix). The blend is shared by both decks (deck A's setting)
- The Buddy has no SAMPLER mode for choosing samples. When using only the Buddy, switch samples with the `F1` – `F8` keys or the `1` – `8` buttons at the bottom of the screen (on the APC40, the grid's 3rd row)
- Holding a hot cue and pressing another hot cue loops between the two (the same on every controller, whatever the mode; exit as described under "Loops" (p. 30) below)

CUE LOOP

When on, pressing a hot cue starts a loop of the current loop length (default 4 beats) from that hot cue's position. It does not go back when you let go. Pressing another hot cue moves the loop there. A stopped deck starts playing too. An empty pad just stores a hot cue as before. Use it to place hot cues on each word of a vocal and play with repeating the words.

- To switch it: the `L` key, a click on the title `HOT CUE` above the pads on the front screen (it shows `CUE LOOP` while on), or choosing the `HOT CUE` mode again while already in `HOT CUE` mode (on controllers with pad mode buttons, such as the Buddy and the APC40)
- Change the loop length with the `J` / `K` keys (shorter / longer), the Buddy's loop knob, or ▲ ▼ on the APC40. Exit as described under "Loops" (p. 30) below
- It is not on the setup screen; it is `CUE LOOP` (slider 179) in REAPER's parameter list

Loops

Action	Key	Reloop Buddy	APC40 mkII
Enter / exit a loop of the current length	`Shift` + `L`	Press the LOOP knob	`SHIFT` + the 1st pad of row 5
Halve / double the length (1/32 to 32 beats, default 4 beats)	`J` / `K`	Turn the LOOP knob	▼ / ▲
Loop of a set length	–	AUTO LOOP mode pads	`SCENE LAUNCH` 1–4 (1, 2, 4, 8 beats; the same one again exits)
Move the loop by 1 beat (outside a loop, jump 1 beat)	–	`SHIFT` + turn the LOOP knob	`SHIFT` + ▲ / ▼
Go back to the last loop (reloop)	–	`SHIFT` + press the LOOP knob	`SCENE LAUNCH` 5 (exits during a loop)

- With quantize (`Q`) on, a loop starts from the nearest beat
- On exit, with SLIP on it jumps to where the track would have been if it had kept running during the loop. With SLIP off it continues from the current position inside the loop
- Switching samples ends the loop

Filter

A DJ filter placed before the volume fader, the crossfader and the headphone Cue. It is controlled by three knobs. Among the sound effects, it comes after the bitcrusher, the vibrato and the others, and before the echo (the cutoff also trims the bitcrusher's sound, and an echo's tail keeps its earlier tone after you turn the filter back).

- **Cutoff**: off in the middle. Turning left gives a low-pass (12 kHz → 100 Hz), turning right a high-pass (25 Hz → 6 kHz)

- **Morph**: the middle is the normal shape above. Turning it changes the filter shape through band-pass, high-pass and notch
- **Resonance**: emphasis at the cutoff (0–1, default 0.4)
- **Type**: 8 types ported from Saike's (Joep Vanlier's) FM Filter 2: `nlSVF`, `Ladder`, `Steiner`, `Pillow`, `Rect`, `Shriek4p`, `MS-20`, `Moog`. Choose with the pads in FILTER TYPE mode (license: the included `LICENSE_Saike_filters.txt`)

The knobs are HIGH = Morph, LOW = Resonance and FILTER = Cutoff on the Buddy, and `TRACK KNOB` 2–4 (deck B: 6–8) on the APC40. On other controllers the CFX or FILTER knob is Cutoff, and the HI and LOW knobs are Morph and Resonance (see "Preset list" (p. 65)).

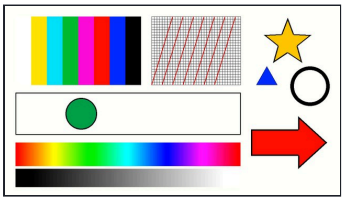
Turning the filter also changes the colour tone of the layered picture (mixed by how loud each deck is).

- Low-pass side (only low sounds) = night: dark parts sink, the colours get a little colder, the brightness slowly sways and the picture ripples. The corners get darker and blurry
- High-pass side (only high sounds) = morning: white light bleeds from bright parts, light rays shine in from above, and the whole picture is washed out white. Blocks and colour bleeding like an old JPEG appear

Sound and video effects

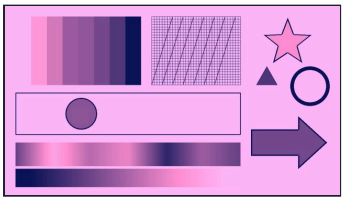
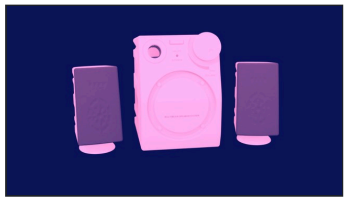
Effects come in pairs of picture and sound. When you apply a video effect, a matching sound effect is applied too. The video effect after the mix also applies its sound effect to both decks.

The images in the table below show each video effect on a deck's picture at amount 100 % (Hue rotate, LoRez and Line art at 50 %): on the left a test pattern, on the right the 3D model shown when a sound-only sample is loaded. Every shot was taken with the deck stopped at the same place, so the moving parts are at the same moment (the 3D model facing the front). Trails is the exception: it shows nothing on a still picture, so it was shot while playing.




— No effect

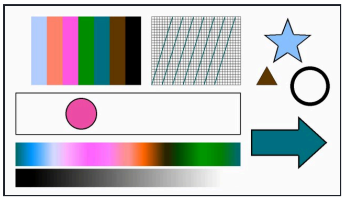
Look The picture before any effect (for comparison)

01 Tint

Look Colours it in two colours (`Tint dark colour` / `Tint light colour` in setup's `FX` tab)

Sound A warmer, muffled tone (cuts the high sounds: above 500 Hz at amount 1)



Amount 50%

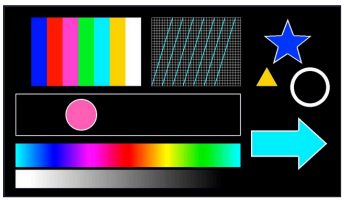


Amount 50%

02 Hue rotate

Look Turns the hue (exactly once round at amount 1, back to the original colours)

Sound Raises the key (amount 1 = one octave)

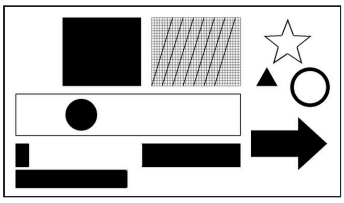


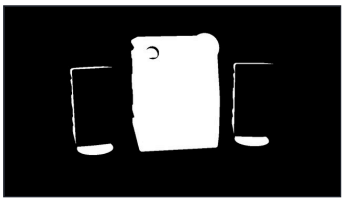


03 Invert

Look Inverts the colours

Sound Ring modulator

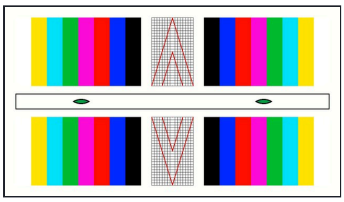


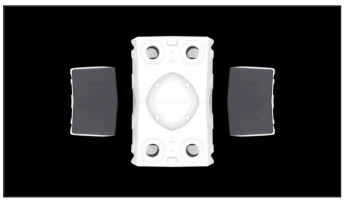


04 Threshold

Look Black and white

Sound Distortion (hard clip; the volume does not go up)

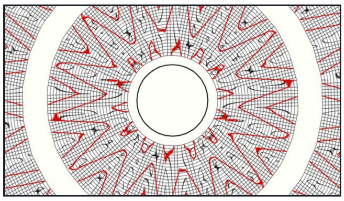


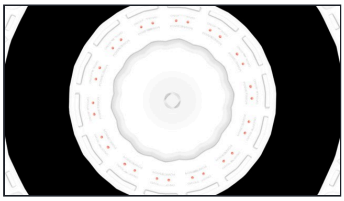


05 Mirror

Look A mirror image

Sound Chorus (as many copies as the picture has parts). Below 0.5: the original plus a copy with left and right swapped, pitched slightly up. From 0.5: also a copy pitched slightly down and a swapped copy pitched slightly down, four in all. The larger the amount, the wider the detune (2–30 cents)

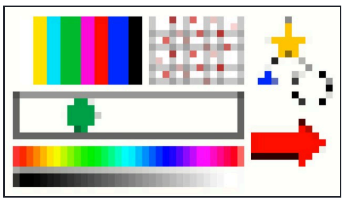




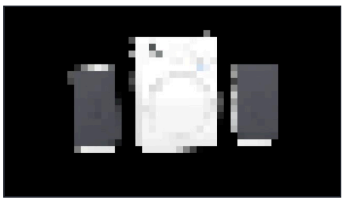
06 Kaleidoscope

Look A kaleidoscope that turns with the beat

Sound Chorus



Amount 50%

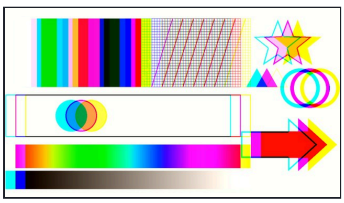


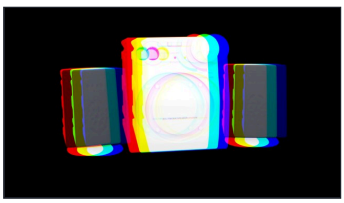
Amount 50%

07 LoRez

Look Mosaic

Sound Bitcrusher

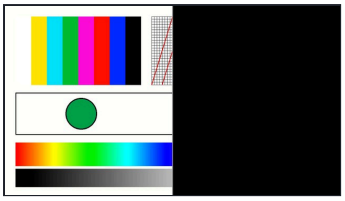




08 Shift RGB

Look Colour offset

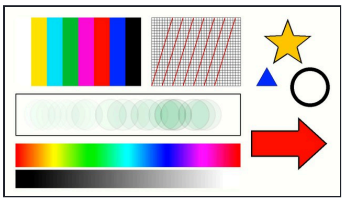
Sound Stereo offset (delays the right channel)




09 Strobe

Look Flashes with the beat (the left half of the image is a lit frame, the right half a black one)

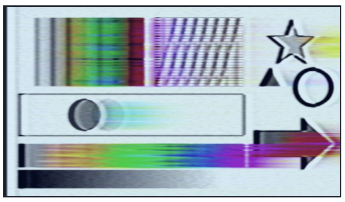
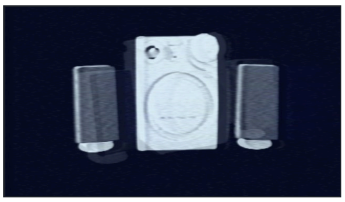
Sound Gate on the beat (same steps as the picture's flashing)




10 Trails

Look Afterimages

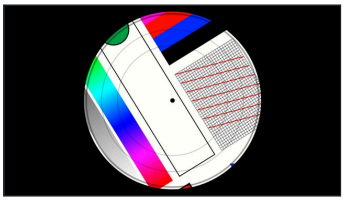
Sound Echo

11 VHS

Look An old videotape look: the picture degraded as an analogue signal, one scan line at a time, ported from zhuker's ntsc: colour smearing to the right, white halos and rainbow edges, noise streaks trailing right, hue wandering and colour dropping out line by line, and irregular sideways shifts. The higher the amount, the slower the tape speed (SP to EP) and the more noise and shifting, with scan lines above 30 % and dubbing-like generation loss above 60 %

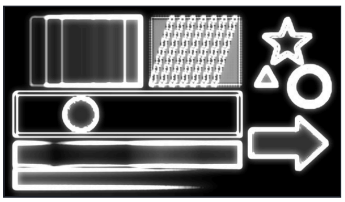
Sound Tape wow (the pitch wavers slowly), the highs rounded, a little hiss

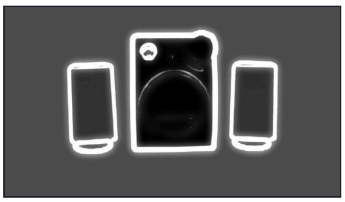



12 Disc

Look Cut out as a record that turns with the platter, and turns in reverse when you scratch

Sound A record's sound (a slightly off phono EQ, a warped record's wobble, crackle). Low amounts are brighter with almost no wobble or crackle; the higher the amount, the darker it gets (fuller lows, duller highs), with more wobble (about 3% at amount 1) and crackle. The tone at maximum amount is set by `Disc tone` in setup's `More` tab → `Video FX`

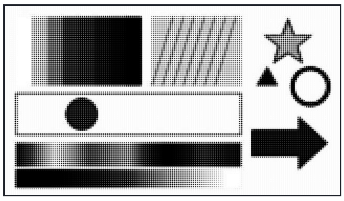

Amount 50%

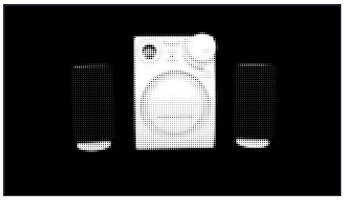

Amount 50%

13 Line art

Look An X-ray-like line drawing: glowing white outlines on black, with a faint image in inverted brightness

Sound A thin sound with the lows cut (below 1.5 kHz at amount 1)

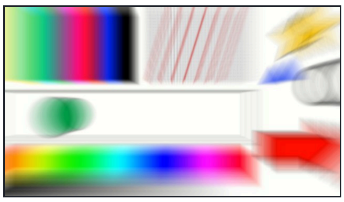


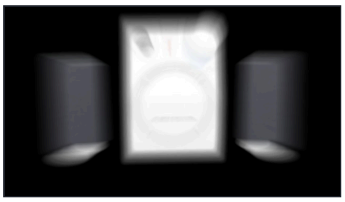


14 Halftone

Look Black dots on white paper

Sound Cuts the sound into small grains and lines them up again (8 grains a beat; 16 a second with no BPM). Each grain plays a random moment from just before, reaching further back at higher amounts (up to 0.125 s). The finer the dots, the more grains overlap (2 at the minimum, 8 at 1)





15 Zoom blur

Look A blur streaming out from the centre

Sound A flanger (a jet-like whoosh; one sweep every 2 beats, or every second with no BPM) and 4 short reflections

Sound and picture are processed in the order below, from top to bottom.

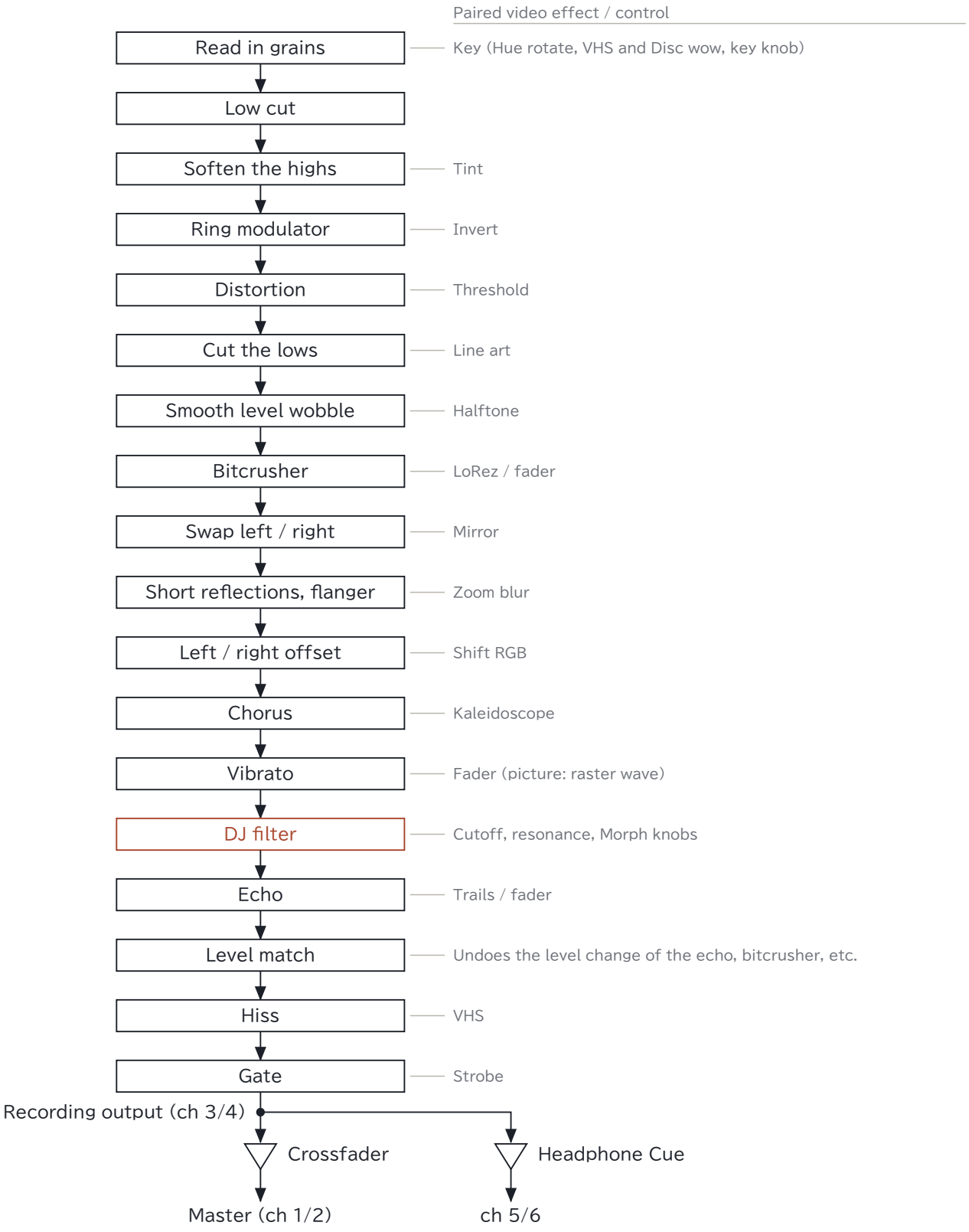


Figure 5 The sound path (per deck)

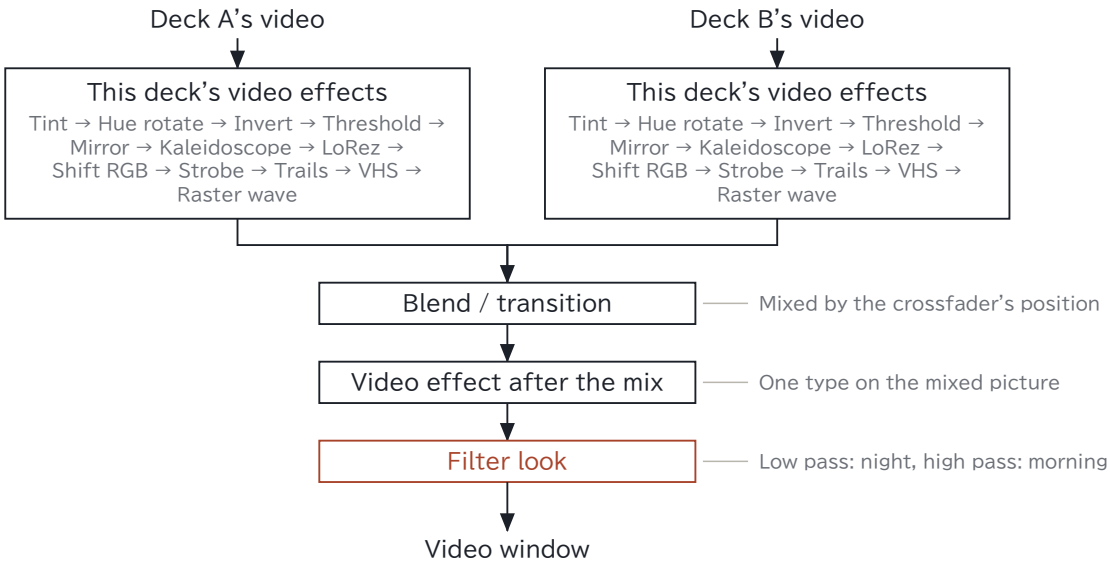


Figure 6 The picture path

Effects are turned on as follows. When the same type comes from several places, the largest amount is used.

- **This deck's video effect (one type)**: choose the type on the setup `FX` tab. The `V` key (on/off) and `Shift` + `V` (amount), the Buddy's FX ON bar (while the bar is on, the FX knob sets the amount, changing the picture and the sound together, and `FX PARAM` ◂ ▸ steps the type back / forward, cycling through Tint to Shift RGB, then Disc, Line art, Halftone, Zoom blur, VHS, Trails, Strobe), `SHIFT` + `CLIP STOP` (columns 1 and 5) on the APC40, or on other controllers the CFX or similar knob (see "12. Video effects" (p. 59) above)
- **8 types at once (APC40)**: the 8 `DEVICE CONTROL` knobs are the amounts of 8 types on the selected deck: by default Tint, Hue rotate, Invert, Threshold, Mirror, Kaleidoscope, Shift RGB and VHS. Choose each knob's type from the 15 in setup's `More` tab → `Video FX` → `DEVICE knob 1 type` to `DEVICE knob 8 type` (per deck, saved with the project). Strobe is knob 8 with `SHIFT` held
- **Echo, bitcrusher, vibrato**: APC40 track faders 2–4 (deck B: 6–8). On the picture they are Trails, LoRez (from the centre of the screen) and a raster wave (the picture ripples up and down). `CLIP STOP` (columns 2–4 and 6–8) turns them on and off while the fader stays up. TOUCH FX pads 1–3 play them too
- **Vibrato sound type**: set by the `pitch — tape` bar on the front screen (`Pitch mode (tape)`). Toward pitch (left), the voice loses its own pitch movements and swings smoothly up and down around a fixed note (C4; about 2.4 semitones each way). The voice is analysed and made again at the new pitch, so it keeps its character (the sound comes about 23 ms later for this). Toward tape (right), the pitch swings smoothly up and down (about ±4 semitones at the end). In between, the two are mixed
  - The key (TRACK KNOB 4 on the `USER` page) moves the centre note; the width of the swing (0–12; the number × 0.48 semitones each way) is changed by TRACK KNOB 3 on the `SENDS` page with SHIFT held. The width is saved with the project
  - Where the voice has no clear pitch (consonants, silence, unpitched sounds), the sound plays as it is
- **Video effect after the mix**: the type with the Buddy's `SHIFT` + `FX PARAM` ◂ ▸ or page 5 of the second APC40; the amount with the Buddy's `SHIFT` + FX knob or the APC40's master fader. Its sound goes on both decks (each following its own "how much the video effects act on the sound")
- **Video blend and transitions**: on the Buddy, pick the blend (Mix to Chroma key) with `FX PARAM` ◂ ▸ while holding the mode (`PERFORMANCE MODES`) button, and the transition with ◂ ▸ alone (hold to scroll fast; the far

left is no transition). Blends and transitions can be combined: turn the combination on with the mode button + `SHIFT` + ◀ ▶ (it starts off; while off, a transition brings in B as usual), and the picture coming in is the blend of A and B instead of B (instead of A when going back), and at the end only that deck shows. With Mix, B comes in as usual. On the APC40, `METRONOME` (each press goes to the next, including transitions) or the pads in mode 8

Sound adjustments:

- **How much the video effects affect the sound** (0–1, default 1; 0 = picture only): APC40 track fader 1 (deck B: 5), on the Buddy, `SHIFT` + turning BROWSE with that deck's FX ON bar on (5% per click). It is not on the setup screen; change it as `Sound of the video FX` in REAPER's parameter list ("13. Controller numbers and other settings" (p. 60)). To move it with a fader of another controller, enter `(channel − 1) × 128 + CC` as `Sound of the video FX fader` in the `Video FX` group of setup's `More` tab. This is that deck's overall sound multiplier
- **Each effect's own sound amount** (defaults: Threshold 30%, bitcrusher (LoRez) 45%, echo (Trails) 50%, the others 100%): on the Buddy, turning the BROWSE knob changes the sound amount of the effect that matches what you hold by 5% per click, shown on the screen as `VHS sound 30 %` (a TOUCH FX pad held = that pad's effect, the FX ON bar on = that deck's effect). With nothing held, `SHIFT` + BROWSE changes the sound of the effect after the mix (turning it without `SHIFT` changes the headphone Cue mix). Each effect keeps its own value, shared by both decks and saved with the project. It is multiplied by the overall sound multiplier above. Echo shares Trails' value and the bitcrusher shares LoRez's
- **Echo length and feedback**: TRACK KNOB 1 and 2 on the APC40's `SENDS` page, `SHIFT` + HIGH and LOW on the Buddy
- **Vibrato and gate steps**: TRACK KNOB 3 and 4 on the APC40's `SENDS` page. The vibrato step is how long each upward and downward swing takes (1, 1/2, 1/3, 1/4, 1/6 or 1/8 beat; 1/4 beat = a 16th note at first). One swing up and down takes two steps and follows the deck's BPM and beat position (with no BPM, 0.2 s a swing). The tape side swings at the same speed
- **Key** (±12 semitones; the tempo does not change): TRACK KNOB 4 on the APC40's `USER` page, `SHIFT` + FILTER on the Buddy

Turning everything off or stopping:

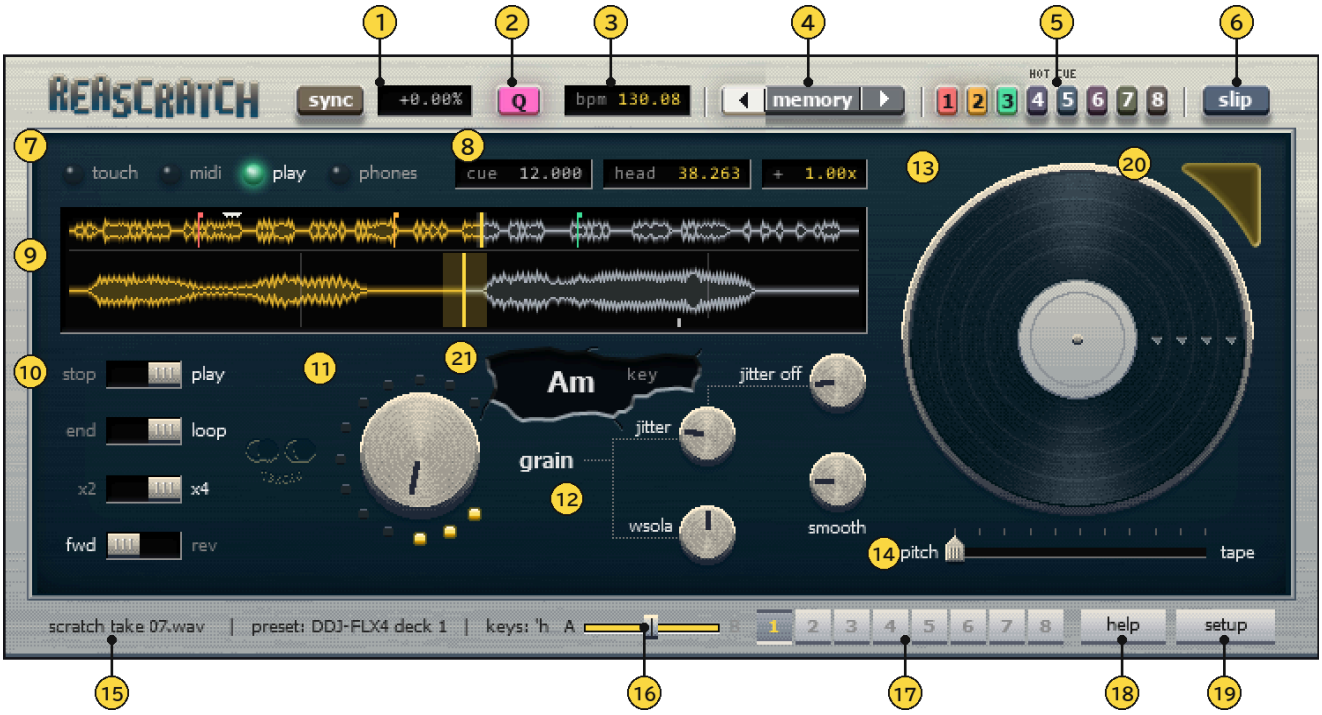
- **Turn all effects off for a moment** (while held; the filter stays): `BANK` on the APC40 (both decks), `SHIFT` + the FX ON bar on the Buddy (that deck)
- **Freeze** (while held, sound and picture stop where they are and the same grain keeps sounding): `CLIP STOP` (columns 1 and 5) on the APC40, `SHIFT` + CUE on the Buddy, the `X` key. On release, with SLIP on it continues from where the track would have been; with SLIP off, from where it stopped
- **Hide only the picture** (the sound stays): `SHIFT` + `ACTIVATOR` on the APC40, `SHIFT` + the headphone CUE on the Buddy

JSFX (ReaScratch) parameter reference

`scratch engine` is **the track's name**; `ReaScratch` (`reascratch.jsfx`) on it is **the plug-in that makes the sound**. Press the **[FX]** button of the `scratch engine` track to open the JSFX window, where you adjust the sound and the feel.

How it works: a "read position (head)" moves through the loaded WAV; grains of a few tens of ms are cut from that position and always played at normal speed, overlapped. The jog only moves the head, so the pitch does not change however fast you move it.

Reading the screen



1. sync and the tempo window
2. Q (quantize)
3. bpm
4. ◂ memory ▸ (memory cues)
5. HOT CUE 1–8
6. slip
7. Lamps
8. Small windows
9. Waveforms (two rows)
10. Switches
11. grain (grain length)
12. jitter, jitter off, wsola, smooth
13. Record
14. pitch — tape
15. Bottom strip
16. Crossfader
17. Samples 1–8
18. help
19. setup
20. Metronome
21. key

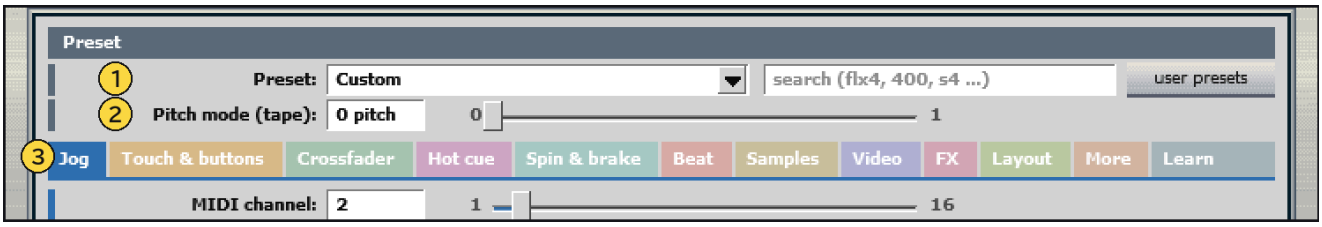
Figure 7 The front screen (for playing). The numbers match the list under the figure

Press the **[FX]** button of the `scratch engine` track to open the ReaScratch screen. It has two sides, **the front (for playing)** and **the back (setup, the controller settings)**; switch with the button at the bottom right (bottom left in the left-handed layout).

Front
(a retro hardware look with a silver frame and a black panel. The light comes from straight above and parts cast their shadows straight down).

- **Lamps** (the row above the waveforms): each has its own colour. `touch` = mustard (counted as touching), `midi` = orange (lights when MIDI arrives), `play` = green (set to run at 1x when your hand is off), `phones` = cerulean blue (this deck is on the headphones; click to toggle)
- **Three small windows at the top right**: `cue` (the CUE point in seconds), `head` (the current read position in seconds), `+1.00x` (how fast the read position moves: 1 = normal, 0 = stopped, negative = backwards)
- **Waveforms** (two rows): the upper row is the whole track; the lower row zooms in on about 1 second around the read position (head). The glowing vertical line is the head; in the lower row the head always stays in the middle and the waveform flows past. The waveform spreads symmetrically above and below its centre line; the played part is mustard and the unplayed part grey. The faint box in the upper row is the range shown in the lower row, the faint band around the head in the lower row is the grain length, and the thin line is the CUE point. **Drag the upper row to search, drag the lower waveform to scratch, and right-click to set the CUE point there**
- **The record on the right** (on the left in the left-handed layout): grab and turn it to scratch (see "Without a controller" above). The white triangles lined up on the record show the angle of the read position, pointing straight up at the start of the track (the triangles point clockwise = forward). The record is always shown. The corner at the top right of the record (top left in the left-handed layout) is the metronome's black window: while the metronome is on its rim glows in the lamp colour, glowing brighter for a moment each time a click sounds on this deck's beats. Click it to turn the metronome on/off (see "Listening to a metronome while you play" (p. 203))
- **The `key` window above the word `grain`**: the track's key (such as `Am`). The window is shaped like a hole where the panel has broken away; the key is shown large towards the left, and `key` in small text at the top right (the other way round in the left-handed layout). With the key knob turned, the key is in the lamp colour and the small text at the top right shows the semitones, like `+2`. Click it for KEY SYNC (match the other deck's key; "Musical key and KEY SYNC" under "Features" (p. 6)), `Shift`+click to go back to the sample's own key
- **The long bar under the record, `pitch` — `tape`**: sets how far the pitch follows the speed. `pitch` on the left (0) = grains whose pitch never changes, as before; `tape` on the right (1) = tape style (like a record: faster is higher, backwards plays in reverse, stopped is silent). Values in between give a proportional mix of the two; for example, at 0.5 double speed sounds half an octave higher (it is still a single sound, not two sounds blended). Drag the handle left/right (hold Shift for fine steps); double-click or right-click resets it to 0. It can also be set with `Pitch mode (tape)` on the setup side
- **Switches on the left** (one column): the words on either side of each switch show its two states. `stop / play` (stops / runs at 1x when your hand is off), `end / loop` (stops at the end / returns to the start), `x2 / x4` (number of overlapping grains), `fwd / rev` (direction of the grains when you move backwards). Hover over a switch to see its full name and description in the strip at the bottom
- **The big knob in the middle, `grain`**: grain length. The dots around it form a C shape, rising clockwise from the bottom right, through the bottom, left and top, to the top right
- **Knobs on the right** (two pairs rising to the right, stacked into a parallelogram; names at each knob's top left): `jitter` and `jitter off` (wobble when stopped), `wsola` (aligning the joins), `smooth` (how smoothly the head follows the jog). The ones joined by lines are the ones related to `grain`

- **Crossfader in the bottom strip** (`A —handle— B`): shows the crossfader position received over MIDI (it cannot be moved on screen). The groove lights up as far as this deck is audible, and the letter on this deck's side (A or B) is shown darker. When an envelope fader (a pattern's cuts, for example) has closed the sound, a faint "shadow handle" moves that far towards the closed side
- **Bottom strip**: normally shows the preset name; hover over a control to see its value and description. While a pattern is playing (triggered from a PATTERN pad or placed on an envelope lane), its name appears right after the deck name, as in `pattern (lane): Baby` . `help` at the bottom right (or the `?` key) shows the key list and notes from the author (click anywhere, or press `Esc` or `?` , to close it; while it is shown, the controls under it do not respond). `setup` to its right switches to the back (setup) side
- **Buttons and small windows at the top (above the frame)**:
  - `slip` : SLIP mode on/off (toggles with each press; while on, it sits slightly lower and glows blue)
  - `Q` : quantize on/off (glows pink while on). With a BPM, the lower waveform shows a line at every beat whether quantize is on or off; each bar's first beat (every 4 beats from the beat grid) is thicker and brighter, and the loop's two ends are in the lamp colour
  - `bpm` : the BPM at the current tempo (hover to see the sample's original BPM in the bottom strip). When the label reads `bpm?` , the sample's BPM has decimals and does not match its length: it is often a little off, so tap the tempo or type it. While playing, clicking the `bpm` box on the beat taps the tempo (like `T`); while stopped, or with Shift+click, you type the BPM
  - Tempo window (`+0.00%`): drag left/right or use the wheel to change it; right-click or double-click resets to 0%
  - `sync` : SYNC on/off (glows orange while on)



1 Preset (controller) 2 Pitch mode (tape) 3 Tabs

Figure 8 The upper part of the setup screen (the back)

Back (setup)

(a screen listing the settings)

- Settings are shown as "name: value box min —slider— max". Colours mark their roles (grey = preset, blue = jog (the part you turn), orange = touch and buttons, green = crossfader, purple = hot cues, teal = spin and brake, red = beat, brown = samples, indigo = video blending, crimson = video effects, yellow-green = layout, reddish brown = the controller's numbers (`More`), dark grey = checking the controller (`Learn`)). The lower part is switched with 12 tabs.
- `Preset` (choosing the controller) and `Pitch mode (tape)` (the same as the `pitch — tape` bar on the front) are here too. `Lamp colour` (the colour of the read position line and the lit numbers: yellow / blue) is on the `Layout` tab.

Controls (front and back):

- Drag knobs up/down and sliders left/right. They move relative to the current value, so a click alone never makes the value jump. **Hold Shift while dragging for 1/10 steps**
- Switches and choices change when you click and release
- Mouse wheel: one step at a time (10 steps with Shift)
- **Double-click or right-click**: resets the control to its default (only right-clicking the waveform sets the CUE point instead)

Screen size: drag the edge of the plug-in window to enlarge it and the screen is scaled up by whole numbers (2x, 3x, ...) and centred in the window (in-between scales are not used, so the pixels stay crisp).

The WAV to play is chosen not on screen but by the import script (`reascratch_import_video.lua`).

The parameters fall into the following 13 groups. The names in the tables are the former parameter names; the setup screen shows shorter ones (for example `Hot cue pad MIDI channel` → `Pad MIDI channel:`). The controller number settings are not sliders, so they do not appear in REAPER's parameter list.

1. Preset

Parameter	Default	What it does
Preset	Custom	Loads a complete set of settings for a controller (jog, touch and buttons, crossfader, pads, tempo, video effect knob and so on) in one go. To keep values you changed by hand when you reopen the project, set it back to `Custom`
Lamp colour	Yellow	Colour of the read position line, the grain ring dots, the loop end lines, the lit numbers and the like (yellow / blue). The lamps keep their own colours. Switched on the setup screen's `Layout` tab
Layout	Right-handed	The layout of the front page. `Left-handed` swaps left and right, with the record on the left (with two decks side by side, the records are on the outside). When you make deck B, deck A starts as `Left-handed` . The setup screen's `Layout` tab

2. Sample and playback

Parameter	Default	What it does
Sample file	(empty).wav	The WAV to play (in `Data/ReaScratch_samples/`). The import script chooses it for you. If it shows `ERR` , import again
Untouched transport	Stop	What happens while your hand is off the jog. `Stop` = stops, silent; `Play` = runs at 1x. Toggled by the controller's PLAY button. Opening a project or loading the FX always starts at `Stop` , whatever value was saved
Cue position (s)	0	Where the CUE button (`C` key) returns to (seconds). Set by pressing CUE while stopped, right-clicking the waveform, or `Shift` + `C` . On import it is set where the sound starts (5 ms before the first sample over −45 dB in the first 30 s); the file and video are not cut

Parameter	Default	What it does
At end of sample	Loop to start	What happens at the end. `Stop (silent)` = silent if you reach the end with your hand off (if you are touching, the last grain keeps sounding and you can pull back); `Loop to start` = go back to the start (pulled back past the start, it carries on from the end). With a BPM, see below the table

At end of sample

: When a BPM is set, it loops from the CUE (or the start of the sample if there is none) to the first bar line after the end of the sample. From the end of the sample to that bar line is silence, and the video holds its last frame (so the loop is a whole number of bars and the beat stays in time). A BPM entered later changes the loop length immediately.

3. Jog

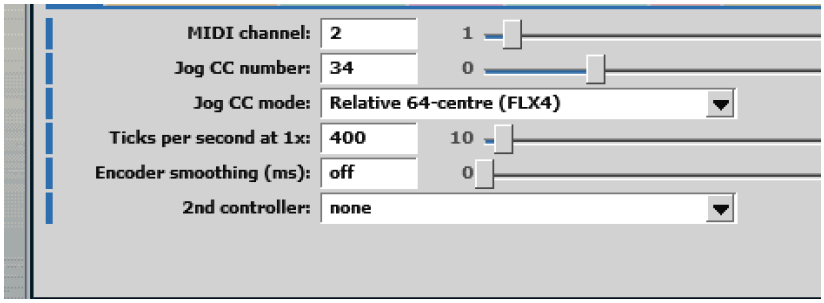


Figure 9 The Jog tab of the setup screen

Parameter	Default	What it does
MIDI channel (deck)	2	The MIDI channel to listen to. On DJ controllers the deck number is often the channel
Jog CC number	34	The CC number of the jog / knob
Jog CC mode	Relative 64-centre (FLX4)	How the jog values are read (8 modes): 3 relative (64-centre / two's complement / sign bit), 2 absolute (7-bit / 14-bit), CDJ speed, Denon platter position, and an absolute knob that can be turned endlessly. For the mode names and details, see "Controller setup and MIDI mapping" (p. 65)
Jog ticks per second at 1x	400	How many jog ticks per second correspond to normal speed. The smaller the value, the further the same turn moves
Encoder smoothing (ms)	0	For rotary encoders. Estimates the turning speed from the spacing of the ticks and moves smoothly at that speed for this long. 0 moves directly with each tick (for DJ jogs)
2nd controller	none	The preset of a second controller (for example Reloop Buddy + APC40 mkII). See "Two controllers at once" (p. 195)

Parameter	Default	What it does
Head smoothing (ms)	15	(Not on the setup screen: the `smooth` knob on the front page.) Smooths the jog input. Larger = the head lags behind more (hides jitter); smaller = more direct. The actual delay is about 1.5 times this value (about 24 ms at the default). In tape mode, a value that is too small makes the pitch flutter

4. Touch and buttons

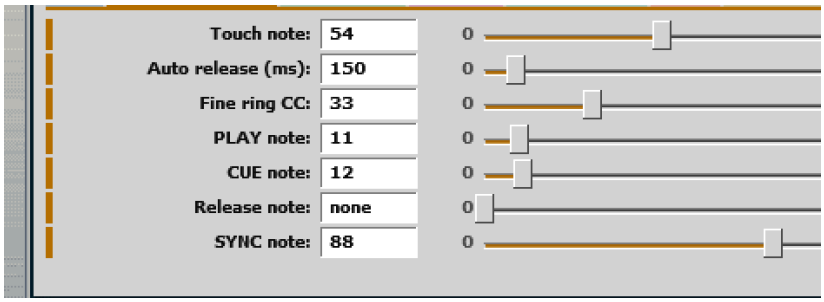


Figure 10 The Touch & buttons tab of the setup screen

Parameter	Default	What it does
Touch note (0=none)	54	Note number of the platter's touch sensor. 0 gives no-touch mode (counted as touching only while turning). With a number set, turning the side without touching the top nudges the pitch while playing (about 0.18 s per turn) and moves the position while stopped. Turning in the 0.15 s after you let go is ignored
Auto release (ms, 0=hold)	150	In no-touch mode, how long after the turning stops before it counts as "hand off". 0 freezes on the spot and keeps sounding (hold) when you stop; stop it with the `Release note` button
Fine ring CC (0=none)	33	A CC for fine adjustment (the outer ring on the FLX4, for example). Moves at 1/4 of the jog's speed. 0 turns it off
PLAY note	11	Note number of the button that toggles `Untouched transport`
CUE note	12	Note number of the CUE button (works like a CDJ; both press and release are read)
Release note (0=none)	0	Note number of a button that ends a hold and goes silent. 0 turns it off
Cue (headphones: this deck on channels 5/6)	Off	Listen to this deck on headphones (Cue). Also toggled by clicking the `phones` lamp or the `H` key. Not on the setup screen; it is in REAPER's parameter list. See "Listening to a deck on headphones (Cue)" (p. 202)
Cue button note (deck channel; -1 = none)	−1	Note number of the button that toggles Cue (on the channel of `Mixer MIDI channel`, where 0 = the deck's channel; −1 = not used). Presets set it to the headphone CUE button (Reloop Buddy: 27; DDJ-FLX4, FLX2 and others: 84; without the button: −1)

5. Sound (how grains are made)

Parameter	Default	What it does
Grain length (ms)	50	Grain length. Shorter = more mechanical and grainy; longer = smoother and closer to the original (and the repeat cycle when stopped gets longer too)
Overlap (grains)	4	How many grains overlap at once (2 or 4). 4 is smoother, 2 is lighter
Grain direction	Always forward	Direction of the grains when moving backwards. `Always forward` = grains always play forward (words stay intelligible when you pull back); `Follow motion` = grains play in reverse too (a real backspin sound)
WSOLA search (ms, 0=off)	10	Searches ±this much for a start point where the new grain's waveform matches the continuation of the previous grain. Reduces phase mismatch at the joins (a flanger-like smear). 0 turns it off
Freeze jitter (ms)	8	While the head stands still, shifts each grain's start point randomly by up to this much. Breaks up the buzzing repeat when stopped
Jitter off above (x)	0.15	Above this speed (1 = normal) the jitter is turned off, so the sound does not get muddy while moving
Pitch mode (0=pitch-locked, 1=tape)	0	0 = grains (the pitch does not change), 1 = tape style (the read speed becomes the pitch; silent when stopped); in between, the proportion by which the pitch follows the speed (0.5 gives half an octave up at double speed). Above 0, moving backwards also plays the sound in reverse. More below the table

Pitch mode

: In between, the grain is read at "speed to the power of this value". Jumps from CUE and hot cues are crossfaded over 5 ms. Above 0, it reads with 6-point interpolation and removes the low rumble from slow movement and direction changes (up to 50 Hz when stopped, 5 Hz at 1x and above). Grain alignment (WSOLA) and jitter get weaker the closer it is to 1, and stop at 1.

6. Crossfader

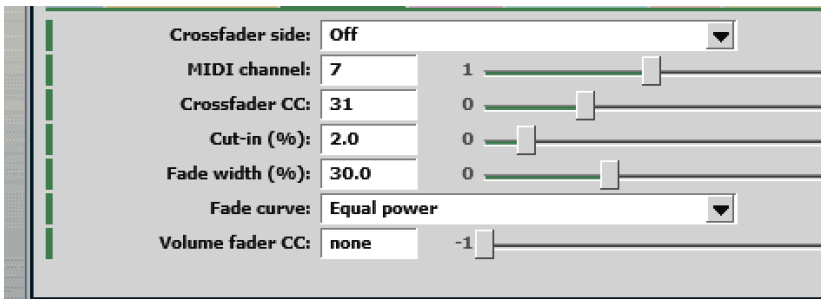


Figure 11 The Crossfader tab of the setup screen

Parameter	Default	What it does
Crossfader side	Off	Which side of the crossfader this deck is on. `Left (deck 1 side)` cuts when the fader goes to the right end; `Right (deck 2 side)` cuts at the left end. `Off` does not use the crossfader
Crossfader MIDI channel	7	MIDI channel of the crossfader (on the DDJ-FLX4 the mixer section uses 7)
Crossfader CC (LSB = CC+32 if sent)	31	CC number of the crossfader (DDJ-FLX4: 31). Controllers that also send CC+32 (14-bit) are read in finer steps
Cut-in (% from closed end)	2	How far from the closed end the sound starts (the gap). Smaller = the sound comes in right next to the end
Fade width (%, 0=instant)	30	Distance from where the sound starts to full volume. 0 opens and closes instantly (a sharp curve for scratching). Larger values change smoothly
Volume fader CC (mixer channel; -1 = none)	-1	CC number of the deck's volume (channel) fader (on the channel of `Mixer MIDI channel`, where 0 = the deck's channel; Reloop Buddy: 28, DDJ-FLX4 and DDJ-FLX2: 19). Pulling it down lowers the sound and, with two decks, fades that deck's picture. The `9` / `0` keys move it too. -1 = not used
Fade curve	Equal power	How the volume changes. `Linear` or `Equal power` (no dip in the middle when mixing two decks). On the setup `Crossfader` tab

Switching between cut and pass uses a crossfade of about 1 ms to avoid clicks. Until crossfader MIDI has been received, it is treated as "open".

7. Hot cues

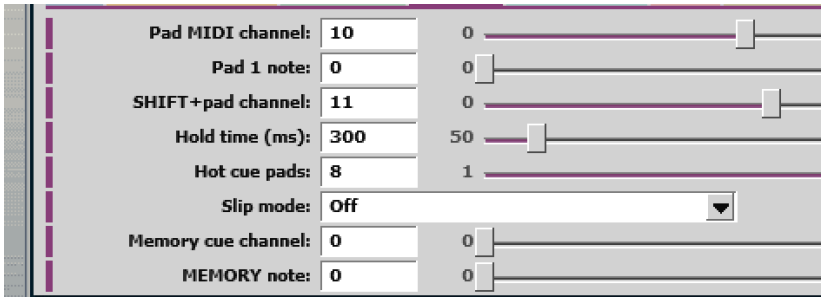


Figure 12 The Hot cue tab of the setup screen

Parameter	Default	What it does
Hot cue pad MIDI channel (0=none)	10	MIDI channel of the pads (DDJ-FLX4 HOT CUE mode: 8 for deck 1, 10 for deck 2). 0 = not used
Hot cue pad 1 note (pads 2-8 = +1..+7)	0	Note number of pad 1. Pads 2–8 are the next 7 numbers (DDJ-FLX4: 0–7, Reloop Buddy: 20–27)

Parameter	Default	What it does
Number of hot cue pads (the pad 1 note and the next ones)	8	How many pads, starting from pad 1, are used for hot cues (1–8). Use fewer on controllers with 4 pads, or where the following numbers belong to other buttons (such as the other deck's pads). Presets set this
Hot cue SHIFT+pad channel (clear, 0=none)	11	MIDI channel of SHIFT + pad (clear) (DDJ-FLX4: 9 for deck 1, 11 for deck 2). 0 = not used
Hot cue hold time (ms)	300	Pressing longer than this counts as "held" (returns to where you were on release). Shorter presses "jump and keep playing"
Slip mode	Off	When on, from the moment you start scratching or press a hot cue while playing, playback keeps running in the background at 1x; when you let go, it jumps to that position and keeps playing. Toggled by the `slip` button at the top right, the `S` key, or the setup `Hot cue` tab. More below the table
Hot cue 1–8 (s, -1=empty)	-1	Stored positions (seconds). -1 = empty. Not shown on screen; saved in the project (1–4 are sliders 42–45, 5–8 are 110–113; remembered per sample)
Memory cue MIDI channel (0=none)	0	MIDI channel of the memory cue buttons. 0 = not used (off by default, because the DDJ-FLX4 has no dedicated buttons)
Memory cue MEMORY note (CALL back / next / DELETE = +1..+3)	0	Note number of MEMORY (store). CALL (back), CALL (next) and DELETE (clear) are the next 3 numbers

Slip mode: Scratching can be with the jog, record, waveform or the `X` key. A hot cue jumps to the background position and keeps playing when released after being held longer than `Hot cue hold time`. The background position is shown as a faint line on the waveform. Starting from a stopped state behaves as when off (a hot cue returns to the position before you pressed and stops; a scratch stays where it is). A short hot cue press, or pressing PLAY while holding, also keeps playing from the hot cue as when off. While on, the spin after letting go (`Spin on release`) does not happen. The CUE button and searching on the upper waveform end the background playback. When switching samples while on (and playing), samples you are not using also run in the background and play from that much further on when you come back (if you switch while scratching, it counts from the background position).

Memory cue positions are not parameters; they are saved in the project as the JSFX's saved data (up to 32). They survive running `reascratch_setup.lua` again (when setup reloads the JSFX, it carries over the parameter values, saved data and envelopes as they are).

8. Spin and brake (setup `Spin & brake` tab)

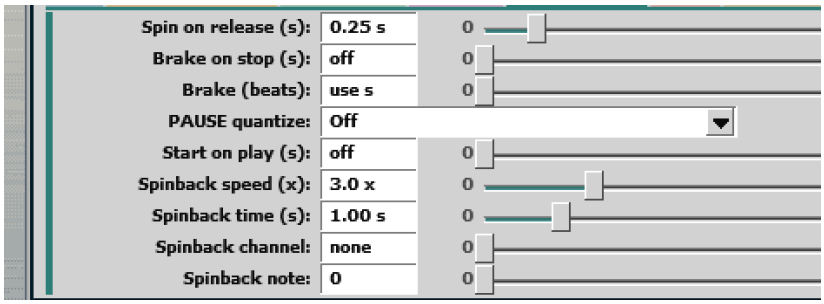


Figure 13 The Spin & brake tab of the setup screen

Parameter	Default	What it does
Spin on release (s to lose 1.8x, 0=off)	0.25	If you let go while turning hard (touch or mouse button), the record keeps spinning with that momentum, then slows at a steady rate back to 1x (or a stop with `stop`); for rewinds. The value is the seconds to lose the momentum of one turn per second (1.8x). It only acts when your hand's speed differs from the destination by 1x or more. Touching again, CUE, a hot cue or a search stops it. 0 turns it off
Brake on stop (s from 1x to still, 0=off)	0	When you stop with the PLAY button (`Space`), it slows at a steady rate instead of stopping at once (a turntable brake). The value is the seconds from 1x to standstill. CUE, hot cues, memory cues and the end of the track stop at once. Pressing PLAY during the brake goes back to playing
Brake on stop in beats	0	Sets the brake length in beats (0–4, in steps of 0.25): the slowdown from the current speed to standstill takes this many of this deck's beats (still in time when tempo or SYNC changes the speed). While 0, or without a BPM, the seconds value above is used. `Brake (beats)` in setup
PAUSE quantize	Off	With quantize (`Q`) on, makes stopping with the PLAY button wait for the beat (`1 beat`, `1/2 beat (8th)`, `1/4 beat (16th)`). The grid is the other deck's beat if that deck is playing, otherwise this deck's. More below the table
Start on play (s from still to 1x, 0=off)	0	When you play with the PLAY button, it speeds up at a steady rate from standstill to 1x (motor start-up). The value is the seconds to reach 1x
Spinback speed (x backward)	3	How fast the spinback (`B` key or the MIDI button below) throws backwards (times normal speed)
Spinback time (s to stop)	1	Seconds for the spinback to slow to a stop. After it stops, the deck is stopped
Spinback MIDI channel (0=none)	0	MIDI channel of the spinback button. 0 = not used (off by default, because the DDJ-FLX4 has no dedicated button)
Spinback note	0	Note number of the spinback button

PAUSE quantize

: A press up to 0.03 s after a grid line stops at once. With a brake, the brake starts at the grid line. Pressing PLAY again while waiting keeps playing without stopping (touching, searching or CUE also cancel it). On a loop shorter than the grid step, it stops when the loop returns to its start.

The pitch follows the bar at the bottom (`pitch — tape`). Towards `pitch` the pitch does not change and the grains slowly advance and stop; for the pitch-dropping slowdown effect, move the bar towards `tape` . Touching again, or CUE, a hot cue or a search, ends the slowdown and the spinback.

9. Beat-quantized jumps (setup `Beat` tab)

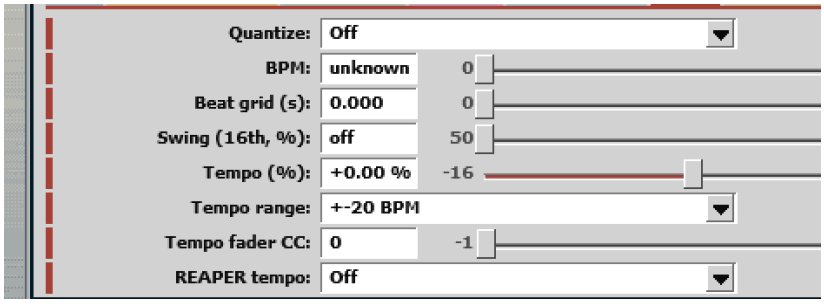


Figure 14 The Beat tab of the setup screen

Parameter	Default	What it does
Quantize (jumps wait for the next beat)	Off	When On, hot cues, memory cue CALL, the return on releasing a held hot cue, and CUE wait for the next beat before jumping while playing. Immediate while stopped, touching or searching. Also toggled by the `Q` key and the `Q` button at the top. More below the table
BPM (0=unknown)	0	The sample's BPM. Quantize does not work while it is 0. Estimated and entered automatically on import. Also entered by tapping along with the beat while playing: the `T` key, or clicks on the front page's `bpm` box (tap tempo, from the 4th tap). More below the table
Beat grid (s, any beat)	0	The position (seconds) of any one beat in the sample. Beats line up from here at the BPM's spacing. `Shift` + `T` sets it to the current read position. In setup, the wheel moves it by 1 ms (10 ms with Shift)
Tempo (%)	0	Playback speed (%). The pitch does not change (even with `Pitch mode` towards tape). The picture stretches at the same rate. Changed by the DDJ-FLX4 tempo slider, the `↑` `↓` `PageUp` `PageDown` keys, or the window at the top of the screen. Changing it by hand turns SYNC off
Tempo range	±20 BPM	How far from the sample's BPM the ends of the tempo slider are, in BPM (7 ranges, ±5 to ±60 BPM) or in % (±8%, ±16%, ±50%). `Shift` + `Y` or `SHIFT` + `BEAT SYNC` on the DDJ-FLX4 cycles within the % ranges or within the BPM ranges. SYNC can go up to ±50%. More below the table

Parameter	Default	What it does
Sync (follow the other deck's BPM)	Off	While on, keeps following the other deck's BPM (including its tempo). The beat positions are lined up when you turn it on and when you start playing with it on (see below the table). Near double or half BPM, it matches at double or half. The `sync` button, the `Y` key, or `BEAT SYNC` on the DDJ-FLX4 (not on the setup screen)
Swing (16th, %; 50 = straight, 67 = triplet)	50	While playing, delays the off-beat 16ths (swing). 50 = straight, 67 = triplet. Based on the BPM and the beat position (`Beat grid`). Envelope patterns placed later and patterns played from the PATTERN pads also follow this swing if it is above 50. See "Fitting the swing" (p. 209)
Tempo fader CC (deck channel; LSB = CC+32; 128 = pitch bend; -1=none)	0	CC number of the tempo slider (on the deck's MIDI channel; CC+32 is read as the low byte for finer steps. DDJ-FLX4: 0). 128 = pitch bend (Reloop controllers; see below the table). -1 = not used. When the slider starts to take effect: also below the table
SYNC note (note+8 = tempo range; 0=none)	88	Note number of the SYNC button (on the deck's MIDI channel; DDJ-FLX4 `BEAT SYNC`: 88). This number + 8 switches the tempo range (`SHIFT` + `BEAT SYNC`: 96). 0 = not used. On the setup `Touch & buttons` tab
REAPER tempo	Off	Sync with REAPER's own tempo (per deck): **SYNC follows REAPER** or **REAPER follows this deck** (the difference is below the table). One deck cannot do both

Quantize

: Jumps land on the same spot within the beat, so the beat is not broken. A press up to 0.03 s after a beat jumps immediately (pressing slightly late does not make you wait a whole beat). If the other deck is playing, pressing PLAY while stopped waits for the start of its next bar (every 4 beats from its beat grid) before playback starts (press again while waiting to cancel; if the other deck is stopped, playback starts immediately). The value order is On = 0, Off = 1 (in older projects this slot held the record-display setting).

Sync

: The beat positions can be off by up to one audio block because of the order in which the decks are processed.

BPM: A clip of 20 seconds or less (a drum loop, for example) gets the tempo that fits a whole number of beats into its length. Since a loop plays at the speed determined by its length, decimal BPMs are kept. When a BPM with decimals does not fit the sample's length, the front page's label reads `bpm?`. In setup, the mouse wheel moves it by 0.1 (by 1 with Shift). Click the front page's `bpm` box while stopped, or Shift+click it, to type it (Enter = set, Esc = cancel; it is the sample's BPM, whatever the tempo).

Tempo range: The BPM ranges are ±5, ±10, ±15, ±20, ±30, ±40 and ±60 BPM (with a 131 BPM sample at ±20 BPM, the ends are exactly 111 and 151 BPM). With the % ranges, for samples with a known BPM the ends are rounded to the nearest whole BPM (131 BPM at ±16% gives ±21 BPM). For samples without a BPM, the BPM ranges act as ±16%. On the DDJ-FLX2, `SHIFT` + `BEAT SYNC` selects the pad mode, so it does not switch the range.

Tempo fader CC

: 128 (pitch bend) is used by the Reloop Mixon 4, Beatmix 2/4, Mixon 8 Pro, Touch, Terminal Mix, Beatpad and Beatpad 2. When the slider position differs from the current tempo (after SYNC, after a key change, after changing the range, right after opening), it has no effect until the slider passes the current tempo's position (soft takeover). From then on the slider takes over and SYNC turns off.

REAPER tempo

:

- **SYNC follows REAPER**: the `SYNC` button follows REAPER's tempo instead of the other deck (half / double time too). While REAPER plays, the deck is put on REAPER's beat when SYNC goes on, when the deck starts and when REAPER starts.
- **REAPER follows this deck**: while this deck plays and is louder than the other on the crossfader, REAPER's tempo is set to this deck's BPM (including its tempo; while the video sync runs. In a project with a beats timebase the kit's tracks are set to time).

10. Samples (setup `Samples` tab)

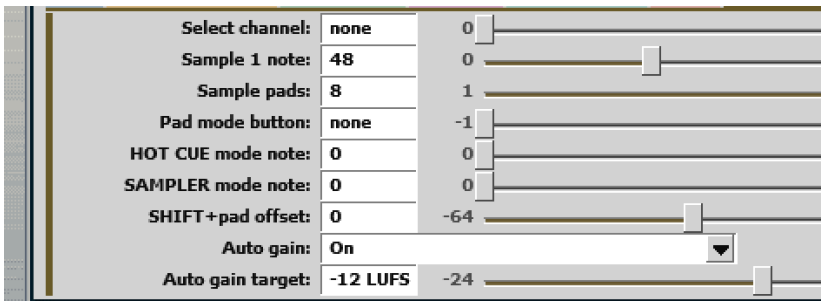


Figure 15 The Samples tab of the setup screen

Parameter	Default	What it does
Sample select MIDI channel (0=none)	0	MIDI channel of the pads that choose samples. 0 = not used (the DDJ-FLX4 preset uses Deck 1 = 8, Deck 2 = 10)
Pad mode button note (deck channel; -1 = no pad modes)	-1	Note number of the pad mode button (on the deck channel; Reloop Buddy: 32). Hold it and press a pad to change mode (8 modes; see "Pad modes" (p. 28)). -1 = no modes (the DDJ-FLX4 and others whose pads send different numbers per mode use `Pad modes by notes`)
HOT CUE mode note / SAMPLER mode note	0 / 0	The note number that arrives on the pad channel for mode button + pad (Reloop Buddy: `HOT CUE` = 0, the `SAMPLER` label = 11). In the Buddy preset the `SAMPLER` label is FILTER TYPE (no mode for choosing samples). Other modes: below the table
SHIFT+pad note offset (0 = the pad notes on the SHIFT channel)	0	How much is added to the pad's number for SHIFT + pad (Reloop Buddy: 8, giving 28–35 on the same channel). 0 = the same numbers as the pads (the DDJ-FLX4 and others where SHIFT changes the channel). Negative values are allowed (APC40 mkI: −1)

Parameter	Default	What it does
Sample select note of slot 1 (2-8 = +1..+7)	48	Note number of the pad for sample 1. Samples 2–8 are the next 7 notes (DDJ-FLX4 `SAMPLER` mode: 48–55)
Number of sample select pads (the sample 1 note and the next ones)	8	How many samples, starting from sample 1, can be chosen with pads (1–8). Same idea as the number of hot cue pads
Auto gain	On	Brings each sample to the target loudness (see below)
Auto gain target	-12 LUFS	The loudness the samples are brought to (-24 to -6 LUFS; higher = louder)
Sample file 2 – Sample file 8	(empty).wav	Audio files of samples 2–8 (sample 1 is `Sample file`). The import script fills them in. Empty samples point to the contentless `(empty).wav`
Sample slot (1-8)	1	The sample in use. Read and written by the import script and the video sync script
Clear sample cues	0	Used by the import script to clear a replaced sample's CUE, hot cues, memory cues and BPM (no need to touch it)

HOT CUE mode note / SAMPLER mode note: The numbers of the other modes are in the `Pad modes` group of the setup screen's `More` tab (group 13 of the parameter reference). The settings keep the names of earlier modes: `AUTO LOOP mode note`, `TOUCH FX mode note`, `SLICER mode note` and `BOUNCE LOOP mode note` pick the mode of that name, `MEMORY mode note` picks PITCH PLAY, `FILTER TYPE mode note` picks NEURAL MIX, and `VIDEO FX mode note` picks a mode that chooses video effects (pads 1–8 keep video effects 1–8 on; pressing the lit one again turns it off).

Auto gain

: Like auto gain in DJ software, each sample's loudness is measured when it is imported or loaded (measured according to the ITU-R BS.1770 / EBU R128 broadcast loudness standards), and while it plays, the deck's gain is adjusted to reach the target loudness. Switching samples keeps about the same loudness. Gain is increased only until the sample's peak reaches -1 dBFS (to prevent clipping), so quiet samples may remain below the target.

- It works before the effects, the channel volume and the crossfader (the monitor (Cue) sound gets the same level)
- Samples imported with an older kit (before auto gain) have not been measured yet. Running setup again measures every sample of both decks (about a second each)
- The values are saved in the project for each sample. In the setup screen's `Samples` tab, you can turn it on / off and set the target for each deck

Each sample's position, CUE, hot cues, memory cues and BPM are saved in the project as the JSFX's saved data.

11. Decks and video blending (setup `Video` tab)

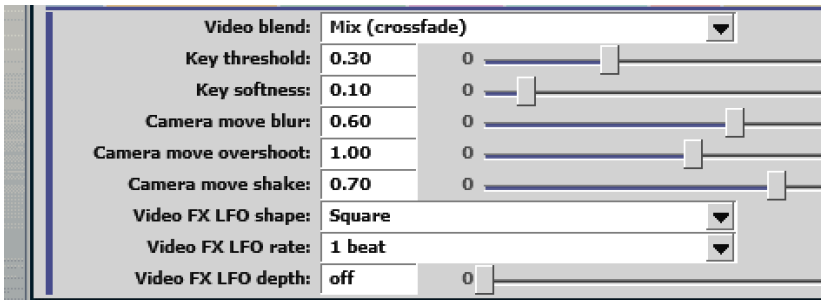


Figure 16 The Video tab of the setup screen

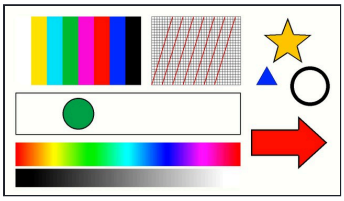
Parameter	Default	What it does
Video blend	Mix	How deck B's picture is layered over deck A's (while the crossfader is between the ends). 0–8 are blends, 9–37 transitions. How each looks: "Blends and transitions" below. Deck A's setting is used
Key threshold	0.30	The key's threshold. With `Luma key`, parts darker than this become see-through. With `Chroma key`, how green a part must be to become see-through (0.3 is standard)
Key softness	0.10	For `Luma key`, the width of the soft edge between see-through and solid
Camera move blur	0.60	How strong the camera moves' motion blur is. It is made by layering pictures from just before and after the move's position. 0 = no blur
Camera move overshoot	1.00	How far a camera move goes past its end and comes back. 1 = normal, 0 = no overshoot, 2 = a big bounce
Camera move shake	0.70	How strong the shake is in `Shake cut` and when `Drop bounce` lands. The colours split with the shake
Video FX LFO shape	Square	The shape of the wave that raises and lowers this deck's video effect amounts with the beat: `Sine`, `Triangle`, `Square` (on for the first half of each cycle), `Saw down` (full on the beat, then fading), `Saw up`. All are full on the beat
Video FX LFO rate	1 beat	The length of one cycle (1/4 to 8 beats). A deck without a BPM runs as if at 120 BPM
Video FX LFO depth	off	How far the amounts go down (0 = off, 1 = all the way to 0). Applies to the chosen effect, the FX knobs, the echo and bitcrusher faders, the vibrato fader's picture and sound, and the pads (not Strobe). The mixed picture's effect uses deck A's setting
Deck	A	Whether this JSFX is deck A or deck B (set by setup; not shown on screen)

The three Camera move values can also be automated; the three Video FX LFO values cannot. All six values are saved with the project.

On the Reloop Buddy, turning the FX knob while holding the mode button changes `Key threshold` (it takes effect when the blend is `Luma key` or `Chroma key`). Turning the FX knob with nothing held changes `Key softness` (this is the same value as the transition edge softness).

Blends and transitions

The images in the table below were taken with the test pattern on deck A and the 3D model on deck B (the picture of a sound-only sample, as a still with a green square added at the bottom left to show the `Chroma key`). The blends are at the crossfader's middle; the transitions are 40 % of the way from A's end. From `Zoom` on, the two pictures are never on screen together at a single place, so each is shown at 40 % (left) and 60 % (right). `Picture in picture` and `PiP to corner` were shot with deck A's and B's pictures swapped, so that the 3D model is in the small window. All other settings are at their defaults (`Key threshold` 0.30, `Key softness` 0.10 and so on).



Deck A

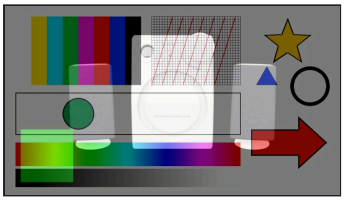


Deck B

— Before blending

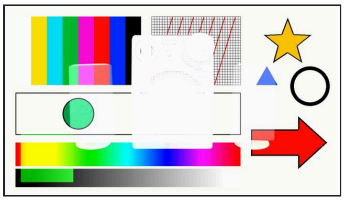
Left: deck A alone; right: deck B alone

Blends



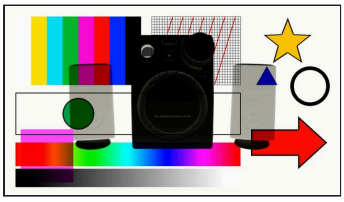
00 Mix

An ordinary crossfade



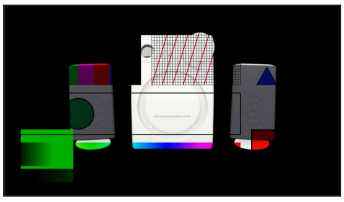
01 Add

Adds the two pictures' brightness (brighter)



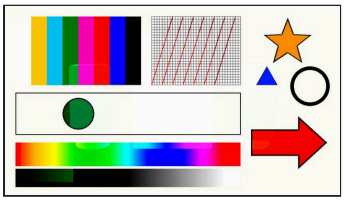
02 Subtract

Subtracts B from A (darker)



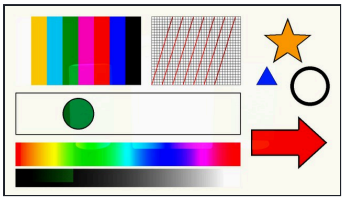
03 Multiply

Multiplies the two (darker; B's black stays black)



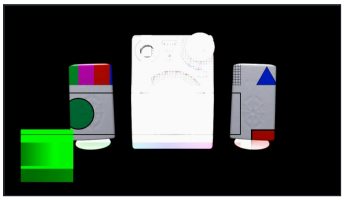
04 Overlay

B's light parts lighten A and its dark parts darken A. A's white and black stay as they are, and B shows only in A's mid-tones (in the sample, A's background is white, so it looks almost like A alone)



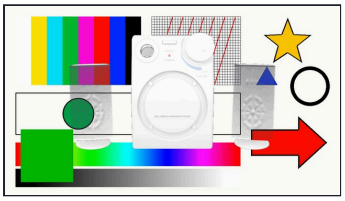
05 Soft light

B gently lightens or darkens A. As with Overlay, A's white and black stay as they are



06 Hard light

B's light parts lighten A and its dark parts darken A (stronger)



07 Luma key

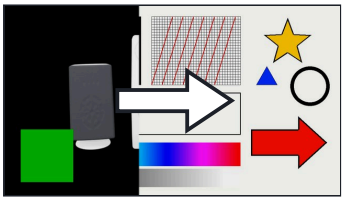
B's dark parts are see-through (`Key threshold`, `Key softness`)



08 Chroma key

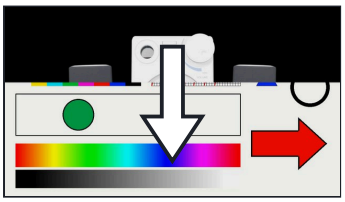
B's green is see-through (`Key threshold`)

Transitions



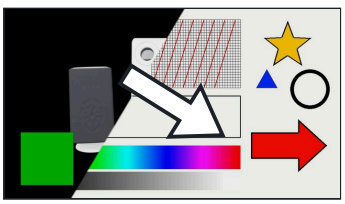
09 Wipe right

An edge crosses from left to right



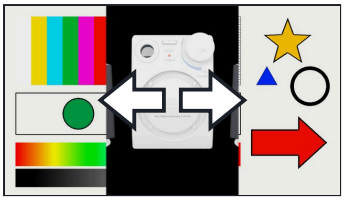
10 Wipe down

An edge crosses from top to bottom



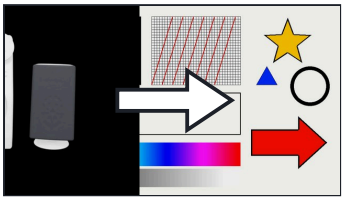
11 Wipe diagonal

An edge crosses diagonally



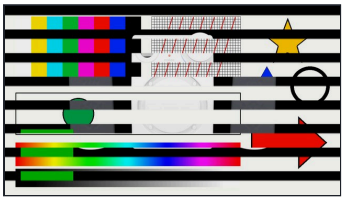
12 Split

Opens from the centre



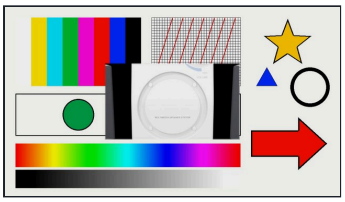
13 Slide

B slides in from the left



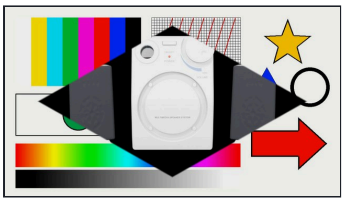
14 Blinds

8 bands open



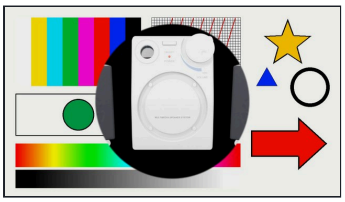
15 Iris box

A square grows



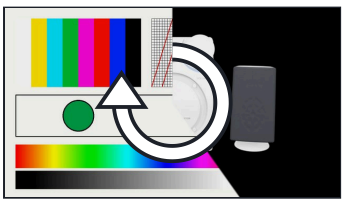
16 Iris diamond

A diamond grows



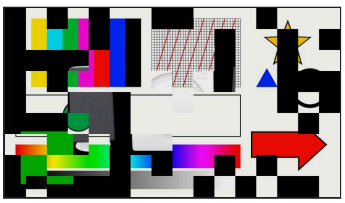
17 Iris circle

A circle grows



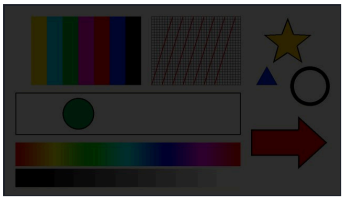
18 Clock

Clockwise from 12 o'clock



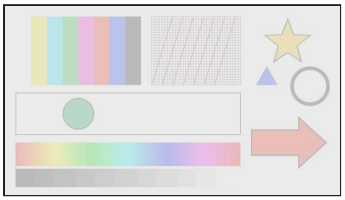
19 Blocks

Blocks swap in at random (not softened)



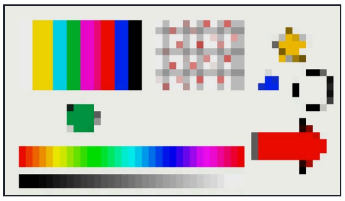
20 Dip to black

Through black



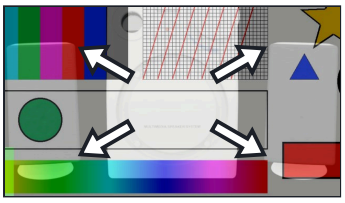
21 Dip to white

Through white

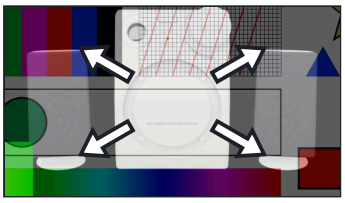


22 Pixelate

Pixelates in the middle



40%

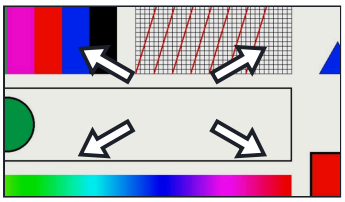


60%

23 Zoom

Zooms in

Camera moves



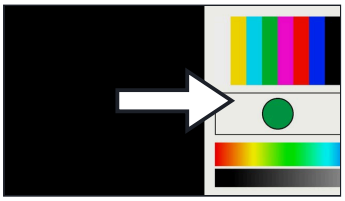
40%



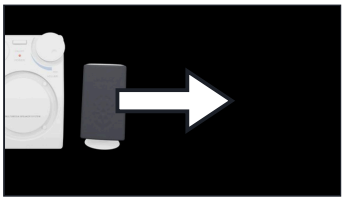
60%

24 Punch zoom

A pulls back a little, then zooms in fast and cuts to B at the closest point; B overshoots normal size slightly and settles



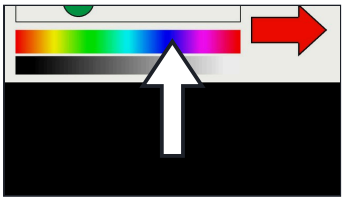
40%



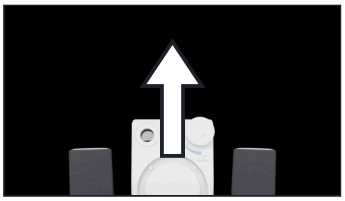
60%

25 Whip right

A flies off the screen to the right and B flies in the same direction



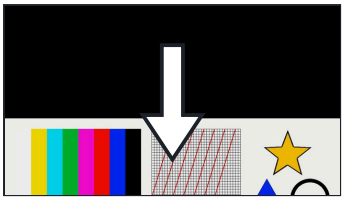
40%



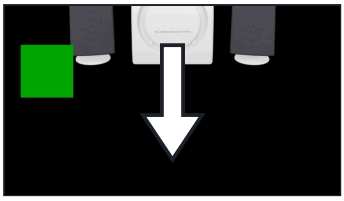
60%

26 Whip up

`Whip right` upward



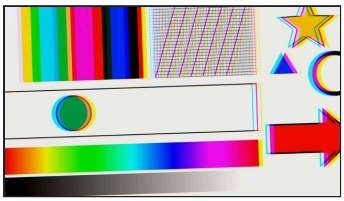
40%



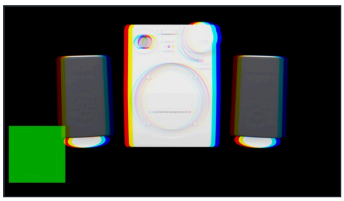
60%

27 Whip down

`Whip right` downward



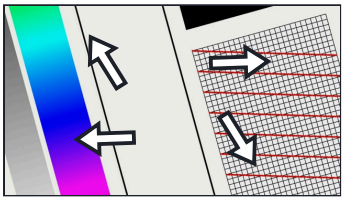
40%



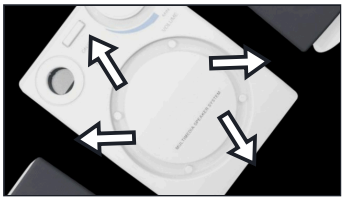
60%

28 Shake cut

The picture shakes more and more toward the middle and cuts there



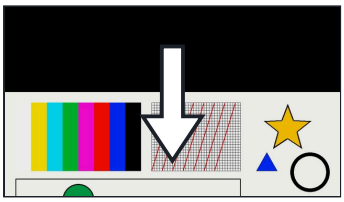
40%



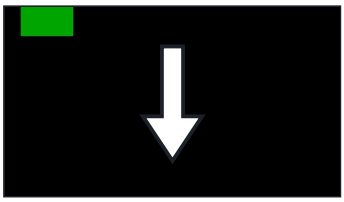
60%

29 Spin zoom

Spins while zooming in, enlarged so the picture's edges stay off the screen



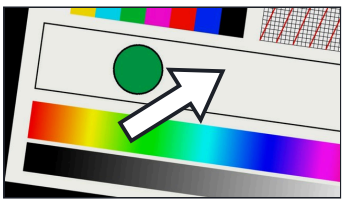
40%



60%

30 Drop bounce

A falls out, B drops in from the top and bounces



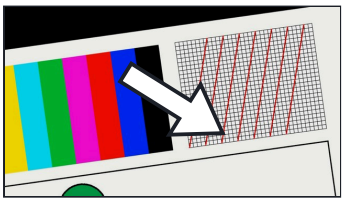
40%



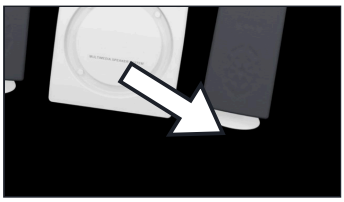
60%

31 Tilt right

A tilts and leaves on a slant, B comes in tilted the other way, from lower left to upper right; enlarged so the top and bottom edges stay off the screen



40%

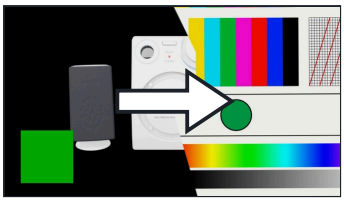


60%

32 Tilt down

`Tilt right` moving from upper left to lower right

Other transitions



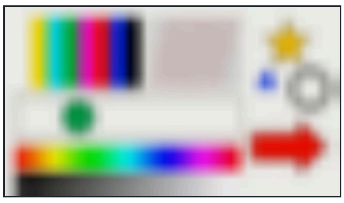
40%



60%

33 Brush wipe

A's picture becomes a card with an irregular, sharp-cornered cut edge and is pulled out to the right like kamishibai (paper theater), revealing B from underneath. It moves in 12 stop-motion steps, tilting slightly at each step. When returning to A, B's card is pulled out to the left



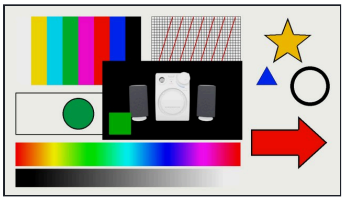
40%



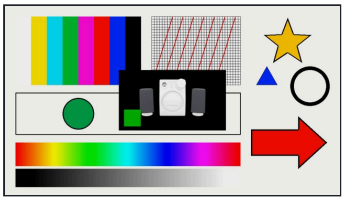
60%

34 Defocus

A goes out of focus, becomes B at the blurriest point, and B comes into focus



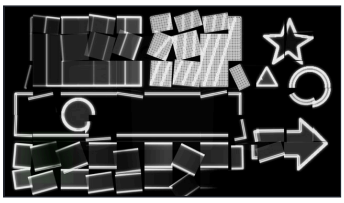
40%



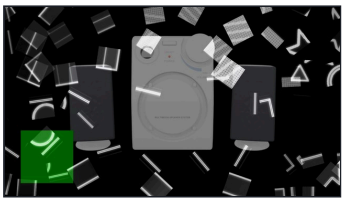
60%

35 Picture in picture

A shrinks into a small window in the centre and keeps shrinking until it disappears; B is behind it from the start



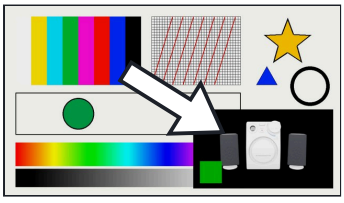
40%



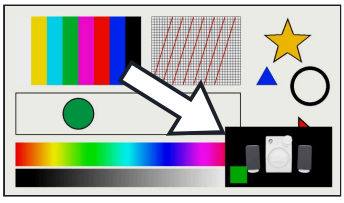
60%

36 Shatter

A turns into a line drawing whose lines break into pieces and fly apart, revealing B



40%



60%

37 PiP to corner

The `Picture in picture` window shrinks into the lower right corner and slides out past it

During transitions, the crossfader position directly controls the progress (end A = A, end B = B). Once the crossfader has gone all the way to the B side, moving back toward A starts a new transition from B to A. `Wipe right`, `Wipe diagonal`, `Slide`, `Clock`, `Whip right`, `Tilt right`, `Tilt down` and `PiP to corner` then run in reverse or mirrored horizontally (with `Wipe right`, A wipes in over B from the right, so the edge moves the same way as the crossfader; `Clock` runs counter-clockwise). The camera moves get a motion blur that follows their speed (along the arc while turning). Shape edges are smoothed based on each pixel's distance to the edge and softened by `Key softness` (default 0.1 is about 5 pixels, 1 is about 40 pixels).

12. Video effects (setup `FX` tab)

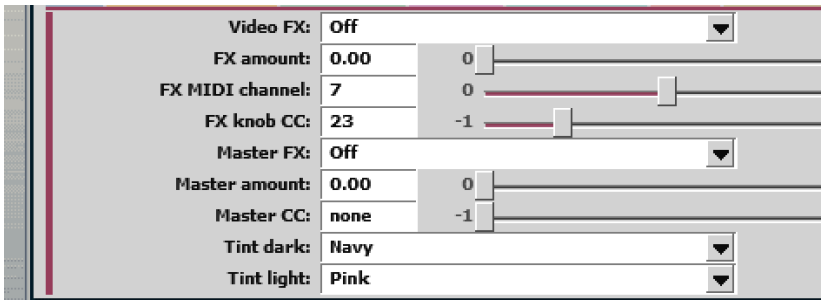


Figure 17 The FX tab of the setup screen

Effects are drawn by the Video processor that setup puts on the `video` track when it creates deck B. So they have no effect in projects without deck B. Neither the effect type nor the amount is shown on the front screen.

Parameter	Default	What it does
Video FX (this deck's picture)	Off	The effect on this deck's picture (15 kinds). How each looks, and the sound it pairs with: the table in "Sound and video effects" (p. 31). Line art and Halftone change colour when combined with `Tint` (examples below the table). These two and Zoom blur are applied before Tint and the others
Video FX amount	0	The amount (0 = off). What it means for each effect is below the table
Video FX MIDI channel (0=none)	7	MIDI channel of the knob that sets the amount
Video FX CC (centre = off; FLX4 CFX knob; -1=none)	23	CC number of the knob that sets the amount. Off in the middle, stronger either way (DDJ-FLX4 `CFX` knob: 23 for deck 1, 24 for deck 2; not yet tested on the real hardware)
Master video FX (after the mix; deck A's is used)	Off	An effect on the picture after the two are layered (same types as above). Deck A's setting is used
Master video FX amount	0	Its amount
Tint dark colour / Tint light colour	Navy / Pink	The Tint's colours: the darks of the picture become the dark colour, the lights the light colour, with a gradient between. 15 colours (below the table). The Tint after the mix uses deck A's colours
Master video FX CC (on the Video FX channel; -1=none)	-1	CC number that sets the master video FX amount (on the `Video FX MIDI channel`; 0–127 is the amount). -1 = not used (an earlier version's 0 = not used becomes -1 when opened). The Reloop Buddy's FX knob chooses the type, so this is -1
Video FX on/off note (Video FX channel; -1 = none)	-1	Note number of a button that turns this deck's video effect on/off (Reloop Buddy FX ON: note 0 on ch 9 for deck 1, ch 10 for deck 2). Each press toggles the amount between 0 and the previous amount (1 the first time; the Buddy's bar is on while engaged). On the setup screen's `More` tab (`Video FX` group)

Video FX
: Colour examples for Line art and Halftone: Tint's dark colour Purple and light colour White give white lines on purple paper; Magenta and White give magenta dots on white.

What

Video FX amount
means for each effect:

- Tint, Invert, Threshold: the blend with the original picture
- Hue rotate: the angle (1 = a full turn)
- Mirror: below 0.5 left-right, from 0.5 left-right and up-down
- Kaleidoscope: the number of mirrors
- LoRez: the coarseness
- Shift RGB: the offset
- Strobe: the flash rate (1 to 1/16 beat; without a BPM, 2–16 times a second; light and dark the same number of frames each, every other frame when faster than the frame rate)
- Trails: the length of the trail
- VHS: how worn it looks
- Disc: nothing on the picture (it is fully on as soon as it is on), only the sound changes
- Line art: line thickness (takes full effect when enabled, not a blend with the original: lower values draw thin lines with fine detail; 1 draws thick lines with only rough outlines)
- Halftone: how fine the dots are (it is fully on as soon as it is on: coarse when low, fine at 1)
- Zoom blur: how far it streams

Tint dark colour / Tint light colour: the 15 colours are Black, White, Lamp colour (the yellow / blue of `Lamp colour`), Red, Orange, Yellow, Lime, Green, Cyan, Sky, Blue, Navy, Purple, Magenta and Pink.

13. Controller numbers and other settings (the setup screen's `More` tab and REAPER's parameter list)

The numbers of the controller's buttons and knobs (most are filled in by presets) are on the setup screen's `More` tab, in 8 groups (a 9th group, `Metronome`, holds the metronome's settings). To use a controller without a preset, enter the numbers there. Pick a group on the left; scroll a long group with the wheel or the bar on the right (the wheel over a value changes the value, as on the other tabs). Hover over an item to see an explanation (such as which channel it reads) in the bottom bar. `none` (−1) = not used. The names are the same as when they were in REAPER's parameter list. For how they work, see the chapter "Pad modes, loops, filter and effects" (p. 28).

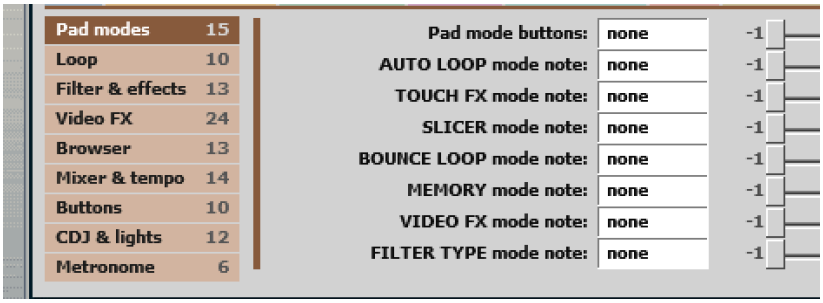


Figure 18 The setup screen's More tab (the groups on the left, the group's items on the right)

Group	Items
Pad modes	`Pad mode buttons`, `AUTO LOOP mode note` – `VIDEO FX mode note`, `FILTER TYPE mode note` (the numbers that choose the mode), `Pad modes by notes` (the list for controllers that send a different number for each mode), `Pad layout`, `Hot cue pad order`, `Hot cues 5–8 row`, `Sample 5–8 notes`, `Sample pads scroll back / on note` (the APC40 grid layout, shifting the sample row)
Loop	`Loop knob MIDI channel` – `Reloop note` (the loop knob: turn, press, move; reloop), `Loop half note` / `Loop double note`, `Auto loop notes`, `RELOOP/EXIT note` (a single button that exits and goes back), `Relative knobs' format` (two's complement / 64-centre)
Filter & effects	`Filter morph knob CC` – `Filter cutoff knob CC`, `Filter knobs' channel`, `Filter cutoff knob channel`, `SHIFT for the filter knobs` (whether SHIFT gives them another job), `Echo / Bitcrusher / Vibrato fader`, `Fader effects on / off channel`, `Freeze note`, `Effects off while held note` and `Effects off while held (deck)` (buttons that turn all effects off for a moment)
Video FX	`Video FX on/off note`, `Video FX knob type`, `Video FX amount - / + button`, `Master video FX amount - / + button`, `Master video FX type knob CC`, `Both decks note` (while held, `PLAY`, `CUE` and `SYNC` act on both decks; the APC40's `MASTER`), `Video blend knob CC` and `Video blend note`, `FX knobs' first CC`, `DEVICE knob 1 type` to `DEVICE knob 8 type` (the 8 DEVICE knobs' types), `VHS colour`, `VHS on-screen text`, `Disc tone`, `Sound of the video FX fader`, `FX ON bar's combos` (VHS colour and three others: below the table)
Browser	`Browser section MIDI channel` – `LOAD replace (any ch)` (the BROWSE knob and the LOAD buttons; see "Loading with the LOAD button (from the Media Explorer)" (p. 23)), `PREVIEW button` and `Mode button + LOAD` (the preview's buttons; see "Previewing a Media Explorer file" (p. 25)), `Key knob CC` (below the table), `Slot back note` (previous / next sample)
Mixer & tempo	`Mixer MIDI channel`, `Deck select note`, `Crossfader assign note`, `Mute note`, `Picture mute note` (mute only the picture), `Cue mix encoder CC`, `Tempo fader resolution` / `direction` / `centre zone` (the width in the middle that gives 0%), `Tempo range note`, `Tempo encoder CC`, `Nudge - / + note`, `Tap tempo note`
Buttons	`Cue button note` (headphone CUE), `Stop note`, `Back to CUE note`, `SHIFT note`, `Search jog CC` (SHIFT + jog = big jumps), `Knob page notes`, `REAPER metronome / record note`, `Extra buttons MIDI channel / note`
CDJ & lights	`Memory cue buttons' order`, `SLIP note`, `QUANTIZE note`, `REVERSE note`, `Held reverse note`, `Next sample note`, `Brake knob CC`, `Start knob CC`, `Needle search CC` (buttons on CDJs and others), `Pad and button lights` (see "Button lights" (p. 194)), `Light on value` (the value that turns on a one-colour light), `Crossfader end value` (the value taken as the crossfader's end; 120 on the APC40 mkII)

Group	Items
Metronome	`Metronome` (on/off), `Metronome follows` (which deck), `Metronome out` (where it sounds), `Stopped / in hand` (while a deck is stopped or in your hand), `Metronome volume`, `METRONOME button` (a MIDI button for on/off). See "Listening to a metronome while you play" (p. 203)

Notes (items in the `Video FX` and `Browser` groups):

- `VHS colour`: the VHS effect's tint. `Pink fringes` = red and blue fringes alike, pinkish; `Blue-white` (the default) = weaker red fringes and reds, with a cold blue-white cast over the whole picture. Per deck, saved with the project
- `VHS on-screen text`: while VHS is on, "PLAY ▶" at the top left and "SP" at the top right of the picture. Off by default. Per deck, saved with the project
- `Disc tone`: the Disc sound's tone at maximum amount. 0 (`dark`, the default) = fuller lows, duller highs; 1 (`bright`) = leaner lows, the upper mids forward. Low amounts always start bright. Per deck, saved with the project
- `FX ON bar's combos` (Reloop Buddy): while the FX ON bar is on, the FX knob sets that deck's video effect amount, `FX PARAM` ◀ ▶ picks that deck's video effect type, and `SHIFT` + the FX knob sets the amount too. Also when on: with nothing held ◀ ▶ and the FX knob pick the transition and soften it, `SHIFT` + the FX knob and ◀ ▶ set the amount and type after the mix, the mode button + ◀ ▶ and the FX knob pick the blend and set the key threshold, BROWSE sets the headphone Cue mix (with the bar or a pad, its sound), and `SHIFT` + BROWSE sets the sound after the mix (with the bar, that deck's overall sound multiplier)
- `Key knob CC`: the knob for each effect's sound amount and the key, on the browser channel. With `FX ON bar's combos` off, turned while pressed = the key threshold, with `SHIFT` too = the softness

Notes (what the values the presets fill in mean):

- `Tempo fader resolution`: `14-bit` = CC+32 is the low byte (DDJ-FLX4); `10-bit` = CC+54 is the low byte (Reloop Buddy)
- `Tempo range note`: the note that switches the tempo range (−1 = SYNC note + 8; Reloop Buddy SHIFT+SYNC: 3; DDJ-FLX2: 0 = not used)
- `Hot cue pad order`: `CDJ` = `Pad 1 note` +4 to +7 are hot cues 1–4 and +0 to +3 are 5–8, matching HOT CUE A–D and E–H on the CDJ-2000NXS2
- `Tempo fader direction`: `Reversed` = 127 is the + side
- `Mixer MIDI channel`: the MIDI channel of the volume faders, headphone Cue buttons and deck select buttons
- `Deck select note`: the deck select button (when present, the shared buttons such as PLAY act only on the selected deck)
- `Video FX knob type`: the shape of the video effect knob (0 in the middle / 0 at the left end)
- `Extra buttons`: the 4 buttons for spinback, video effect on/off, SLIP and Q
- `Pad layout`: `Across channels` = one row is the same note on channels 1–8, as on the APC40 mkI
- `Pad and button lights`: the controller to light the buttons for (`APC40 mkII` / `APC40 mkI` and others)

The values on the `More` tab are not sliders, so automation cannot move them. They are saved with the project (like the controller settings on the other tabs).

The following are not on the setup screen; view or change them in REAPER's FX parameter list (automation can move them too).

Parameter	What it does
CUE LOOP	CUE LOOP on/off (same as the `L` key). Default Off
Filter morph / Filter / Filter type / Filter resonance	The filter values (Morph, Cutoff, one of the 8 types, Resonance). Changed by the knobs and the FILTER TYPE pads. Cutoff has a dead zone of ±0.05 around the centre, with no filter
Echo / Bitcrusher / Vibrato	Amounts of echo, bitcrusher and vibrato (APC40 track faders 2–4)
Echo length / Echo feedback / Vibrato step / Gate / Strobe step	Echo length (1/8 to 2 beats) and feedback, the vibrato step (1 to 1/8 beat), and the steps of the gate and Strobe
Key	Pitch (±12 semitones; the tempo does not change)
Sound of the video FX	How much the video effects affect the sound (0–1, default 1; 0 = picture only)
Video opacity / Mute	This deck's picture strength (the sound stays) / mute
FX knob 1 – FX knob 8, FX knob: Strobe	Amounts of the video effects set by the APC40's `DEVICE CONTROL` knobs (each knob's type is chosen in setup's `More` tab → `Video FX`; Strobe is `SHIFT` + knob 8). Below 0.05 is a dead zone with no effect
Envelope scratch, Envelope fader, Envelope vfx …, Envelope filter …, Envelope pad FX, Envelope master …	The envelope lanes (patterns, recorded performances). No need to touch them ("Scratching from envelopes (patterns)" (p. 207)). A recorded performance is written to them when you stop recording
New file in slot – Slot command B	Used by scripts (such as LOAD) to hand samples to the JSFX (no need to touch them)

Display only (values to look at; they appear in REAPER's parameter list, and the small windows and lamps at the top right of the screen show the same information)

Parameter	What it shows
Head position (s)	The current read position (seconds). The video sync script reads this to move the edit cursor
Head speed (x)	The current head speed (1 = normal, negative = backwards)
Touched	1 when counted as touching
MIDI messages received	How many MIDI messages arrived on this channel (on screen, the `midi` lamp lights with each one). If it does not go up when you use the controller, MIDI is not arriving

Parameter	What it shows
Blocks processed	Number of audio blocks processed (for checking that it runs)
Output peak (linear, pre-limit)	The highest output since loading (1.0 = 0 dBFS; anything above is clipped)
File sample rate / File length (frames)	Sample rate and length of the loaded WAV
Sample memory (values)	How much audio can be held (number of values across all channels). Depends on the REAPER version
Sample cut (s of file, 0=whole)	When a track was too long to load completely, the length of the whole file (seconds). 0 if it all loaded
Crossfader gain	The volume set by the crossfader (0–1). While it is below 0.5, the video sync script mutes the `video` track to hide the picture
Crossfader position	The received crossfader position (0 = left end, 1 = right end)
GUI scale	Screen scale (1x, 2x, …)
CPU worst block / CPU average	How heavy this JSFX is. Worst = the share of its time slot (the buffer length) used by the heaviest single block in the last second; average = the average share of real time. Read by the load test script
Sample load time	CPU time used by the last sample load (milliseconds, the total of all the pieces)
Load test	The switch used by the load test script (while on, the deck scratches by itself). Always off when a project is opened
Dropout threshold / Dropouts	Dropouts counted by ReaScratch itself. One is counted whenever audio processing falls behind real time by the threshold (ms; 0 = one block's length + 3 ms) or more. The load test script sets the threshold and reads the count

Controller setup and MIDI mapping

Preset list

Table 19 What each preset can do (Numbers: left = deck 1, right = deck 2. Setup notes are below the table. Presets without a green line and ✓ are based on official MIDI lists or Mixxx mappings and have not yet been tested on actual hardware)

● Yes ○ No (use the keyboard or the screen) ▲ [1]Partial (numbers refer to notes below the table)

✓ Tested on real hardware

Controller		Deck						Mixer				Pads			Video
		Jog	PLAY/CUE		Tempo			Crossfader		Phones		Hot cues		Modes	
No.	Preset		Touch	SYNC		Loop			Volume		Filter		Samples		Effects
Akai															
1·2	**APC40 mkI**	●	○	●	●	○	●	●	●	●	●	●	●	●	●
3·4	**APC40 mkII** ✓	●	○	●	●	●	●	●	●	●	●	●	●	●	●
Behringer															
5·6	**CMD Micro**	●	●	●	●	●	●	●	●	●	○	○	○	○	○
Denon DJ															
7·8	**SC5000 / SC6000**	●	▲ [1]	●	●	●	●	○	○	○	○	●	○	●	○
9·10	**Prime 4**	●	●	●	●	●	●	●	●	●	●	●	○	●	●
Gemini															
11·12	**FirstMix**	●	●	●	●	○	○	●	○	●	○	○	○	○	○
Hercules															
13·14	**DJ Control MP3 e2**	●	▲ [1]	●	●	○	●	●	●	●	○	●	○	○	○
15·16	**DJControl Compact**	●	▲ [1]	●	●	○	●	●	○	○	○	●	●	○	●
17·18	**DJControl Starlight**	●	●	●	●	○	○	●	●	●	●	●	●	●	○
19·20	**DJControl MIX**	●	●	●	●	○	○	●	●	●	●	●	●	●	○
21·22	**DJ Control Instinct**	●	●	●	●	○	●	●	●	●	○	●	○	○	●
23·24	**DJ Control AIR**	●	▲ [1]	●	●	●	●	●	●	●	○	○	●	○	●
25·26	**DJ Console Mk4**	●	▲ [1]	●	●	○	●	●	●	●	○	●	○	○	●
27·28	**DJ Console 4-Mx**	●	●	●	●	●	●	●	●	●	○	●	●	○	○
29·30	**DJ 4Set**	●	●	●	●	●	○	●	●	●	○	●	○	○	○
31·32	**DJControl Jogvision**	●	●	●	●	●	●	●	●	●	●	●	●	●	●
33·34	**Inpulse 200 / 300 (MK2)**	●	●	●	●	●	●	●	●	●	●	●	●	●	●
35·36	**Inpulse 500**	●	●	●	●	●	●	●	●	●	●	●	●	●	○
ION															

Controller		Deck						Mixer				Pads			Video
No.	Preset	Jog	PLAY/CUE		SYNC	Tempo	Loop	Crossfader	Volume	Phones	Filter	Hot cues	Samples	Modes	Effects
			Touch												
37-38	**Discover DJ**	●	●	●	●	○	○	●	○	○	○	○	○	○	○
39-40	**Discover DJ Pro**	●	▲ 1	●	●	●	●	●	●	●	○	○	○	○	○
Korg															
41-42	**KAOSS DJ**	●	●	●	●	●	●	●	●	●	○	●	○	○	●
Native Instruments															
43-44	**HID bridge (S2 / S3 / S4)**	●	●	●	●	●	○	●	●	●	○	●	●	○	○
Numark															
45-46	**Party Mix**	●	▲ 1	●	●	●	○	●	●	●	○	●	●	●	○
47-48	**Party Mix II / Live**	●	●	●	●	●	○	●	●	●	●	●	●	●	○
49-50	**DJ2GO2**	●	▲ 1	●	●	●	○	●	○	●	○	●	●	●	○
51-52	**DJ2GO2 Touch**	●	●	●	●	●	○	●	○	●	○	●	●	●	○
53-54	**Mixtrack**	●	●	●	●	●	●	●	●	●	○	○	○	○	●
55-56	**Mixtrack Pro**	●	●	●	●	●	●	●	●	●	○	●	○	○	●
57-58	**Mixtrack 2**	●	●	●	●	●	●	●	●	●	○	●	○	○	●
59-60	**Mixtrack (Pro) 3**	●	●	●	●	●	●	●	●	○	●	●	●	●	●
61-62	**Mixtrack Platinum**	●	●	●	●	●	●	●	●	●	●	●	○	●	●
63-64	**Mixtrack Pro / Platinum FX**	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Pioneer DJ / AlphaTheta															
65-66	**DDJ-FLX2**	●	●	●	●	●	○	●	●	●	●	●	●	●	○
67-68	**DDJ-FLX4**	●	●	●	●	●	●	●	●	●	●	●	●	●	●
69-70	**DDJ-FLX6**	●	●	●	●	●	●	●	●	●	●	●	●	●	●
71-72	**DDJ-FLX10**	●	●	●	●	●	●	●	●	●	●	●	●	●	●
73-74	**DDJ-REV1**	●	●	●	●	●	●	●	●	●	●	●	●	●	●
75-76	**DDJ-REV5**	●	●	●	●	●	●	●	●	●	●	●	○	●	●
77-78	**DDJ-200**	●	●	●	●	●	○	●	●	●	●	●	○	○	●
79-80	**DDJ-RB**	●	●	●	●	●	○	●	●	●	●	●	●	○	●
81-82	**DDJ-400**	●	●	●	●	●	●	●	●	●	●	●	●	●	●
83-84	**DDJ-800**	●	●	●	●	●	●	●	●	●	●	●	●	●	●
85-86	**DDJ-1000**	●	●	●	●	●	●	●	●	●	●	●	●	●	●
87-88	**DDJ-SB**	●	●	●	●	●	○	●	●	●	●	●	●	○	●
89-90	**DDJ-SB2**	●	●	●	●	●	○	●	●	●	●	●	●	○	●
91-92	**DDJ-SB3 / SX**	●	●	●	●	●	●	●	●	●	●	●	●	●	●

Controller		Deck						Mixer				Pads			Video
No.	Preset	Jog	PLAY/CUE		Tempo			Crossfader		Phones		Hot cues		Modes	
			Touch	SYNC		Loop			Volume		Filter		Samples		Effects
93·94	**DDJ-SX3**	●	●	●	●	●	●	●	●	●	●	●	●	●	●
95·96	**CDJ-2000NXS2**	●	●	●	●	●	●	▲ 2	○	○	○	●	○	○	○
97·98	**CDJ-3000 / 3000X**	●	●	●	●	●	●	▲ 2	○	○	○	●	○	○	○
99·100	**XDJ-RX3 / XZ**	●	●	●	●	●	●	●	○	○	●	●	○	●	○
Reloop															
101·102	**Reloop Buddy** ✓	●	●	●	●	●	●	●	●	●	●	●	○	●	●
103·104	**Reloop Ready**	●	●	●	●	●	●	●	●	●	●	●	○	●	●
105·106	**Reloop Mixage**	●	●	●	●	▲ 3	●	●	●	●	●	●	○	○	●
107·108	**Reloop Mixon 4**	●	●	●	●	●	●	●	●	●	●	●	●	●	●
109·110	**Reloop Mixon 8 Pro**	●	●	●	●	●	●	●	●	●	●	●	●	●	○
111·112	**Reloop Touch**	●	●	●	●	●	●	●	●	●	●	●	○	●	●
113·114	**Reloop Beatpad**	●	●	●	●	●	●	●	●	●	●	●	●	●	●
115·116	**Reloop Beatpad 2**	●	●	●	●	●	○	●	●	●	●	●	●	●	○
117·118	**Reloop Beatmix 2/4**	●	●	●	●	●	●	●	●	●	○	●	●	●	●
119·120	**Reloop Terminal Mix 2/4**	●	●	●	●	●	●	●	●	○	●	●	●	○	●
121·122	**Reloop Terminal Mix 8**	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Other															
0	**Custom**	Controllers not in the list. Set the numbers yourself ("Manual mapping (Custom)" (p. 192))													
123	**FLX4 Emu test**	For development; not for normal use													

▲[1] Controllers without a touch signal (SC5000M, Party Mix (original), DJ2GO2, Compact, MP3 e2, AIR, DJ Console Mk4) and with touch sent as a CC (Discover DJ Pro): "touching" means while turning

▲[2] CDJ: the unit has no crossfader, so only the deck's side is set (use the keyboard or another controller for the fader)

▲[3] Mixage: tempo on deck 1 only (deck 2's pitch bend arrives on MIDI channel 2)

Setup notes by controller

Akai

APC40 mkI: two decks. Jog with TRACK KNOB 1 / 5. Buttons light (3 colours)
APC40 mkII: two decks. Jog with TRACK KNOB 1 / 5. Buttons light in hot cue colours

Behringer

CMD Micro: Both decks ch 1

Denon DJ

SC5000 / SC6000: Computer Mode. Two units: split by MIDI input
Prime 4: Put the unit in Computer Mode

Gemini

FirstMix: Both decks ch 1

Hercules

DJ Control MP3 e2: MP3 LE, Glow. Needs the Hercules driver
DJControl Compact: Both decks ch 1. 4 pads
DJControl Starlight: 4 pads

DJControl MIX: Connect over USB. 4 pads
DJ Control Instinct: Both decks ch 1. 2 hot cues
DJ Control AIR: SCRATCH button on. Both decks ch 1
DJ Console Mk4: Both decks ch 1. 4 hot cues
DJ Console 4-Mx: Needs the driver. SCRATCH button on
DJ 4Set: Needs the driver. 3 hot cues
Inpulse 200 / 300 (MK2): VINYL mode (200 MK2: SHIFT+LOOP)
Inpulse 500: VINYL mode

ION

Discover DJ: Both decks ch 1
Discover DJ Pro: Both decks ch 1

Korg

KAOSS DJ: 3 hot cues

Native Instruments

HID bridge (S2 / S3 / S4): Traktor Kontrol S4 MK3 / MK2, S3, S2 MK3 / MK2 / MK1. Keep the zip's HID bridge running

Numark

Party Mix: original. 4 pads (SHIFT+pad does not clear)
Party Mix II / Live: 4 pads
DJ2GO2: no touch. 4 pads. Ticks per turn provisional
DJ2GO2 Touch: 4 pads
Mixtrack: original. Both decks ch 1
Mixtrack Pro: original. Both decks ch 1. 3 pads
Mixtrack 2: Both decks ch 1. 3 pads. Volume numbers doubtful
Mixtrack (Pro) 3: Headphone CUE arrives on ch 1 and cannot be read
Mixtrack Platinum: without FX. 4 pads

Pioneer DJ / AlphaTheta

DDJ-FLX2: Tempo range with the Shift+Y keys
DDJ-FLX4: Turn the jog's VINYL button on
DDJ-REV1: Tempo range with the Shift+Y keys
DDJ-RB: Tempo range with the Shift+Y keys
DDJ-SB: Tempo range with SHIFT+KEYLOCK
DDJ-SB2: Tempo range with SHIFT+KEYLOCK. DECK button back to decks 1/2
DDJ-SB3 / SX: Tempo range with SHIFT+KEYLOCK
DDJ-SX3: Tempo range with SHIFT+KEYLOCK
CDJ-2000NXS2: one unit = one deck. Put the unit in USB-MIDI mode
CDJ-3000 / 3000X: one unit = one deck. Put the unit in SOFTWARE CONTROL
XDJ-RX3 / XZ: Put the unit in PC Control mode

Reloop

Reloop Buddy: Windows: Reloop ASIO Driver
Reloop Ready: Windows probably needs Reloop's driver
Reloop Mixage: Both decks ch 1. Hot cues with SHIFT+button
Reloop Beatpad: Jog in Vinyl mode. 4 pads
Reloop Beatpad 2: Assumed same numbers as the Beatpad (no official list)
Reloop Terminal Mix 8: Turn the VINYL button on

Choose from `Preset` in the settings list opened with `setup` at the bottom right of the JSFX window. The menu goes manufacturer → model → `deck 1` / `deck 2` (`(ch 1)` / `(ch 2)` for CDJs; Pioneer DJ / AlphaTheta has an extra `DDJ` / `CDJ / XDJ` level in between). Within each manufacturer, models are listed together by series (DDJ-FLX, DDJ-REV and so on), cheaper models first. The current model is marked. Click the search box to the right of `Preset` and type part of a name (`flx4`, `ddj 400`, `3000`, `s4`, `reloop` and so on; case and `-` are ignored, and with several words separated by spaces, only entries containing all of them are listed. Adding a number, as in `beatpad 2`, also matches `deck 2`, so pick from the list of candidates) and the candidates appear below. Choose with ↑↓ and Enter, or click; Esc closes it. While you type, keys do not reach the deck. Choosing a preset fills in the

jog, touch and buttons, crossfader, hot cue and sample pads, tempo and SYNC, the video effect knob, the channel fader and the headphone CUE in one go. Presets marked ⚠ have not been tested on the real hardware. The numbers in the table below are the `Preset` values, in the order of the menu.

In “Notes by model”, "pad modes" are written as the controller's mode label → the ReaScratch mode (see “Pad modes, loops, filter and effects” (p. 28) above). Following the Reloop Buddy, the EQ and filter knobs are HI = filter Morph, LOW = Resonance, CFX / FILTER / COLOR = Cutoff, and MID = how much the video effects act on the sound (on controllers with SHIFT, holding SHIFT gives echo length, feedback and key). LOAD does the same job as on the Buddy. The BROWSE knob: turn = how much the video effects act on the sound, push = switch the deck the knob works on, SHIFT + turn = the headphone cue mix (the Reloop Buddy and Ready work as their notes say).

Most controllers with an FX section read it as the Reloop Buddy's FX controls ("Buddy-style" in “Notes by model”). The FX ON button = the Buddy's FX ON bar (each press turns it on or off; while it is lit, the bar counts as engaged), the FX section's knob (LEVEL/DEPTH, DRY/WET and so on) = the FX knob, and BEAT ◂ ▸ or an unused FX button = `FX PARAM` ◂ ▸. What they act on depends on what you hold, as on the Buddy (see the Reloop Buddy notes). On controllers without a separate SHIFT number, the unit's SHIFT is used, and a knob or ◂ ▸ that reaches only deck B is passed to deck A. On Buddy-style controllers, turning BROWSE = the headphone Cue mix (while FX ON is on, that effect's sound amount).

Preset name	Controller / mode	Preparation
1 · 2 **APC40 mkI deck 1 / deck 2** 3 · 4 **APC40 mkII deck 1 / deck 2**	Akai APC40 (original, mkI) and APC40 mkII. Two decks on one unit	—
5 · 6 **Behringer CMD Micro deck 1 / deck 2**	Behringer CMD Micro	—
7 · 8 **Denon SC5000/SC6000 deck 1 / deck 2**	Denon DJ SC5000 and SC6000 (M versions too)	Set the unit to `Computer Mode`. With two units, set each track's MIDI input to its unit
9 · 10 **Denon Prime 4 deck 1 / deck 2**	Denon DJ Prime 4 (Computer Mode)	Put the unit in Computer Mode
11 · 12 **Gemini FirstMix deck 1 / deck 2**	Gemini FirstMix	—
13 · 14 **Hercules DJ Control MP3 e2 deck 1 / deck 2**	Hercules DJ Control MP3 e2, MP3 LE and Glow	Needs the Hercules driver (MIDI channel 1)
15 · 16 **Hercules DJControl Compact deck 1 / deck 2**	Hercules DJControl Compact	—
17 · 18 **Hercules DJControl Starlight deck 1 / deck 2** 19 · 20 **Hercules DJControl MIX deck 1 / deck 2**	Hercules DJControl Starlight and MIX (connect the MIX over USB; Bluetooth is for phones)	—
21 · 22 **Hercules DJ Control Instinct deck 1 / deck 2**	Hercules DJ Control Instinct	—

Preset name	Controller / mode	Preparation
23 · 24 **Hercules DJ Control AIR deck 1 / deck 2**	Hercules DJ Control AIR	—
25 · 26 **Hercules DJ Console Mk4 deck 1 / deck 2**	Hercules DJ Console Mk4	Not known whether the Hercules driver is needed
27 · 28 **Hercules DJ Console 4-Mx deck 1 / deck 2**	Hercules DJ Console 4-Mx	Needs the Hercules driver (deck mode: 2 decks)
29 · 30 **Hercules DJ 4Set deck 1 / deck 2**	Hercules DJ 4Set	Needs the Hercules driver (`Deck mode` : `2-deck basic`)
31 · 32 **Hercules DJControl Jogvision deck 1 / deck 2**	Hercules DJControl Jogvision	—
33 · 34 **Hercules Inpulse 200/300 (MK2) deck 1 / deck 2** 35 · 36 **Hercules Inpulse 500 deck 1 / deck 2**	Hercules DJControl Inpulse 200 and 300 (the MK2s use the same numbers) and 500. On the 200 MK2, VINYL is `SHIFT` + `LOOP`	—
37 · 38 **ION Discover DJ deck 1 / deck 2**	ION Discover DJ	—
39 · 40 **ION Discover DJ Pro deck 1 / deck 2**	ION Discover DJ Pro	—
41 · 42 **Korg KAOSS DJ deck 1 / deck 2**	Korg KAOSS DJ	Not known whether the message Korg's software sends at start-up is needed
43 · 44 **HID bridge (NI S2/S3/S4) deck 1 / deck 2**	Native Instruments Traktor Kontrol S4 MK3, S4 MK2, S3, S2 MK3, S2 MK2 and S2 MK1 (controllers connected over HID)	Run the bridge program (the `HID bridge` folder in the zip)
45 · 46 **Numark Party Mix deck 1 / deck 2**	Numark Party Mix (original)	—
47 · 48 **Numark Party Mix II/Live deck 1 / deck 2**	Numark Party Mix II and Party Mix Live	—
49 · 50 **Numark DJ2GO2 deck 1 / deck 2** 51 · 52 **Numark DJ2GO2 Touch deck 1 / deck 2**	Numark DJ2GO2 and DJ2GO2 Touch	—

Preset name	Controller / mode	Preparation
53 · 54 **Numark Mixtrack deck 1 / deck 2** 55 · 56 **Numark Mixtrack Pro deck 1 / deck 2** 57 · 58 **Numark Mixtrack 2 deck 1 / deck 2**	Numark Mixtrack (original), Mixtrack Pro (original) and Mixtrack 2	—
59 · 60 **Numark Mixtrack (Pro) 3 deck 1 / deck 2**	Numark Mixtrack 3 and Mixtrack Pro 3 (same numbers)	—
61 · 62 **Numark Mixtrack Platinum deck 1 / deck 2**	Numark Mixtrack Platinum (without FX)	—
63 · 64 **Numark Mixtrack Pro/Platinum FX deck 1 / deck 2**	Numark Mixtrack Pro FX and Mixtrack Platinum FX	—
65 · 66 **DDJ-FLX2 deck 1 / deck 2**	AlphaTheta DDJ-FLX2	—
67 · 68 **DDJ-FLX4 deck 1 / deck 2**	Pioneer DDJ-FLX4	Turn the jog's VINYL button ON
69 · 70 **DDJ-FLX6 deck 1 / deck 2** 71 · 72 **DDJ-FLX10 deck 1 / deck 2** 73 · 74 **DDJ-REV1 deck 1 / deck 2** 81 · 82 **DDJ-400 deck 1 / deck 2** 83 · 84 **DDJ-800 deck 1 / deck 2** 85 · 86 **DDJ-1000 deck 1 / deck 2**	Pioneer DJ / AlphaTheta DDJs (same numbering as the FLX4)	—
75 · 76 **DDJ-REV5 deck 1 / deck 2**	AlphaTheta DDJ-REV5	—
77 · 78 **DDJ-200 deck 1 / deck 2**	Pioneer DJ DDJ-200	—
79 · 80 **DDJ-RB deck 1 / deck 2** 87 · 88 **DDJ-SB deck 1 / deck 2** 89 · 90 **DDJ-SB2 deck 1 / deck 2**	Pioneer DJ DDJ-SB, DDJ-SB2 and DDJ-RB	—
91 · 92 **DDJ-SB3 / SX deck 1 / deck 2** 93 · 94 **DDJ-SX3 deck 1 / deck 2**	Pioneer DJ DDJ-SB3, DDJ-SX and DDJ-SX3 (Serato layout)	—
95 · 96 **CDJ-2000NXS2 (MIDI ch 1) / (MIDI ch 2)**	Pioneer DJ CDJ-2000NXS2 (one player = one deck; ch 1 for deck A, ch 2 for deck B)	USB-MIDI mode, MIDI channel (1st player 1, 2nd 2); Windows needs the driver
97 · 98 **CDJ-3000/3000X (MIDI ch 1) / (MIDI ch 2)**	Pioneer DJ CDJ-3000 and AlphaTheta CDJ-3000X (one player = one deck; ch 1 for deck A, ch 2 for deck B)	`SOFTWARE CONTROL`, MIDI channel (1st player 1, 2nd 2); Windows needs the driver
99 · 100 **XDJ-RX3 / XZ deck 1 / deck 2**	Pioneer DJ XDJ-RX3 and XDJ-XZ (both checked against the official MIDI lists)	Put the unit in MIDI mode (XDJ-XZ: PC Control mode)

Preset name	Controller / mode	Preparation
101 · 102 **Reloop Buddy deck 1 / deck 2**	Reloop Buddy (deck 1 = left deck, deck 2 = right deck)	Windows needs the driver ("Troubleshooting" (p. 219))
103 · 104 **Reloop Ready deck 1 / deck 2**	Reloop Ready	Windows probably needs Reloop's driver
105 · 106 **Reloop Mixage deck 1 / deck 2**	Reloop Mixage (Interface Edition, MK2 and Controller Edition; one mapping in Mixxx)	—
107 · 108 **Reloop Mixon 4 deck 1 / deck 2**	Reloop Mixon 4	—
109 · 110 **Reloop Mixon 8 Pro deck 1 / deck 2**	Reloop Mixon 8 Pro	—
111 · 112 **Reloop Touch deck 1 / deck 2**	Reloop Touch	—
113 · 114 **Reloop Beatpad deck 1 / deck 2** 115 · 116 **Reloop Beatpad 2 deck 1 / deck 2**	Reloop Beatpad and Beatpad 2	Set the jog mode to Vinyl
117 · 118 **Reloop Beatmix 2/4 deck 1 / deck 2**	Reloop Beatmix 2 and Beatmix 4 (MK2)	—
119 · 120 **Reloop Terminal Mix 2/4 deck 1 / deck 2** 121 · 122 **Reloop Terminal Mix 8 deck 1 / deck 2**	Reloop Terminal Mix 2, 4 and 8	—
0 **Custom**	Generic / manual setup	Set up by hand using the table below
123 **FLX4 Emu test (loopMIDI deck 2)**	Virtual MIDI for development and testing	Connect `tools/flx4_emu.py` (for development)

Notes by model

1 · 2 APC40 mkI deck 1 / deck 2

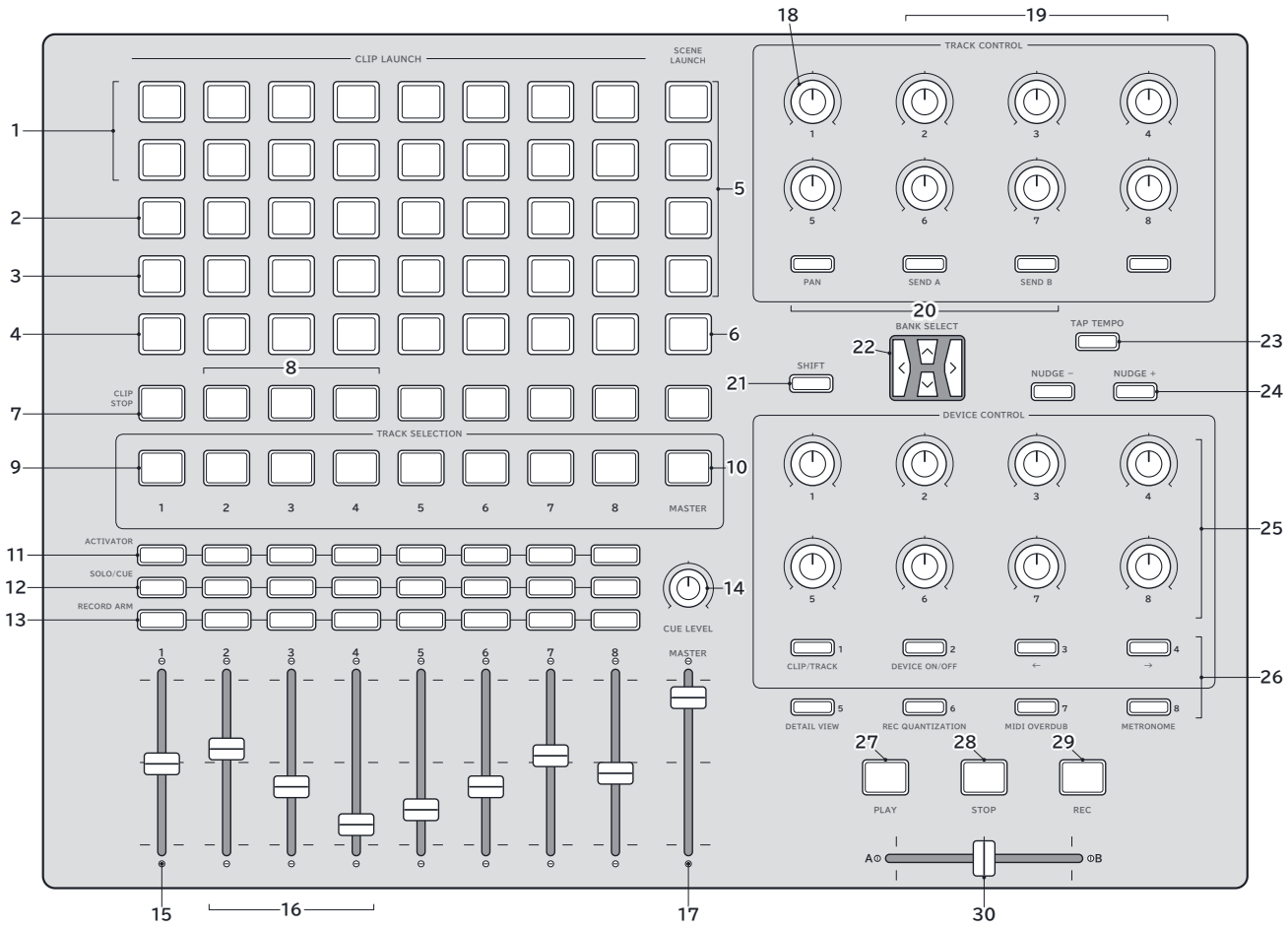


Figure 20 The APC40 mkI's controls (numbered on columns 1–4 = deck A; columns 5–8 = deck B are the same)

Columns 1–4 are deck A, columns 5–8 deck B. The mouse works too for the jog. The `TRACK CONTROL` knobs are in the top right, not above the grid, and `CUE LEVEL` is in the master column. The grid is the 5 rows of clips, the columns are MIDI channels 1–8.

Left (the grid and the rows of buttons)

1 **Grid rows 1 and 2**
Pads 1–8 (in `HOT CUE` mode, hot cues 1–8; SHIFT clears)

2 **Grid row 3**
Samples (22's ◂ ▸ shifts it to 5–8; with SHIFT, LOAD into that sample)

3 **Grid row 4**
Memory cues (MEMORY, CALL ◂, CALL ▸, DELETE)

4 **Grid row 5**
While held, fast spinback, reverse, half speed, fast forward (with SHIFT, LOOP, SYNC, Q and SLIP on/off)

5 **SCENE LAUNCH 1–4**
Auto loops of 1, 2, 4, 8 beats (with SHIFT, jump −4, −1, +1, +4 beats)

6 **SCENE LAUNCH 5**
Exit / return to the loop

7 **CLIP STOP 1 (5)**
Freeze (with SHIFT, that deck's video effect on/off)

8 **CLIP STOP 2–4 (6–8)**
Toggle the effect of the fader below (16)

9 **TRACK SELECTION 1 / 5**
Select deck A / B. 5, 6 and the right side (22–29) act on the selected deck

10 **MASTER**
While held, `PLAY`, `CUE` and `SYNC` act on both decks (`PLAY` plays / stops both together)

11 **ACTIVATOR**
Mute (with SHIFT, picture only)

12 **SOLO/CUE**
Headphone Cue

13 **RECORD ARM**
LOAD (with SHIFT, replace). `RECORD ARM` of column 2 (6) previews (the same as the `P` key; it lights while previewing), and `RECORD ARM` of column 3 (7) turns the metronome on/off (the same as the `N` key; it lights while on)

14 **CUE LEVEL**
Headphone Cue mix

15 **Track fader 1 (5)**
Amount of the video effects applied to the sound

16 **Track faders 2–4 (6–8)**
Echo, bitcrusher, vibrato

17 **Master fader**
Video effect amount after the mix

Right side

18 **TRACK CONTROL knob 1 (5 for deck B)**
Jog (stopping for 0.15 s counts as releasing it)

19 **TRACK CONTROL knobs 2–4 (6–8)**
Filter Morph, Resonance, Cutoff. The `SEND A` page is the mkII's `SENDS` page, the `SEND B` page its `USER` page

20 **PAN, SEND A, SEND B**
The pages of the knobs (19) (`PAN` goes back; while SHIFT is held, `SEND A`)

21 **SHIFT**
Hold with another control to use its second function

22 **BANK SELECT**
◂ ▸ = shift the samples of grid row 3 to 5–8. ▲ ▼ = double / halve the loop (with SHIFT, move it 1 beat)

23 **TAP TEMPO**
Tap tempo

24 **NUDGE − / +**
A little slower / faster while held (beat matching)

25 **DEVICE CONTROL knobs 1–8**
Amounts of the 8 video effects (knob 8 is VHS, SHIFT + knob 8 is Strobe)

26 **DEVICE CONTROL buttons 1–8**
Pad mode (see "Pad modes" (p. 28)). Their labels differ from the mkII's: CLIP/TRACK, DEVICE ON/OFF, ←, →, DETAIL VIEW, REC QUANTIZATION, MIDI OVERDUB, METRONOME. The video blend is chosen with the pads in mode 8

27 **PLAY**
Play / stop (with SHIFT, both decks at once)

28 **STOP**
Tempo range

29 **REC**
CUE

30 **Crossfader**
Left = deck A, right = deck B

Controls without a number (STOP ALL CLIPS, SEND C) are not used. The mkI has no `TEMPO` knob, `CROSSFADER A/B`, `BANK` or `SESSION` (which the mkII has).

3 · 4 APC40 mkII deck 1 / deck 2

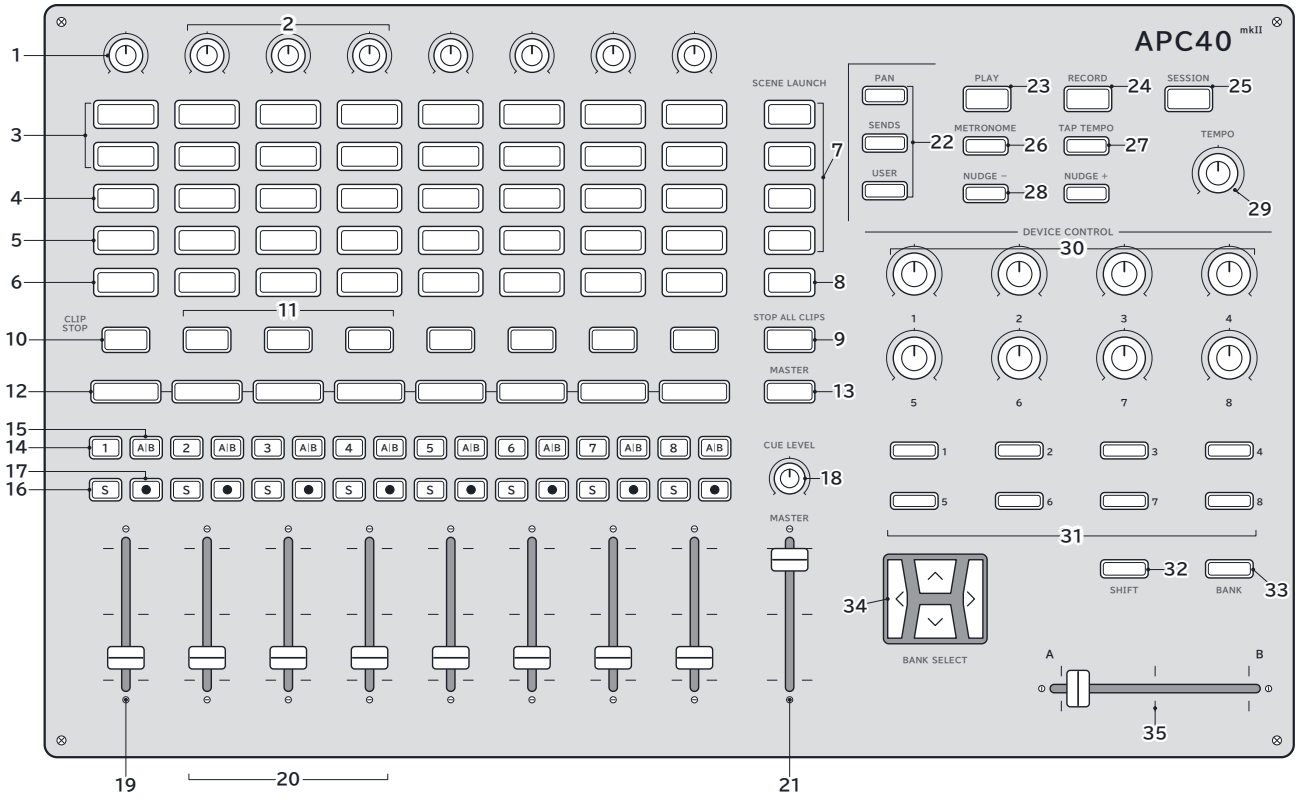


Figure 21 The APC40 mkII's controls (numbered on columns 1–4 = deck A; columns 5–8 = deck B are the same)

Same mapping as the mkI, laid out differently (the mkI's `TRACK CONTROL` knobs are the mkII's `TRACK KNOB`s above the grid). Columns 1–4 are deck A, columns 5–8 deck B. The mouse works too for the jog.

Top (knobs, grid, SCENE LAUNCH)

1 **TRACK KNOB 1 (5 for deck B)**
Jog (stopping for 0.15 s counts as releasing it)

2 **TRACK KNOB 2–4 (6–8)**
Filter Morph, Resonance, Cutoff. On the `SENDS` page: echo length and feedback, vibrato step (with SHIFT, the width of the swing), gate steps; on the `USER` page: volume, picture strength, grain length, key

3 **Grid rows 1 and 2**
Pads 1–8 (in `HOT CUE` mode, hot cues 1–8; SHIFT clears)

4 **Grid row 3**
Samples (34's ◂ ▸ shifts it to 5–8; with SHIFT, LOAD into that sample)

5 **Grid row 4**
Memory cues (MEMORY, CALL ◂, CALL ▸, DELETE)

6 **Grid row 5**
While held, fast spinback, reverse, half speed, fast forward (with SHIFT, LOOP, SYNC, Q and SLIP on/off)

7 **SCENE LAUNCH 1–4**
Auto loops of 1, 2, 4, 8 beats (with SHIFT, jump −4, −1, +1, +4 beats)

8 **SCENE LAUNCH 5**
Exit / return to the loop

9 **STOP ALL CLIPS**
Tempo range

Bottom (buttons and faders)

10 CLIP STOP 1 (5)
Freeze (with SHIFT, that deck's video effect on/off)

11 CLIP STOP 2–4 (6–8)
Toggle the effect of the fader below (20)

12 TRACK SELECT 1 / 5
Select deck A / B. 7–9 and the right side (23–31, 34) act on the selected deck

13 MASTER
While held, `PLAY`, `CUE` and `SYNC` act on both decks (`PLAY` plays / stops both together)

14 ACTIVATOR
Mute (with SHIFT, picture only)

15 CROSSFADER A/B
Crossfader assign (A / B)

16 SOLO
Headphone Cue

17 REC ARM
LOAD (with SHIFT, replace). `REC ARM` of column 2 (6) previews (the same as the `P` key; it lights while previewing), and `REC ARM` of column 3 (7) turns the metronome on/off (the same as the `N` key; it lights while on)

18 CUE LEVEL
Headphone Cue mix

19 Track fader 1 (5)
Amount of the video effects applied to the sound

20 Track faders 2–4 (6–8)
Echo, bitcrusher, vibrato

21 Master fader
Video effect amount after the mix

Right side

22 PAN, SENDS, USER
The pages of TRACK KNOB (2) (`PAN` goes back; while SHIFT is held, `SENDS`)

23 PLAY
Play / stop (with SHIFT, both decks at once)

24 RECORD
CUE

25 SESSION
REAPER recording

26 METRONOME
Video blend (with SHIFT, one back)

27 TAP TEMPO
Tap tempo

28 NUDGE – / +
A little slower / faster while held (beat matching)

29 TEMPO
Tempo

30 DEVICE CONTROL knobs 1–8
Amounts of the 8 video effects (knob 8 is VHS, SHIFT + knob 8 is Strobe)

31 DEVICE CONTROL buttons 1–8
Pad mode (see "Pad modes" (p. 28))

32 SHIFT
Hold with another control to use its second function

33 BANK
All effects off while held

34 BANK SELECT
◂ ▸ = shift the samples of grid row 3 to 5–8. ▲ ▼ = double / halve the loop (with SHIFT, move it 1 beat)

35 Crossfader
Left = deck A, right = deck B

Buttons light up (hot cues in their colours, the sample in use in green and so on). For this, enable the APC40 as a MIDI output in REAPER too (see "Button lights" (p. 194) below).

From the official MIDI lists (mkII from the spec sheet at hand, mkI from a version preserved on the Internet Archive). The mkII was tested on real hardware (jog, hot cues, lights, LOAD, crossfader, two controllers at once and more; the SHIFT combinations, the `SENDS` and `USER` pages, `MASTER`, `METRONOME`, `BANK`, `SESSION` and others have not been tested on real hardware yet). The mkI has not been tested on real hardware yet.

5 · 6 Behringer CMD Micro deck 1 / deck 2

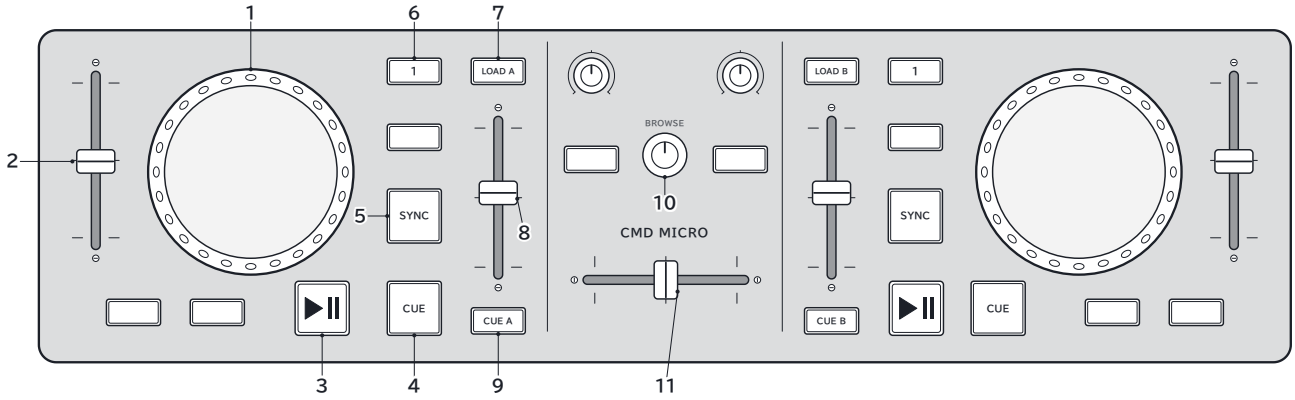


Figure 22 The CMD Micro's controls (numbered on the left deck and the centre; the right deck is the same)

Both decks on MIDI channel 1. No pads.

Deck

1 Jog
Top = touch scratch

2 Outer fader
Tempo (7-bit)

3 PLAY/PAUSE

4 CUE

5 SYNC

6 1
Loop (toggles with each press; the 4-beat beat loop button in Mixxx)

Mixer and centre

7 LOAD A
Loads the file chosen in the Media Explorer into this deck (see "Loading with the LOAD button (from the Media Explorer)" (p. 23))

8 Inner fader
That deck's volume and picture strength (channel fader)

9 CUE A
Listen to this deck on headphones (toggles with each press)

10 BROWSE
Turn = how much the video effects act on the sound

11 Crossfader
Deck 1 on the left side, deck 2 on the right

Controls without a number (2, PITCH BEND − / +, LEFT, RIGHT, MAIN LEVEL, CUE LEVEL) are not used by ReaScratch. From Mixxx's mapping; not yet tested on the real hardware.

7 · 8 Denon SC5000/SC6000 deck 1 / deck 2

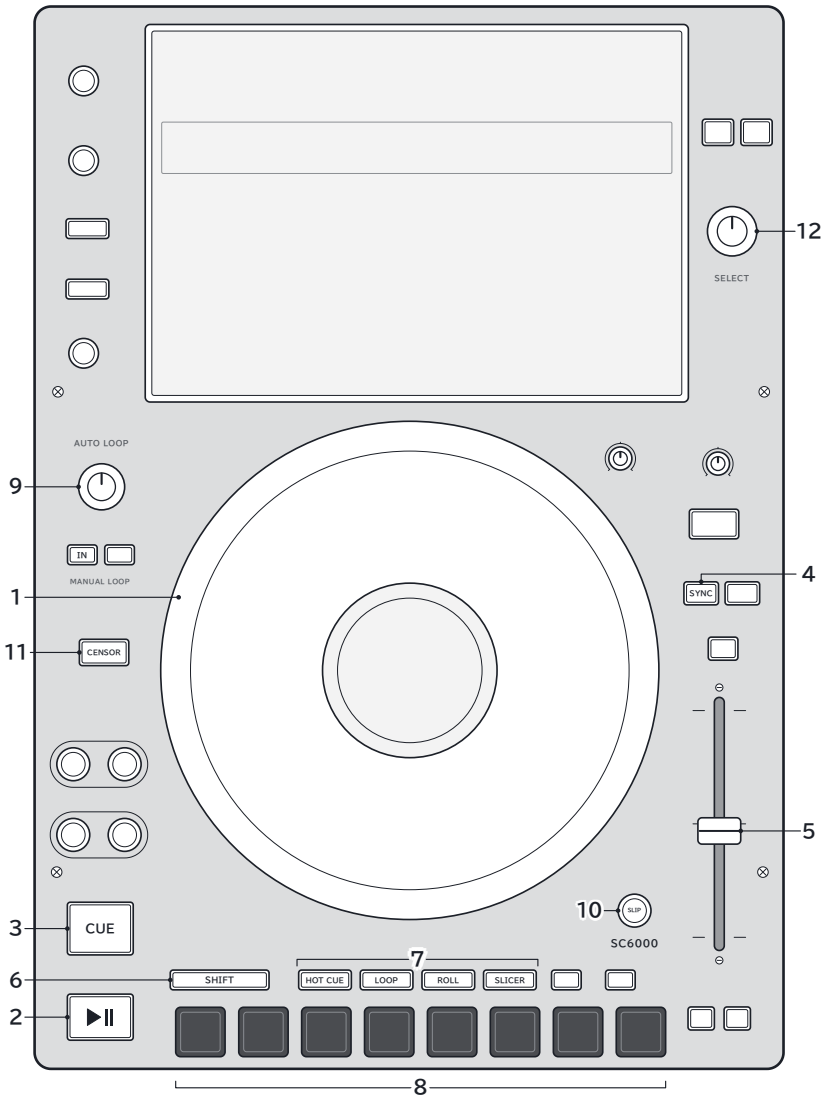


Figure 23 The SC6000's controls (one unit = one deck; the SC5000 is laid out the same)

Set the unit to `Computer Mode` in its Control Center (SHORTCUTS). Every unit sends on MIDI channel 1, so with two units, set the MIDI inputs of the `scratch engine` and `scratch engine B` tracks to the name of each unit (with "All MIDI inputs", both decks move).

Deck

1. **Jog**
   Top scratch (3683 steps per turn, measured on an SC5000M). The SC5000M sends no touch signal, so "touching" means while turning
2. **PLAY**
3. **CUE**
4. **SYNC**
5. **Tempo (pitch fader)**
   The tempo direction is unconfirmed
6. **SHIFT**
   Hold it with another control for its second function
7. **Pad mode buttons**
   `HOT CUE` = HOT CUE, `LOOP` → AUTO LOOP, `ROLL` → BOUNCE LOOP, `SLICER` = SLICER
8. **Pads**
   The 8 pads of the current mode. `SHIFT` + pad clears a hot cue
9. **AUTO LOOP**
   The loop knob (turn = length, push = loop)
10. **SLIP**
11. **CENSOR**
    Reverse while held
12. **SELECT**
    BROWSE

LOOP IN (`IN` of `MANUAL LOOP`) also makes a loop. Controls without a number (OUT, TRACK SKIP, BEAT JUMP, VINYL, PITCH BEND, the buttons round the screen and so on) are not used by ReaScratch.

The numbers come from a third party's measurements on a real SC5000M (the same numbers as Denon's official list for the LC6000). Whether the SC6000 is the same, and the tempo direction, are unconfirmed. Not yet tested on the real hardware.

9 · 10 Denon Prime 4 deck 1 / deck 2

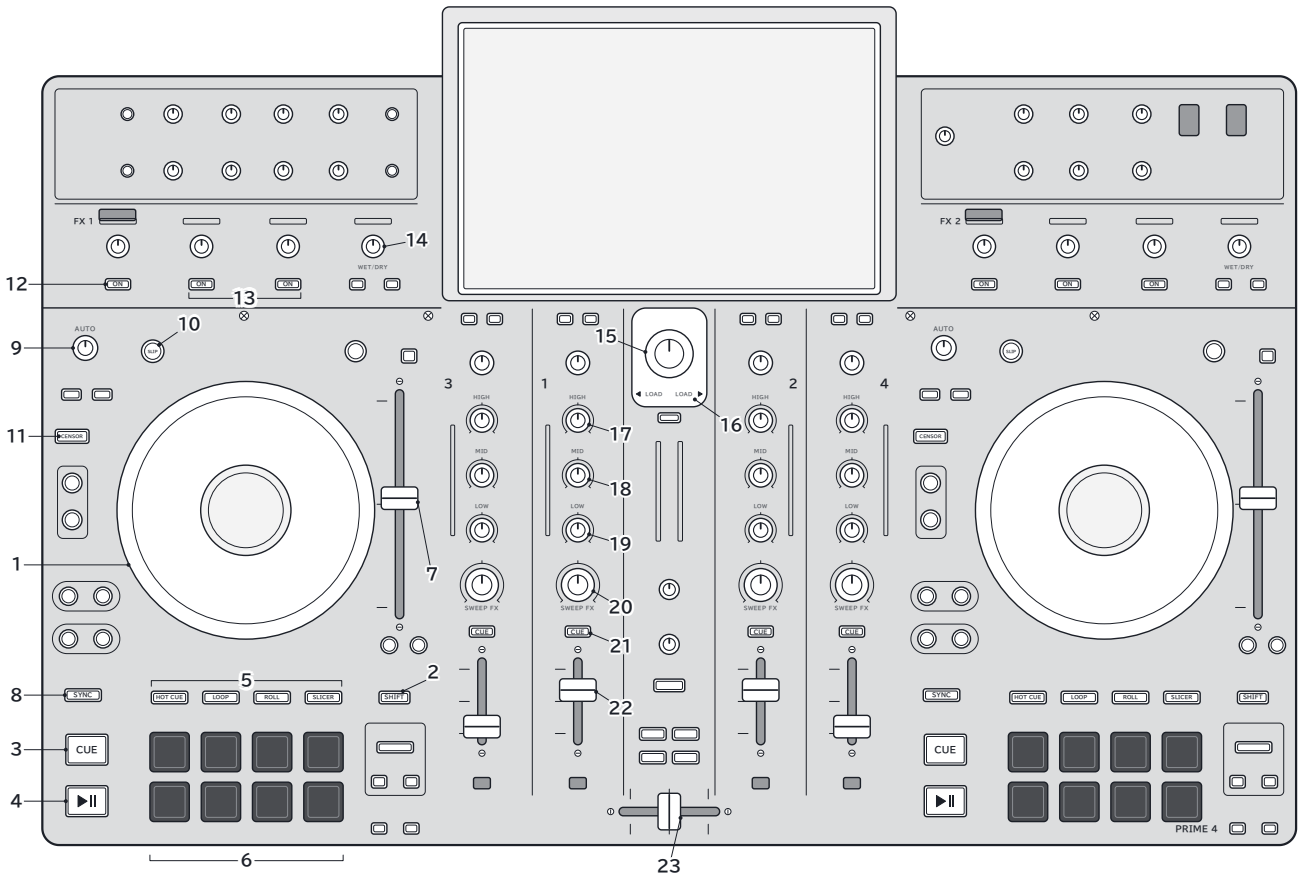


Figure 24 The Prime 4's controls (numbered on the left deck, FX 1, the mixer's channel 1 and the centre; the right deck, FX 2 and channel 2 are the same)

Put the unit in Computer Mode. Left deck on ch 5, right deck on ch 6, the mixer's channels 1 and 2 on ch 1 and ch 2 (deck 1 and deck 2), crossfader on ch 16. From a Mixxx mapping draft that was never merged.

Deck

1 **Jog**
Scratch from the platter position (jog mode `Platter position (Denon)`), touch. The ticks per platter turn are not documented, so Mixxx's value (1000 ticks per turn = `556`) is used provisionally

2 **SHIFT**
Hold it with another control for its second function

3 **CUE**

4 **PLAY**

5 **PERFORMANCE MODES buttons**
`HOT CUE` = HOT CUE, `LOOP` → AUTO LOOP, `ROLL` → BOUNCE LOOP, `SLICER` = SLICER

6 **Pads**
The 8 pads of the current mode. `SHIFT` + pad clears a hot cue

7 **Tempo (pitch fader)**

8 **SYNC**

9 **AUTO**
The loop knob (turn = length, push = loop)

10 **SLIP**

11 **CENSOR**
Reverse while held

FX 1 (Buddy-style)

12 ON (left)
The Buddy's FX ON bar (that deck's video effect on / off)

13 ON (middle, right)
The Buddy's `FX PARAM` ◂ ▸ (effect type; see the note below)

14 WET/DRY
The Buddy's FX knob (effect amount; see the note below)

Mixer

15 BROWSE
Turning and pushing work as on the DDJ-FLX4's BROWSE

16 LOAD ◂ ▸
◂ loads into deck 1, ▸ into deck 2 (as the Reloop Buddy's LOAD A / B)

17 HIGH
Filter Morph (echo length with SHIFT)

18 MID
How much the video effects act on the sound

19 LOW
Filter Resonance (echo feedback with SHIFT)

20 SWEEP FX
Filter Cutoff (key with SHIFT)

21 Headphone CUE
Listen to this deck on headphones (toggles with each press)

22 Channel fader
That deck's volume and picture strength

23 Crossfader
Deck 1 on the left side, deck 2 on the right

Note (13 · 14): What they act on depends on what you hold (as in the Reloop Buddy's note).

Controls without a number (IN and OUT of MANUAL LOOP, DECK, TRACK SKIP, BEAT JUMP, VINYL, KEY LOCK, PITCH BEND, BEAT GRID, PARAMETER, the knobs above the FX buttons, the MIC and MASTER sections, the SWEEP FX buttons, channels 3 and 4 and so on) are not used by ReaScratch. Not yet tested on the real hardware.

11 · 12 Gemini FirstMix deck 1 / deck 2

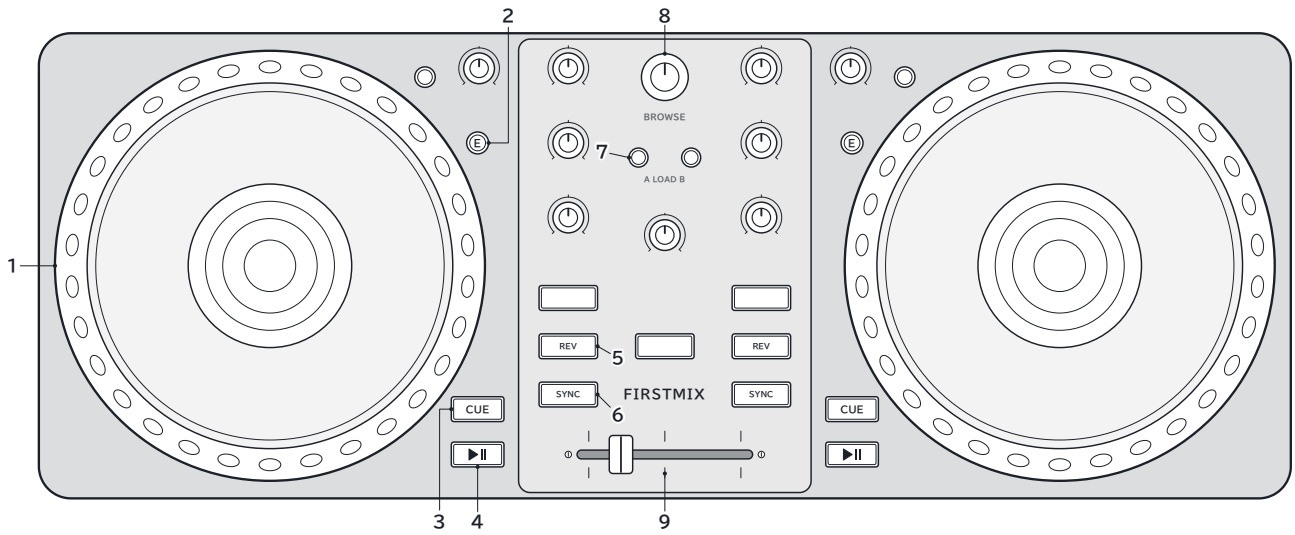


Figure 25 The FirstMix's controls (numbered on the left deck and the left channel; the right side is the same, mirrored)

Both decks on MIDI channel 1. No tempo or channel faders, no pads.

Deck

1 Jog
Top = touch scratch

2 E (the EFX button)
Listen to this deck on headphones (toggles with each press; the same as Mixxx's mapping)

3 CUE

4 PLAY/PAUSE

Mixer

5 REV
Reverse play while held

6 SYNC

7 LOAD A
(the left of `A LOAD B`): Loads the file chosen in the Media Explorer into this deck

8 BROWSE
Turn = how much the video effects act on the sound

9 Crossfader
Deck 1 on the left side, deck 2 on the right

Controls without a number (the EFX knob, SEL, GAIN, TREBLE, BASS, SCRATCH, MASTER, PREVIEW) are not used by ReaScratch. From Mixxx's mapping; not yet tested on the real hardware.

13 · 14 Hercules DJ Control MP3 e2 deck 1 / deck 2

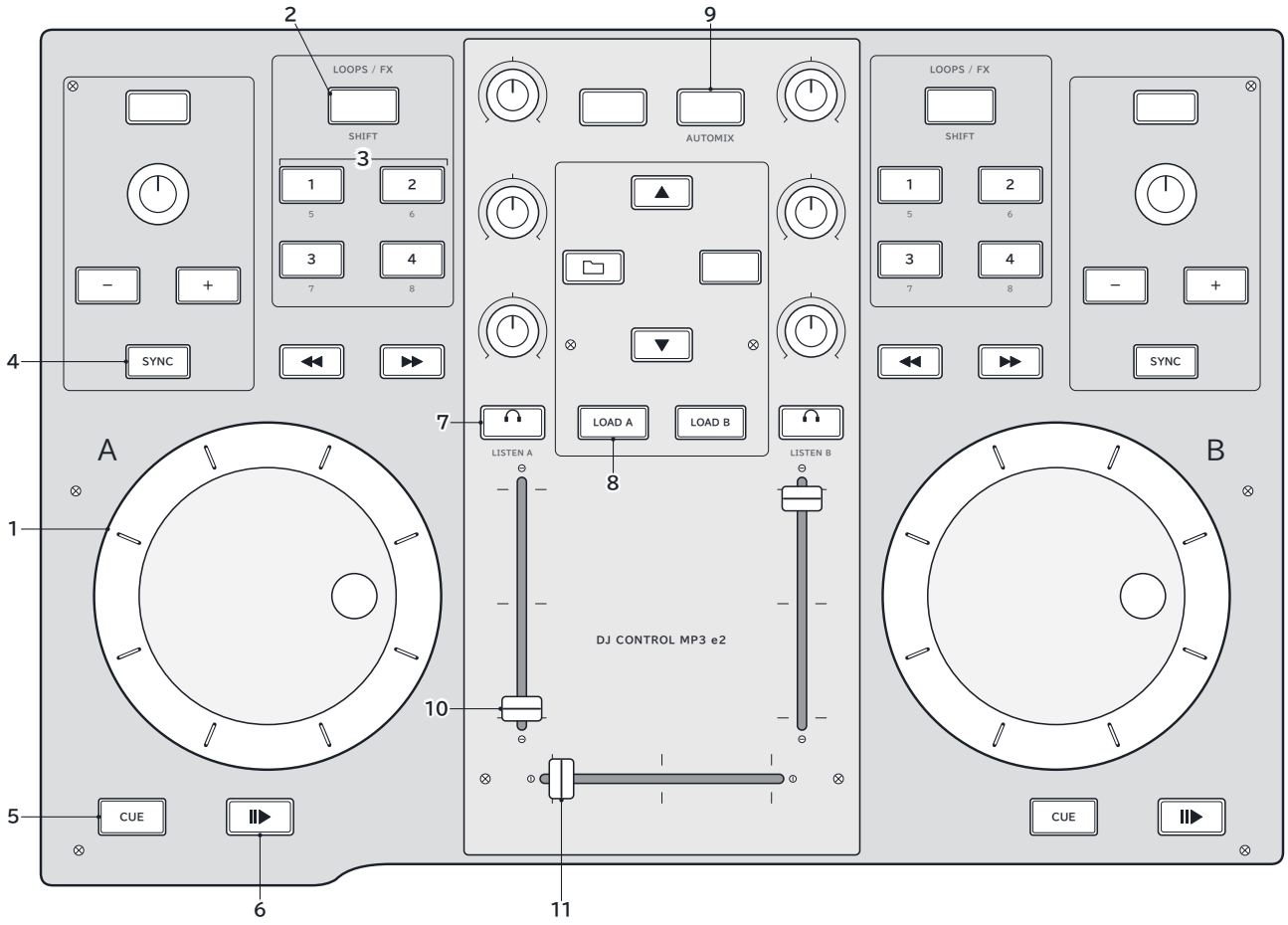


Figure 26 The DJ Control MP3 e2's controls (numbered on the left deck and the centre; the right deck is the same, mirrored)

Needs the Hercules driver (the MIDI channel can be changed in the driver's settings; the preset uses 1).

Deck

1 Jog
Scratch. No touch signal, so "touching" means while turning

2 LOOPS / FX (SHIFT)
Switches the keys 1–4 (while lit, they are the printed 5–8 = K5–K8)

3 Keys 1–4
K5–K8 are hot cues 1–4 (hold AUTOMIX to clear), K1 a loop, K3 reloop

4 SYNC

5 CUE

6 PLAY/PAUSE

Mixer and centre

7 LISTEN A
Headphone CUE (listen to this deck on headphones)

8 LOAD A
LOAD

9 AUTOMIX
Acts as SHIFT while held (clears hot cues and so on)

10 Volume fader
Channel fader (that deck's volume and picture strength)

11 Crossfader

Controls without a number (MASTER TEMPO, PITCH, PITCH BEND − / +, ◀◀ ▶▶, TREBLE, MEDIUM, BASS, SCRATCH, the browser buttons) are not used by ReaScratch. Tempo is an endless knob (PITCH), so it is not read. From Hercules' official VirtualDJ mapping; not yet tested on the real hardware.

15 · 16 Hercules DJControl Compact deck 1 / deck 2

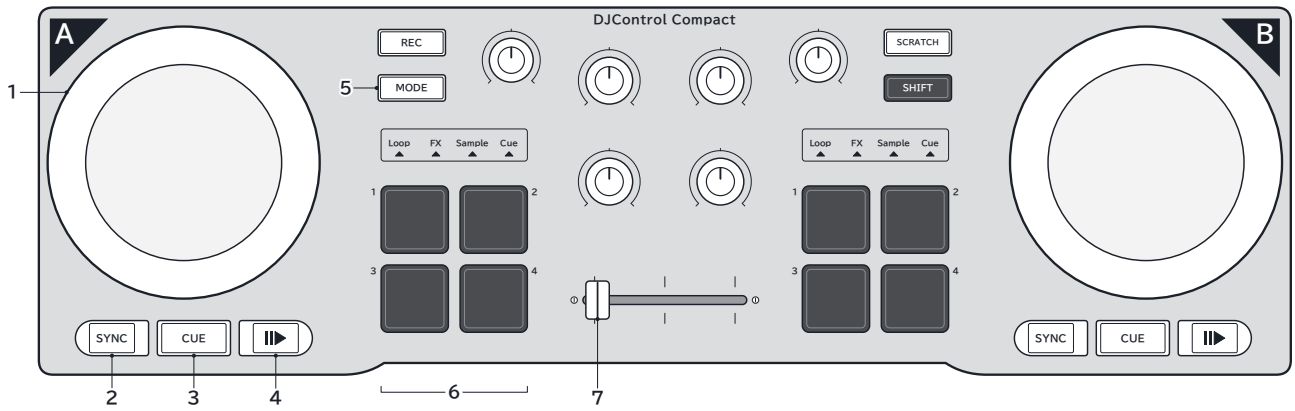


Figure 27 The DJControl Compact's controls (numbered on the left deck and the centre; the pads and the jog are the same on the right deck; MODE and REC are only on the left, SCRATCH and SHIFT only on the right)

Both decks on MIDI channel 1. No tempo or channel faders.

Deck

1 **Jog**
Scratch. No touch signal, so "touching" means while turning

2 **SYNC**

3 **CUE**

4 **PLAY/PAUSE**

5 **MODE**
Switches the pad mode (there is only one MODE, on the left; whether it works on both decks or on one is not confirmed. The lamps over the printed `Loop`, `FX`, `Sample` and `Cue` show the current one)

6 **Pads**
The 4 pads of the current mode. `Cue` = hot cues (cleared with the neighbouring button), `Sample` = samples, `FX` = Buddy-style (pad 9 = FX ON, 10 / 11 = ◀ ▶; no knob), `Loop` = the loop buttons (reloop, 8-beat loop, halve, double)

Centre

7 **Crossfader**

Controls without a number (REC, VOLUME, MEDIUM, BASS, SCRATCH, SHIFT) are not used by ReaScratch. The official manual and MIDI list say SHIFT + pad clears a hot cue (notes 0x05-0x08), which differs from "cleared with the neighbouring button" in 6 (not confirmed on the real hardware). From Mixxx's mapping; not yet tested on the real hardware.

17 · 18 Hercules DJControl Starlight deck 1 / deck 2

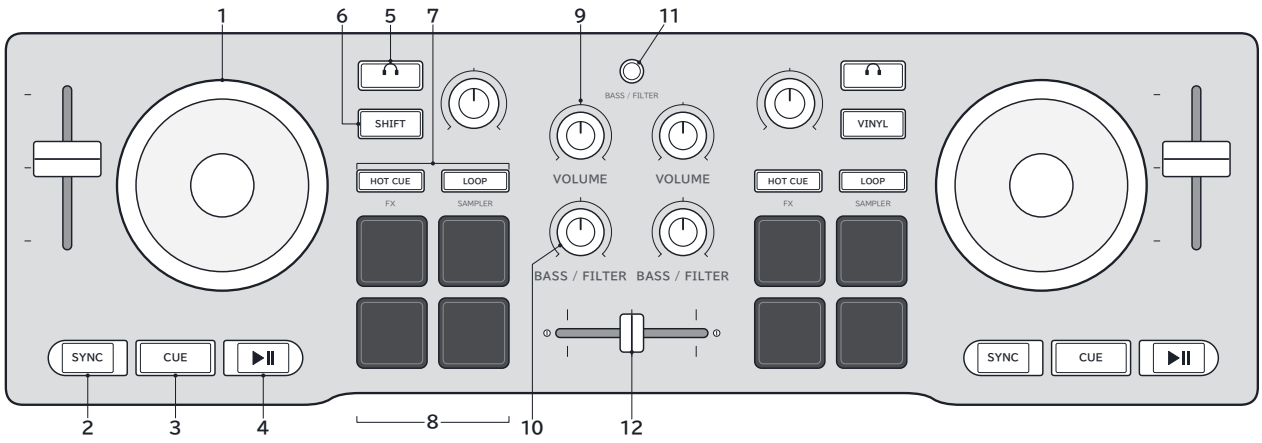


Figure 28 The DJControl Starlight's controls (numbered on the left deck and the centre; the right deck has the same layout, but SHIFT is only on the left and VINYL only on the right)

Same numbers as the Inpulse 200. Decks on ch 2 / ch 3 (pads on ch 7 / ch 8). There is no tempo fader on the unit.

Deck

1 **Jog**
Top scratch (touch), outer ring = nudge

2 **SYNC**

3 **CUE**

4 **PLAY/PAUSE**

5 **Headphone button**
Headphone CUE (listen to this deck on headphones)

6 **SHIFT**
`SHIFT` + pad clears a hot cue

7 **Mode buttons**
The mode of the 4 pads (HOT CUE, LOOP → AUTO LOOP, SHIFT + HOT CUE for FX → TOUCH FX, SHIFT + LOOP for SAMPLER → samples)

8 **Pads**
The 4 pads of the current mode

Mixer

9 **VOLUME**
Works as the channel fader (that deck's volume and picture strength)

10 **BASS / FILTER**
Set to FILTER = the FILTER knob (filter Cutoff), set to BASS = the LOW knob (Resonance)

11 **BASS / FILTER switch**
What knob 10 does (lit = FILTER, off = BASS)

12 **Crossfader**

Controls without a number (TEMPO, MASTER, HEADPHONES, VINYL) are not used by ReaScratch. From Mixxx's mapping; not yet tested on the real hardware.

19 · 20 Hercules DJControl MIX deck 1 / deck 2

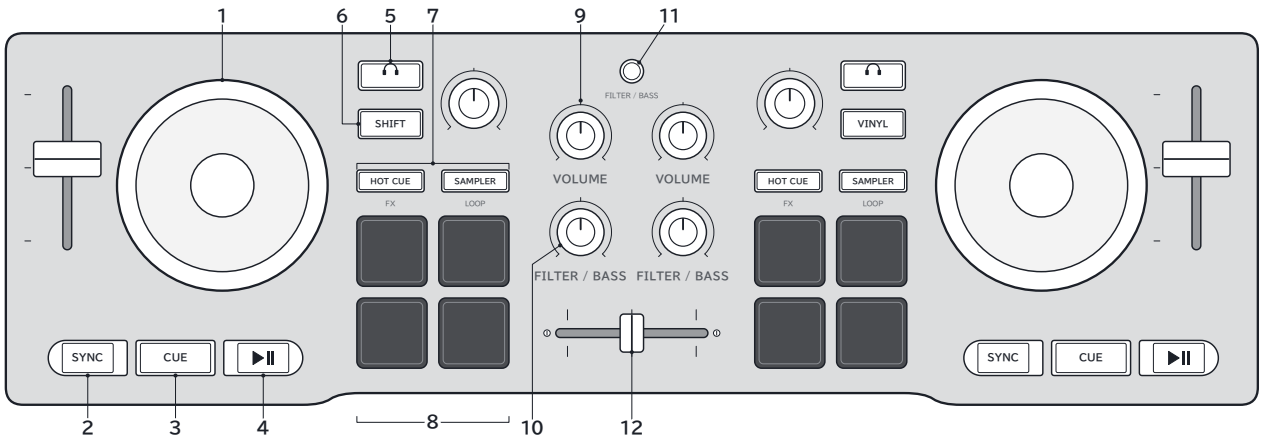


Figure 29 The DJControl MIX's controls (the same layout as the Starlight; numbered on the left deck and the centre; the right deck has the same layout, but SHIFT is only on the left and VINYL only on the right)

Same numbers and layout as the Starlight (only the mode buttons are printed differently). Connect it by USB (Bluetooth is for smartphones). Decks on ch 2 / ch 3 (pads on ch 7 / ch 8). There is no tempo fader on the unit.

Deck

1 Jog
Top scratch (touch), outer ring = nudge

2 SYNC

3 CUE

4 PLAY/PAUSE

5 Headphone button
Headphone CUE (listen to this deck on headphones)

6 SHIFT
`SHIFT` + pad clears a hot cue

7 Mode buttons
The mode of the 4 pads (HOT CUE, SAMPLER → samples, SHIFT + HOT CUE for FX → TOUCH FX, SHIFT + SAMPLER for LOOP → AUTO LOOP)

8 Pads
The 4 pads of the current mode

Mixer

9 VOLUME
Works as the channel fader (that deck's volume and picture strength)

10 FILTER / BASS
Set to FILTER = the FILTER knob (filter Cutoff), set to BASS = the LOW knob (Resonance)

11 FILTER / BASS switch
What knob 10 does (off = FILTER, lit = BASS)

12 Crossfader

Controls without a number (TEMPO, MASTER, HEADPHONES, VINYL) are not used by ReaScratch. From Mixxx's mapping; not yet tested on the real hardware.

21 · 22 Hercules DJ Control Instinct deck 1 / deck 2

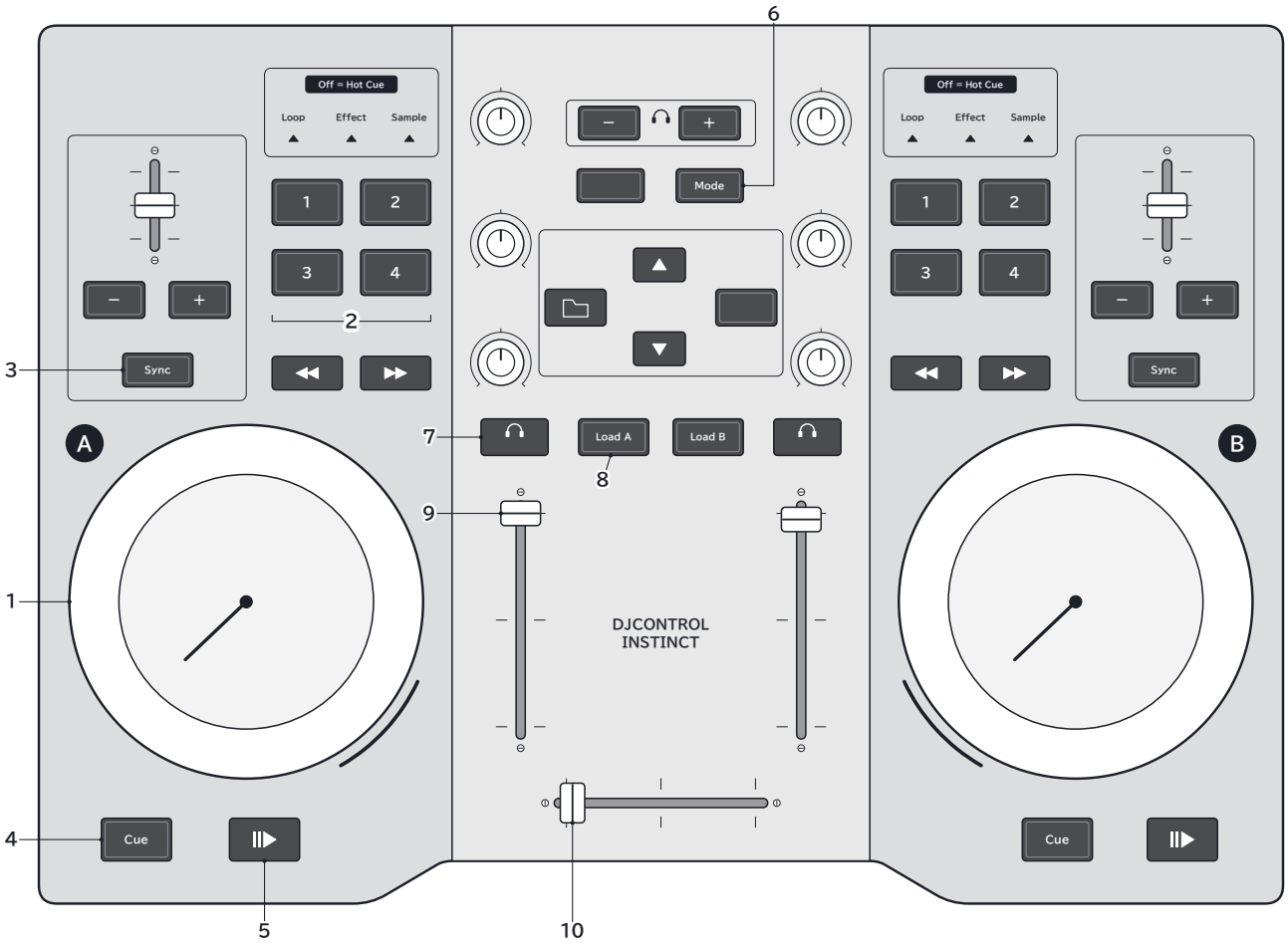


Figure 30 The DJ Control Instinct's controls (numbered on the left deck and the centre; the right deck is the same, mirrored)

Both decks on MIDI channel 1.

Deck

1 Jog
Top scratch (touch)

2 Keys 1–4
What they do depends on the mode chosen with `Mode`. `Hot Cue` (the `Loop`, `Effect` and `Sample` lamps off) = 1 / 2 are hot cues, 3 / 4 clear them. `Loop` = 1 a loop (IN), 3 reloop, 4 halve. `Effect` = Buddy-style (1 = FX ON, 2 / 3 = ◂ ▸)

3 Sync

4 Cue

5 PLAY/PAUSE

6 **Mode**
Switches the mode of the keys 1–4

7 **Headphone button**
Headphone CUE (listen to this deck on headphones)

8 **Load A**
LOAD

9 **Volume fader**
Channel fader (that deck's volume and picture strength)

10 **Crossfader**

Controls without a number (Pitch, Pitch − / +, ◀◀ ▶▶, Treble, Mid, Bass, the headphone volume − / +, Vinyl, the browser buttons) and the keys 1–4 in the `Sample` mode are not used by ReaScratch. Tempo is ± buttons only, so it is not read. From Mixxx's mapping; not yet tested on the real hardware.

23 · 24 Hercules DJ Control AIR deck 1 / deck 2

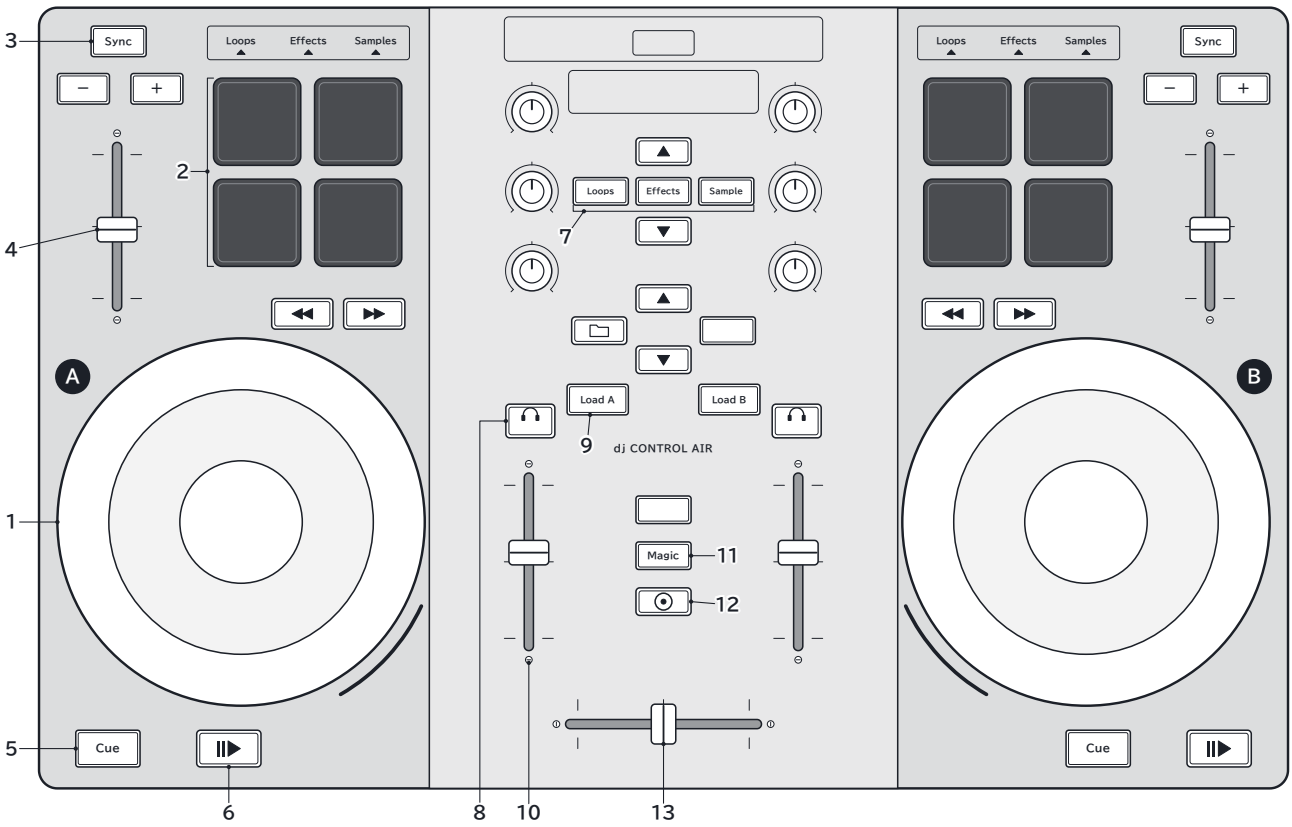


Figure 31 The DJ Control AIR's controls (numbered on the left deck and the centre; the right deck is the same, mirrored)

Both decks on MIDI channel 1.

Deck

1 Jog
With the `SCRATCH` button (12) on, the top scratches (no touch signal, so "touching" means while turning); with it off, the outer ring nudges

2 Pads
The 4 pads of the mode chosen with the mode buttons in the centre (7). `Loops` = 1 a loop (IN), 3 double, 4 reloop. `Effects` = 1 is the Buddy-style FX ON (the Flanger button). `Samples` = the 4 sample buttons

3 Sync

4 Tempo fader
Tempo (7-bit)

5 Cue

6 PLAY/PAUSE

Mixer and centre

7 **Loops · Effects · Sample**
Switch the pad mode

8 **Headphone button**
Headphone CUE (listen to this deck on headphones)

9 **Load A**
LOAD

10 **Volume fader**
Channel fader (that deck's volume and picture strength)

11 **Magic**
Acts as SHIFT while held

12 **SCRATCH**
Top scratch on the jog on / off (1)

13 **Crossfader**

The FX knob of FX (Buddy-style) is CC 0x3F, which Mixxx's mapping calls the Flanger depth (which control on the unit sends it is not confirmed).

Controls without a number (– / +, ◀◀ ▶▶, Treble, Mid, Bass, the browser buttons, Rec) are not used by ReaScratch. From Mixxx's mapping; not yet tested on the real hardware.

25 · 26 Hercules DJ Console Mk4 deck 1 / deck 2

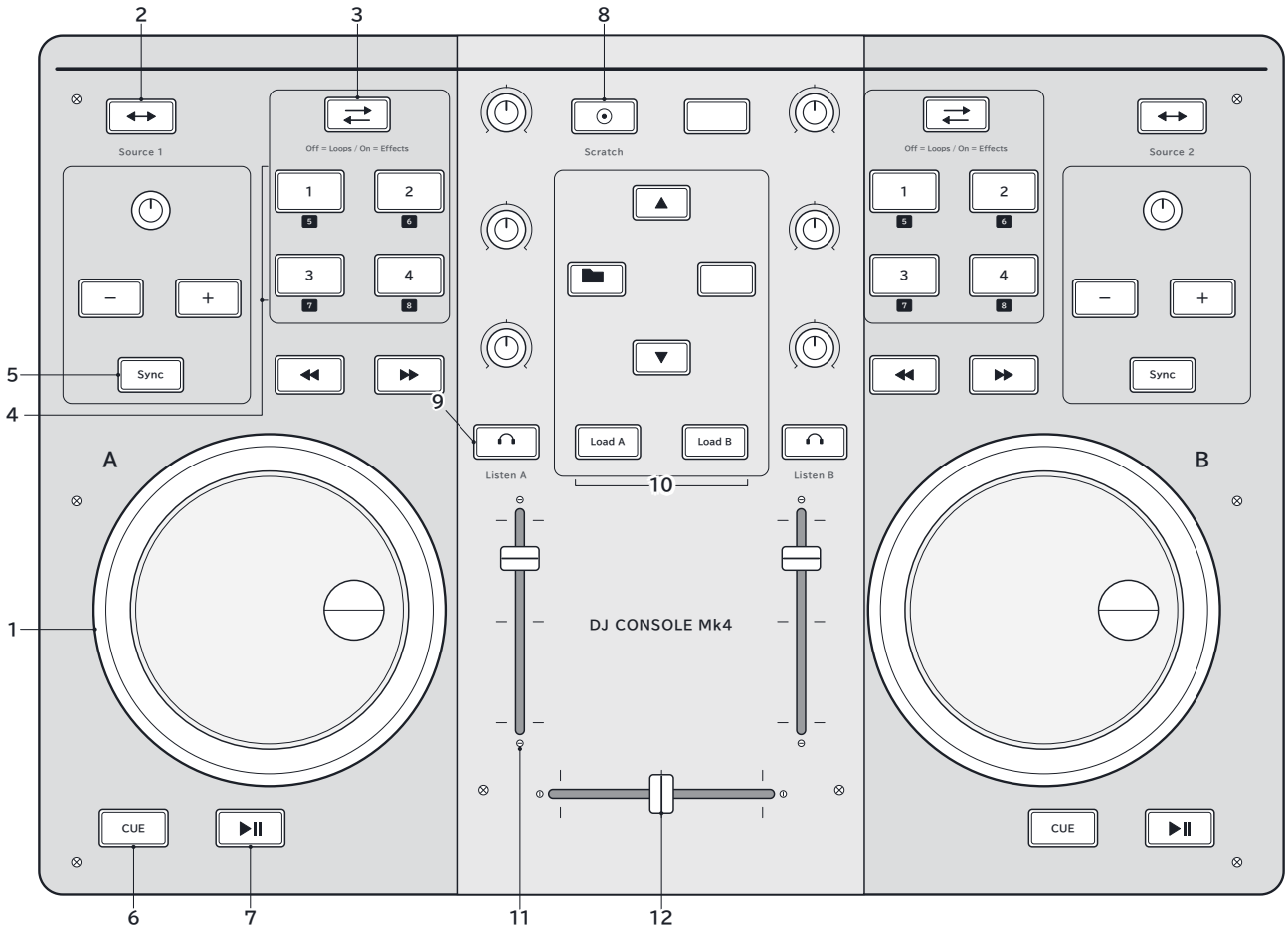


Figure 32 The DJ Console Mk4's controls (numbered on the left deck and the centre; the right deck is the same, mirrored)

Deck

1 Jog
Scratch. No touch signal, so "touching" means while turning

2 Source 1
Video effect on/off (Buddy-style FX ON). Mixxx's mapping calls it the `Flanger` button (notes 0x13 / 0x27); taken to be the unit's `Source` button from the order of the numbers (not checked on the real hardware)

3 Off = Loops / On = Effects
When lit, the keys 1–4 (4) become K5–K8

4 Keys 1–4
Lit (K5–K8) = hot cues 1–4 (hold `Scratch` to clear). Unlit (K1–K4) = key 1 loops, key 3 reloops

5 Sync

6 CUE

7 PLAY/PAUSE

Mixer

8 **Scratch**
Acts as SHIFT while held

9 **Listen A**
Listen to this deck on headphones (headphone CUE)

10 **Load A / B**
Load into deck A / B (as on the Reloop Buddy)

11 **Channel fader**
That deck's volume and picture strength

12 **Crossfader**
Deck 1 on the left side, deck 2 on the right

The DJ Control MP3 e2's numbers (both decks on MIDI channel 1). Tempo (the `Pitch` knob) arrives as relative values, so it is not read. Whether the Hercules driver is needed is not known. The controls without a number (Pitch, Pitch Bend, ◀◀ ▶▶, Treble / Medium / Bass, the browser buttons, Mic On/Off) are not used by ReaScratch. From Mixxx's mapping; not yet tested on the real hardware.

27 · 28 Hercules DJ Console 4-Mx deck 1 / deck 2

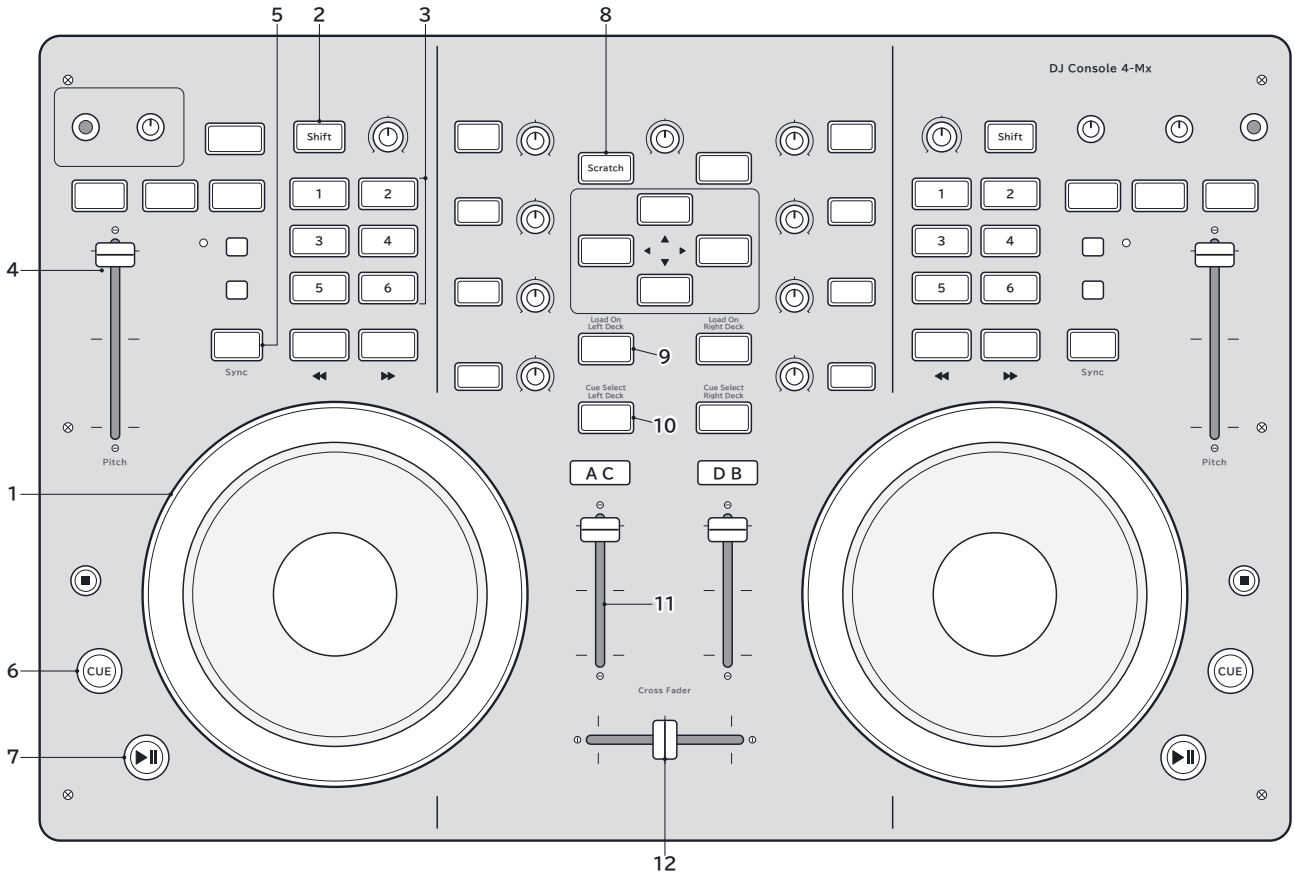


Figure 33 The DJ Console 4-Mx's controls (numbered on the left deck and the centre; the right deck is the same, mirrored)

Deck

1 Jog
With the unit's `Scratch` (8) on, the top scratches (touch); turning without touching works as the outer ring

2 Shift
When lit, the buttons 1–6 (3) become 7–12 (the unit's user guide). ReaScratch also reads it as SHIFT while held

3 Buttons 1–6 (the FX buttons)
With Shift unlit = button 2 reloops, 5 and 6 pick samples. 1–4 turned into 7–10 with Shift = hot cues 1–4 (as in Mixxx)

4 Pitch
Tempo (7-bit)

5 Sync

6 Cue (round button)

7 Play (round button)

8 **Scratch**
Top scratch on the jog on / off (1)

9 **Load On Left Deck**
LOAD (as on the Reloop Buddy)

10 **Cue Select Left Deck**
Listen to this deck on headphones (headphone CUE)

11 **Channel fader**
That deck's volume and picture strength

12 **Crossfader**
Deck 1 on the left side, deck 2 on the right

Needs the Hercules driver (set its deck mode to 2 decks). Both decks on MIDI channel 1. The controls without a number (Stop, P. Bend, Pitch Scale, Deck C / D, Ctrl, ◀◀ ▶▶, Source, Kill, the EQ knobs, the browser buttons, Auto) are not used by ReaScratch. From Mixxx's mapping; not yet tested on the real hardware.

29 · 30 Hercules DJ 4Set deck 1 / deck 2

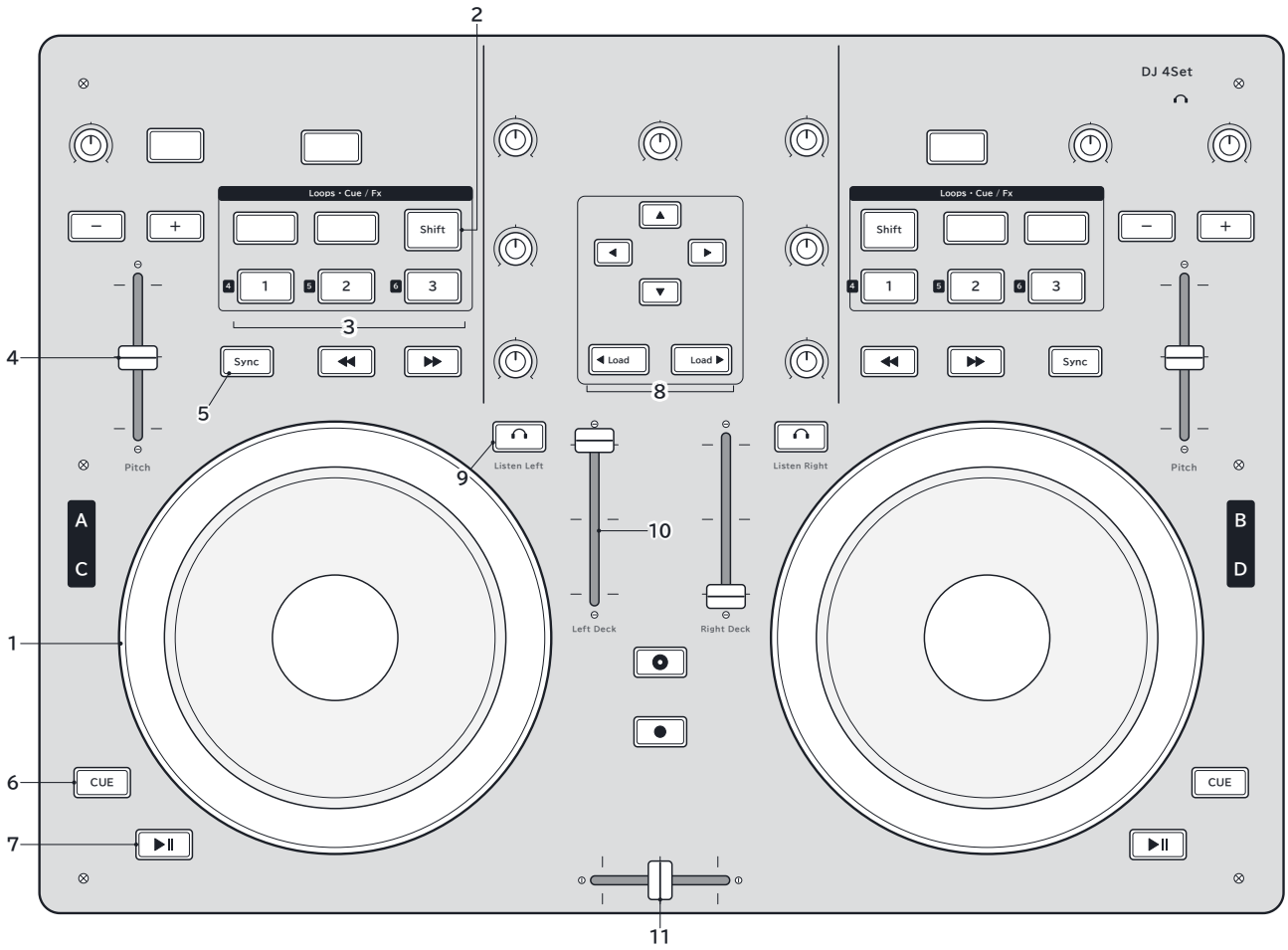


Figure 34 The DJ 4Set's controls (numbered on the left deck and the centre; the right deck is the same, mirrored)

Deck

1 Jog
Top scratch (touch); turning without touching works as the outer ring

2 Shift
When lit, the keys (3) become 4–6 and clear that hot cue

3 Keys 1–3 (K1–K3)
Hot cues 1–3

4 Pitch
Tempo (7-bit)

5 Sync

6 CUE

7 PLAY/PAUSE

Mixer

8 Load ◂ ▸
◂ = load into deck A, ▸ = deck B (as on the Reloop Buddy)

9 Listen Left
Listen to this deck on headphones (headphone CUE)

10 Channel fader
That deck's volume and picture strength

11 Crossfader
Deck 1 on the left side, deck 2 on the right

The DJ Console 4-Mx's numbers (from the MIDI list at the end of Hercules' official user guide, via the Internet Archive). Needs the Hercules driver (set `Deck mode` in its control panel to `2-deck basic`). Both decks on MIDI channel 1. The controls without a number (Mic, DECK C / D, Modulate, Pitch Bend, ◀◀ ▶▶, Hi / Med / Low, Main, the browser buttons, Scratch, Record) are not used by ReaScratch. Not yet tested on the real hardware.

31 · 32 Hercules DJControl Jogvision deck 1 / deck 2

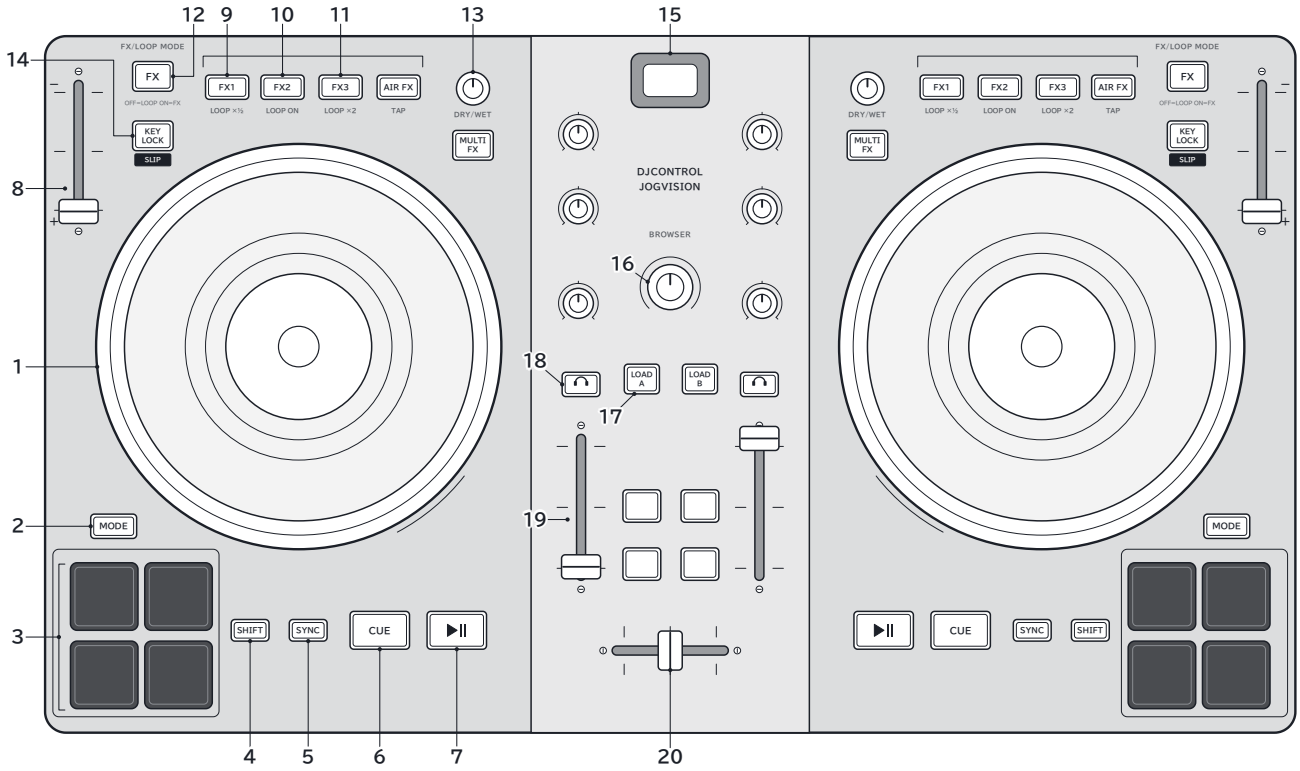


Figure 35 The DJControl Jogvision's controls (numbered on the left deck and the centre; the right deck is the same, mirrored)

Deck

1 Jog
Top scratch (touch), outer ring

2 MODE
Hold it and press pad 1–4 to pick the pad mode (HOT CUE, SAMPLER → samples)

3 Pads
The 4 pads of the current mode. In `HOT CUE`, SHIFT + pad clears a hot cue

4 SHIFT
Hold it with another control for its second function

5 SYNC

6 CUE

7 PLAY/PAUSE

8 Tempo fader
Tempo (14-bit)

9 FX1 (LOOP ×½)
In FX mode = the Buddy's FX ON bar (note 0x04; toggles with each press); in loop mode = halve the loop

10 FX2 (LOOP ON)
In FX mode = the Buddy's `FX PARAM ◂` (note 0x05); in loop mode = loop

11 FX3 (LOOP ×2)
In FX mode = `FX PARAM ▸` (note 0x06); in loop mode = double the loop

12 FX/LOOP MODE
Switches 9–11 and 13 between FX and loop

13 DRY/WET (LOOP SIZE)
In FX mode = the Buddy's FX knob; in loop mode = the loop knob (length)

14 KEY-LOCK / SLIP
SHIFT + KEY-LOCK = SLIP

15 AIR FX sensor
Filter Cutoff (CC 0x0F; the value Mixxx's mapping calls Filter; switch it on with the AIR FX button)

16 **BROWSER**
Turn = the headphone Cue mix (while FX ON is on, that effect's sound amount), push = switch decks

17 **LOAD A / B**
As on the Reloop Buddy (LOAD A = deck 1, LOAD B = deck 2; LOAD B is on ch2 81)

18 **CUE-SELECT (the headphone button)**
Listen to this deck on headphones (headphone CUE)

19 **Channel fader**
That deck's volume and picture strength

20 **Crossfader**
Deck 1 on the left side, deck 2 on the right

Decks on ch 1 / ch 2. The FX section is read Buddy-style (9–11 and 13 above). Besides press, halve, double and the loop knob, the loop can also be moved (Mixxx's `Move Loop By N Beats`; which control on the unit sends it is not known). The controls without a number (AIR FX, MULTI FX, TREBLE / MEDIUM / BASS, VINYL, MIC, VIEW, LOAD PREPARE) are not used by ReaScratch. From Mixxx's mapping; not yet tested on the real hardware.

33 · 34 Hercules Inpulse 200/300 (MK2) deck 1 / deck 2

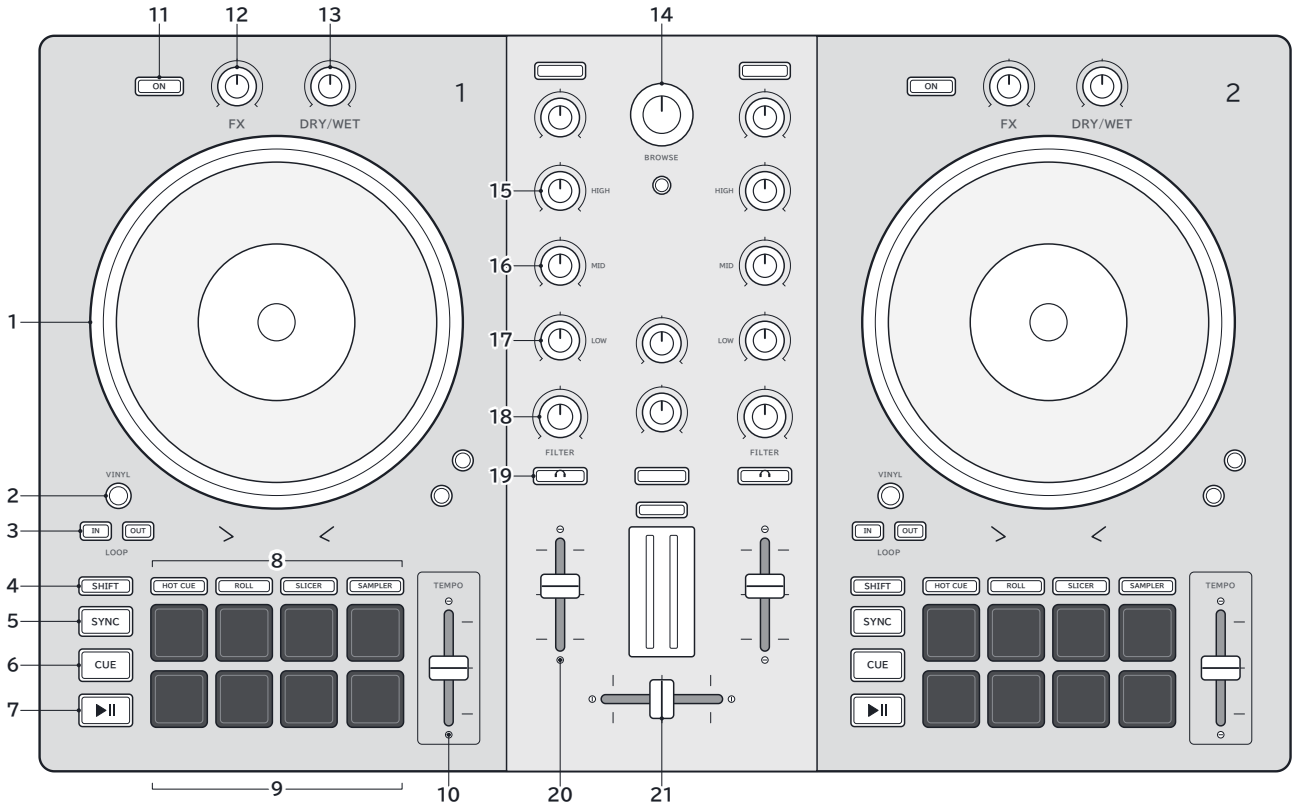


Figure 36 The DJControl Inpulse 300's controls (numbered on the left deck and the centre; the right deck is the same. The 300 MK2 is taken to have nearly the same layout; not checked)

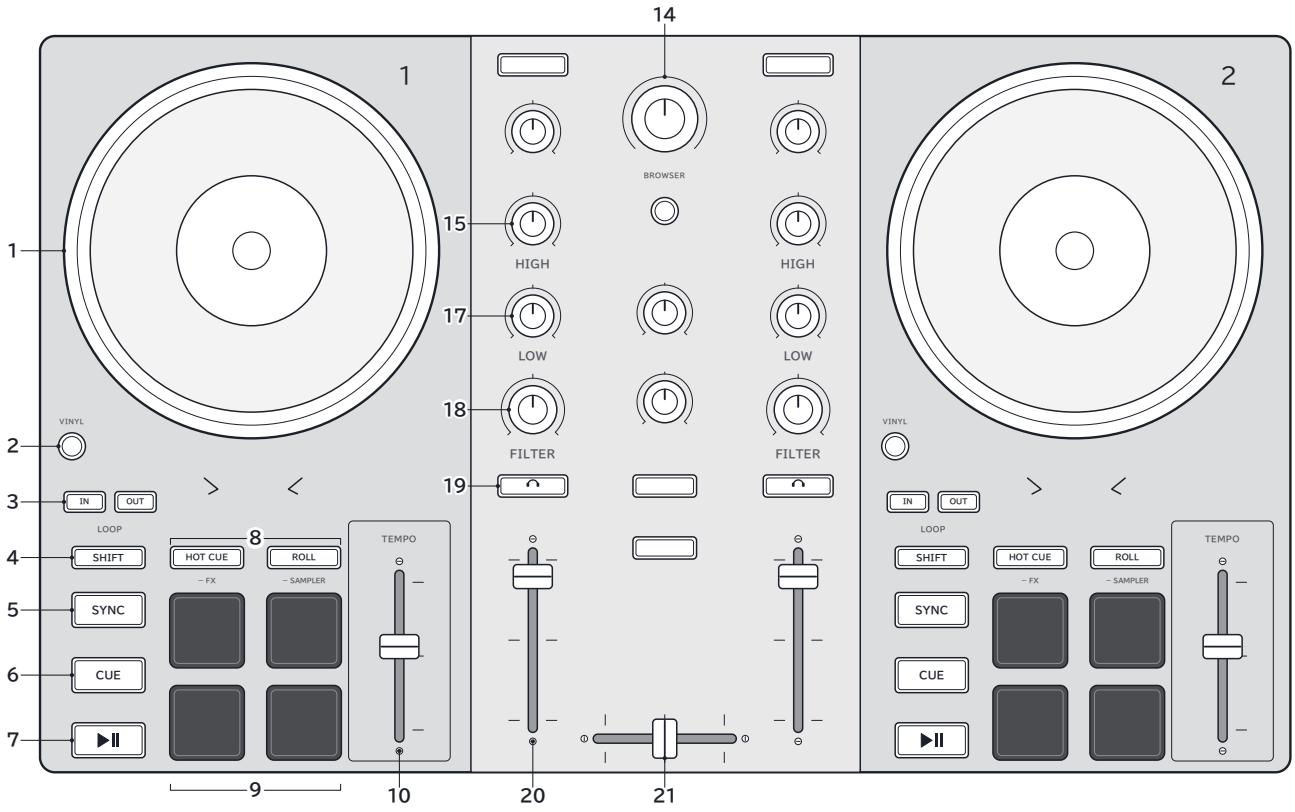


Figure 37 The DJControl Inpulse 200's controls (the MK2 has the same layout; the numbers are those of the 300's list; 11–13 and 16 are on the 300 only)

Deck

1 Jog
Top scratch (VINYL mode), outer ring, touch

2 VINYL
Top scratch on/off (`SHIFT` + `LOOP` on the 200 MK2)

3 IN · OUT
IN (LOOP IN) = loop, OUT (LOOP OUT) = reloop

4 SHIFT
Hold it with another control for its second function

5 SYNC

6 CUE

7 PLAY/PAUSE

8 Mode buttons
300: `HOT CUE` = HOT CUE, `ROLL` → BOUNCE LOOP, `SLICER` → SLICER, `SAMPLER` → samples. The second modes (a 0.5-second long press): Toneplay → PITCH PLAY, FX → TOUCH FX, Slicer loop → AUTO LOOP, Beatjump → NEURAL MIX. For the 200, see the note below

9 Pads
The 8 pads of the current mode (4 on the 200). SHIFT + pad clears a hot cue

10 TEMPO
Tempo fader

11 ON (300 only)
The Buddy's FX ON bar (toggles with each press; with SHIFT, turns this deck's effects off while held)

12 FX (300 only)
The amount of the video effect after the mix (FX LVL)

13 DRY/WET (300 only)
The Buddy's FX knob

14 **BROWSE**
Turn = the headphone Cue mix (while FX ON is on, that effect's sound amount), push = switch decks

15 **HIGH**
Filter Morph (echo length with SHIFT)

16 **MID (300 only)**
How much the video effects act on the sound

17 **LOW**
Filter Resonance (echo feedback with SHIFT)

18 **FILTER**
Filter Cutoff (key with SHIFT)

19 **Headphone CUE**
Listen to this deck on headphones

20 **Channel fader**
That deck's volume and picture strength

21 **Crossfader**
Deck 1 on the left side, deck 2 on the right

Decks on ch 2 / ch 3 (pads on ch 7 / ch 8), crossfader on ch 1. The FX section (300) is read Buddy-style. PADFX SELECT (notes 0x25 / 0x26 in the official MIDI list) is set as ◂▸, but the list does not say which control on the unit sends it, so it has no number in the drawing. The 200 has 2 mode buttons (`HOT CUE`, `ROLL`; on the MK2 `HOT CUE`, `STEMS`), and holding SHIFT gives the second modes (FX, SAMPLER). Its pads use the 300's numbers: HOT CUE = HOT CUE, ROLL (STEMS) → BOUNCE LOOP, FX → SLICER, SAMPLER → samples (read from the numbers; not checked on the real hardware). LOAD (ch2 13 / ch3 13) loads the file chosen in the Media Explorer into an empty sample slot. SHIFT + LOAD (ch5 13 / ch6 13) replaces the slot in use. Hold a mode button (8) and press LOAD to preview (the same as the `P` key). The controls without a number (Q, SLIP, GAIN, ASSISTANT, the HEADPHONES / MASTER knobs, MASTER, BEATMATCH GUIDE) are not used by ReaScratch. From Hercules' official MIDI list and Mixxx's mapping; not yet tested on the real hardware.

35 · 36 Hercules Inpulse 500 deck 1 / deck 2

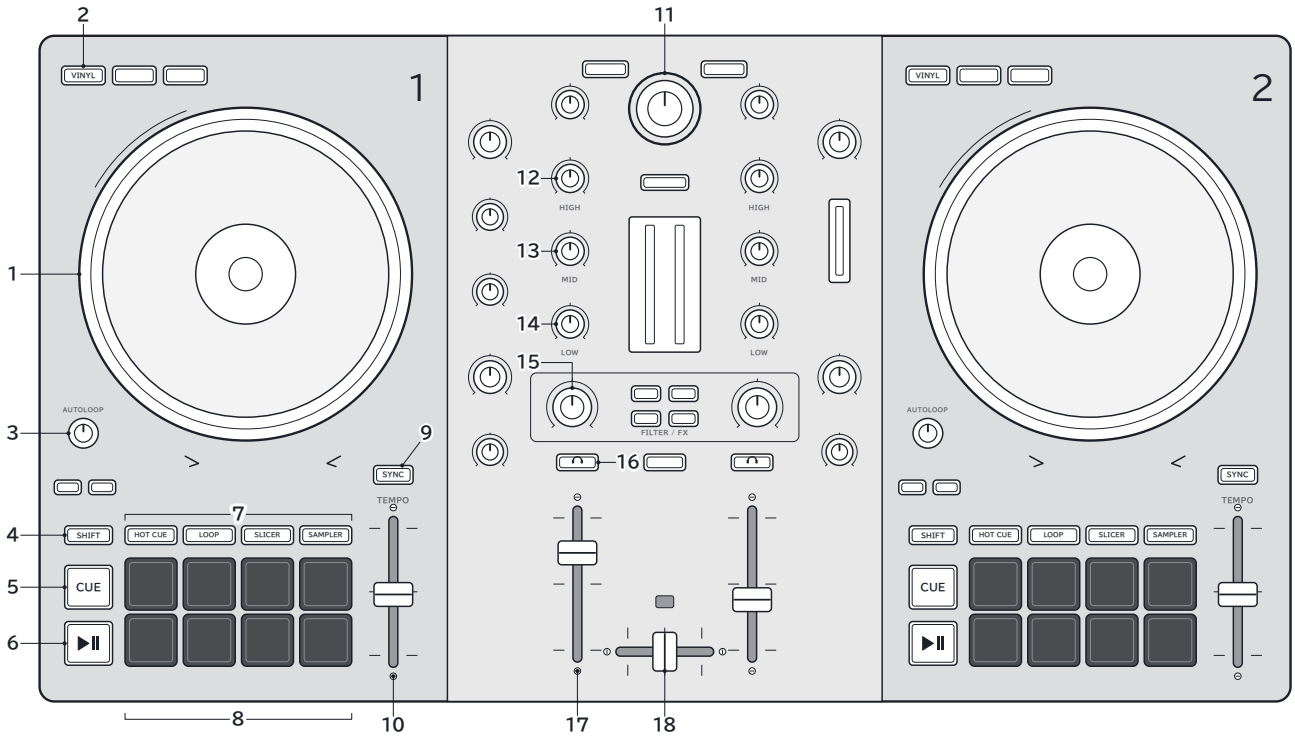


Figure 38 The DJControl Inpulse 500's controls (numbered on the left deck and the centre; the right deck is the same)

Deck

1 Jog
Top scratch (VINYL mode), outer ring, touch

2 VINYL
Top scratch on/off

3 Loop knob (AUTOLOOP)
Turn = length, press = loop

4 SHIFT
Hold it with another control for its second function

5 CUE

6 PLAY/PAUSE

7 Mode buttons
`HOT CUE` = HOT CUE, `LOOP` → AUTO LOOP, `SLICER` → SLICER, `SAMPLER` → samples. The second modes (MODE 5–8): Pitch → PITCH PLAY, Roll → BOUNCE LOOP, Effect → TOUCH FX, BeatJump → NEURAL MIX

8 Pads
The 8 pads of the current mode. SHIFT + pad clears a hot cue

9 SYNC

10 TEMPO
Tempo fader

Mixer

11 **BROWSE**
Turn = how much the video effects act on the sound, push = switch the deck the knob works on, SHIFT + turn = the headphone Cue mix

12 **HIGH**
Filter Morph (echo length with SHIFT)

13 **MID**
How much the video effects act on the sound

14 **LOW**
Filter Resonance (echo feedback with SHIFT)

15 **FILTER**
Filter Cutoff (key with SHIFT)

16 **Headphone CUE**
Listen to this deck on headphones

17 **Channel fader**
That deck's volume and picture strength

18 **Crossfader**
Deck 1 on the left side, deck 2 on the right

Decks on ch 2 / ch 3 (pads on ch 7 / ch 8), crossfader on ch 1. The jog's tick count differs from the 200/300 (`138` on the 200/300, `400` on the 500). LOAD (ch2 13 / ch3 13) loads the file chosen in the Media Explorer into an empty sample slot. SHIFT + LOAD (ch5 13 / ch6 13) replaces the slot in use. Hold a mode button (7) and press LOAD to preview (the same as the `P` key). The controls without a number (SLIP, QUANT, IN, OUT, ASSISTANT, the MIC / AUX knobs, GAIN, MASTER VOL, HEADPHONES, CUE / MASTER, FX1–FX4, BEATMATCH GUIDE) are not used by ReaScratch. From Hercules' official MIDI list and Mixxx's mapping; not yet tested on the real hardware.

37 · 38 ION Discover DJ deck 1 / deck 2

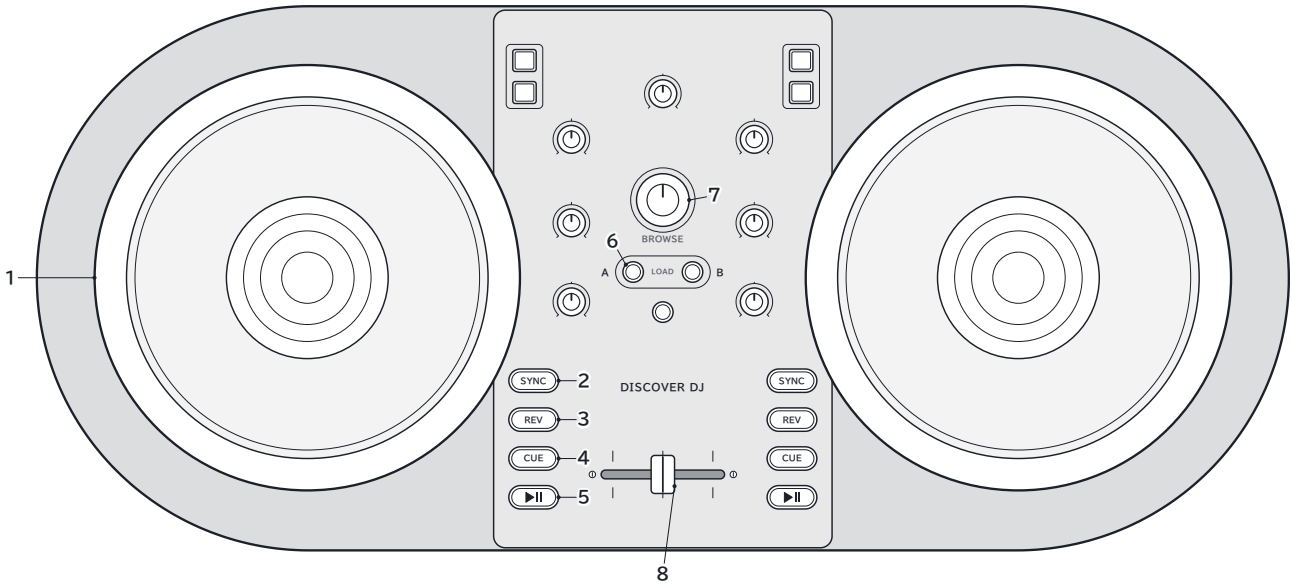


Figure 39 The Discover DJ's controls (numbered on the left deck and the centre; the right deck is the same, mirrored)

Both decks on MIDI channel 1. No tempo or channel faders, no pads.

Deck

1 Jog
Top = touch scratch

2 SYNC

3 REV
Reverse play while held

4 CUE

5 PLAY/PAUSE

Centre

6 LOAD A
Loads the file chosen in the Media Explorer into this deck

7 BROWSE
Turn = how much the video effects act on the sound

8 Crossfader
Deck 1 on the left side, deck 2 on the right

Controls without a number (PITCH −% / +%, MASTER VOLUME, TREBLE, BASS, VOLUME, SCRATCH/SEARCH) are not used by ReaScratch. From Mixxx's mapping; not yet tested on the real hardware.

39 · 40 ION Discover DJ Pro deck 1 / deck 2

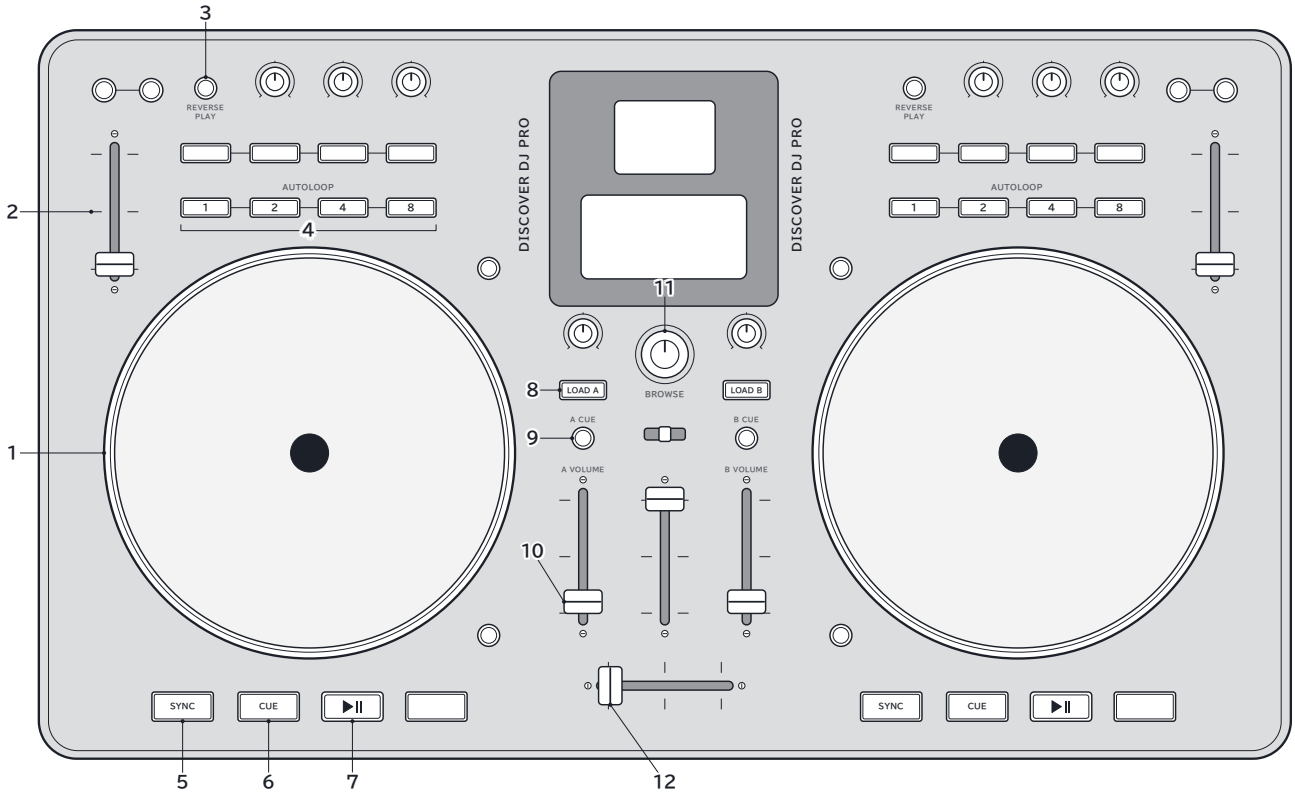


Figure 40 The Discover DJ Pro's controls (numbered on the left deck and the centre; the right deck is the same)

Both decks on MIDI channel 1. No pads.

Deck

1 Jog
Top scratch. Touch arrives as a CC rather than a note, so "touching" means while turning

2 Outer slider
Tempo (7-bit)

3 REVERSE PLAY
Each press turns reverse play on/off

4 AUTOLOOP buttons
Loop (toggles with each press) and reloop. Taken from the notes in Mixxx's mapping (deck 1: 0x53 / 0x55, deck 2: 0x56 / 0x58); which of the buttons 1, 2, 4 and 8 sends which is unconfirmed (in Mixxx 0x53 = loop in, 0x54 = loop out, 0x55 = reloop, 0x61 = manual loop toggle)

5 SYNC

6 CUE

7 PLAY/PAUSE

Centre

8 LOAD A
Loads the file chosen in the Media Explorer into this deck

9 A CUE
Listen to this deck on headphones (toggles with each press)

10 A VOLUME
That deck's volume and picture strength (channel fader)

11 BROWSE
Turn = how much the video effects act on the sound

12 Crossfader
Deck 1 on the left side, deck 2 on the right

Controls without a number (PITCH BEND, BASS, MID, TREBLE, MUTE, KEYLOCK, SCRATCH, STUTTER, CUE VOLUME, MASTER VOLUME, CUE SOURCE, iPod VOLUME) are not used by ReaScratch. From Mixxx's mapping; not yet tested on the real hardware.

41 · 42 Korg KAOSS DJ deck 1 / deck 2

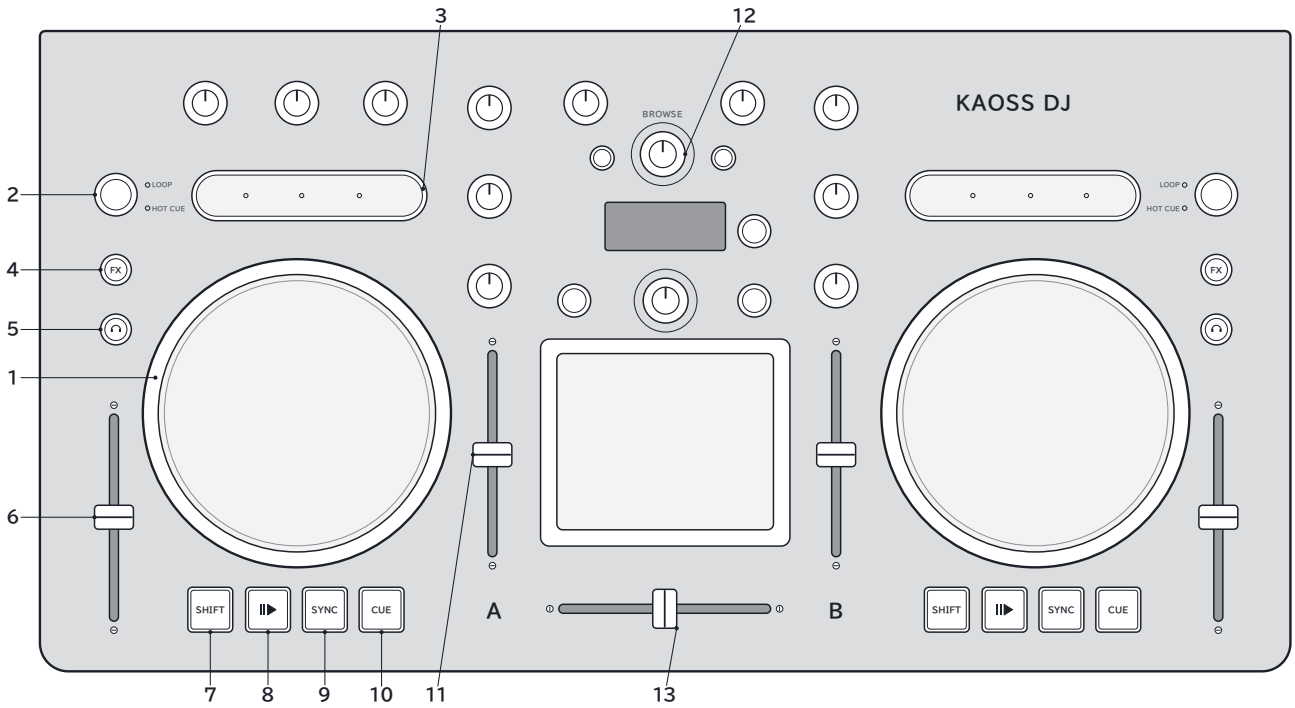


Figure 41 The KAOSS DJ's controls (numbered on the left deck and the centre; the right deck is the same)

Decks on ch 8 / ch 9, the crossfader and BROWSE on ch 7. From Mixxx's mapping (the main numbers also match Korg's official MIDI list; the numbers for SHIFT + touch slider are not in the official list, only in Mixxx's mapping).

Deck

1 **Touch wheel**
Touch and turn = touch scratch. What the controller sends while its scratch mode is off is read as the outer ring (nudge). The scratch mode switches when you touch the centre of the touch slider in its Normal mode (Korg's owner's manual). `SHIFT` + turn makes big jumps

2 **Touch slider mode button**
Switches the touch slider's mode (Normal, Loop, Hot Cue; inside the controller)

3 **Touch slider**
In Hot Cue mode, left / centre / right = 3 hot cues (`SHIFT` + touch clears). In Loop mode, left = halve the loop, centre = loop on/off (reloop with SHIFT), right = double

4 **FX**
That deck's video effect on/off

5 **Headphone CUE**
Listen to this deck on headphones (toggles with each press)

6 **Pitch fader**
Tempo (7-bit)

7 **SHIFT**
Hold it with another control for its second function

8 **PLAY/PAUSE**

9 **SYNC**

10 **CUE**

Mixer and centre

11 Level fader
That deck's volume and picture strength

12 BROWSE
Turn = how much the video effects act on the sound of the selected deck (usually deck A)

13 Crossfader
Deck 1 on the left side, deck 2 on the right

LOAD (ch8 14 / ch9 14) loads the file chosen in the Media Explorer into an empty sample slot. Controls without a number (headphone level, BALANCE, MASTER, GAIN, HI, MID, LO, the display, the touchpad and its mode button, the program/value knob, TAP, HOLD) are not used by ReaScratch. It is not known whether the message Korg's software sends at start-up is needed. Not yet tested on the real hardware.

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43 · 44 HID bridge (NI S2/S3/S4) deck 1 / deck 2

ReaScratch can only read MIDI, so a separate bridge program (the `HID bridge` folder in the zip; see "HID controllers (NI Traktor Kontrol S4, S3, S2)" (p. 197)) reads the HID data and sends MIDI with the DDJ-FLX4's numbers to a virtual MIDI port. The numbers come from Mixxx's mappings; not yet tested on the real hardware.

The drawings per model follow. On every model the left deck is deck 1 and the right deck is deck 2 (the S4's and S3's DECK SELECT C / D and the mixer's C / D columns are not used).

Traktor Kontrol S4 MK3

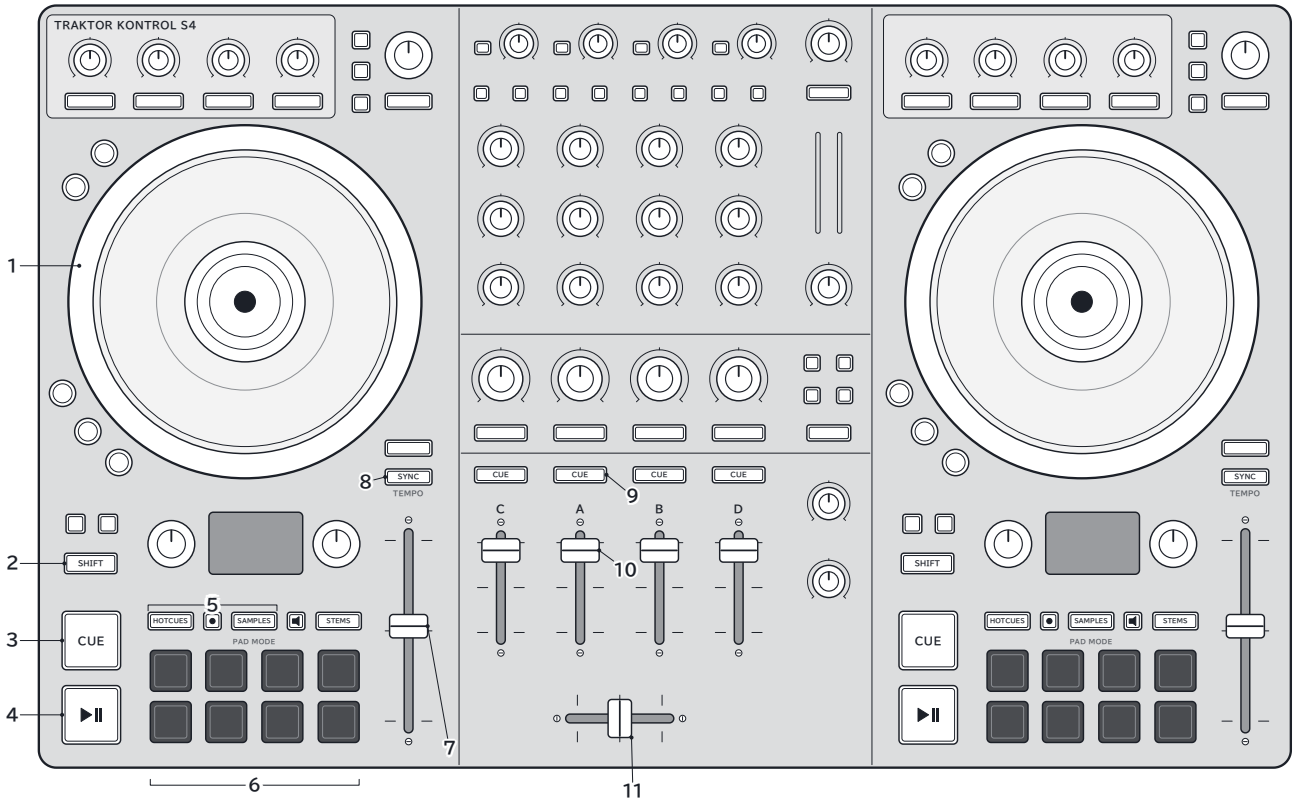


Figure 42 The Traktor Kontrol S4 MK3's controls (numbered on the left deck and the mixer's A column; the right deck and the B column are the same)

Deck

1 Jog
Touch the top and turn = touch scratch (the bridge program stops the jog motors at start-up)

2 SHIFT
Hold it with another control for its second function

3 CUE

4 PLAY/PAUSE

5 Pad mode buttons
`HOTCUES` = HOT CUE, `SAMPLES` → sample switching (the ● between them, mute and `STEMS` are not used)

6 Pads
The 8 pads of the current mode. In `HOTCUES`, SHIFT + pad clears a hot cue

7 TEMPO
Tempo fader

8 SYNC
SHIFT switches the tempo range

Mixer

9 Headphone CUE (A column)
Listen to this deck on headphones (toggles with each press)

10 Channel fader (A column)
That deck's volume and picture strength

11 Crossfader
Deck 1 on the left side, deck 2 on the right

Controls without a number (the FX unit, the browse controls, FLX, REV, GRID, TT, JOG, MASTER, DECK SELECT, MOVE, LOOP, EQ, MIXER FX, the C and D columns and so on) are not used by ReaScratch.

Traktor Kontrol S3

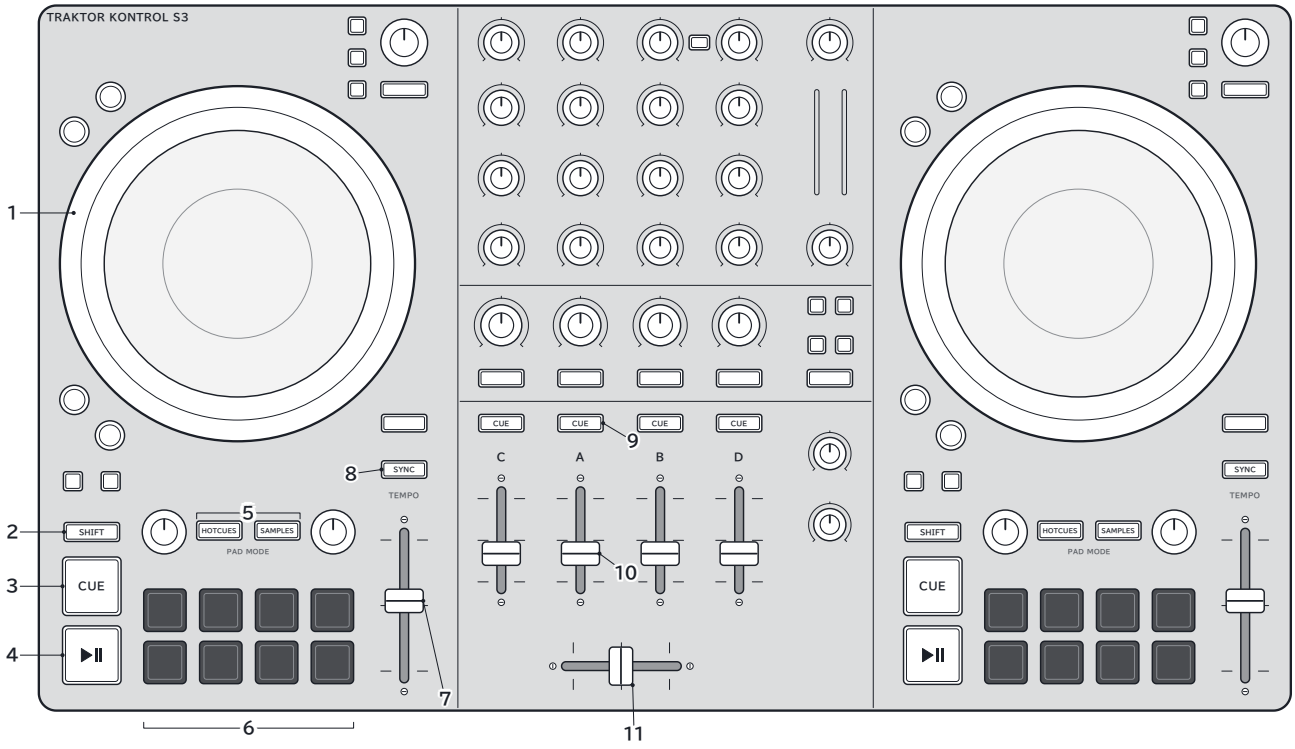


Figure 43 The Traktor Kontrol S3's controls (numbered on the left deck and the mixer's A column; the right deck and the B column are the same)

Deck

1 Jog
Touch the top and turn = touch scratch

2 SHIFT
Hold it with another control for its second function

3 CUE

4 PLAY/PAUSE

5 Pad mode buttons
`HOTCUES` = HOT CUE, `SAMPLES` → sample switching

6 Pads
The 8 pads of the current mode. In `HOTCUES`, SHIFT + pad clears a hot cue

7 TEMPO
Tempo fader

8 SYNC
SHIFT switches the tempo range

Mixer

9 Headphone CUE (A column)
Listen to this deck on headphones (toggles with each press)

10 Channel fader (A column)
That deck's volume and picture strength

11 Crossfader
Deck 1 on the left side, deck 2 on the right

Controls without a number (the browse controls, FLX, REV, GRID, JOG, KEYLOCK, DECK SELECT, MOVE, LOOP, EQ, MIXER FX, the C and D columns and so on) are not used by ReaScratch.

Traktor Kontrol S2 MK3

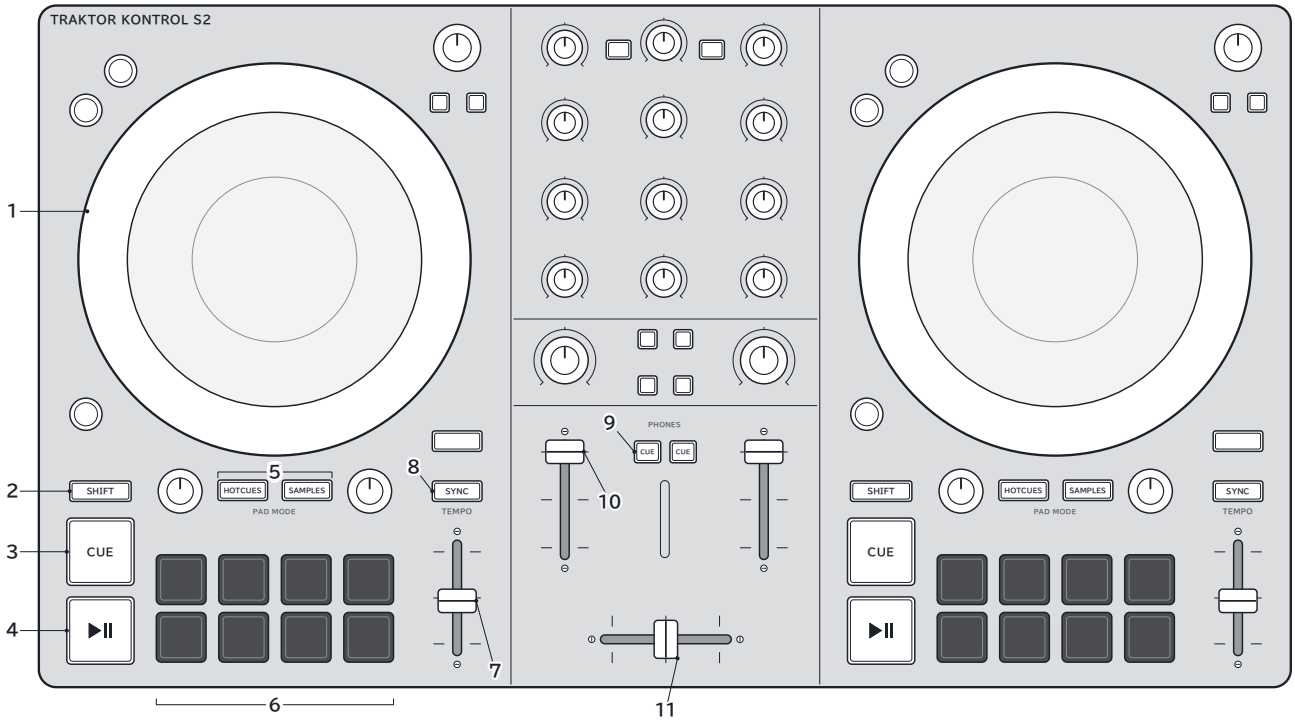


Figure 44 The Traktor Kontrol S2 MK3's controls (numbered on the left deck and the mixer's left side; the right deck is the same)

Deck

1 **Jog**
Touch the top and turn = touch scratch

2 **SHIFT**
Hold it with another control for its second function

3 **CUE**

4 **PLAY/PAUSE**

5 **Pad mode buttons**
`HOTCUES` = HOT CUE, `SAMPLES` → sample switching

6 **Pads**
The 8 pads of the current mode. In `HOTCUES`, SHIFT + pad clears a hot cue

7 **TEMPO**
Tempo fader

8 **SYNC**
SHIFT switches the tempo range

Mixer

9 **Headphone CUE (left)**
Listen to this deck on headphones (toggles with each press)

10 **Channel fader (left)**
That deck's volume and picture strength

11 **Crossfader**
Deck 1 on the left side, deck 2 on the right

Controls without a number (the browse controls, FLX, REV, GRID, KEYLOCK, MOVE, LOOP, EQ, the FX knobs, FX SELECT and so on) are not used by ReaScratch.

Traktor Kontrol S4 MK2

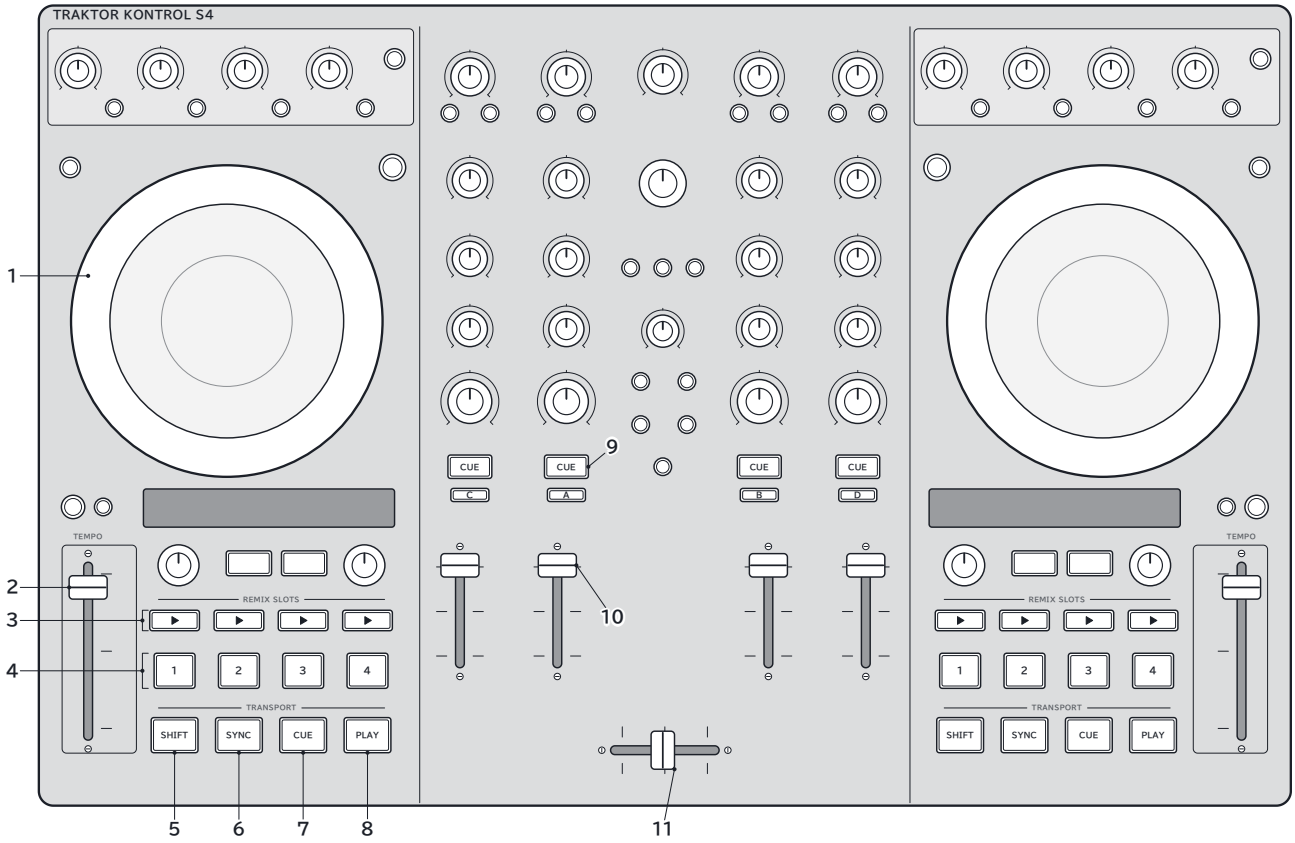


Figure 45 The Traktor Kontrol S4 MK2's controls (numbered on the left deck and the mixer's A column; the right deck and the B column are the same)

Deck

1 Jog
Touch the top and turn = touch scratch

2 TEMPO
Tempo fader

3 REMIX SLOTS row
4 buttons for sample switching (1–4)

4 HOTCUE row (1–4)
Hot cues. SHIFT + button clears a hot cue

5 SHIFT
Hold it with another control for its second function

6 SYNC
SHIFT switches the tempo range

7 CUE

8 PLAY

Mixer

9 Headphone CUE (A column)
Listen to this deck on headphones (toggles with each press)

10 Channel fader (A column)
That deck's volume and picture strength

11 Crossfader
Deck 1 on the left side, deck 2 on the right

Controls without a number (the FX unit, DECK C, LOAD, FLUX, RESET, the loop knobs, IN, OUT, EQ, FILTER, BROWSE, LOOP RECORDER, the C and D columns and so on) are not used by ReaScratch.

Traktor Kontrol S2 MK2 · S2 MK1

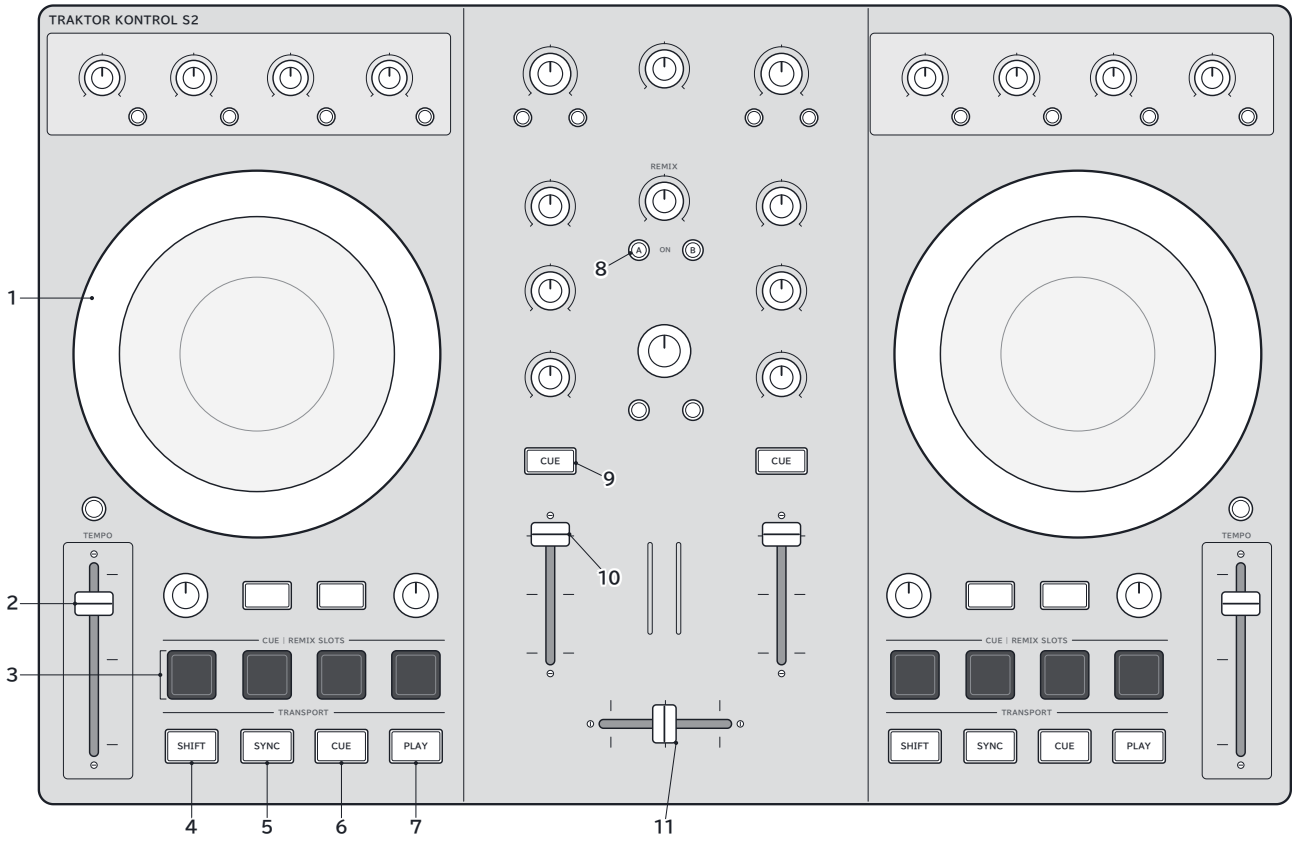


Figure 46 The Traktor Kontrol S2 MK2's controls (numbered on the left deck and the mixer's left side; the right deck is the same. The S2 MK1 is laid out almost the same)

Deck

1 Jog
Touch the top and turn = touch scratch (on the S2 MK1, how hard the top is pressed decides "touching"; see "HID controllers (NI Traktor Kontrol S4, S3, S2)" (p. 197))

2 TEMPO
Tempo fader

3 Pads (CUE | REMIX SLOTS 1–4)
The 4 pads of the current mode. In the hot cue mode, SHIFT + pad clears a hot cue

4 SHIFT
Hold it with another control for its second function

5 SYNC
SHIFT switches the tempo range

6 CUE

7 PLAY

8 **REMIX ON A**
Each press flips the left deck's pads between hot cues and sample switching (B for the right deck; on the S2 MK1, `SAMPLES` ON A / B)

9 **Headphone CUE (left)**
Listen to this deck on headphones (toggles with each press)

10 **Channel fader (left)**
That deck's volume and picture strength

11 **Crossfader**
Deck 1 on the left side, deck 2 on the right

Controls without a number (the FX unit, FLUX (RESET on the S2 MK1), the loop knobs, IN, OUT, GAIN | FILTER, EQ, the REMIX knob, BROWSE, LOAD and so on) are not used by ReaScratch.

45 · 46 Numark Party Mix deck 1 / deck 2

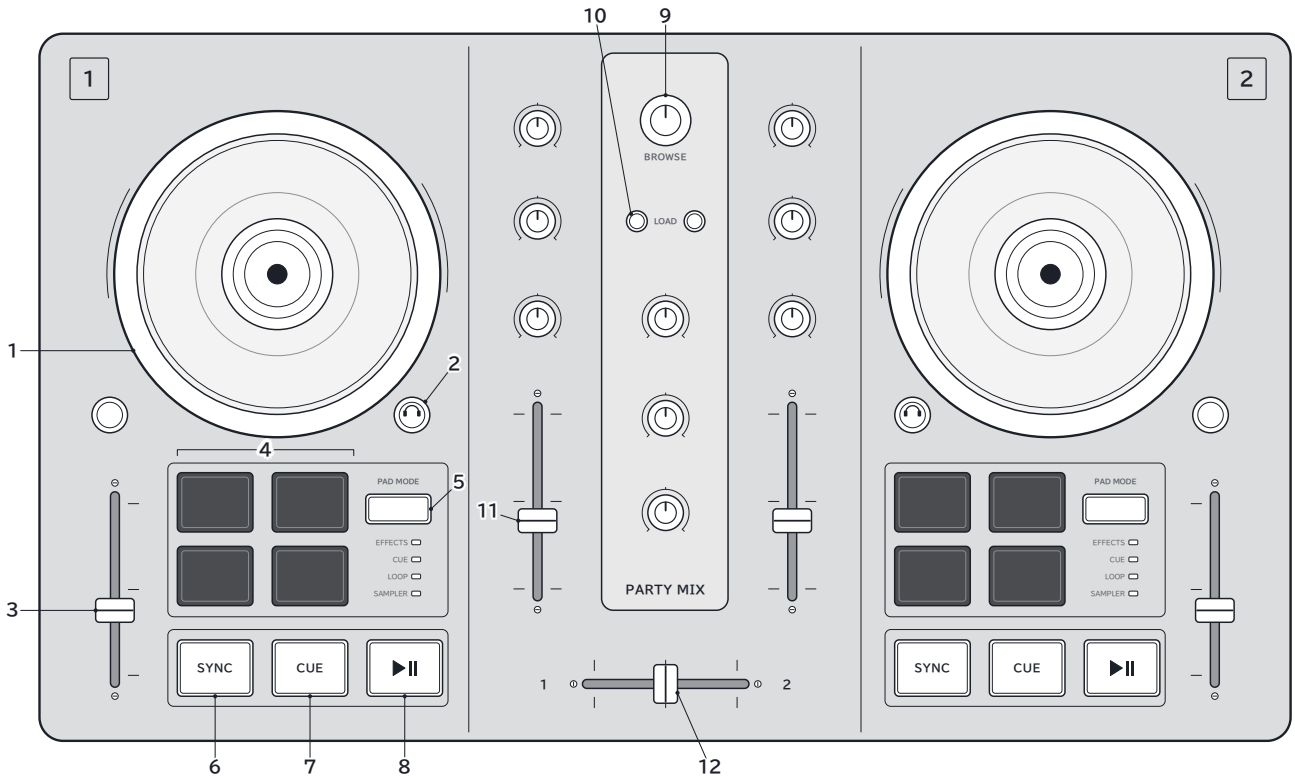


Figure 47 The original Numark Party Mix's controls (numbered on the left deck and the centre; the right deck has the same controls, with the jog's surroundings and the tempo fader mirrored)

Decks on ch 1 / ch 2, crossfader on ch 16. From Mixxx's mapping (its mode numbers differ from the Party Mix II's; two mappings made by owners of the original agree).

Deck

1 Jog
Top scratch. There is no touch signal, so "touching" means while turning (340 ticks per turn)

2 Headphone CUE (PFL)
Listen to this deck on headphones (toggles with each press)

3 Tempo fader
7-bit

4 Pads
The 4 pads of the current mode. Clear a hot cue by right-clicking its line on the waveform

5 PAD MODE
Each press changes the mode (Cue → HOT CUE, Loop → AUTO LOOP, Sampler → samples, Effects → TOUCH FX)

6 SYNC

7 CUE

8 PLAY/PAUSE

9 **BROWSE**
Turn = how much the video effects act on the sound, push = switch the deck the knob works on

10 **LOAD**
Same as the Reloop Buddy (left = deck 1, right = deck 2)

11 **Channel fader**
That deck's volume and picture strength

12 **Crossfader**
Deck 1 on the left side, deck 2 on the right

Controls without a number (SCRATCH, GAIN, TREBLE, BASS, MASTER GAIN, CUE MIX, CUE GAIN) are not used by ReaScratch. There is no SHIFT button. The message Mixxx sends to the unit at start-up is "send back the current knob and fader values", which is probably not needed to receive MIDI (since it is not sent, fader positions are not reflected until you move them). Not yet tested on the real hardware.

47 · 48 Numark Party Mix II/Live deck 1 / deck 2

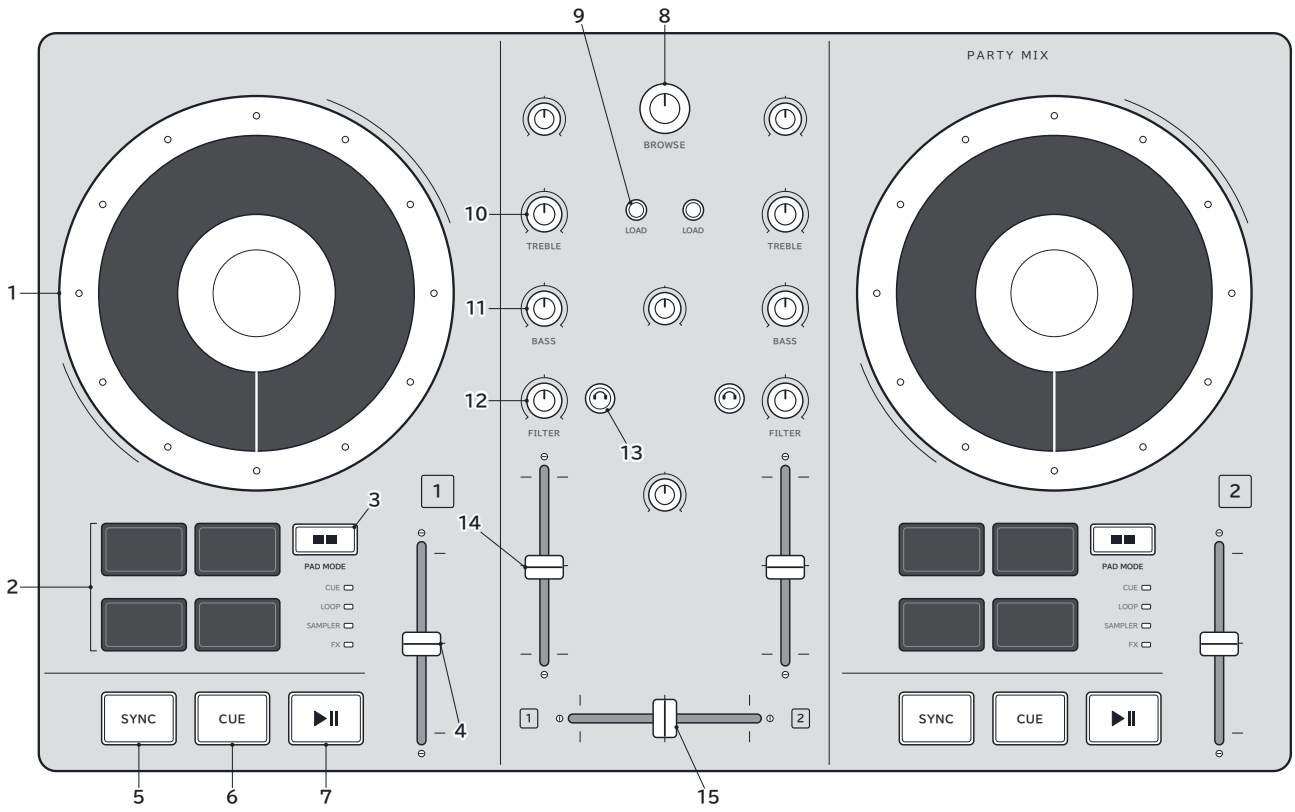


Figure 48 The Numark Party Mix II's controls (numbered on the left deck and the centre; the right deck is the same. The Party Mix Live adds a speaker on top, with the same controls)

Same numbers as the Mixtrack (the Party Mix Live is a II with speakers, assumed to use the same numbers). From Mixxx's mapping (a draft not yet merged).

Deck

1 Jog
Top = touch scratch (300 ticks per turn)

2 Pads
The 4 pads of the current mode

3 PAD MODE
Each press changes the mode (Cue → HOT CUE, Loop → AUTO LOOP, Sampler → samples, Effects → TOUCH FX)

4 Tempo fader
7-bit

5 SYNC

6 CUE

7 PLAY/PAUSE

8 **BROWSE**
Turn = how much the video effects act on the sound, push = switch the deck the knob works on

9 **LOAD**
Same as the Reloop Buddy (left = deck 1, right = deck 2)

10 **TREBLE**
Filter Morph (HIGH on other controllers)

11 **BASS**
Filter Resonance (LOW on other controllers)

12 **FILTER**
Filter Cutoff

13 **Headphone buttons**
Listen to this deck on headphones (toggles with each press; left = deck 1, right = deck 2)

14 **Channel fader**
That deck's volume and picture strength

15 **Crossfader**
Deck 1 on the left side, deck 2 on the right

Controls without a number (LEVEL, HEADPHONE, MAIN) are not used by ReaScratch. There is no SHIFT button or MID knob. Mixxx keeps sending messages to the unit while running, but that is for the unit's lights and is probably not needed to receive MIDI (from the discussion of the Mixxx mapping draft). Not yet tested on the real hardware.

49 · 50 Numark DJ2GO2 deck 1 / deck 2, 51 · 52 Numark DJ2GO2 Touch deck 1 / deck 2

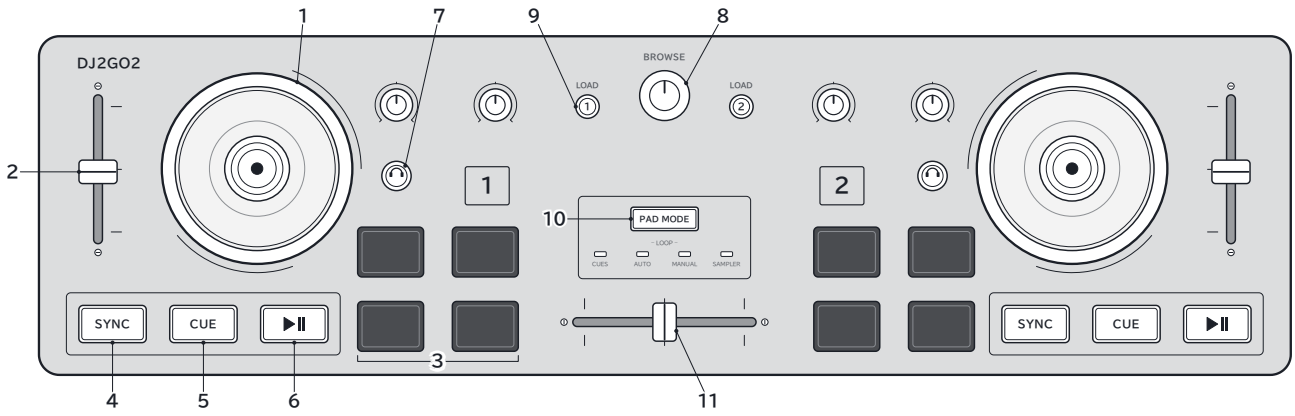


Figure 49 The Numark DJ2GO2's and DJ2GO2 Touch's controls (both laid out the same; numbered on the left deck and the centre; the right deck has the same controls, mirrored, except that SYNC / CUE / PLAY keep the same order. On the Touch, GAIN is printed LEVEL)

Decks on ch 1 / ch 2 (pads on ch 5 / ch 6), crossfader on ch 16. The Touch is from Mixxx's mapping, the DJ2GO2 from Numark's official Traktor mapping (Internet Archive).

Deck

1 Jog
Top scratch (by touch on the Touch; 236 ticks per turn). The DJ2GO2 (without touch) has no touch signal, so "touching" means while turning (its tick count is assumed to be the Touch's)

2 Tempo fader
7-bit

3 Pads
The 4 pads of the current mode. On the Touch, hold `PAD MODE` and press a pad to clear a hot cue

4 SYNC

5 CUE

6 PLAY/PAUSE

7 Headphone CUE (PFL)
Listen to this deck on headphones (toggles with each press)

Centre

8 BROWSE
Turn = how much the video effects act on the sound, push = switch the deck the knob works on

9 LOAD 1 · 2
Same as the Reloop Buddy (1 = deck 1, 2 = deck 2)

10 PAD MODE
The pad mode of both decks. Each press changes it (Cue → HOT CUE, Auto Loop → AUTO LOOP, Manual Loop → BOUNCE LOOP, Sampler → samples)

11 Crossfader
Deck 1 on the left side, deck 2 on the right

Controls without a number (GAIN, the Touch's LEVEL, MASTER GAIN, CUE GAIN) are not used by ReaScratch. No channel faders or FILTER knob (the Touch's knob in the FILTER position is gain and its MIDI is not used). There is no SHIFT button either. Not yet tested on the real hardware.

53 · 54 Numark Mixtrack deck 1 / deck 2

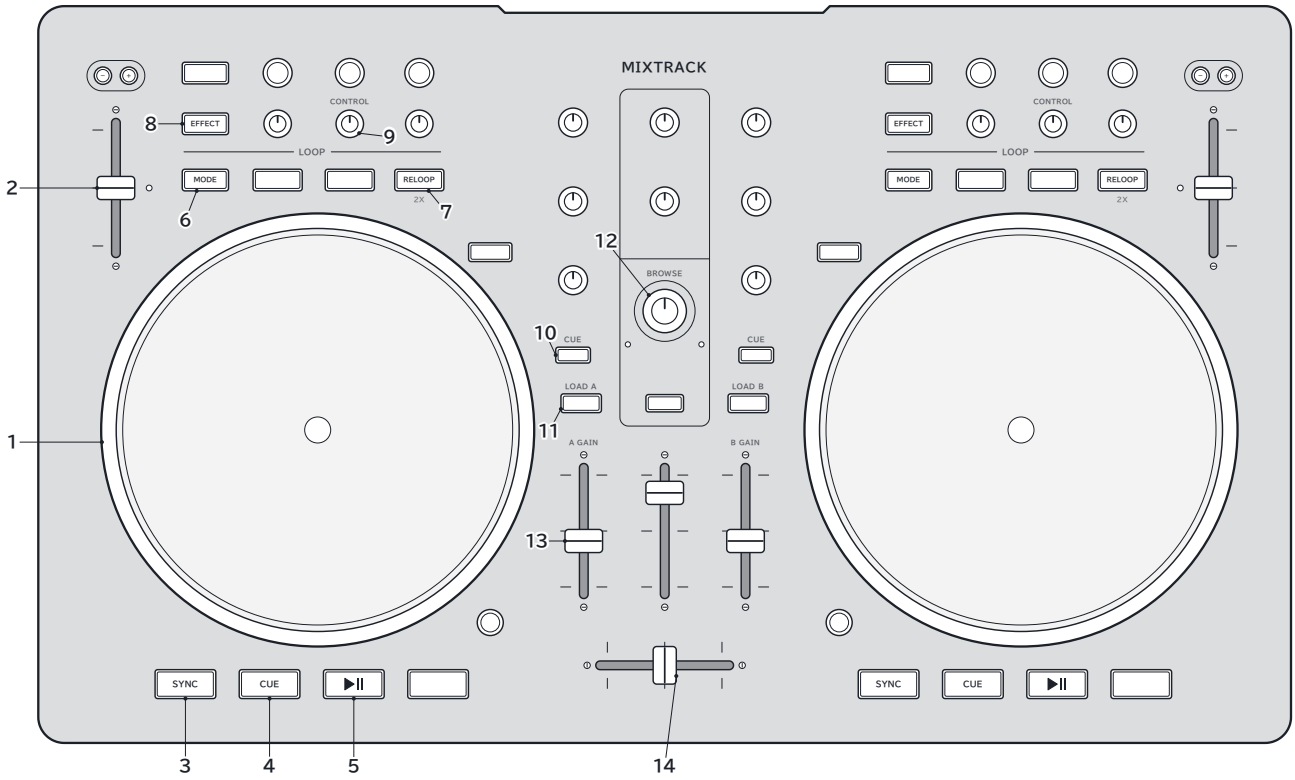


Figure 50 The Numark Mixtrack's (original) controls (numbered on the left deck and the centre; the right deck is the same)

Both decks on MIDI channel 1, told apart by number. No pads. From Mixxx's mapping.

Deck

1 Jog
Touch scratch (touch the top and turn; 512 ticks per turn)

2 PITCH fader
Tempo (7-bit)

3 SYNC

4 CUE

5 PLAY/PAUSE

6 MODE
(left end of the LOOP row): Loop

7 RELOOP
(2X): Reloop

FX (Buddy-style)

8 EFFECT
The Buddy's FX ON bar (toggles with each press)

9 CONTROL knob
The Buddy's FX knob (effect amount). Mixxx's mapping gives it the Flanger's LFO DEPTH number (CC 0x1C); both decks' presets read this knob on the left deck. The original's sources do not say which of the three knobs has this number; from the Mixtrack Pro's mapping notes (the same numbers 0x1B–0x1D, 0x1B = the first knob), it is taken to be the second knob, the first CONTROL

Note (9): What it acts on depends on what you hold (as in the Reloop Buddy's note). While EFFECT is on = that deck's effect amount; nothing held = edge softness. There is no SHIFT and no button for ◀ ▶.

Mixer

10 Headphone CUE
Listen to this deck on headphones (toggles with each press)

11 LOAD A / B
Load the file selected in the Media Explorer into deck A / B (as on the Reloop Buddy)

12 BROWSE
Turn = the headphone Cue mix (while EFFECT is on, that effect's sound amount)

13 Channel fader
That deck's volume and picture strength

14 Crossfader
Deck 1 on the left side, deck 2 on the right

Controls without a number (PITCH BEND, KEYLOCK, SCRATCH, STUTTER, VIEW, TICK, EQ KILL, SELECT, IN, OUT, TREBLE, MID, BASS, CUE GAIN, CUE MIX, BACK, MASTER GAIN) are not used by ReaScratch. Not yet tested on the real hardware.

55 · 56 Numark Mixtrack Pro deck 1 / deck 2

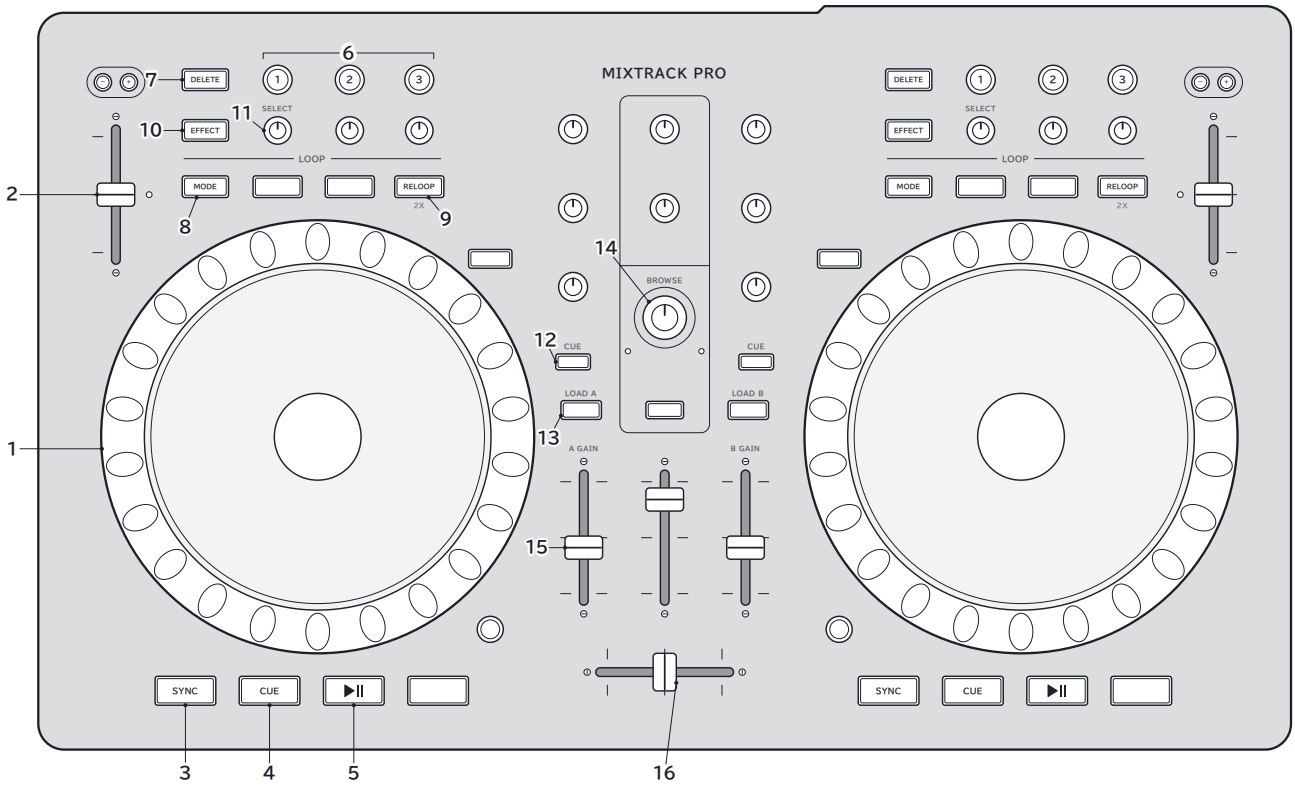


Figure 51 The Numark Mixtrack Pro's (original) controls (numbered on the left deck and the centre; the right deck is the same)

Almost the same numbers as the original Mixtrack, both decks on MIDI channel 1, with 3 hot cue buttons. From Mixxx's mapping.

Deck

1 Jog
Touch scratch (touch the top and turn; 600 ticks per turn)

2 PITCH fader
Tempo (7-bit)

3 SYNC

4 CUE

5 PLAY/PAUSE

6 HOT CUE 1–3
Hot cues

7 DELETE
Hold it and press a HOT CUE button to clear that hot cue

8 MODE
(left end of the LOOP row): Loop

9 RELOOP
(2X): Reloop

FX (Buddy-style)

10 EFFECT
The Buddy's FX ON bar (toggles with each press)

11 SELECT knob
The Buddy's FX knob (that deck's effect amount). Mixxx's mapping gives it the LFO DEPTH number (deck 1 = CC 0x1B, deck 2 = 0x1E). Mixxx's mapping notes say the first knob (SELECT) has this number

Note (11): What it acts on depends on what you hold (as in the Reloop Buddy's note). While EFFECT is on = that deck's effect amount; nothing held = edge softness. There is no SHIFT and no button for ◂ ▸.

Mixer

12 Headphone CUE
Listen to this deck on headphones (toggles with each press)

13 LOAD A / B
Load the file selected in the Media Explorer into deck A / B (as on the Reloop Buddy)

14 BROWSE
Turn = the headphone Cue mix (while EFFECT is on, that effect's sound amount)

15 Channel fader
That deck's volume and picture strength

16 Crossfader
Deck 1 on the left side, deck 2 on the right

Controls without a number (PITCH BEND, KEYLOCK, SCRATCH, STUTTER, CONTROL, IN, OUT, TREBLE, MID, BASS, CUE GAIN, CUE MIX, BACK, MASTER GAIN) are not used by ReaScratch. Not yet tested on the real hardware.

57 · 58 Numark Mixtrack 2 deck 1 / deck 2

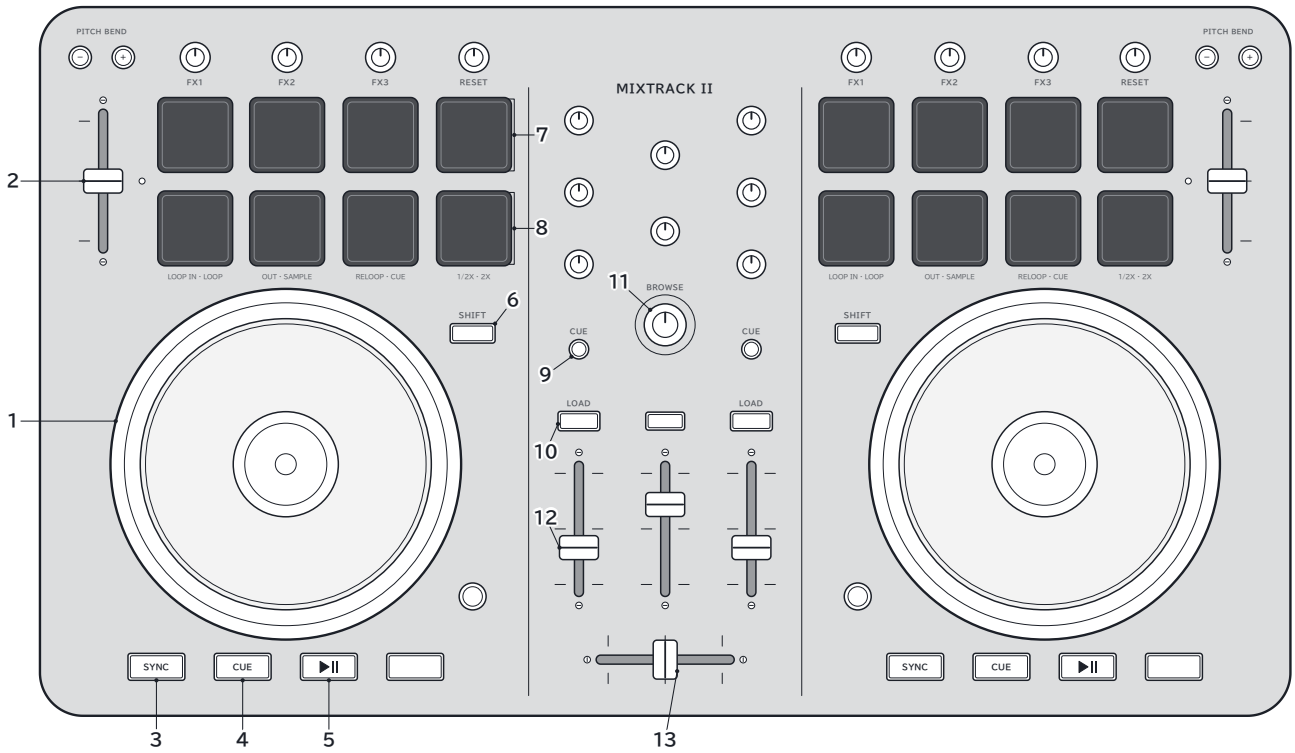


Figure 52 The Numark Mixtrack II's and Mixtrack Pro II's controls (the same layout; numbered on the left deck and the centre; the right deck is the same. The Pro II has BEATS where FILTER is and TAP where RESET is)

Both decks on MIDI channel 1, told apart by number. From Mixxx's mapping (MixTrack (Pro) II).

Deck

1. **Jog**
   Touch scratch (touch the top and turn; 600 ticks per turn)
2. **PITCH fader**
   Tempo (7-bit)
3. **SYNC**
4. **CUE**
5. **PLAY/PAUSE**
6. **SHIFT**
   Set to clear a hot cue when held while pressing its pad (on the unit, SHIFT + a bottom-row pad also switches the pad mode)
7. **Top row of pads**
   (FX, Buddy-style): `FX1` = the Buddy's FX ON bar (toggles with each press), `FX2` / `FX3` = the Buddy's `FX PARAM` ◂ ▸ (effect type). `RESET` (`TAP` on the Pro II) is not used
8. **Bottom row of pads**
   What they do depends on the unit's pad mode. LOOP mode = `LOOP IN` loops, `RELOOP` reloops, `1/2X` halves the loop. CUE mode = the left three are hot cues 1–3

9 **Headphone CUE**
Listen to this deck on headphones (toggles with each press)

10 **LOAD**
Load the file selected in the Media Explorer into this deck (as on the Reloop Buddy)

11 **BROWSE**
Turn = the headphone Cue mix (while FX1 is on, that effect's sound amount)

12 **Channel fader**
That deck's volume and picture strength. The numbers follow Mixxx's mapping (deck 1 = CC 0x16, deck 2 = 0x07), but their order differs from the other models, so they are doubtful

13 **Crossfader**
Deck 1 on the left side, deck 2 on the right

Controls without a number (PITCH BEND, SCRATCH, STUTTER, SELECT, FILTER (BEATS), the bottom row's `OUT`, the SAMPLE mode, TREBLE, MID, BASS, CUE GAIN, CUE MIX, BACK, MASTER) are not used by ReaScratch. The FX knobs (SELECT) are endless and not read. Not yet tested on the real hardware.

59 · 60 Numark Mixtrack (Pro) 3 deck 1 / deck 2

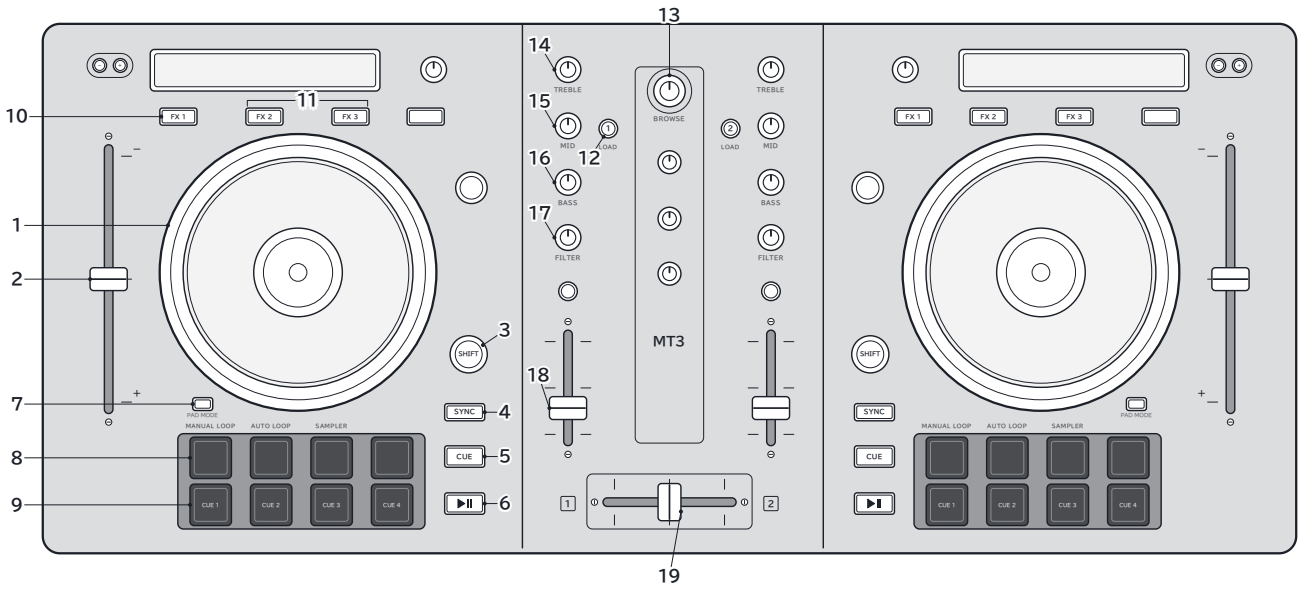


Figure 53 The Numark Mixtrack 3's and Mixtrack Pro 3's controls (the same layout; numbered on the left deck and the centre; the right deck is the same)

Decks on ch 2 / ch 3, crossfader on ch 1. From Mixxx's mapping (the Mixtrack 3 and Pro 3 use the same numbers).

Deck

1 Jog
Touch scratch (touch the top and turn; 1200 ticks per turn)

2 PITCH fader
Tempo (14-bit)

3 SHIFT
Hold it with another control for its second function

4 SYNC

5 CUE

6 PLAY/PAUSE

7 PAD MODE
Switches the top row's pad mode (MANUAL LOOP, AUTO LOOP, SAMPLER)

8 Top row of pads
MANUAL LOOP = `IN` loops, `OFF` reloops, `1/2` halves the loop. AUTO LOOP → AUTO LOOP, SAMPLER → sample switching

9 Bottom row of pads
(CUE 1–4): Hot cues (SHIFT + pad clears)

FX (Buddy-style)

10 FX 1
The Buddy's FX ON bar (toggles with each press)

11 FX 2 / FX 3
The Buddy's `FX PARAM` ◂ ▸ (effect type)

12 **LOAD 1 / 2**
Load the file selected in the Media Explorer into deck 1 / deck 2 (as on the Reloop Buddy)

13 **BROWSE**
Turn = the headphone Cue mix (while FX 1 is on, that effect's sound amount), push = switch decks

14 **TREBLE**
Filter Morph (echo length with SHIFT)

15 **MID**
How much the video effects act on the sound

16 **BASS**
Filter Resonance (echo feedback with SHIFT)

17 **FILTER**
Filter Cutoff (key with SHIFT)

18 **Channel fader**
That deck's volume and picture strength

19 **Crossfader**
Deck 1 on the left side, deck 2 on the right

Controls without a number (PITCH BEND, the touch strip, BEATS, TAP, WHEEL, the headphone CUE, MASTER GAIN, CUE MIX, CUE GAIN) are not used by ReaScratch. The BEATS knobs are endless and not read. Headphone CUE arrives on channel 1 and cannot be read. There is no knob for the FX knob. Not yet tested on the real hardware.

61 · 62 Numark Mixtrack Platinum deck 1 / deck 2

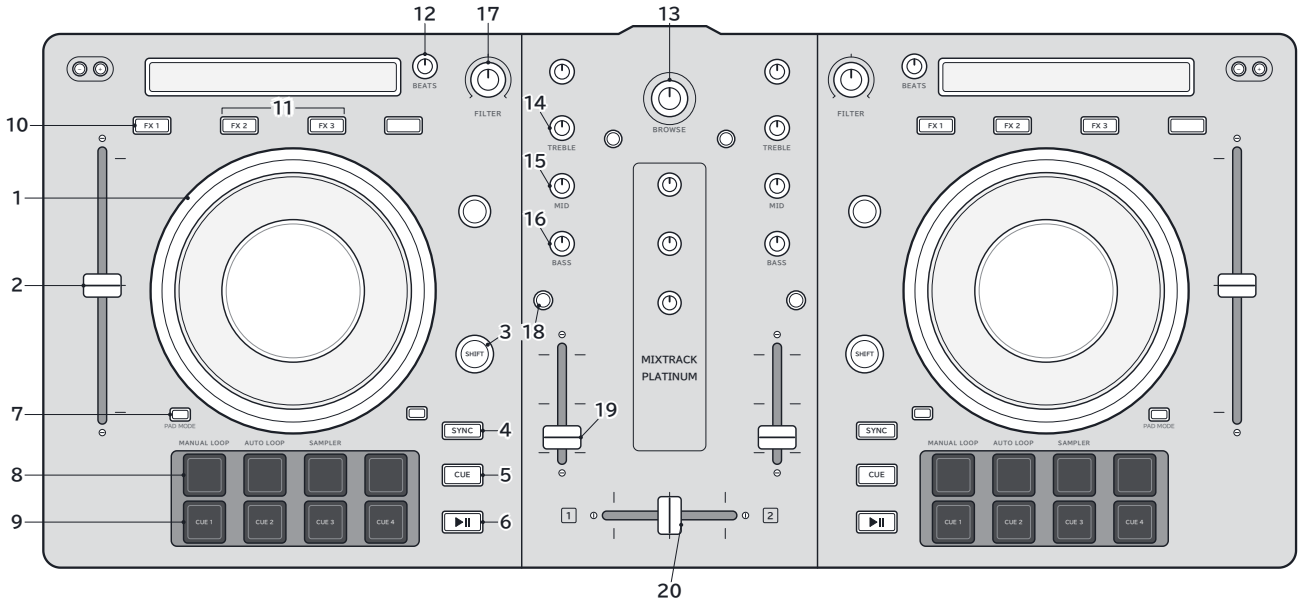


Figure 54 The Numark Mixtrack Platinum's controls (numbered on the left deck and the centre; the right deck is the same)

Almost the same numbers as the Mixtrack Pro/Platinum FX; the hot cue pad numbers and the ticks per turn differ.

Deck

1 Jog
Touch scratch (touch the top and turn; 1240 ticks per turn)

2 PITCH fader
Tempo. Only the upper 7 bits are read (the low-byte number in Mixxx's mapping is doubtful)

3 SHIFT
Hold it with another control for its second function

4 SYNC
SHIFT switches the tempo range

5 CUE

6 PLAY/PAUSE

7 PAD MODE
Hold it and press a top-row pad to choose the top row's mode

8 Top row of pads
AUTO LOOP → AUTO LOOP. In MANUAL LOOP the third pad (`OFF`) loops. SAMPLER is not used (the sample buttons are shared by both decks)

9 Bottom row of pads
(CUE 1–4): Hot cues (SHIFT + pad clears)

FX (Buddy-style)

10 FX 1
The Buddy's FX ON bar (toggles with each press)

11 FX 2 / FX 3
The Buddy's `FX PARAM` ◂ ▸ (effect type; see the note below)

12 BEATS
The Buddy's FX knob (effect amount; see the note below). Mixxx's mapping has it as Dry/Wet (CC 0x03)

Note (11 · 12): What they act on depends on what you hold (as in the Reloop Buddy's note). While FX 1 is on = that deck's effect amount and type, SHIFT = the amount and type after the mix. Nothing held = edge softness

and transition.

Mixer

13 **BROWSE**
Turn = the headphone Cue mix (while FX 1 is on, that effect's sound amount), push = switch decks

14 **TREBLE**
Filter Morph (echo length with SHIFT)

15 **MID**
How much the video effects act on the sound

16 **BASS**
Filter Resonance (echo feedback with SHIFT)

17 **FILTER**
Filter Cutoff (key with SHIFT)

18 **Headphone CUE**
Listen to this deck on headphones (toggles with each press)

19 **Channel fader**
That deck's volume and picture strength

20 **Crossfader**
Deck 1 on the left side, deck 2 on the right

LOAD (ch1 16 / ch2 16) loads the file chosen in the Media Explorer into an empty sample slot. Hold PAD MODE (7) and press LOAD to preview (the same as the `P` key). Controls without a number (PITCH BEND, the touch strip, TAP, WHEEL, LAYER, GAIN, MASTER GAIN, CUE MIX, CUE GAIN) are not used by ReaScratch. Not yet tested on the real hardware.

63 · 64 Numark Mixtrack Pro/Platinum FX deck 1 / deck 2

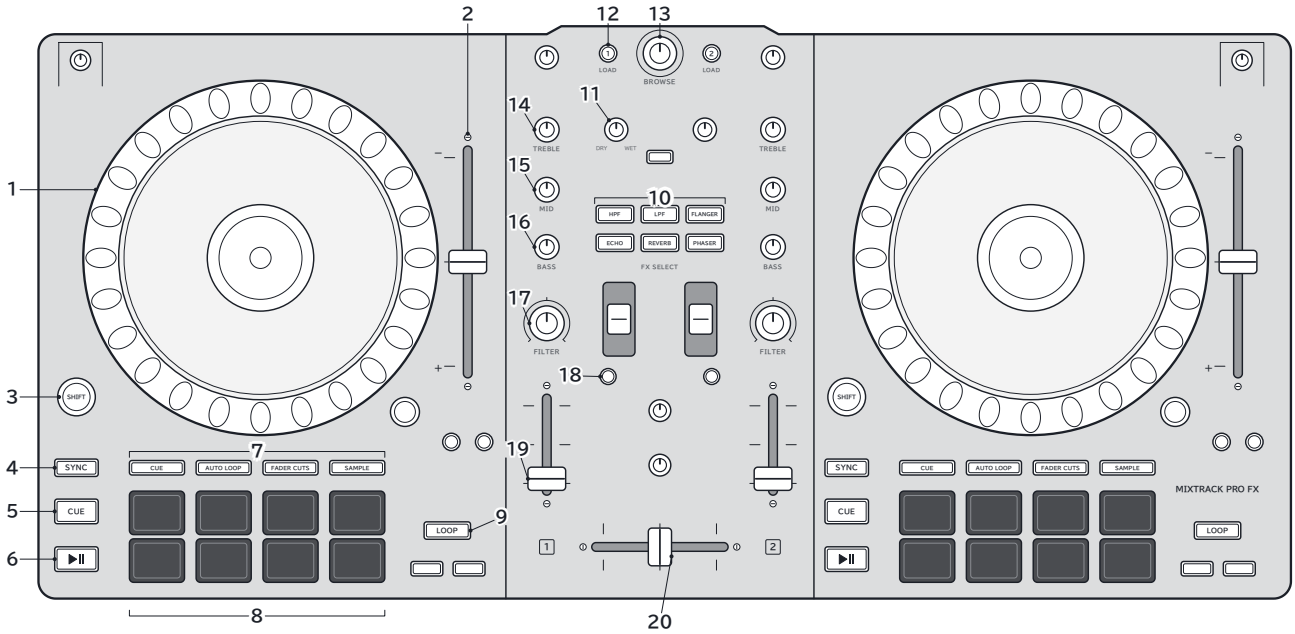


Figure 55 The Numark Mixtrack Pro FX's and Mixtrack Platinum FX's controls (the same layout; numbered on the left deck and the centre; the right deck is the same. The Platinum FX has a display inside the jog)

Decks on ch 1 / ch 2, crossfader on ch 16. From Mixxx's mapping.

Deck

1 Jog
Touch scratch (touch the top and turn; 1024 ticks per turn)

2 PITCH fader
Tempo

3 SHIFT
Hold it with another control for its second function

4 SYNC
SHIFT switches the tempo range

5 CUE

6 PLAY/PAUSE

7 Pad mode buttons
`CUE` = HOT CUE, `AUTO LOOP` = AUTO LOOP, `SAMPLE` → sample switching. With SHIFT held, `BEATJUMP` → SLICER, `SAMPLE2` → TOUCH FX. `FADER CUTS` is not used

8 Pads
The pads of the current mode. In HOT CUE, SHIFT + pad clears a hot cue

9 LOOP
Loop

FX (Buddy-style)

10 The six FX SELECT buttons
Deck 1 reads the three on ch 9, deck 2 the three on ch 10. Of each three, the first = the Buddy's FX ON bar, the second and third = the Buddy's `FX PARAM` ◂ ▸ (effect type; see the note below). Which three are on ch 9 has not been confirmed in the sources

11 DRY/WET
The Buddy's FX knob (effect amount; see the note below). Deck 1's preset only

Note (10 · 11): What they act on depends on what you hold (as in the Reloop Buddy's note). While FX ON is on = that deck's effect amount and type, SHIFT = the amount and type after the mix. Nothing held = edge softness

and transition.

Mixer

12 LOAD 1 / 2
Load the file selected in the Media Explorer into deck 1 / deck 2 (as on the Reloop Buddy)

13 BROWSE
Turn = the headphone Cue mix (while FX ON is on, that effect's sound amount), push = switch decks

14 TREBLE
Filter Morph (echo length with SHIFT)

15 MID
How much the video effects act on the sound

16 BASS
Filter Resonance (echo feedback with SHIFT)

17 FILTER
Filter Cutoff (key with SHIFT)

18 Headphone CUE
Listen to this deck on headphones (toggles with each press)

19 Channel fader
That deck's volume and picture strength

20 Crossfader
Deck 1 on the left side, deck 2 on the right

Controls without a number (SCRATCH, PITCH BEND, LEVEL, BEATS, TAP, the FX paddles, the loop 1/2 and x2, MIC, MASTER, CUE GAIN, CUE MIX) are not used by ReaScratch. Not yet tested on the real hardware.

65 · 66 DDJ-FLX2 deck 1 / deck 2

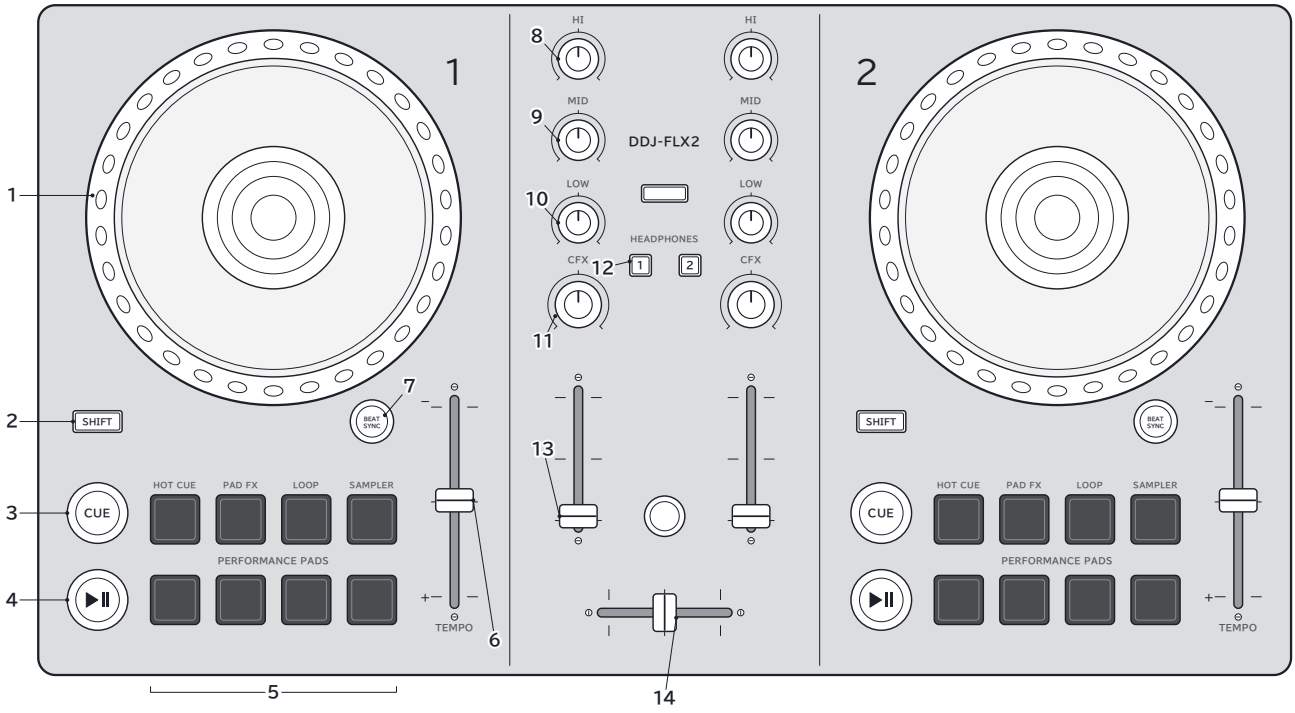


Figure 56 The DDJ-FLX2's controls (numbered on the left deck and the centre; the right deck is the same)

Same as the DDJ-FLX4 (same MIDI numbers). There are no loop, BEAT FX, BROWSE or LOAD buttons.

Deck

1 Jog
Top = touch scratch, outer ring = nudge (1/4 speed). `SHIFT` + jog makes big jumps. There is no VINYL button (the controller is in VINYL mode from the moment it is connected). The jog is smaller, so the tick count (`Jog ticks per second at 1x`) is `256` (calculated as 460 ticks per turn)

2 SHIFT
Hold it with another control for its second function

3 CUE

4 PLAY/PAUSE

5 Pads
The 8 pads of the current mode. To change the mode, press `SHIFT` + `BEAT SYNC`, then the pad with that label (`HOT CUE` = HOT CUE, `PAD FX` → TOUCH FX, `LOOP` → AUTO LOOP, `SAMPLER` → sample switching). In `HOT CUE`, SHIFT + pad clears a hot cue

6 TEMPO
Tempo slider

7 BEAT SYNC
`SHIFT` + `BEAT SYNC` selects the controller's pad mode, so it does not switch the tempo range (use `Shift` + `Y` on the keyboard)

Mixer

8 **HI**
Filter Morph (echo length with SHIFT)

9 **MID**
How much the video effects act on the sound

10 **LOW**
Filter Resonance (echo feedback with SHIFT)

11 **CFX**
Filter Cutoff (key with SHIFT)

12 **HEADPHONES 1 · 2**
Listen to this deck on headphones (toggles with each press)

13 **Channel fader**
That deck's volume and picture strength

14 **Crossfader**
Deck 1 on the left side, deck 2 on the right

The MIDI numbers are taken from AlphaTheta's official MIDI message list; not yet tested on the real hardware.

67 · 68 DDJ-FLX4 deck 1 / deck 2

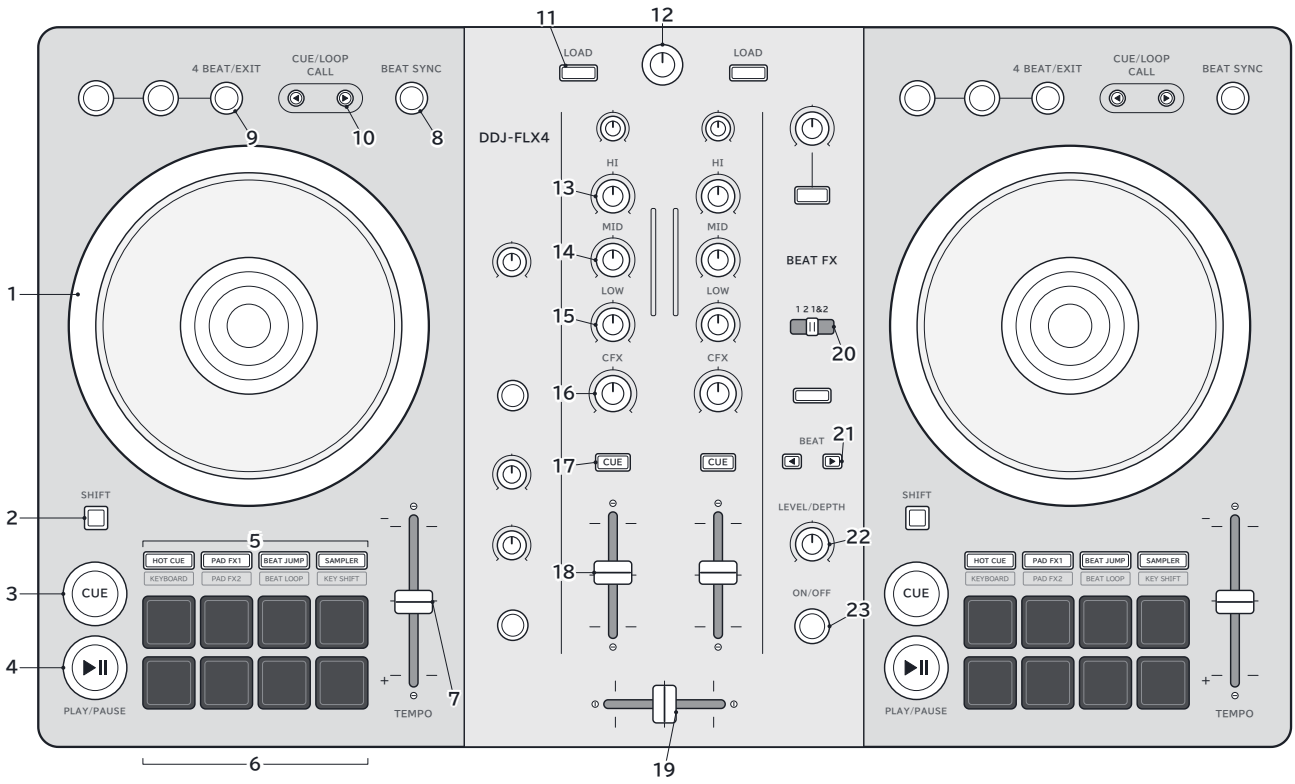


Figure 57 The DDJ-FLX4's controls (numbered on the left deck and the centre; the right deck is the same)

Deck

1 Jog
Top = touch scratch (switch the controller's VINYL on), outer ring = nudge (1/4 speed). `SHIFT` + jog makes big jumps

2 SHIFT
Hold it with another control for its second function

3 CUE

4 PLAY/PAUSE

5 Pad mode buttons
`HOT CUE` = HOT CUE, `PAD FX1` → TOUCH FX, `BEAT JUMP` → SLICER, `SAMPLER` → sample switching. With SHIFT held, `KEYBOARD` → PITCH PLAY, `PAD FX2` → BOUNCE LOOP, `BEAT LOOP` → AUTO LOOP, `KEY SHIFT` → NEURAL MIX

6 Pads
The 8 pads of the current mode. In `HOT CUE`, SHIFT + pad clears a hot cue

7 TEMPO
Tempo slider

8 BEAT SYNC
SHIFT switches the tempo range

9 4 BEAT/EXIT
Loop (reloop with SHIFT)

10 CUE/LOOP CALL ◂ ▸
Halve / double the loop

Mixer

11 LOAD
Same as the Reloop Buddy (SHIFT + LOAD replaces the current sample)

12 BROWSE
Turn = the headphone Cue mix (while BEAT FX is on, that effect's sound amount), push = switch decks

13 HI
Filter Morph (echo length with SHIFT)

14 MID
How much the video effects act on the sound

15 LOW
Filter Resonance (echo feedback with SHIFT)

16 CFX
Filter Cutoff (key with SHIFT)

17 Headphone CUE
Listen to this deck on headphones (toggles with each press; SHIFT hides only the picture)

18 Channel fader
That deck's volume and picture strength

19 Crossfader
Deck 1 on the left side, deck 2 on the right

BEAT FX (Buddy-style)

20 CH SELECT (1 · 2 · 1&2)
Which deck BEAT FX applies to

21 BEAT ◂ ▸
The Buddy's `FX PARAM` ◂ ▸ (effect type; see the note below)

22 LEVEL/DEPTH
The Buddy's FX knob (effect amount; see the note below)

23 ON/OFF
The Buddy's FX ON bar (toggles with each press; SHIFT + ON/OFF turns all effects off while held)

Note (21 · 22): What they act on depends on what you hold (as in the Reloop Buddy's note). While ON/OFF is on = that deck's effect amount and type, SHIFT = the amount and type after the mix. Nothing held = edge softness and transition, holding a pad mode button = the video blend.

Controls without a number (IN, OUT, TRIM, MIC LEVEL, SMART CFX, SMART FADER, MASTER LEVEL, FX SELECT and so on) are not used by ReaScratch. Tested on real hardware only early in development, not yet with current features (⚠ until tested again).

69 · 70 DDJ-FLX6 deck 1 / deck 2

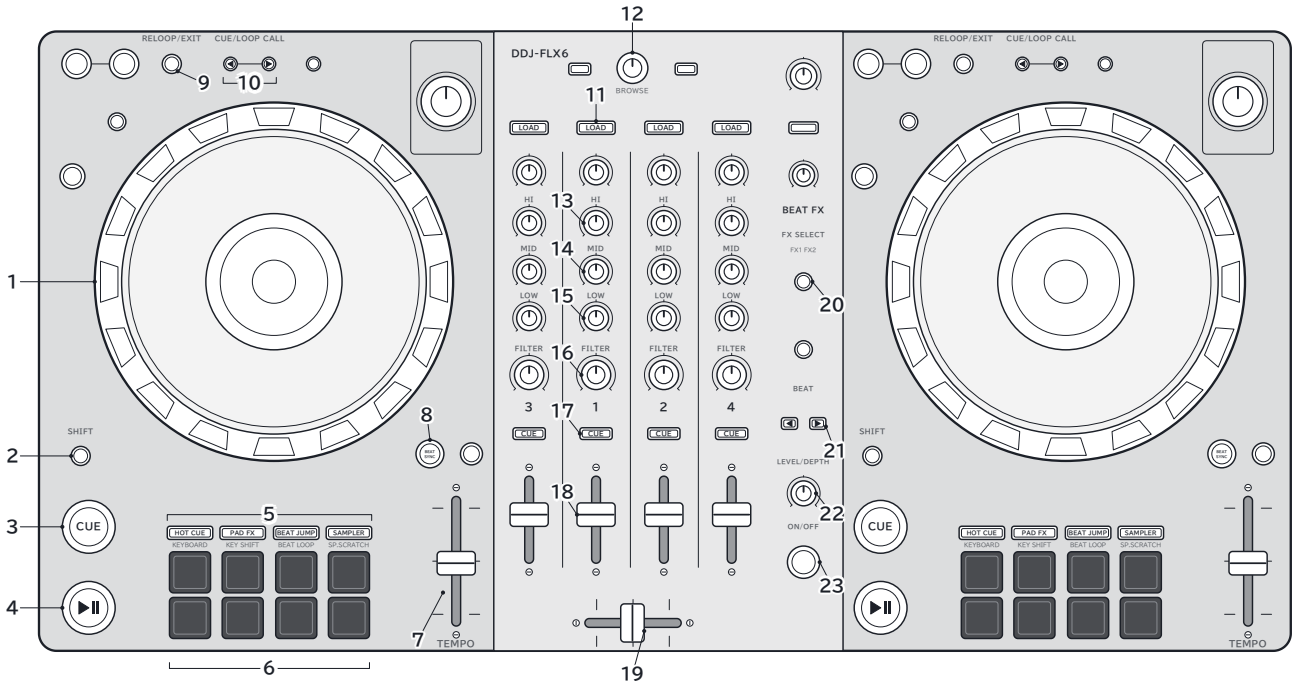


Figure 58 The DDJ-FLX6's controls (numbered on the left deck, mixer channel 1 and BEAT FX; the right deck is the same)

Almost the same as the DDJ-FLX4. The mixer has 4 channels (3, 1, 2, 4 from the left); deck 1 uses channel 1 and deck 2 channel 2.

Deck

1 Jog
Top = touch scratch (switch the controller's VINYL on), outer ring = nudge (1/4 speed). `SHIFT` + jog makes big jumps. The manufacturer does not publish the tick count (`Jog ticks per second at 1x`), so it is `4000`, from Mixxx's mapping (if the jog is too fast or too slow, change this value)

2 SHIFT
Hold it with another control for its second function

3 CUE

4 PLAY/PAUSE

5 Pad mode buttons
`HOT CUE` = HOT CUE, `PAD FX` → TOUCH FX, `BEAT JUMP` → SLICER, `SAMPLER` → sample switching. With SHIFT held, `KEYBOARD` → PITCH PLAY, `KEY SHIFT` → BOUNCE LOOP, `BEAT LOOP` → AUTO LOOP, `SP. SCRATCH` → NEURAL MIX (`KEY SHIFT` and `SP. SCRATCH` sit where the FLX4 has `PAD FX2` and `KEY SHIFT`)

6 Pads
The 8 pads of the current mode. In `HOT CUE`, SHIFT + pad clears a hot cue

7 TEMPO
Tempo slider

8 BEAT SYNC
SHIFT switches the tempo range (note 93)

9 RELOOP/EXIT
Loop (reloop with SHIFT; the same number as the FLX4's 4 BEAT/EXIT)

10 CUE/LOOP CALL ◂ ▸
Halve / double the loop

Mixer

11 LOAD
(above channel 1): Same as the Reloop Buddy (SHIFT + LOAD replaces the current sample)

12 BROWSE
Turn = the headphone Cue mix (while BEAT FX is on, that effect's sound amount), push = switch decks

13 HI
Filter Morph (echo length with SHIFT)

14 MID
How much the video effects act on the sound

15 LOW
Filter Resonance (echo feedback with SHIFT)

16 FILTER
Filter Cutoff (key with SHIFT; the FLX4's CFX)

17 Headphone CUE
Listen to this deck on headphones (toggles with each press; SHIFT hides only the picture)

18 Channel fader
That deck's volume and picture strength

19 Crossfader
Deck 1 on the left side, deck 2 on the right

BEAT FX (Buddy-style)

20 FX SELECT
Which deck BEAT FX applies to (the first effect of FX1 = deck 1, of FX2 = deck 2; it does not work on the second and third effects)

21 BEAT ◂ ▸
The Buddy's `FX PARAM` ◂ ▸ (effect type; as in the DDJ-FLX4's note)

22 LEVEL/DEPTH
The Buddy's FX knob (effect amount; as in the DDJ-FLX4's note)

23 ON/OFF
The Buddy's FX ON bar (toggles with each press; SHIFT + ON/OFF turns all effects off while held)

Controls without a number (MERGE FX, IN, OUT, MEMORY, JOG CUTTER, DECK, MASTER, TRIM, CH SELECT, MASTER LEVEL, BOOTH, channels 3 and 4 and so on) are not used by ReaScratch. The MIDI numbers were checked against the official MIDI list (the version preserved on the Internet Archive); not yet tested on the real hardware.

71 · 72 DDJ-FLX10 deck 1 / deck 2

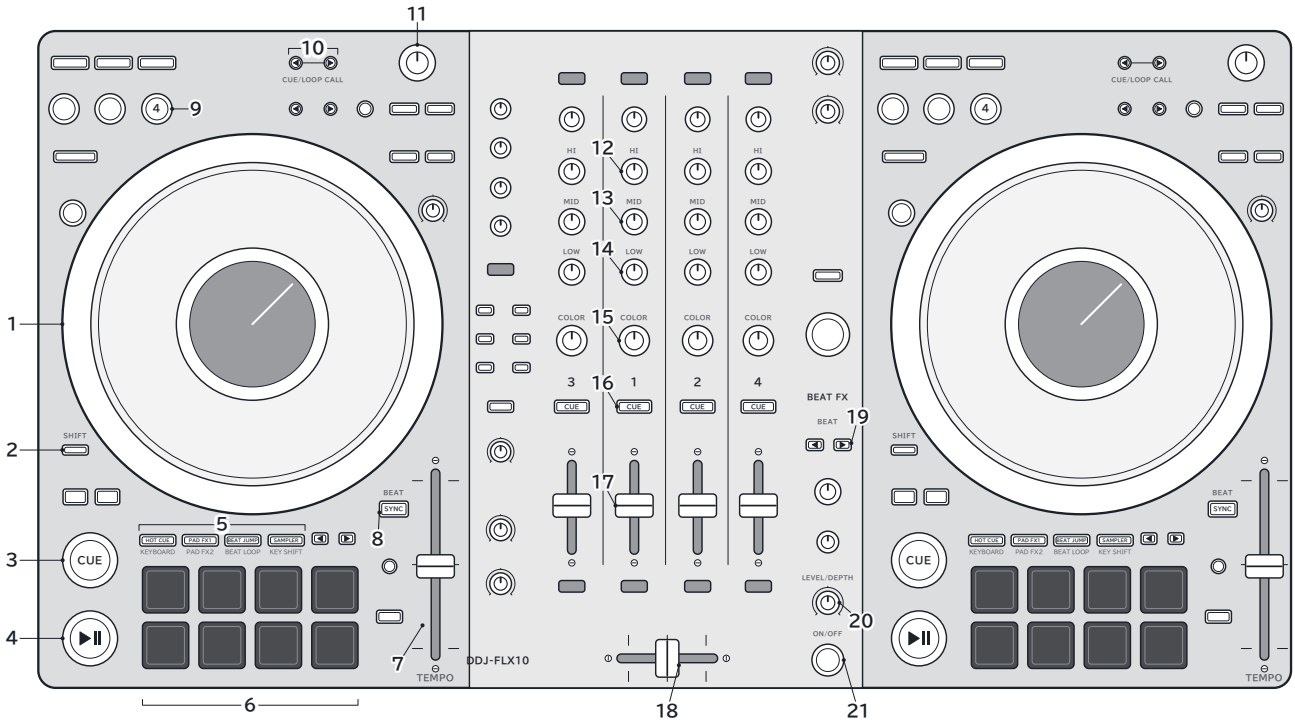


Figure 59 The DDJ-FLX10's controls (numbered on the left deck, mixer channel 1 and BEAT FX; the right deck is the same)

Almost the same as the DDJ-FLX4. The mixer has 4 channels (3, 1, 2, 4 from the left); deck 1 uses channel 1 and deck 2 channel 2. Each deck has its own BROWSE knob at its top; pushing it is LOAD.

Deck

1 Jog
Top = touch scratch (switch the controller's VINYL on), outer ring = nudge (1/4 speed). `SHIFT` + jog makes big jumps. The manufacturer does not publish the tick count (`Jog ticks per second at 1x`), so it is `7167`, the value measured (in Mixxx) on a real DDJ-1000, whose jog is the same size. If the jog is too fast or too slow, change this value

2 SHIFT
Hold it with another control for its second function

3 CUE

4 PLAY/PAUSE

5 Pad mode buttons
`HOT CUE` = HOT CUE, `PAD FX1` → TOUCH FX, `BEAT JUMP` → SLICER, `SAMPLER` → sample switching. With SHIFT held, `KEYBOARD` → PITCH PLAY, `PAD FX2` → BOUNCE LOOP, `BEAT LOOP` → AUTO LOOP, `KEY SHIFT` → NEURAL MIX

6 Pads
The 8 pads of the current mode. In `HOT CUE`, SHIFT + pad clears a hot cue

7 TEMPO
Tempo slider

8 BEAT SYNC
SHIFT switches the tempo range (note 92)

9 4 BEAT/EXIT
Loop (reloop with SHIFT)

10 CUE/LOOP CALL ◂ ▸
Halve / double the loop

11 BROWSE
(the knob at the top of the deck): Turn = the headphone Cue mix (while BEAT FX is on, that effect's sound amount; both decks' knobs send the same number, so turning either one does the same), push = LOAD (same as the Reloop Buddy; pushing it with SHIFT held replaces the current sample). There is no push to switch decks

Mixer

12 HI
Filter Morph (echo length with SHIFT)

13 MID
How much the video effects act on the sound

14 LOW
Filter Resonance (echo feedback with SHIFT)

15 COLOR
Filter Cutoff (key with SHIFT; the FLX4's CFX)

16 Headphone CUE
Listen to this deck on headphones (toggles with each press; SHIFT hides only the picture)

17 Channel fader
That deck's volume and picture strength

18 Crossfader
Deck 1 on the left side, deck 2 on the right

BEAT FX (Buddy-style)

19 BEAT ◂ ▸
The Buddy's `FX PARAM` ◂ ▸ (effect type; as in the DDJ-FLX4's note)

20 LEVEL/DEPTH
The Buddy's FX knob (effect amount; as in the DDJ-FLX4's note). Works on the deck 1 preset only

21 ON/OFF
The Buddy's FX ON bar (toggles with each press; SHIFT + ON/OFF turns all effects off while held). Works on the deck 1 preset only

Controls without a number (IN, OUT, ACTIVE PART, MIX POINT, SLIP, QUANTIZE, 4 BEAT JUMP, PAGE, KEY SYNC, TEMPO RESET, TRIM, the COLOR FX buttons, FX PART, FX SELECT, CH SELECT, channels 3 and 4 and so on) are not used by ReaScratch. The MIDI numbers were checked against the official MIDI list (the version preserved on the Internet Archive); not yet tested on the real hardware.

73 · 74 DDJ-REV1 deck 1 / deck 2

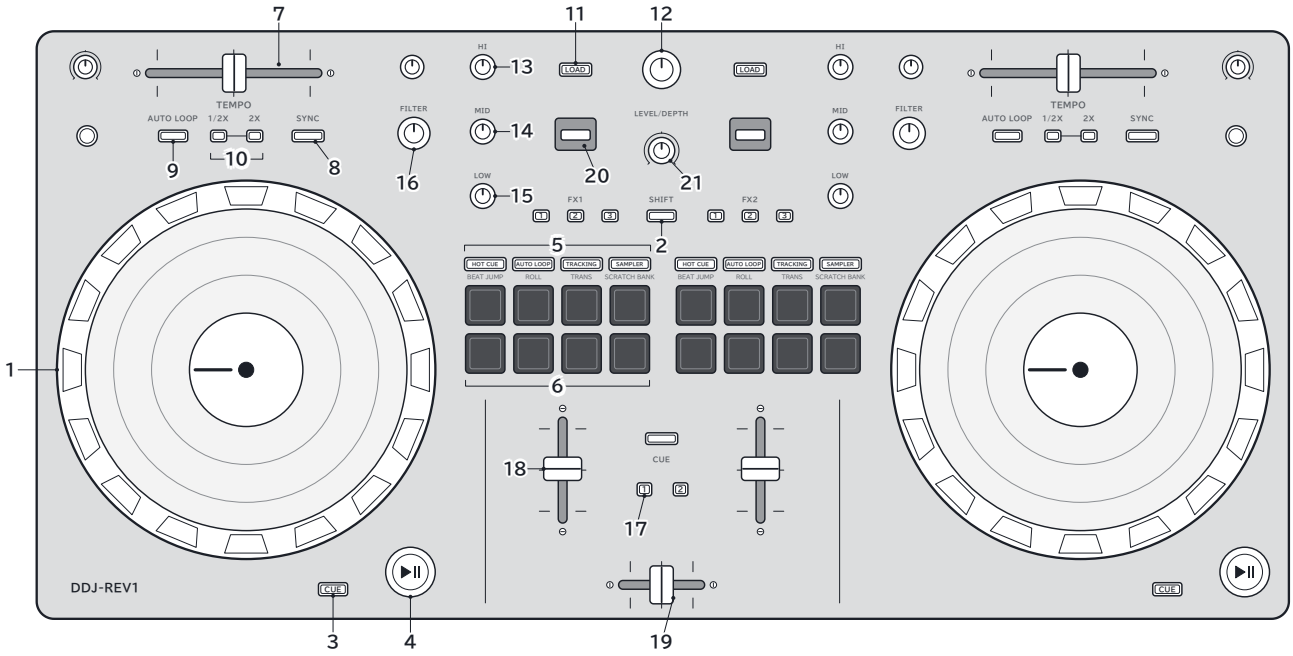


Figure 60 The DDJ-REV1's controls (numbered on the left deck and the centre; the right deck is the same; its DECK button and the EQ are mirrored)

Many controls have the DDJ-FLX4's numbers; SHIFT is in the centre (MIDI ch 7). The tempo sliders lie across the top of the decks, and both decks' pads are in the centre (the left 8 are deck 1's).

Deck

1 **Jog**
Top = touch scratch (switch the controller's VINYL on), outer ring = nudge (1/4 speed). `SHIFT` + jog makes big jumps. The manufacturer does not publish the tick count (`Jog ticks per second at 1x`), so it is `400`, from Mixxx's mapping (if the jog is too fast or too slow, change this value)

2 **SHIFT**
(centre): Hold it with another control for its second function

3 **CUE**

4 **PLAY/PAUSE**

5 **Pad mode buttons**
`HOT CUE` = HOT CUE, `AUTO LOOP` = AUTO LOOP, `TRACKING` (TRACKING SCRATCH) → PITCH PLAY, `SAMPLER` → sample switching. With SHIFT held, `BEAT JUMP` → SLICER, `ROLL` → BOUNCE LOOP, `TRANS` → TOUCH FX, `SCRATCH BANK` → NEURAL MIX

6 **Pads**
The 8 pads of the current mode. In `HOT CUE`, SHIFT + pad clears a hot cue

7 **TEMPO**
Tempo slider (horizontal)

8 **SYNC**
BEAT SYNC. SHIFT + SYNC does not change the tempo range (use `Shift` + `Y` on the keyboard)

9 **AUTO LOOP**
Loop (reloop with SHIFT)

10 **1/2X · 2X**
Halve / double the loop

Mixer

11 LOAD
Same as the Reloop Buddy (SHIFT + LOAD replaces the current sample)

12 BROWSE
Turn = the headphone Cue mix (while BEAT FX is on, that effect's sound amount), push = switch decks

13 HI
Filter Morph (echo length with SHIFT)

14 MID
How much the video effects act on the sound

15 LOW
Filter Resonance (echo feedback with SHIFT)

16 FILTER
Filter Cutoff (key with SHIFT; the FLX4's CFX)

17 Headphone CUE (1)
Listen to this deck on headphones (toggles with each press; SHIFT hides only the picture)

18 Channel fader
That deck's volume and picture strength

19 Crossfader
Deck 1 on the left side, deck 2 on the right

FX (Buddy-style)

20 FX1 paddle
The Buddy's FX ON bar (toggles with each flick). The FX1 paddle = deck 1, the FX2 paddle = deck 2. Select SLOT 1 with the FX's `1` button first

21 LEVEL/DEPTH
The Buddy's FX knob (effect amount; as in the DDJ-FLX4's note). It is sent on the number of the FX whose SLOT 1 is on (FX1 = deck 1, FX2 = deck 2)

The ◂ ▸ for the effect type (SHIFT + the FX's `2` and `3` buttons) and SHIFT + paddle to turn all effects off while held are not mapped. Controls without a number (TRIM, the headphone LEVEL, MASTER LEVEL, DECK, the FX's `2` and `3` buttons, MASTER's CUE and so on) are not used by ReaScratch. The MIDI numbers were checked against the official MIDI list (the version preserved on the Internet Archive); not yet tested on the real hardware.

81 · 82 DDJ-400 deck 1 / deck 2

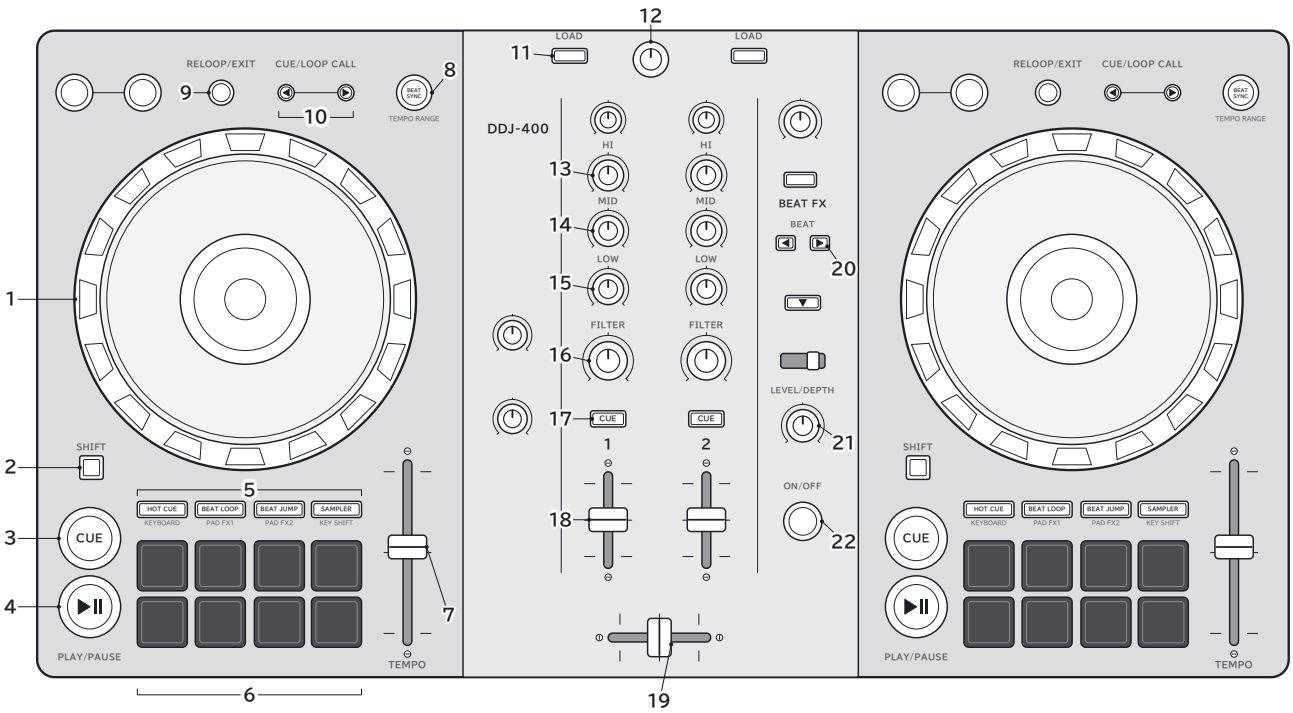


Figure 61 The DDJ-400's controls (numbered on the left deck and the centre; the right deck is the same)

The same MIDI numbers as the DDJ-FLX4 (all of them matched in the official MIDI list (the version preserved on the Internet Archive)). The printed mode names and the mixer's layout differ.

Deck

1 Jog
Top = touch scratch, outer ring = nudge (1/4 speed). `SHIFT` + jog makes big jumps. The manufacturer does not publish the tick count (`Jog ticks per second at 1x`), so it is `400`, from Mixxx's mapping (if the jog is too fast or too slow, change this value)

2 SHIFT
Hold it with another control for its second function

3 CUE

4 PLAY/PAUSE

5 Pad mode buttons
`HOT CUE` = HOT CUE, `BEAT LOOP` → AUTO LOOP, `BEAT JUMP` → SLICER, `SAMPLER` → sample switching. With SHIFT held, `KEYBOARD` → PITCH PLAY, `PAD FX1` → TOUCH FX, `PAD FX2` → BOUNCE LOOP, `KEY SHIFT` → NEURAL MIX

6 Pads
The 8 pads of the current mode. In `HOT CUE`, SHIFT + pad clears a hot cue

7 TEMPO
Tempo slider

8 BEAT SYNC
SHIFT switches the tempo range (printed TEMPO RANGE)

9 RELOOP/EXIT
Loop (reloop with SHIFT; the same number as the FLX4's 4 BEAT/EXIT)

10 CUE/LOOP CALL ◂ ▸
Halve / double the loop

Mixer

11 LOAD
Same as the Reloop Buddy (SHIFT + LOAD replaces the current sample)

12 BROWSE
Turn = the headphone Cue mix (while BEAT FX is on, that effect's sound amount), push = switch decks

13 HI
Filter Morph (echo length with SHIFT)

14 MID
How much the video effects act on the sound

15 LOW
Filter Resonance (echo feedback with SHIFT)

16 FILTER
Filter Cutoff (key with SHIFT; the FLX4's CFX)

17 Headphone CUE
Listen to this deck on headphones (toggles with each press; SHIFT hides only the picture)

18 Channel fader
That deck's volume and picture strength

19 Crossfader
Deck 1 on the left side, deck 2 on the right

BEAT FX (Buddy-style)

20 BEAT ◂ ▸
The Buddy's `FX PARAM` ◂ ▸ (effect type; as in the DDJ-FLX4's note)

21 LEVEL/DEPTH
The Buddy's FX knob (effect amount; as in the DDJ-FLX4's note). Works on the deck 1 preset only

22 ON/OFF
The Buddy's FX ON bar (toggles with each press; SHIFT + ON/OFF turns all effects off while held). Works on the deck 1 preset only

BEAT FX sends the same numbers whatever CH SELECT is set to, so it cannot pick a deck. Controls without a number (IN/4BEAT, OUT, TRIM, HEADPHONES MIXING and LEVEL, MASTER LEVEL and CUE, FX SELECT, CH SELECT and so on) are not used by ReaScratch. Not yet tested on the real hardware.

83 · 84 DDJ-800 deck 1 / deck 2

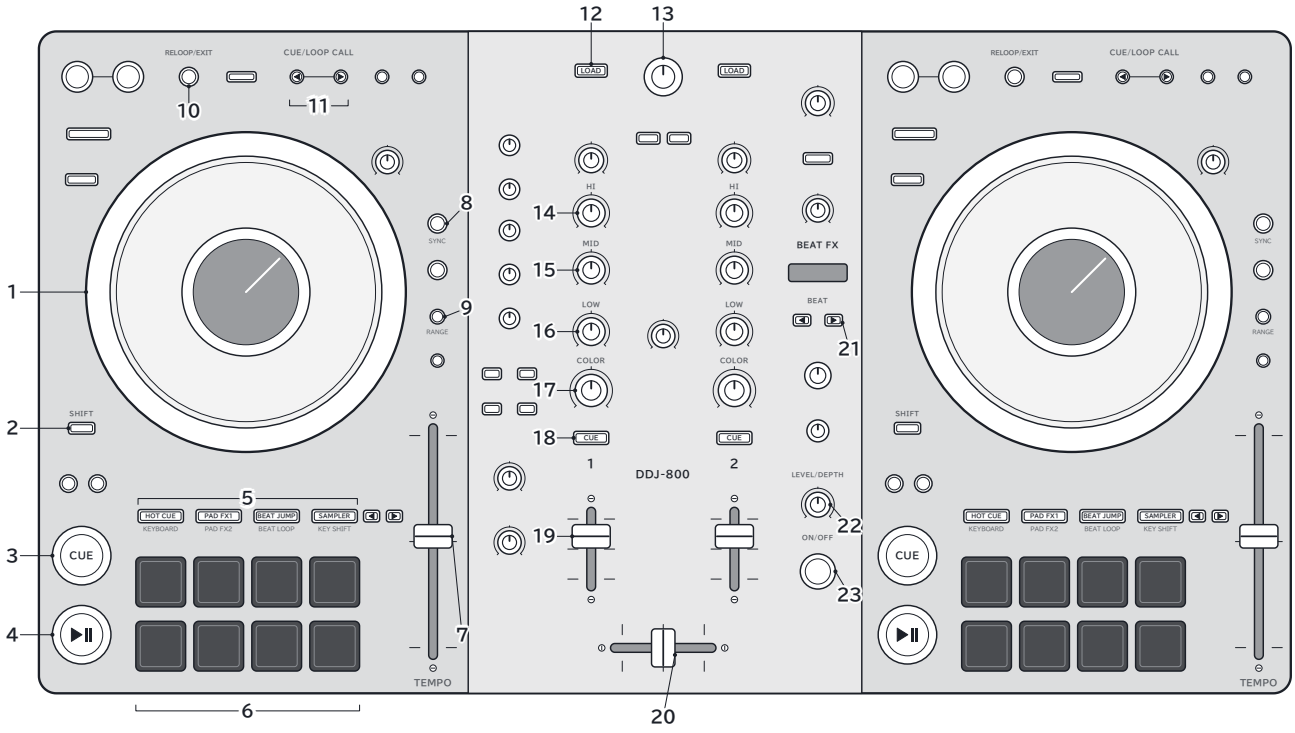


Figure 62 The DDJ-800's controls (numbered on the left deck and the centre; the right deck is the same)

Almost the same as the DDJ-FLX4. The tempo range is switched with its own TEMPO RANGE button.

Deck

1 Jog
Top = touch scratch (switch the controller's VINYL on), outer ring = nudge (1/4 speed). `SHIFT` + jog makes big jumps. The manufacturer does not publish the tick count (`Jog ticks per second at 1x`), so it is `7167`, assumed to be the same as the DDJ-1000's. If the jog is too fast or too slow, change this value

2 SHIFT
Hold it with another control for its second function

3 CUE

4 PLAY/PAUSE

5 Pad mode buttons
`HOT CUE` = HOT CUE, `PAD FX1` → TOUCH FX, `BEAT JUMP` → SLICER, `SAMPLER` → sample switching. With SHIFT held, `KEYBOARD` → PITCH PLAY, `PAD FX2` → BOUNCE LOOP, `BEAT LOOP` → AUTO LOOP, `KEY SHIFT` → NEURAL MIX

6 Pads
The 8 pads of the current mode. In `HOT CUE`, SHIFT + pad clears a hot cue

7 TEMPO
Tempo slider

8 BEAT SYNC

9 TEMPO RANGE
Switches the tempo range (note 96)

10 RELOOP/EXIT
Loop (reloop with SHIFT; the same number as the FLX4's 4 BEAT/EXIT)

11 CUE/LOOP CALL ◂ ▸
Halve / double the loop

Mixer

12 LOAD
Same as the Reloop Buddy (SHIFT + LOAD replaces the current sample)

13 BROWSE
Turn = the headphone Cue mix (while BEAT FX is on, that effect's sound amount), push = switch decks

14 HI
Filter Morph (echo length with SHIFT)

15 MID
How much the video effects act on the sound

16 LOW
Filter Resonance (echo feedback with SHIFT)

17 COLOR
Filter Cutoff (key with SHIFT; the FLX4's CFX)

18 Headphone CUE
Listen to this deck on headphones (toggles with each press; SHIFT hides only the picture)

19 Channel fader
That deck's volume and picture strength

20 Crossfader
Deck 1 on the left side, deck 2 on the right

BEAT FX (Buddy-style)

21 BEAT ◂ ▸
The Buddy's `FX PARAM` ◂ ▸ (effect type; as in the DDJ-FLX4's note)

22 LEVEL/DEPTH
The Buddy's FX knob (effect amount; as in the DDJ-FLX4's note). Works on the deck 1 preset only

23 ON/OFF
The Buddy's FX ON bar (toggles with each press; SHIFT + ON/OFF turns all effects off while held). Works on the deck 1 preset only

Controls without a number (IN, OUT, QUANTIZE, DELETE, MEMORY, SLIP, SLIP REVERSE, JOG ADJUST, KEY SYNC, MASTER TEMPO, SEARCH, PAGE, TRIM, MIC, the SOUND COLOR FX buttons, HEADPHONES, MASTER, BOOTH, FX SELECT, CH SELECT and so on) are not used by ReaScratch. The MIDI numbers were checked against the official MIDI list (the version preserved on the Internet Archive); not yet tested on the real hardware.

85 · 86 DDJ-1000 deck 1 / deck 2

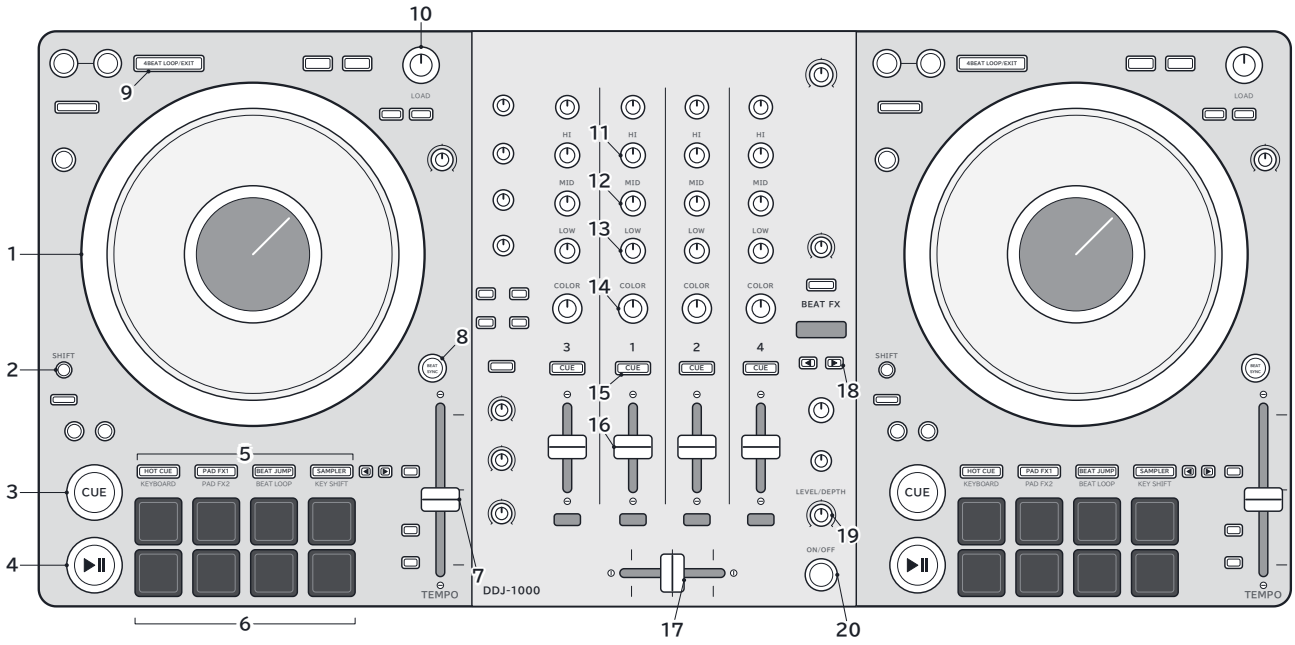


Figure 63 The DDJ-1000's controls (numbered on the left deck, mixer channel 1 and BEAT FX; the right deck is the same)

Almost the same as the DDJ-FLX4. The mixer has 4 channels (3, 1, 2, 4 from the left); deck 1 uses channel 1 and deck 2 channel 2. Each deck has its own BROWSE knob at its top; pushing it is LOAD.

Deck

1 Jog
Top = touch scratch (switch the controller's VINYL on), outer ring = nudge (1/4 speed). `SHIFT` + jog makes big jumps. The manufacturer does not publish the tick count (`Jog ticks per second at 1x`), so it is `7167`, from Mixxx's mapping (this value was measured on real hardware). If the jog is too fast or too slow, change this value

2 SHIFT
Hold it with another control for its second function

3 CUE

4 PLAY/PAUSE

5 Pad mode buttons
`HOT CUE` = HOT CUE, `PAD FX1` → TOUCH FX, `BEAT JUMP` → SLICER, `SAMPLER` → sample switching. With SHIFT held, `KEYBOARD` → PITCH PLAY, `PAD FX2` → BOUNCE LOOP, `BEAT LOOP` → AUTO LOOP, `KEY SHIFT` → NEURAL MIX

6 Pads
The 8 pads of the current mode. In `HOT CUE`, SHIFT + pad clears a hot cue

7 TEMPO
Tempo slider

8 BEAT SYNC
SHIFT switches the tempo range (note 92)

9 4 BEAT LOOP/EXIT
Loop (reloop with SHIFT)

10 BROWSE
(the knob at the top of the deck): Turn = the headphone Cue mix (while BEAT FX is on, that effect's sound amount; both decks' knobs send the same number, so turning either one does the same), push = LOAD (same as the Reloop Buddy; pushing it with SHIFT held replaces the current sample). There is no push to switch decks

Mixer

11 HI
Filter Morph (echo length with SHIFT)

12 MID
How much the video effects act on the sound

13 LOW
Filter Resonance (echo feedback with SHIFT)

14 COLOR
Filter Cutoff (key with SHIFT; the FLX4's CFX)

15 Headphone CUE
Listen to this deck on headphones (toggles with each press; SHIFT hides only the picture)

16 Channel fader
That deck's volume and picture strength

17 Crossfader
Deck 1 on the left side, deck 2 on the right

BEAT FX (Buddy-style)

18 BEAT ◂ ▸
The Buddy's `FX PARAM` ◂ ▸ (effect type; as in the DDJ-FLX4's note)

19 LEVEL/DEPTH
The Buddy's FX knob (effect amount; as in the DDJ-FLX4's note). Works on the deck 1 preset only

20 ON/OFF
The Buddy's FX ON bar (toggles with each press; SHIFT + ON/OFF turns all effects off while held). Works on the deck 1 preset only

Loop halve and double are not mapped. Controls without a number (IN, OUT, QUANTIZE, SLIP, SLIP REVERSE, DECK, BACK, VIEW, JOG ADJUST, MEMORY, SEARCH, PAGE, MASTER TEMPO, KEY SYNC, KEY RESET, TRIM, the SOUND COLOR FX buttons, FX SELECT, CH SELECT, channels 3 and 4 and so on) are not used by ReaScratch. The MIDI numbers were checked against the official MIDI list (the version preserved on the Internet Archive); not yet tested on the real hardware.

75 · 76 DDJ-REV5 deck 1 / deck 2

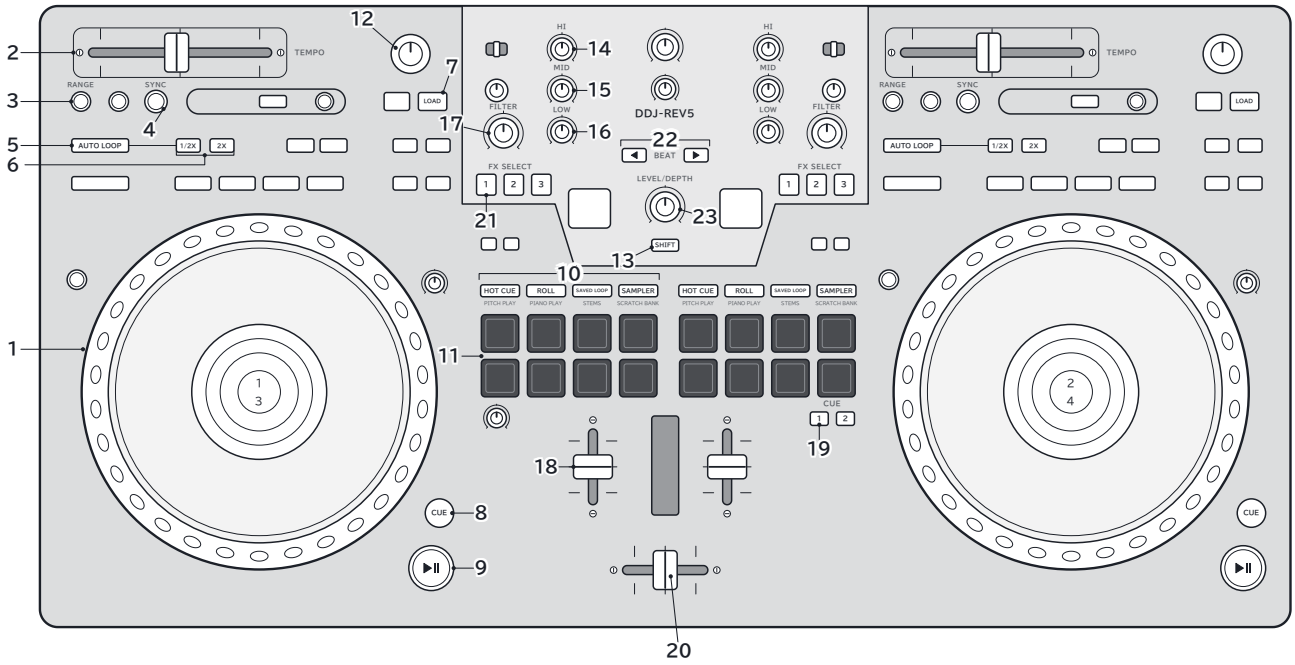


Figure 64 The DDJ-REV5's controls (numbered on the left deck and the centre; the right deck is the same)

Almost the same as the DDJ-FLX4. Top touch (note 13), SYNC (note 10), TEMPO RANGE (note 8) and headphone CUE (note 7) differ. The pads sit in the lower part of the mixer, and there is one SHIFT, in the middle of the mixer.

Deck

1 Jog
Top = touch scratch (switch the controller's VINYL on with SHIFT + KEY LOCK), outer ring = nudge (1/4 speed). `SHIFT` + jog makes big jumps

2 TEMPO
Tempo slider (lying across the top of the deck)

3 RANGE (TEMPO RANGE)
Switches the tempo range

4 SYNC

5 AUTO LOOP
Loop (reloop with SHIFT)

6 1/2X · 2X
Halve / double the loop

7 LOAD
Same as the Reloop Buddy (SHIFT + LOAD replaces the current sample)

8 CUE

9 PLAY/PAUSE

Pads

10 Pad mode buttons
`HOT CUE` = HOT CUE, `ROLL` → BOUNCE LOOP, `SAVED LOOP` → AUTO LOOP, `SAMPLER` (in this mode the pads do nothing). With SHIFT held, `PITCH PLAY` = PITCH PLAY, `PIANO PLAY` → SLICER, `STEMS` → TOUCH FX, `SCRATCH BANK` → NEURAL MIX

11 Pads
The 8 pads of the current mode. In `HOT CUE`, SHIFT + pad clears a hot cue

Mixer

12 BROWSE
Turn = the headphone Cue mix (while an effect is on, that effect's sound amount), push = switch decks (the knob is at the deck's top right)

13 SHIFT
Hold it with another control for its second function (shared by both decks)

14 HI
Filter Morph (echo length with SHIFT)

15 MID
How much the video effects act on the sound

16 LOW
Filter Resonance (echo feedback with SHIFT)

17 FILTER
Filter Cutoff (key with SHIFT)

18 Channel fader
That deck's volume and picture strength

19 Headphone CUE (1)
Listen to this deck on headphones (toggles with each press; SHIFT hides only the picture)

20 Crossfader
Deck 1 on the left side, deck 2 on the right

FX (Buddy-style)

21 FX SELECT 1
The Buddy's FX ON bar (FX slot 1 on / off; toggles with each press)

22 BEAT ◂ ▸
The Buddy's `FX PARAM` ◂ ▸ (effect type)

23 LEVEL/DEPTH
The Buddy's FX knob (slot 1's LEVEL; effect amount)

Note (Buddy-style FX): deck 2 uses the same controls on the right (FX2). What they act on is as in the DDJ-FLX4's note: it depends on what you hold.

Controls without a number (AUTO BPM TRANSITION, BACK, SLIP, CENSOR, KEY − / +, STEMS, VOCAL to DRUMS, DECK, JOG FEEL, TRIM, LOCK ON, PARAMETER 1, SAMPLER VOLUME and so on) are not used by ReaScratch. From the official MIDI list; not yet tested on the real hardware.

77 · 78 DDJ-200 deck 1 / deck 2

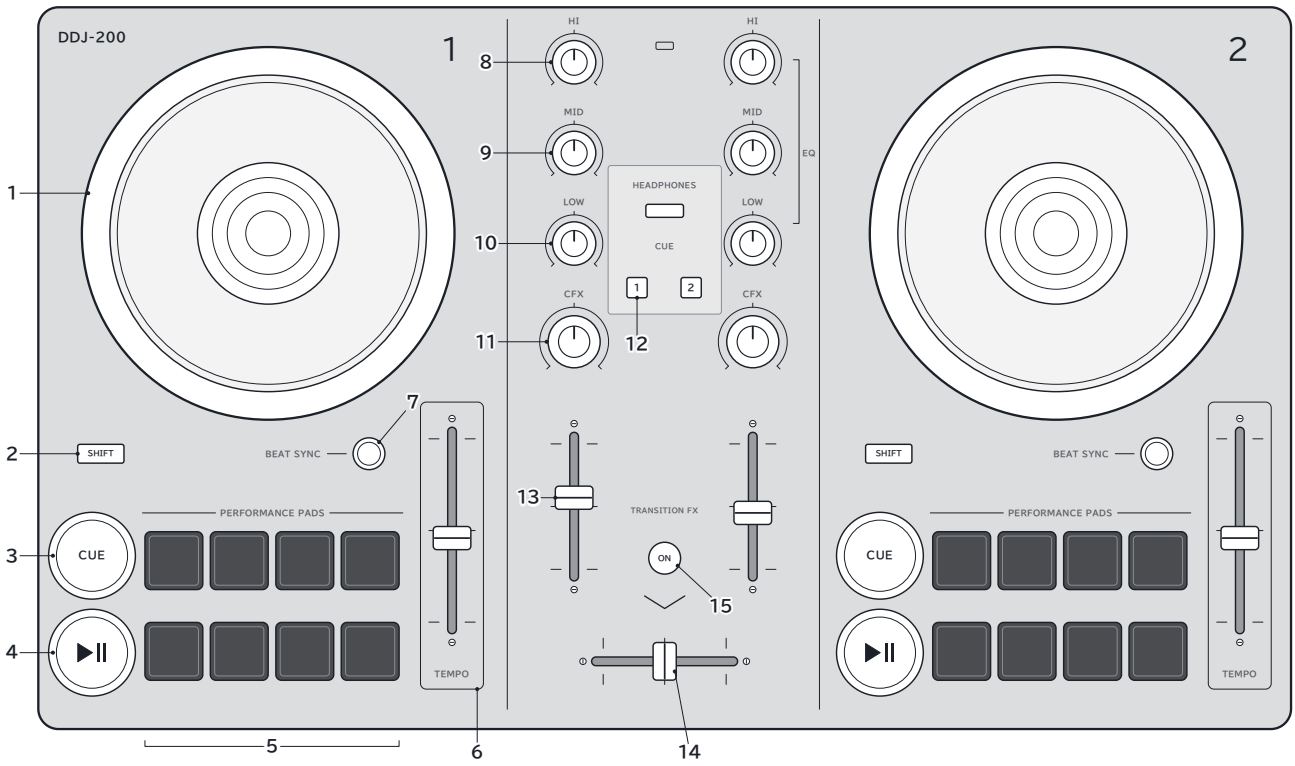


Figure 65 The DDJ-200's controls (numbered on the left deck and the centre; the right deck is the same)

Same numbers as the DDJ-FLX4 (official MIDI list, version preserved on the Internet Archive). There are no loop, BEAT FX, BROWSE or LOAD buttons.

Deck

1 Jog
Top = touch scratch, outer ring = nudge (1/4 speed). `SHIFT` + jog makes big jumps. The jog is small, so the tick count (`Jog ticks per second at 1x`) is `71` (from Mixxx's 128 ticks per turn)

2 SHIFT
Hold it with another control for its second function

3 CUE

4 PLAY/PAUSE

5 Pads
8 pads, HOT CUE only (no SAMPLER). SHIFT + pad clears a hot cue

6 TEMPO
Tempo slider

7 BEAT SYNC
SHIFT switches the tempo range (`TEMPO RANGE`)

8 **HI**
Filter Morph (echo length with SHIFT)

9 **MID**
How much the video effects act on the sound

10 **LOW**
Filter Resonance (echo feedback with SHIFT)

11 **CFX**
Filter Cutoff (key with SHIFT)

12 **HEADPHONES CUE 1**
Listen to this deck on headphones (toggles with each press; SHIFT hides only the picture; 2 for deck 2)

13 **Channel fader**
That deck's volume and picture strength

14 **Crossfader**
Deck 1 on the left side, deck 2 on the right

15 **TRANSITION FX ON**
Video effect on/off

Controls without a number (HEADPHONES MASTER) are not used by ReaScratch. Not yet tested on the real hardware.

79 · 80 DDJ-RB deck 1 / deck 2

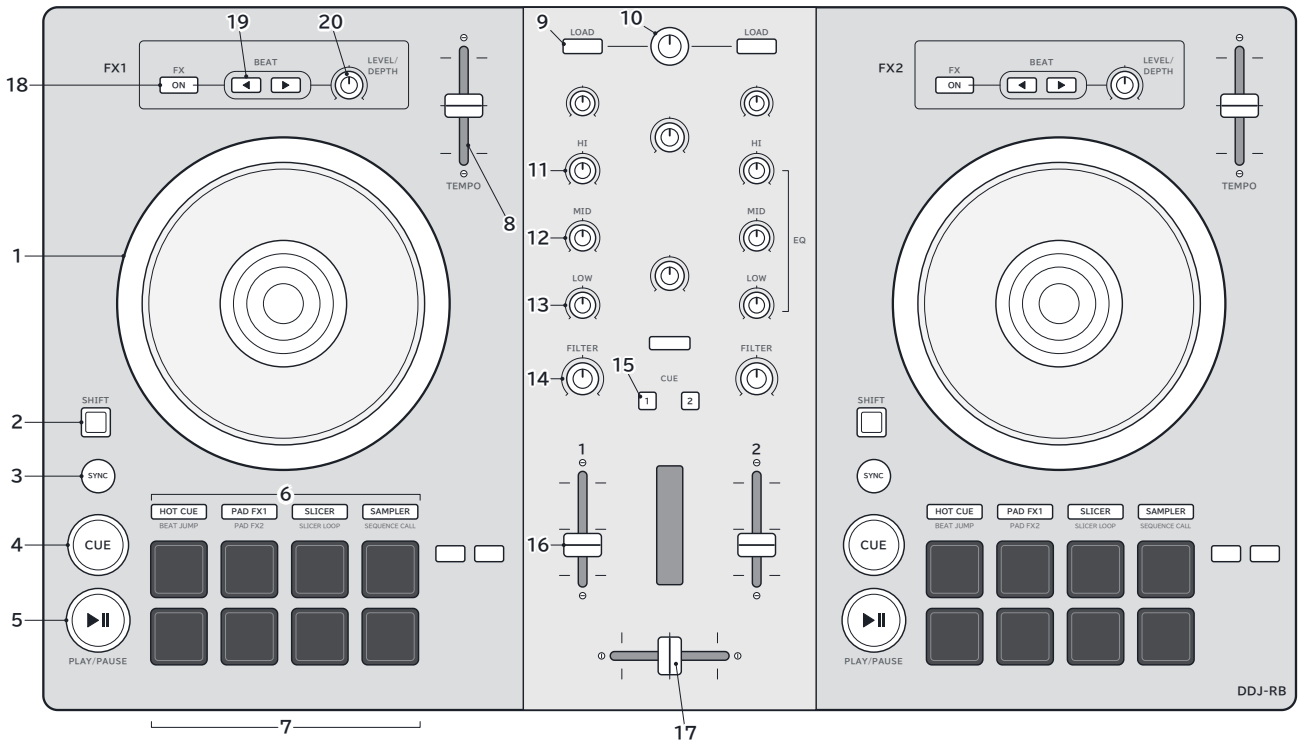


Figure 66 The DDJ-RB's controls (numbered on the left deck and the centre; the right deck is the same)

Same numbers as the DDJ-SB3 (official MIDI list, version preserved on the Internet Archive). Pad modes and loops are not added. Not yet tested on the real hardware.

Deck

1 Jog
Top = touch scratch, outer ring = nudge (1/4 speed).
`SHIFT` + jog makes big jumps

2 SHIFT
Hold it with another control for its second function

3 SYNC

4 CUE

5 PLAY/PAUSE

6 Pad mode buttons
Switch the controller's own mode. ReaScratch reads only the pads of the `HOT CUE` and `SAMPLER` modes (ReaScratch's pad modes are not added)

7 Pads
In `HOT CUE`, hot cues (SHIFT + pad clears); in `SAMPLER`, sample switching

8 TEMPO
Tempo slider. There is no KEYLOCK, so switch the tempo range with `Shift` + `Y` on the keyboard

Mixer

9 LOAD
Same as the Reloop Buddy (SHIFT + LOAD replaces the current sample)

10 BROWSE
Turn = the headphone Cue mix (while an effect is on, that effect's sound amount). Pushing it is not read

11 HI
Filter Morph (echo length with SHIFT)

12 MID
How much the video effects act on the sound

13 LOW
Filter Resonance (echo feedback with SHIFT)

14 FILTER
Filter Cutoff (key with SHIFT)

15 Headphone CUE (1)
Listen to this deck on headphones (toggles with each press; SHIFT hides only the picture)

16 Channel fader
That deck's volume and picture strength

17 Crossfader
Deck 1 on the left side, deck 2 on the right

FX1 (Buddy-style)

18 FX (ON)
The Buddy's FX ON bar (toggles with each press; SHIFT turns all effects off while held)

19 BEAT ◂ ▸
The Buddy's `FX PARAM` ◂ ▸ (effect type)

20 LEVEL/DEPTH
The Buddy's FX knob (effect amount)

Note (Buddy-style FX): deck 2 uses the same controls on the right (FX2). What they act on is as in the DDJ-FLX4's note: it depends on what you hold.

Controls without a number (TRIM, MASTER LEVEL, HEADPHONES LEVEL, CUE MASTER, 4 BEAT, IN, OUT and so on) are not used by ReaScratch.

87 · 88 DDJ-SB deck 1 / deck 2, 89 · 90 DDJ-SB2 deck 1 / deck 2

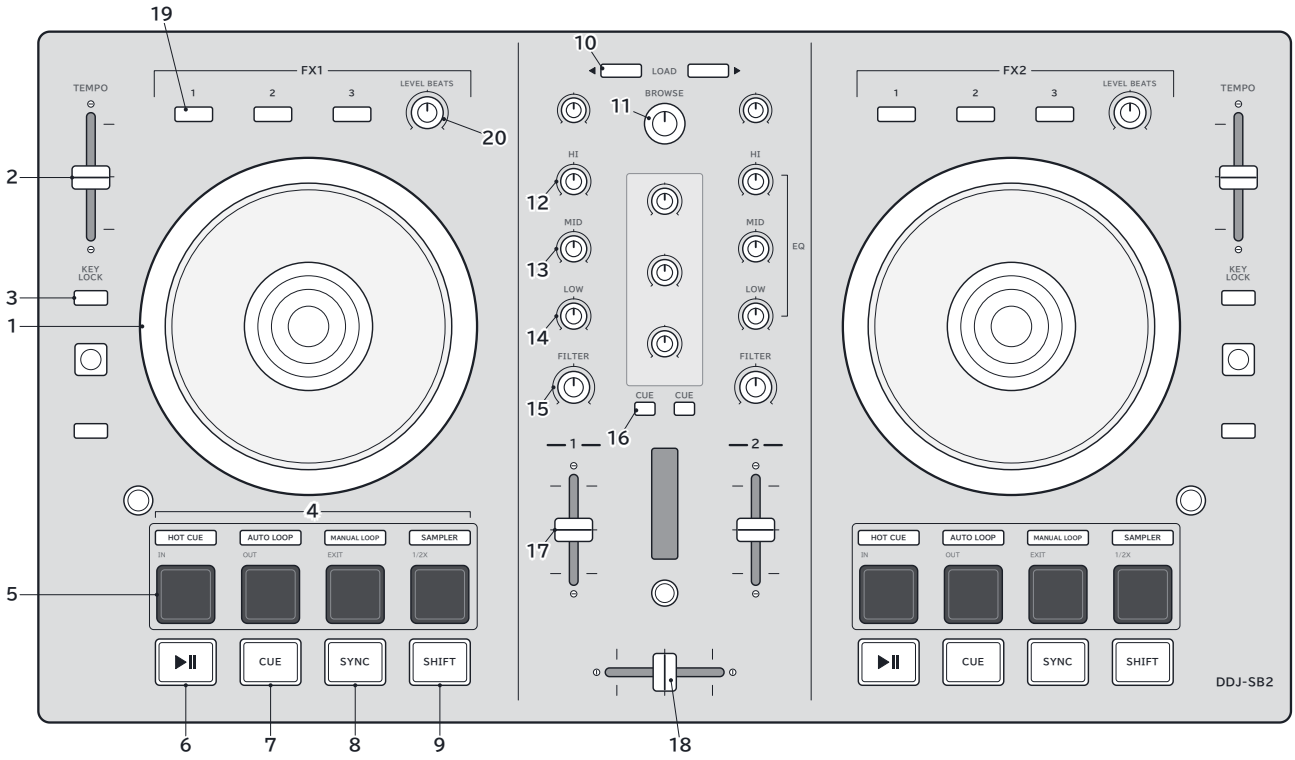


Figure 67 The DDJ-SB2's controls (the DDJ-SB is laid out the same, without TRIM, DECK and PAD TRANS; numbered on the left deck and the centre; the right deck is the same, with its TEMPO column on the right edge)

Same numbers as the DDJ-SB3 (official MIDI lists, versions preserved on the Internet Archive). Pad modes and loops are not added. If decks 3/4 are selected on the DDJ-SB2, switch back to decks 1/2 with the `DECK` button. Not yet tested on the real hardware.

Deck

1 **Jog**
Top = touch scratch (switch the controller's VINYL on), outer ring = nudge (1/4 speed). `SHIFT` + jog makes big jumps

2 **TEMPO**
Tempo slider

3 **KEY LOCK**
SHIFT + KEY LOCK switches the tempo range

4 **Pad mode buttons**
Switch the controller's own mode. ReaScratch reads only the pads of the `HOT CUE` and `SAMPLER` modes (ReaScratch's pad modes are not added)

5 **Pads**
4 pads. In `HOT CUE`, hot cues (SHIFT + pad clears); in `SAMPLER`, sample switching

6 **PLAY/PAUSE**

7 **CUE**

8 **SYNC**

9 **SHIFT**
Hold it with another control for its second function

Mixer

10 **LOAD**
Same as the Reloop Buddy (SHIFT + LOAD replaces the current sample). The left button is deck 1, the right one deck 2

11 **BROWSE**
Turn = the headphone Cue mix (while an effect is on, that effect's sound amount), push = switch decks

12 **HI**
Filter Morph (echo length with SHIFT)

13 **MID**
How much the video effects act on the sound

14 **LOW**
Filter Resonance (echo feedback with SHIFT)

15 **FILTER**
Filter Cutoff (key with SHIFT)

16 **Headphone CUE**
Listen to this deck on headphones (toggles with each press; SHIFT hides only the picture)

17 **Channel fader**
That deck's volume and picture strength

18 **Crossfader**
Deck 1 on the left side, deck 2 on the right

FX1 (Buddy-style)

19 **FX1 button 1**
The Buddy's FX ON bar (toggles with each press; SHIFT turns all effects off while held)

20 **LEVEL**
The Buddy's FX knob (effect amount). There is no ◂ ▸

Note (Buddy-style FX): deck 2 uses the same controls on the right (FX2). What they act on is as in the DDJ-FLX4's note: it depends on what you hold.

Controls without a number (TRIM, DECK, PAD TRANS, FX1 buttons 2 · 3, MASTER LEVEL, HEADPHONES MIX, HEADPHONES LEVEL, FILTER FADE and so on) are not used by ReaScratch.

91 · 92 DDJ-SB3 / SX deck 1 / deck 2

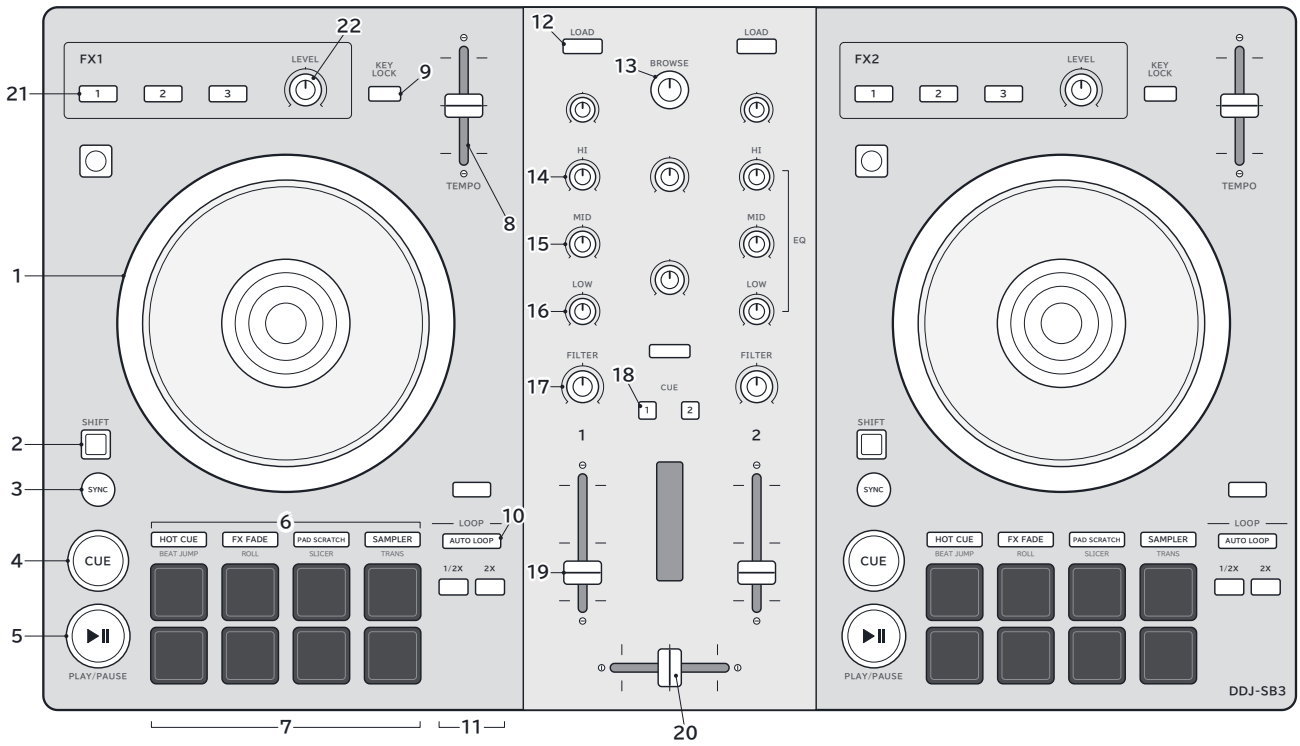


Figure 68 The DDJ-SB3's controls (numbered on the left deck and the centre; the right deck is the same)

Almost the same as the DDJ-FLX4. Pads are on ch 8 for deck 1 and ch 9 for deck 2; SHIFT + pad is the number + 8 (clear). Tempo range is SHIFT + KEYLOCK. SB3 from the official MIDI list (version preserved on the Internet Archive), SX from Mixxx's mapping. Tick count: `400` for SB3 / SX, same as the FLX4. Not yet tested on the real hardware.

Deck

1 Jog
Top = touch scratch (switch the controller's VINYL on), outer ring = nudge (1/4 speed). `SHIFT` + jog makes big jumps

2 SHIFT
Hold it with another control for its second function

3 SYNC

4 CUE

5 PLAY/PAUSE

6 Pad mode buttons
`HOT CUE` = HOT CUE, `FX FADE` (FADE MIX in the MIDI list) → FILTER TYPE, `PAD SCRATCH` → NEURAL MIX, `SAMPLER` → sample switching. With SHIFT held, `BEAT JUMP` → AUTO LOOP, `ROLL` → BOUNCE LOOP, `SLICER` = SLICER, `TRANS` → TOUCH FX

7 Pads
The 8 pads of the current mode. In `HOT CUE`, SHIFT + pad clears a hot cue

8 TEMPO
Tempo slider

9 KEY LOCK
SHIFT + KEY LOCK switches the tempo range

10 AUTO LOOP
Loop (reloop with SHIFT)

11 1/2X · 2X
Halve / double the loop

Mixer

12 LOAD
Same as the Reloop Buddy (SHIFT + LOAD replaces the current sample)

13 BROWSE
Turn = the headphone Cue mix (while an effect is on, that effect's sound amount), push = switch decks

14 HI
Filter Morph (echo length with SHIFT)

15 MID
How much the video effects act on the sound

16 LOW
Filter Resonance (echo feedback with SHIFT)

17 FILTER
Filter Cutoff (key with SHIFT)

18 Headphone CUE
Listen to this deck on headphones (toggles with each press; SHIFT hides only the picture)

19 Channel fader
That deck's volume and picture strength

20 Crossfader
Deck 1 on the left side, deck 2 on the right

FX1 (Buddy-style)

21 FX1 button 1
The Buddy's FX ON bar (toggles with each press; SHIFT turns all effects off while held)

22 FX1 LEVEL
The Buddy's FX knob (effect amount). There is no ◂ ▸

Note (Buddy-style FX): deck 2 uses the same controls on the right (FX2). What they act on is as in the DDJ-FLX4's note: it depends on what you hold.

Controls without a number (DECK, FX1 buttons 2 · 3, TRIM, MASTER LEVEL, HEADPHONES LEVEL, MASTER and so on) are not used by ReaScratch.

DDJ-SX

(the same preset)

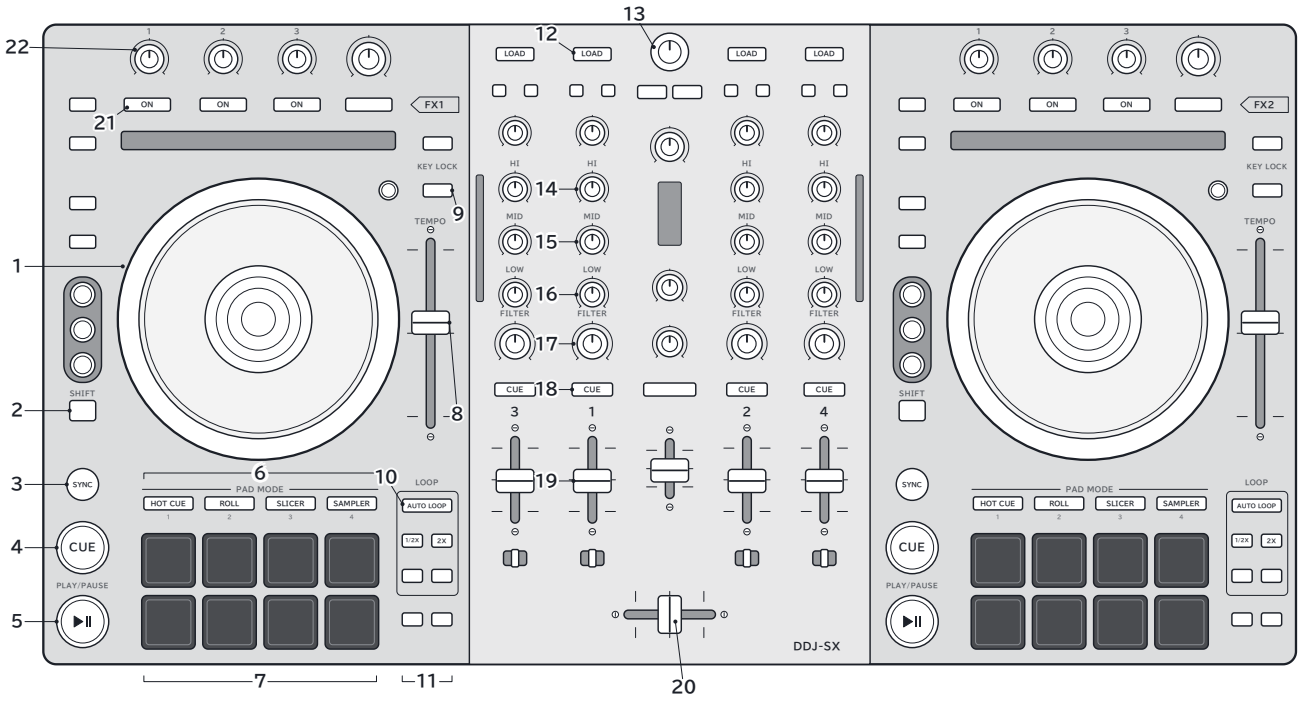


Figure 69 The DDJ-SX's controls (numbered on the left deck (DECK 1) and mixer channel 1; the right deck (DECK 2) and channel 2 are the same)

The DDJ-SX uses the same preset and is read with the SB3's numbers. Select DECK 1 on the left deck and DECK 2 on the right.

Deck

1. **Jog**
   Top = touch scratch (set the controller's JOG MODE to VINYL), outer ring = nudge (1/4 speed). `SHIFT` + jog makes big jumps
2. **SHIFT**
   Hold it with another control for its second function
3. **SYNC**
4. **CUE**
5. **PLAY/PAUSE**
6. **Pad mode buttons**
   Same numbers as the SB3's buttons in the same places (the SX's `HOT CUE`, `ROLL`, `SLICER`, `SAMPLER` = the SB3's `HOT CUE`, `FX FADE`, `PAD SCRATCH`, `SAMPLER`; what they do is the SB3's item 6 above). But SHIFT + SAMPLER sends a different number and does not switch to TOUCH FX
7. **Pads**
   The 8 pads of the current mode. In `HOT CUE`, SHIFT + pad clears a hot cue
8. **TEMPO**
   Tempo slider
9. **KEY LOCK**
   SHIFT + KEY LOCK switches the tempo range
10. **AUTO LOOP**
    Loop (reloop with SHIFT)
11. **1/2X · 2X**
    Halve / double the loop

Mixer

12. **LOAD**
    Same as the Reloop Buddy (SHIFT + LOAD replaces the current sample)
13. **BROWSE**
    Turn = the headphone Cue mix (while an effect is on, that effect's sound amount), push = switch decks
14. **HI**
    Filter Morph (echo length with SHIFT)
15. **MID**
    How much the video effects act on the sound
16. **LOW**
    Filter Resonance (echo feedback with SHIFT)
17. **FILTER**
    Filter Cutoff (key with SHIFT)
18. **Headphone CUE**
    Listen to this deck on headphones (toggles with each press; SHIFT hides only the picture)
19. **Channel fader**
    That deck's volume and picture strength
20. **Crossfader**
    Deck 1 on the left side, deck 2 on the right

FX1 (Buddy-style)

21. **FX1's first ON**
    The Buddy's FX ON bar (toggles with each press; SHIFT turns all effects off while held)
22. **FX1's first knob**
    The Buddy's FX knob (effect amount). There is no ◂ ▸

Note (Buddy-style FX): deck 2 uses the same controls on the right (FX2). What they act on is as in the DDJ-FLX4's note: it depends on what you hold.

Controls without a number (NEEDLE SEARCH, GRID ADJUST, GRID SLIDE, SLIP, BEATS, TAP, PARAMETER1, channels 3 · 4 and so on) are not used by ReaScratch.

93 · 94 DDJ-SX3 deck 1 / deck 2

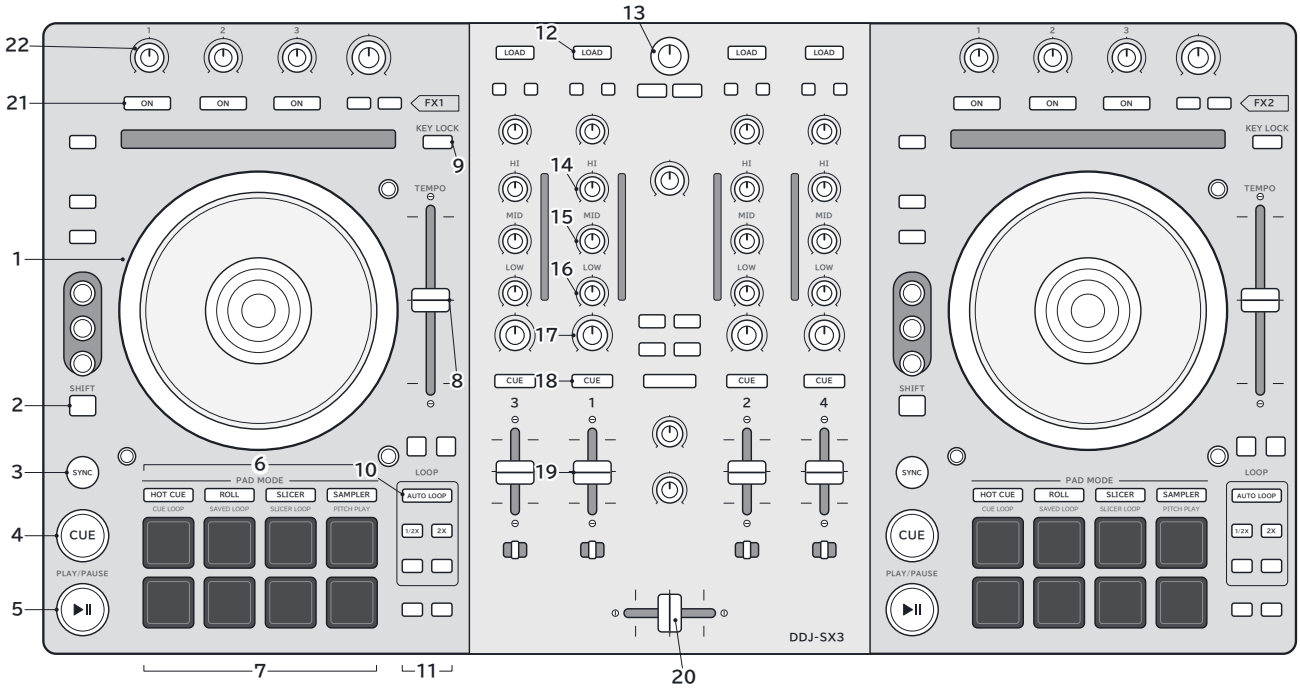


Figure 70 The DDJ-SX3's controls (numbered on the left deck (DECK 1) and mixer channel 1; the right deck (DECK 2) and channel 2 are the same)

Almost the same numbers as the DDJ-SB3 / SX; the SYNC number differs (93). From the official MIDI list (version preserved on the Internet Archive). Tick count `1138`, from Mixxx's 2048 ticks per turn for the DDJ-SX. Not yet tested on the real hardware.

Deck

1 **Jog**
Top = touch scratch (switch the controller's VINYL on), outer ring = nudge (1/4 speed). `SHIFT` + jog makes big jumps

2 **SHIFT**
Hold it with another control for its second function

3 **SYNC**

4 **CUE**

5 **PLAY/PAUSE**

6 **Pad mode buttons**
`HOT CUE` = HOT CUE, `ROLL` → BOUNCE LOOP, `SLICER` = SLICER, `SAMPLER` → sample switching. With SHIFT held, `CUE LOOP` → AUTO LOOP, `SAVED LOOP` → NEURAL MIX, `SLICER LOOP` → TOUCH FX, `PITCH PLAY` = PITCH PLAY

7 **Pads**
The 8 pads of the current mode. In `HOT CUE`, SHIFT + pad clears a hot cue

8 **TEMPO**
Tempo slider

9 **KEY LOCK**
SHIFT + KEY LOCK switches the tempo range

10 **AUTO LOOP**
Loop (reloop with SHIFT)

11 **1/2X · 2X**
Halve / double the loop

Mixer

12 LOAD
Same as the Reloop Buddy (SHIFT + LOAD replaces the current sample)

13 BROWSE
Turn = the headphone Cue mix (while an effect is on, that effect's sound amount), push = switch decks

14 HI
Filter Morph (echo length with SHIFT)

15 MID
How much the video effects act on the sound

16 LOW
Filter Resonance (echo feedback with SHIFT)

17 CFX
Filter Cutoff (key with SHIFT)

18 Headphone CUE
Listen to this deck on headphones (toggles with each press; SHIFT hides only the picture)

19 Channel fader
That deck's volume and picture strength

20 Crossfader
Deck 1 on the left side, deck 2 on the right

FX1 (Buddy-style)

21 FX1's first ON
The Buddy's FX ON bar (toggles with each press; SHIFT turns all effects off while held)

22 FX1's first knob
The Buddy's FX knob (effect amount). There is no ◂ ▸

Note (Buddy-style FX): deck 2 uses the same controls on the right (FX2). What they act on is as in the DDJ-FLX4's note: it depends on what you hold.

Controls without a number (NEEDLE SEARCH, CENSOR, KEY UP / DOWN, SLIP, QUANTIZE, FLIP, BEATS, CH ASSIGN, PARAMETER1, the SOUND COLOR FX buttons, channels 3 · 4 and so on) are not used by ReaScratch.

95 · 96 CDJ-2000NXS2 (MIDI ch 1) / (MIDI ch 2)

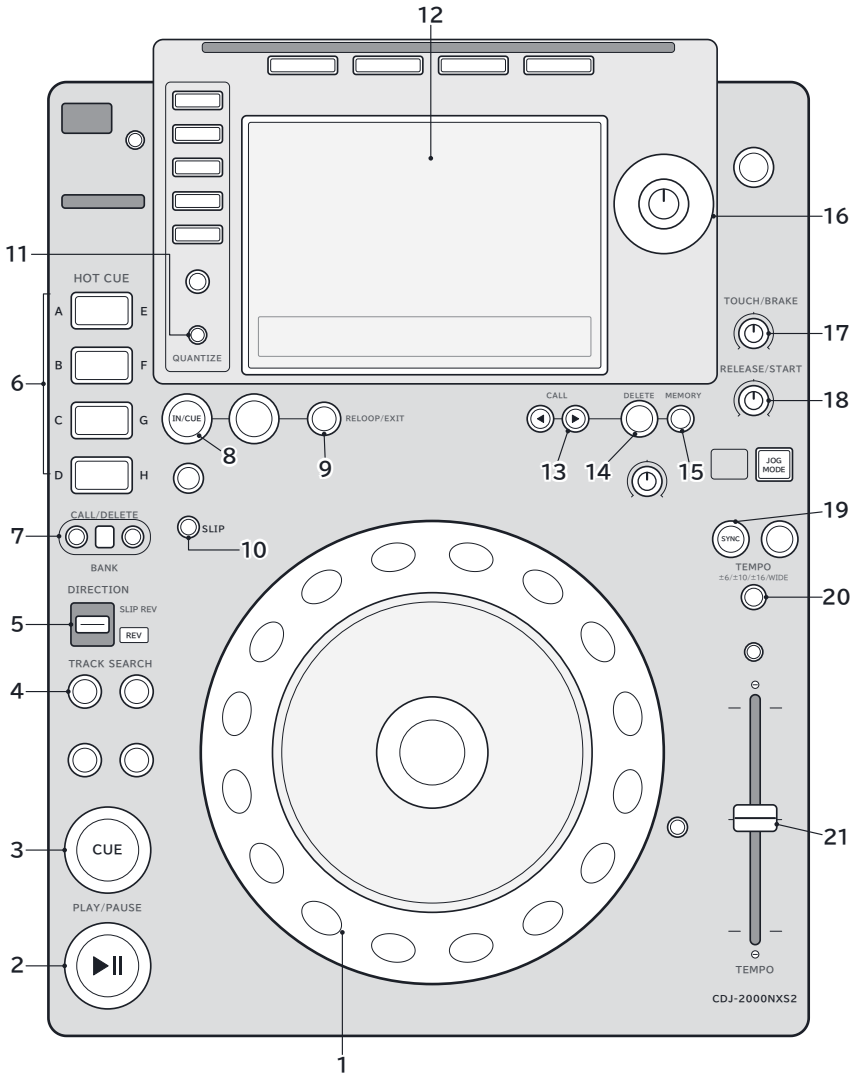


Figure 71 The CDJ-2000NXS2's controls (one player = one deck)

Connect the CDJ to the computer over USB and put it in MIDI mode with `BROWSE/SEARCH` → `LINK` → `CONTROL MODE` → `USB-MIDI`. Hold `MENU/UTILITY` → `MIDI CHANNEL` and set the first player to 1 and the second to 2. Windows needs the player's driver.

Left side and jog

1. **Jog**
   Touch the top and turn to scratch (set `JOG MODE` to VINYL). Turning the side nudges while playing and moves the position while stopped
2. **PLAY/PAUSE**
3. **CUE**
4. **TRACK SEARCH**
   Previous / next sample
5. **DIRECTION**
   `REV` = `REVERSE` (each press toggles reverse play), `SLIP REV` = `SLIP REVERSE` (reverse while held)
6. **HOT CUE**
   A–H are hot cues 1–8 (store if empty)
7. **CALL/DELETE and BANK**
   Hold `HOT CUE CALL/DELETE` and press a hot cue to clear it. `BANK` switches between A–D and E–H
8. **IN/CUE**
   `LOOP IN` = loop
9. **RELOOP/EXIT**
   Exit / go back to the loop
10. **SLIP**
11. **QUANTIZE**

Round the screen and right side

12. **Screen (waveform)**
    `NEEDLE SEARCH` = while stopped, go to the touched position
13. **CUE/LOOP CALL ◂ ▸**
    Recall memory cues
14. **DELETE**
    Delete a memory cue
15. **MEMORY**
    Store a memory cue
16. **Rotary selector**
    How much the video effects act on the sound
17. **TOUCH/BRAKE**
    Brake length
18. **RELEASE/START**
    Start-up length
19. **SYNC**
20. **TEMPO ±6/±10/±16/WIDE**
    TEMPO RANGE (tempo range)
21. **TEMPO**
    Tempo slider

Controls without a number (4/8 BEAT, OUT, SEARCH, TIME MODE, MASTER, MASTER TEMPO, TEMPO RESET, the buttons above the screen and so on) are not used by ReaScratch. There is no crossfader or volume fader, so use the keyboard or another controller (the preset only sets the crossfader side and leaves the channel and CC as they were).

In MIDI mode the jog sends "turning speed" as 7 bits (steps of about 0.06x), so the movement is coarser than on the FLX4 or Buddy. The numbers come from the MIDI message list in the player's manual; not yet tested on the real hardware.

97 · 98 CDJ-3000/3000X (MIDI ch 1) / (MIDI ch 2)

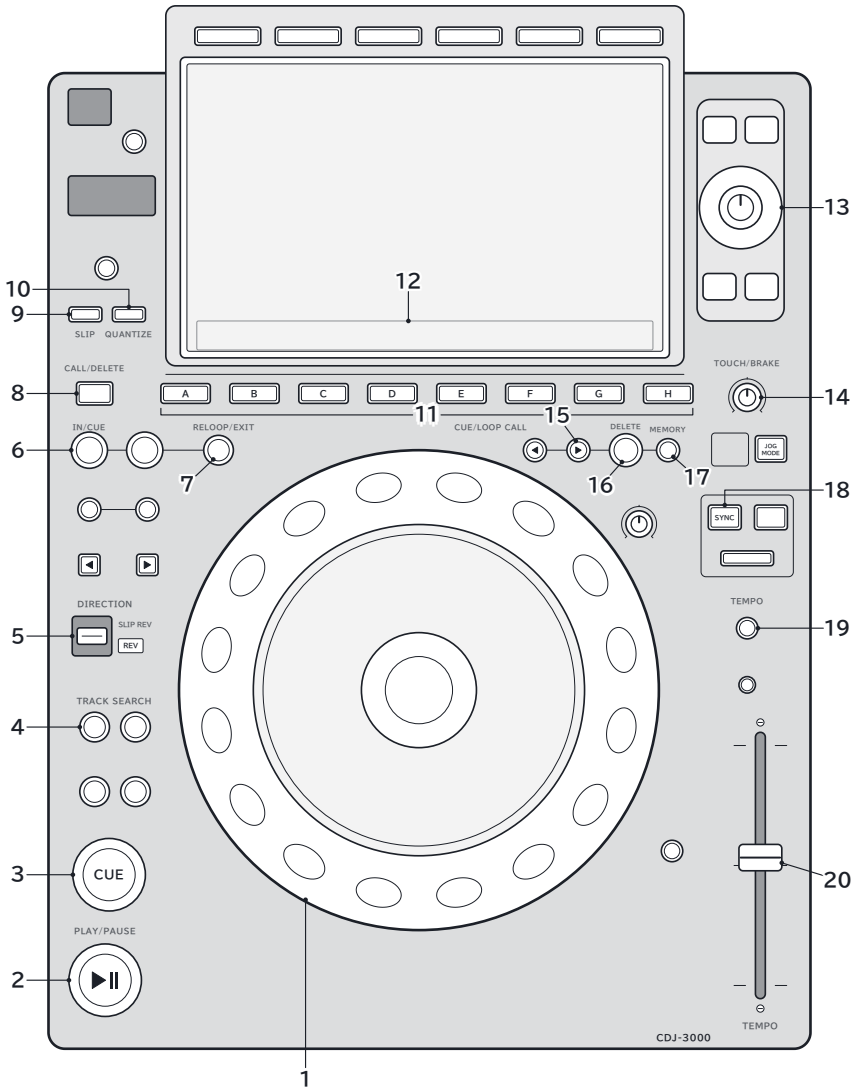


Figure 72 The CDJ-3000's controls (one player = one deck; the CDJ-3000X has the same controls in the same order)

Connect the CDJ to the computer over USB and put it in MIDI mode with `SOURCE` → `SOFTWARE CONTROL`. Hold `MENU` (`UTILITY`) → `MIDI Channel` and set the first player to 1 and the second to 2. Windows needs the player's driver.

Same mapping as the CDJ-2000NXS2 (including loops, memory cues, SLIP, REVERSE and so on; there is no `RELEASE/START` knob). The MIDI numbers differ (top touch, hot cues, SYNC, TEMPO RANGE), so it is a separate preset.

Left side and jog

1 **Jog**
Touch the top and turn to scratch (set `JOG MODE` to VINYL). Turning the side nudges while playing and moves the position while stopped

2 **PLAY/PAUSE**

3 **CUE**

4 **TRACK SEARCH**
Previous / next sample

5 **DIRECTION**
`REV` = `REVERSE` (each press toggles reverse play), `SLIP REV` = `SLIP REVERSE` (reverse while held)

6 **IN/CUE**
`LOOP IN` = loop

7 **RELOOP/EXIT**
Exit / go back to the loop

8 **CALL/DELETE**
Hold it and press a hot cue to clear it

9 **SLIP**

10 **QUANTIZE**

11 **HOT CUE A–H**
Hot cues 1–8 (store if empty)

Round the screen and right side

12 **Screen (waveform)**
`NEEDLE SEARCH` = while stopped, go to the touched position

13 **Rotary selector**
How much the video effects act on the sound

14 **TOUCH/BRAKE**
Brake length

15 **CUE/LOOP CALL ◂ ▸**
Recall memory cues

16 **DELETE**
Delete a memory cue

17 **MEMORY**
Store a memory cue

18 **BEAT SYNC**
SYNC

19 **TEMPO ±6/±10/±16/WIDE**
TEMPO RANGE (tempo range)

20 **TEMPO**
Tempo slider

Controls without a number (OUT, BEAT LOOP, BEAT JUMP, SEARCH, TIME/A.CUE, JOG ADJUST, MASTER, KEY SYNC, MASTER TEMPO, TEMPO RESET, the buttons above the screen and so on) are not used by ReaScratch. There is no crossfader or volume fader, so use the keyboard or another controller.

The numbers come from the official MIDI message lists of the CDJ-3000 and the CDJ-3000X (the numbers used are the same on both); not yet tested on the real hardware.

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99 · 100 XDJ-RX3 / XZ deck 1 / deck 2

Put the unit in MIDI mode (on the XDJ-XZ, PC Control mode via `BROWSE` → `MIDI` → `LOAD`). The jog works like a DDJ, the buttons use the CDJ numbers (PLAY/PAUSE, CUE, SYNC, TEMPO RANGE). LOAD (ch1 81 / ch2 81) loads the file chosen in the Media Explorer into an empty sample slot. Hold a PAD MODE button (5) and press LOAD to preview (the same as the `P` key). From the official MIDI list (RX3); not yet tested on the real hardware.

XDJ-RX3

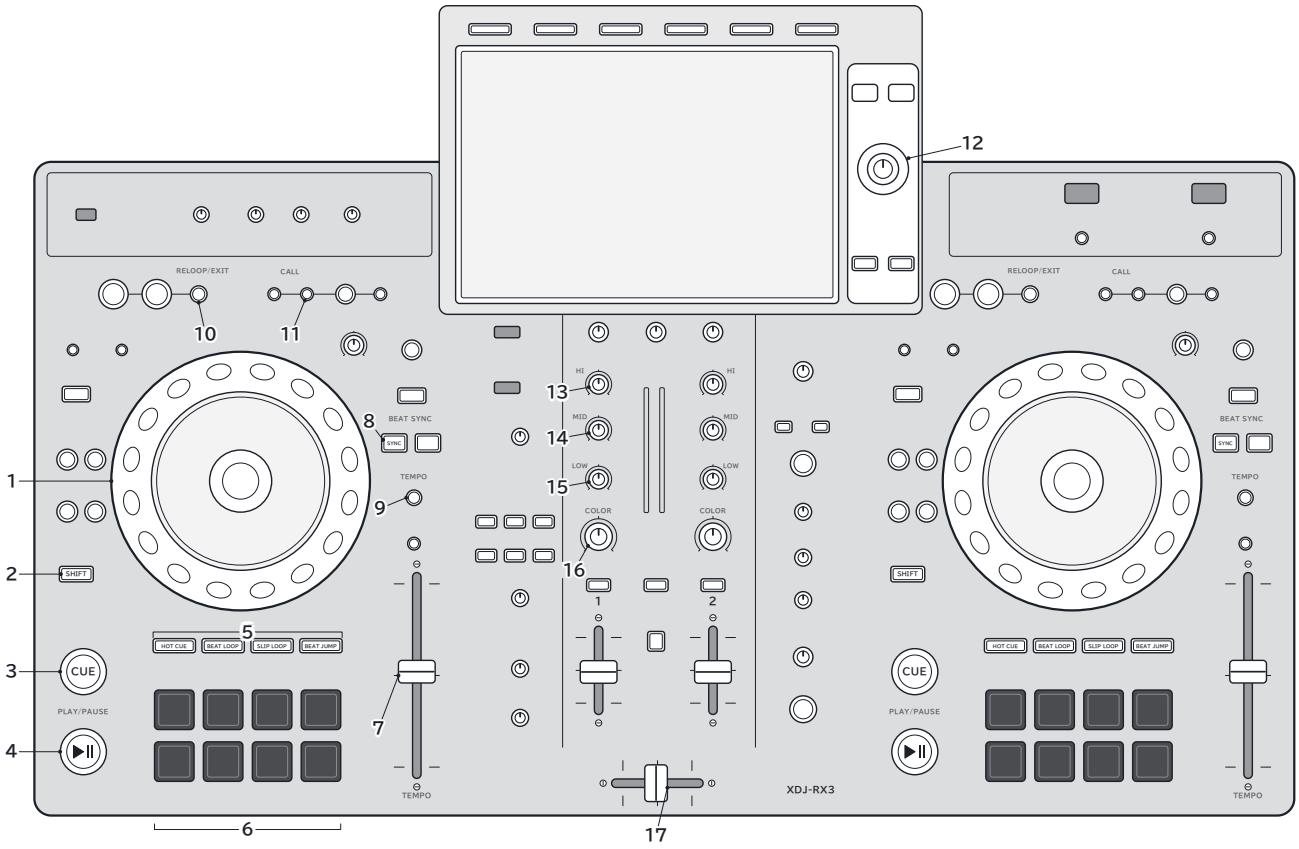


Figure 73 The XDJ-RX3's controls (numbered on the left deck and the centre; the right deck is the same)

Deck

1 Jog
Top = touch scratch, outer ring = nudge. `SHIFT` + jog makes big jumps

2 SHIFT
Hold it with another control for its second function

3 CUE

4 PLAY/PAUSE

5 PAD MODE buttons
`HOT CUE` = HOT CUE, `BEAT LOOP` → AUTO LOOP, `SLIP LOOP` → BOUNCE LOOP, `BEAT JUMP` → SLICER. With SHIFT held (EXTENSION 1–4) → TOUCH FX, PITCH PLAY, NEURAL MIX, FILTER TYPE

6 Pads
The 8 pads of the current mode. In `HOT CUE`, SHIFT + pad clears a hot cue

7 TEMPO
Tempo slider

8 BEAT SYNC
SYNC

9 TEMPO ±6/±10/±16/WIDE
TEMPO RANGE (tempo range)

10 RELOOP/EXIT
Exit / go back to the loop

11 CUE/LOOP CALL ◂ ▸
Halve / double the loop

12 Rotary selector
BROWSE

13 HI
Filter Morph (echo length with SHIFT)

14 MID
How much the video effects act on the sound

15 LOW
Filter Resonance (echo feedback with SHIFT)

16 COLOR
Filter Cutoff (key with SHIFT)

17 Crossfader
Deck 1 on the left side, deck 2 on the right (ch 5)

4 BEAT (loop) is set to the number of the XDJ-XZ's 4/8 BEAT button (10 in the XDJ-XZ drawing below). Controls without a number (IN, OUT, DELETE, MEMORY, SLIP, QUANTIZE, DIRECTION, TRACK SEARCH, SEARCH, JOG MODE, TRIM, the headphone CUE, the channel faders, SOUND COLOR FX, BEAT FX and so on) are not used by ReaScratch.

XDJ-XZ

The same mapping as the XDJ-RX3, plus the 4/8 BEAT button (10).

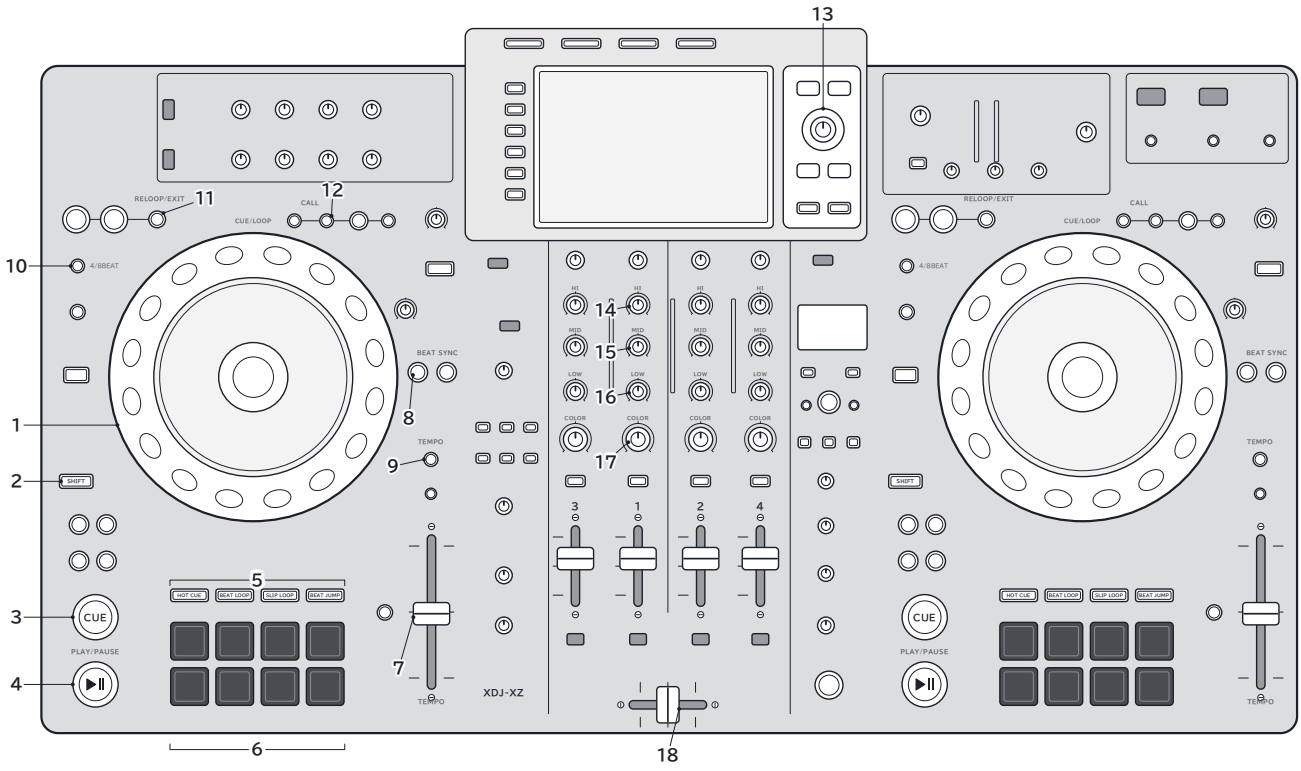


Figure 74 The XDJ-XZ's controls (numbered on the left deck and the centre; the right deck is the same. Deck 1 is the mixer's channel 1)

Deck

1 **Jog**
Top = touch scratch, outer ring = nudge. `SHIFT` + jog makes big jumps

2 **SHIFT**
Hold it with another control for its second function

3 **CUE**

4 **PLAY/PAUSE**

5 **PAD MODE buttons**
`HOT CUE` = HOT CUE, `BEAT LOOP` → AUTO LOOP, `SLIP LOOP` → BOUNCE LOOP, `BEAT JUMP` → SLICER. With SHIFT held (EXTENSION 1–4) → TOUCH FX, PITCH PLAY, NEURAL MIX, FILTER TYPE

6 **Pads**
The 8 pads of the current mode. In `HOT CUE`, SHIFT + pad clears a hot cue

7 **TEMPO**
Tempo slider

8 **BEAT SYNC**
SYNC

9 **TEMPO ±6/±10/±16/WIDE**
TEMPO RANGE (tempo range)

10 **4/8 BEAT**
Loop

11 **RELOOP/EXIT**
Exit / go back to the loop

12 **CUE/LOOP CALL ◂ ▸**
Halve / double the loop

Mixer

13 **Rotary selector**
BROWSE

14 **HI**
Filter Morph (echo length with SHIFT)

15 **MID**
How much the video effects act on the sound

16 **LOW**
Filter Resonance (echo feedback with SHIFT)

17 **COLOR**
Filter Cutoff (key with SHIFT)

18 **Crossfader**
Deck 1 on the left side, deck 2 on the right (ch 5)

Controls without a number (IN/CUE, OUT, DELETE, MEMORY, SLIP, DIRECTION, TRACK SEARCH, SEARCH, JOG MODE, TRIM, the headphone CUE, the channel faders, channels 3 and 4, SOUND COLOR FX, BEAT FX, MIC and so on) are not used by ReaScratch.

101 · 102 Reloop Buddy deck 1 / deck 2

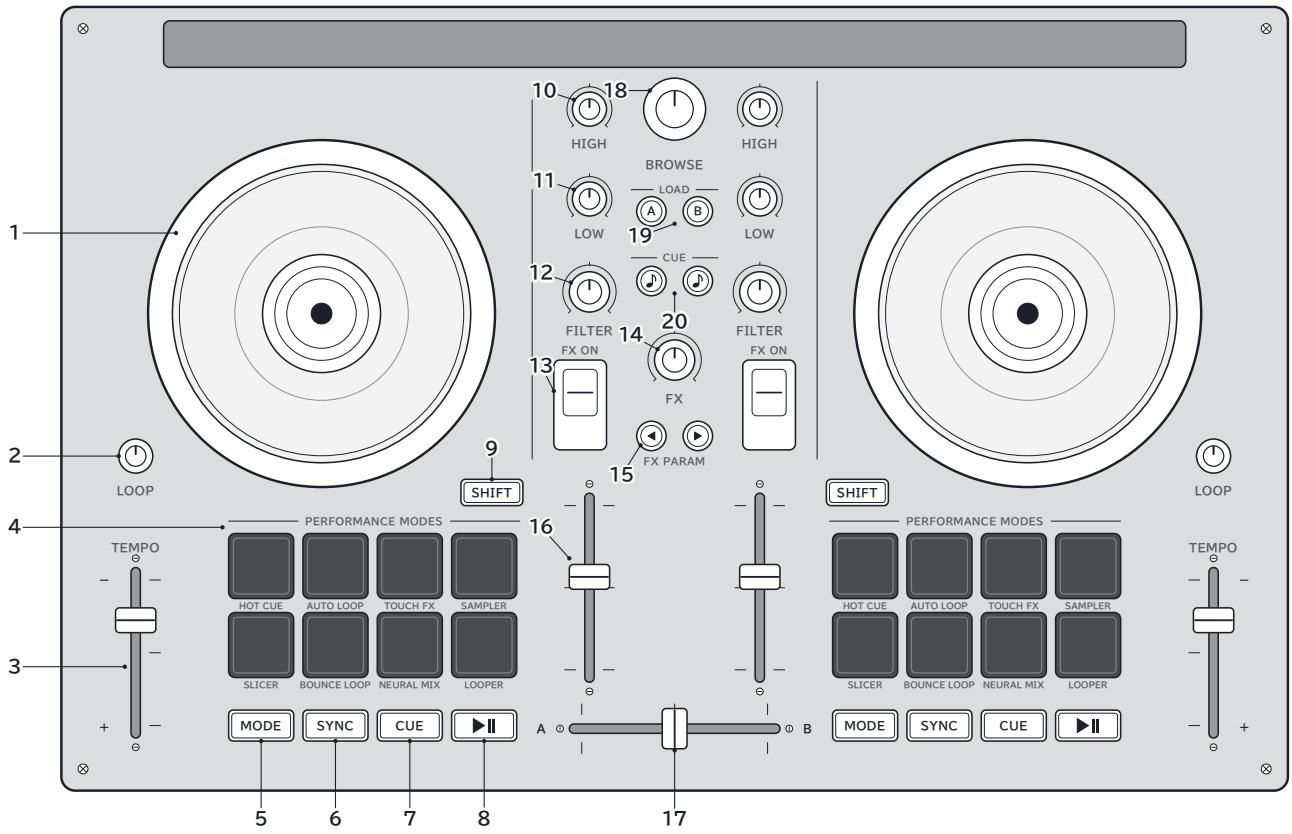


Figure 75 The Reloop Buddy's controls (the left deck and the centre; the right deck is the same, mirrored)

Deck

1 Jog
Touch scratch (press the top and turn). Turning the side without touching the top nudges to match beats while playing (pitch bend) and moves the position while stopped. Coasting right after you let go is ignored. `SHIFT` + jog makes big jumps

2 LOOP knob
Turn = length, press = loop (with SHIFT, move 1 beat / reloop)

3 Tempo fader
Bottom = +

4 Pads
Eight pads for the current mode. Starts in `HOT CUE` when the project opens. There is no mode for choosing samples (use the `F1` – `F8` keys or the `1` – `8` buttons at the bottom of the screen)

5 PERFORMANCE MODES (the mode button)
Hold it and press the pad with the mode's label to choose one of 8 modes (the `SAMPLER` label = FILTER TYPE, the `NEURAL MIX` label = PITCH PLAY, the `LOOPER` label = NEURAL MIX; see "Pad modes" (p. 28) above)

6 SYNC
With SHIFT, changes the tempo range

7 CUE
With SHIFT, freeze

8 PLAY
With SHIFT, plays / stops both decks at once

9 SHIFT
Hold with another control to use its second function

10 **HIGH**
Filter Morph (with SHIFT, echo length)

11 **LOW**
Filter Resonance (with SHIFT, echo feedback)

12 **FILTER**
Filter Cutoff (with SHIFT, key). After switching SHIFT on or off, 10–12 take effect only once the knob reaches the current value

13 **FX ON bar**
That deck's video effect on/off (with SHIFT, temporarily turns off all effects)

14 **FX knob**
The effect amount (see the note below)

15 **FX PARAM ◂ ▸**
The effect type (see the note below)

16 **Volume fader**
That deck's volume and picture strength

17 **Crossfader**
Deck 1 on the left side, deck 2 on the right

18 **BROWSE**
Turn = the headphone Cue mix (with the bar or a pad held, that effect's sound amount). Turn with SHIFT = the sound of the effect after the mix (with the bar on, that deck's overall sound multiplier)

19 **LOAD A / B**
Load the file chosen in the Media Explorer into deck A / B (SHIFT + LOAD replaces the current sample; see "Loading with the LOAD button (from the Media Explorer)" (p. 23)). Hold the mode button (5) and press LOAD to preview (see "Previewing a Media Explorer file (P key)" (p. 25))

20 **Headphone button**
Listen to this deck on headphones (toggles with each press; with SHIFT, hides only the picture)

Note (14, 15): The FX knob sets an amount and `FX PARAM` ◂ ▸ picks a type. The FX knob takes effect only when turned near or past the current value, so the value never jumps (until then, the screen shows where to turn it). The amounts of the deck's effect and the effect after the mix are remembered for each type: changing the type restores that type's amount (when REAPER starts: Mirror 25%, Kaleidoscope 20%, Shift RGB 30%, Strobe 74%, VHS 30%, the others 100%). What they act on depends on what you hold: nothing = the transition (◂ ▸ picks, hold to scroll fast; the FX knob softens its edges); SHIFT = the video effect after the mix; the mode button = the video blend (◂ ▸ picks, the FX knob sets the key threshold, SHIFT + ◂ ▸ turns the blend-with-transitions combination on or off); the bar on = that deck's video effect; a TOUCH FX pad held = that pad's strength (◂ ▸ too).

On Windows it needs the driver described under "Troubleshooting" (p. 219) below. The buttons light up in the hot cue colours and so on (see "Button lights" (p. 194) below). The jog, pads, LOOP knob, filter, FX and LOAD were tested on real hardware. Many SHIFT combinations (CUE, PLAY, jog, FX ON, headphones, HIGH / LOW / FILTER) and the TOUCH FX, PITCH PLAY and FILTER TYPE modes have not been tested on real hardware yet.

103 · 104 Reloop Ready deck 1 / deck 2

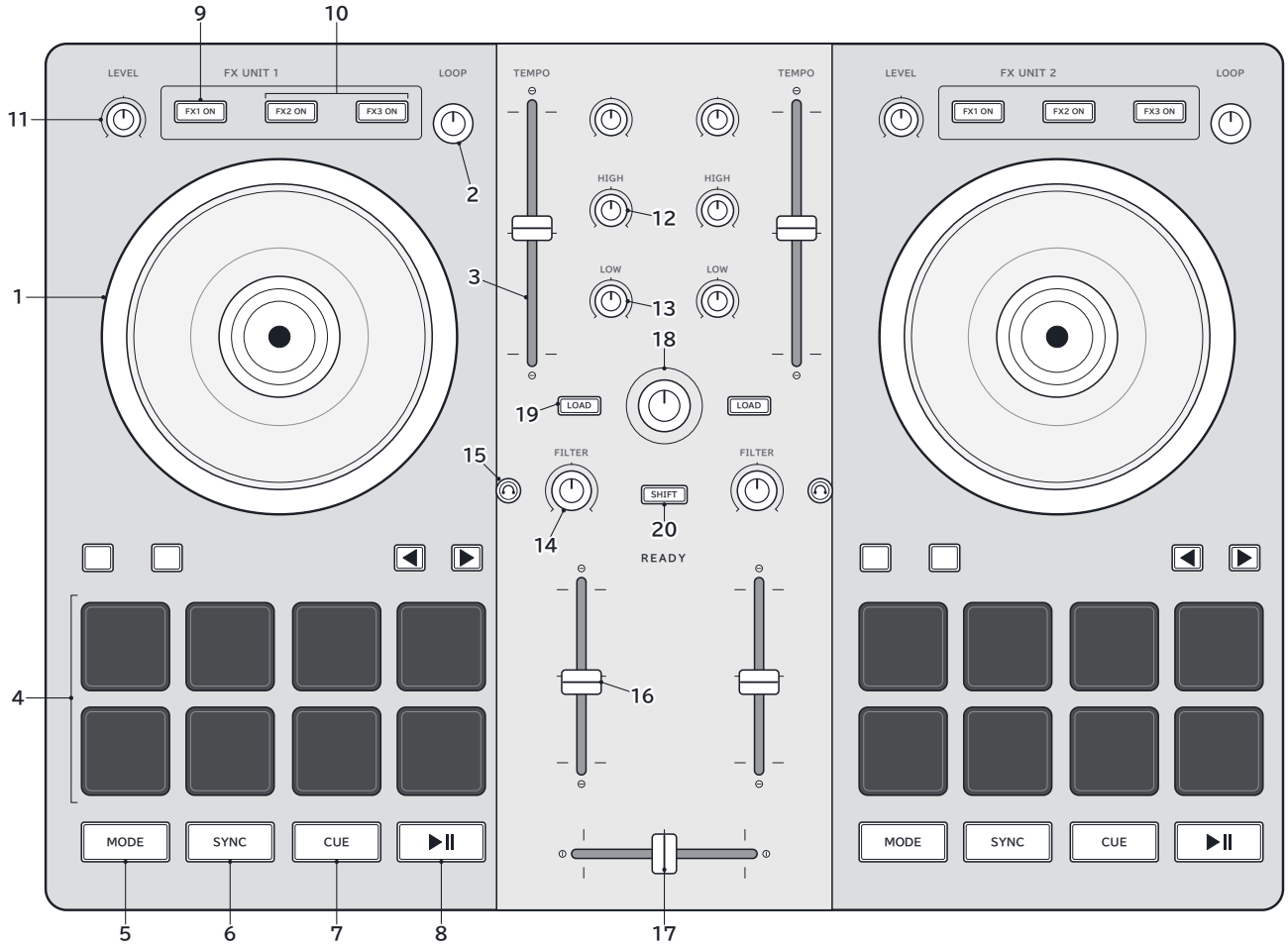


Figure 76 The Reloop Ready's controls (numbered on the left deck and the centre; the right deck is the same, except that its TEMPO is on the other side, both next to the mixer)

Deck

1 Jog
Touch scratch (as on the Reloop Buddy). `SHIFT` + jog makes big jumps

2 LOOP knob
Turn = length, press = loop (with SHIFT, move 1 beat / reloop; the SHIFT side arrives on the pad channel, ch 5 / 6)

3 TEMPO
Like the Buddy, + (faster) at the bottom, at its maximum value

4 Pads
The 8 pads of the current mode. Starts in `HOT CUE` when the project opens

5 MODE
Hold it and press a pad to choose one of 8 modes (Hotcue → HOT CUE, AutoLoop → AUTO LOOP, ManualLoop → SLICER, Sampler → FILTER TYPE, Pitch → PITCH PLAY, ScratchBank → NEURAL MIX, LoopRoll → BOUNCE LOOP, BeatGrid → TOUCH FX)

6 SYNC
With SHIFT, changes the tempo range

7 CUE
With SHIFT, freeze

8 PLAY
With SHIFT, plays / stops both decks at once

FX UNIT (Buddy-style)

9 FX1 ON
The Buddy's FX ON bar (that deck's video effect on/off; with SHIFT, temporarily turns off all effects). For the right deck, FX1 ON of the right FX UNIT 2

10 FX2 ON · FX3 ON
The Buddy's `FX PARAM` ◂ ▸ (the effect type; ch 9, notes 1 / 2 = the second and third buttons of the left FX UNIT 1; the right deck uses these two as well)

11 LEVEL
The Buddy's FX knob (the effect amount). SHIFT + FX knob has no number of its own, so the unit's SHIFT is used

Mixer and centre

12 HIGH
Filter Morph (with SHIFT, echo length)

13 LOW
Filter Resonance (with SHIFT, echo feedback)

14 FILTER
Filter Cutoff (with SHIFT, key)

15 Headphone CUE
Listen to this deck on headphones (toggles with each press; with SHIFT, hides only the picture)

16 Channel fader
That deck's volume and picture strength

17 Crossfader
Deck 1 on the left side, deck 2 on the right

18 BROWSE
As on the Reloop Buddy (turn = the headphone Cue mix, turn with SHIFT = the sound of the effect after the mix)

19 LOAD
Load the file chosen in the Media Explorer into deck A / B (SHIFT + LOAD replaces the current sample)

20 SHIFT
Hold with another control to use its second function (one, in the centre)

Same mapping as the Reloop Buddy, with the same MIDI numbers (checked against Mixxx's mapping). The differences are the SHIFT side of 2 and items 5, 10 and 11 above (from an unmerged Mixxx mapping draft). What 9–11 act on depends on what you hold (as in the Reloop Buddy's note). The tempo fader's direction was checked on real hardware by the author of the Mixxx mapping for the Ready, and by ear on real hardware for the Buddy. On Windows it probably needs Reloop's driver like the Buddy. Not yet tested on the real hardware.

Controls without a number (GAIN, VINYL, KEY LOCK, PARAMETER ◂ ▸, FX2 ON and FX3 ON of the right FX UNIT 2 and so on) are not used by ReaScratch.

105 · 106 Reloop Mixage deck 1 / deck 2

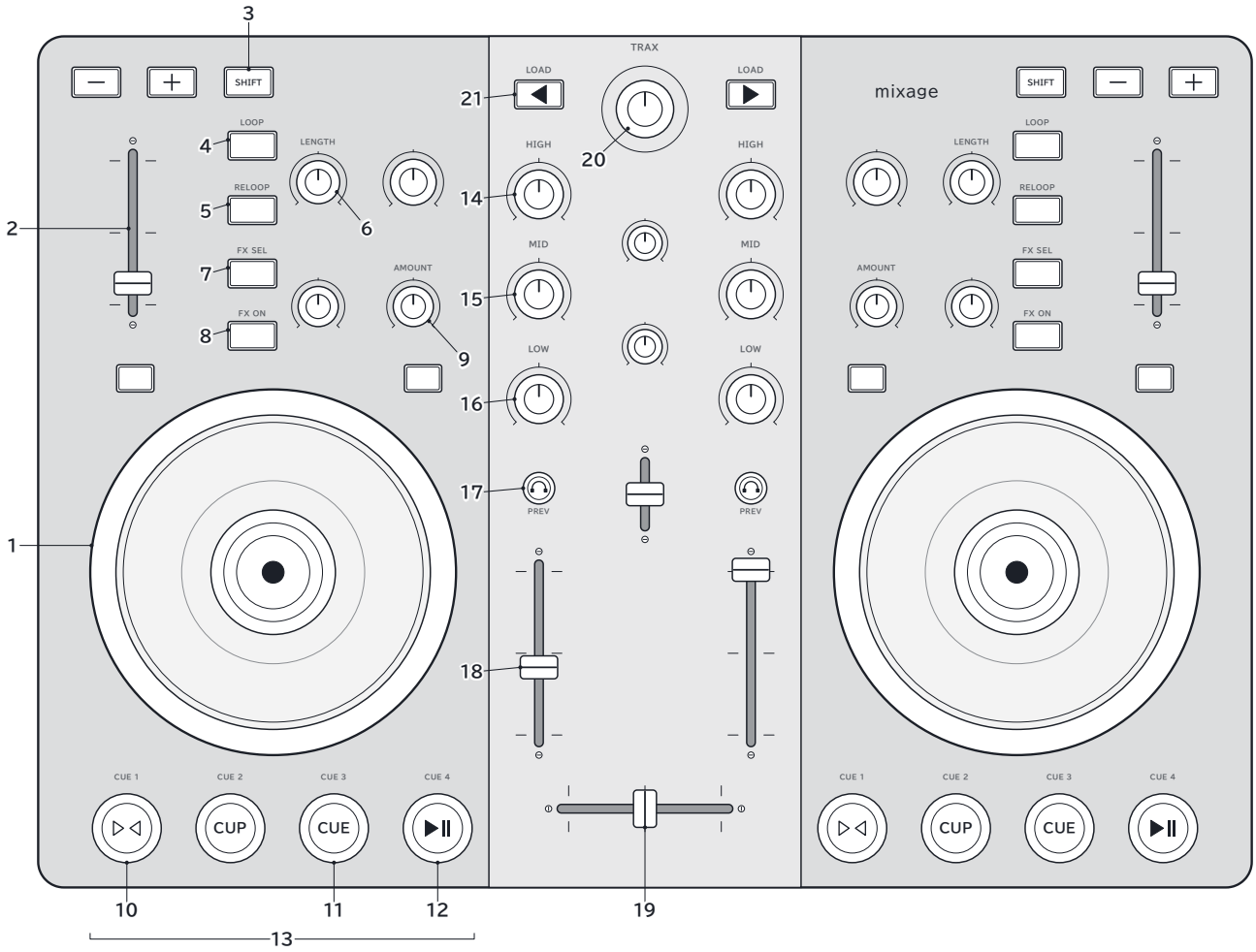


Figure 77 The Reloop Mixage's controls (numbered on the left deck and the centre; the right deck is the same, mirrored)

Deck

1 Jog
Top scratch (touch)

2 TEMPO
Tempo (pitch bend). Deck 1 only (deck 2's pitch bend arrives on channel 2 and cannot be read)

3 SHIFT
Hold it with another control for its second function

4 LOOP
Loop

5 RELOOP
Reloop

6 LENGTH knob
The loop length. Turn with SHIFT = move the loop

7 FX SEL
The Buddy's `FX PARAM` ◂ (the right deck's FX SEL is ▸).
`SHIFT` + press = the type of the effect after the mix

8 FX ON
The Buddy's FX ON (that deck's video effect on/off).
`SHIFT` + FX ON = all effects off while held (that deck)

9 AMOUNT
The Buddy's FX knob (the effect amount). `SHIFT` + turn = filter Cutoff

10 SYNC

11 CUE

12 PLAY

13 SHIFT + SYNC, CUP, CUE, PLAY
Hot cues 1–4

14 **HIGH**
Filter Morph

15 **MID**
How much the video effects act on the sound

16 **LOW**
Filter Resonance

17 **Headphone CUE**
Listen to this deck on headphones

18 **Channel fader**
That deck's volume and picture strength

19 **Crossfader**
Deck 1 on the left side, deck 2 on the right

20 **TRAX**
BROWSE (turn, push)

21 **LOAD ◂ ▸**
LOAD (◂ = deck 1, ▸ = deck 2)

Both decks on MIDI channel 1, told apart by number. What 7–9 act on depends on what you hold (as in the Reloop Buddy's note). From Mixxx's mapping (one mapping for the Interface Edition, MK2 and Controller Edition); not yet tested on the real hardware.

Controls without a number (GAIN, DRY/WET, pitch − / +, the two buttons above the jog, MASTER, PHONES, CUEMIX and so on) are not used by ReaScratch.

107 · 108 Reloop Mixon 4 deck 1 / deck 2

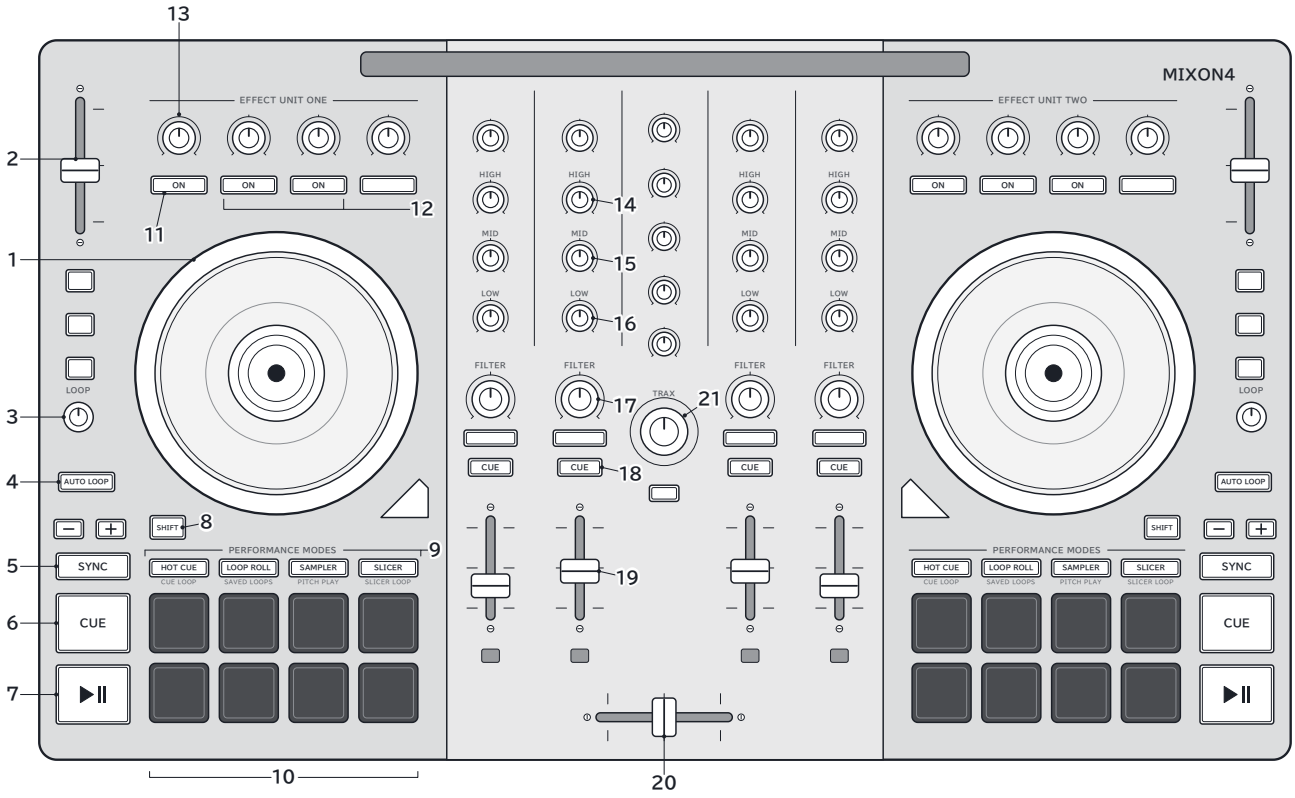


Figure 78 The Reloop Mixon 4's controls (numbered on the left deck, the mixer's CHANNEL 1 and the centre; the right deck and CHANNEL 2 are the same, with the TEMPO and the SYNC / CUE / PLAY column mirrored to the outer edge)

Deck

1 Jog
Top scratch (CC 6) and touch

2 TEMPO
Tempo (pitch bend)

3 LOOP knob
Turn = length, push = AUTO LOOP

4 AUTO LOOP
The auto loop button

5 SYNC

6 CUE

7 PLAY/PAUSE

8 SHIFT
SHIFT + pad clears a hot cue

9 Pad mode buttons
8 modes (the second ones with SHIFT). HOT CUE, LOOP ROLL → BOUNCE LOOP, SAMPLER → samples, SLICER, CUE LOOP → AUTO LOOP, SAVED LOOPS → NEURAL MIX, PITCH PLAY, SLICER LOOP → TOUCH FX

10 Pads
The 8 pads of the current mode (ch 1 / ch 2)

EFFECT UNIT (Buddy-style)

11 ON (the first)
The Buddy's FX ON

12 ON (the second and third)
The Buddy's `FX PARAM` ◂ ▸ (FX ON 2 / 3)

13 The first knob
The Buddy's FX knob (the effect amount)

Mixer and centre

14 **HIGH**
Filter Morph

15 **MID**
How much the video effects act on the sound

16 **LOW**
Filter Resonance

17 **FILTER**
Filter Cutoff

18 **Headphone CUE**
Listen to this deck on headphones

19 **Channel fader**
That deck's volume and picture strength

20 **Crossfader**
ch 11. Deck 1 on the left side, deck 2 on the right

21 **TRAX**
BROWSE (turn, push)

Decks on ch 5 / ch 6 (pads on ch 1 / ch 2). What 11–13 act on depends on what you hold (as in the Reloop Buddy's note). Numbers from Reloop's official MIDI map (version preserved on the Internet Archive). The jog tick count and tempo direction are not in the documents and are provisional. LOAD (ch5 2 / ch6 2) loads the file chosen in the Media Explorer into an empty sample slot (there is no SHIFT + LOAD). Hold a pad mode button (9) and press LOAD to preview (the same as the `P` key). The SHIFT combinations other than 8 and 9 above are not mapped. Not yet tested on the real hardware.

Controls without a number (GAIN, the effect unit's other knobs, TAP, PARAM − / +, the three buttons left of the jog, CHANNEL 3 / 4, the MASTER column's knobs and so on) are not used by ReaScratch.

109 · 110 Reloop Mixon 8 Pro deck 1 / deck 2

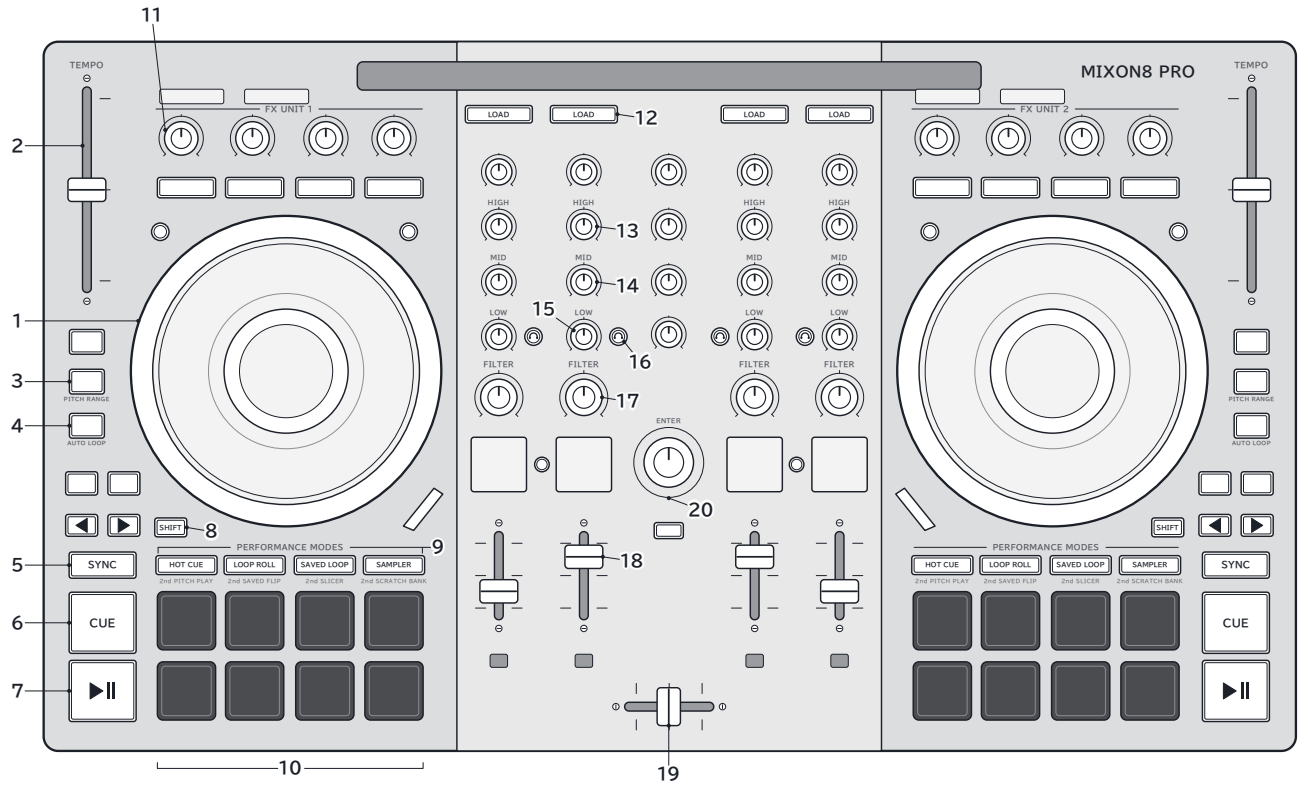


Figure 79 The Reloop Mixon 8 Pro's controls (numbered on the left deck, the mixer's CHANNEL 1 and the centre; the right deck and CHANNEL 2 are the same, with the TEMPO and the SYNC / CUE / PLAY column mirrored to the outer edge)

Deck

1 Jog
Top scratch (CC 6, 64-centre values) and touch

2 TEMPO
Tempo (pitch bend)

3 PITCH RANGE
Changes the tempo range

4 AUTO LOOP
The auto loop button

5 SYNC

6 CUE

7 PLAY/PAUSE

8 SHIFT
Hold it with another control for its second function (SHIFT + pad clears a hot cue; turn BROWSE with SHIFT = the headphone Cue mix, push = which deck BROWSE works on, A / B)

9 Pad mode buttons
hot cue → HOT CUE, roll → BOUNCE LOOP, slicer → SLICER, sampler → samples, saved loop → AUTO LOOP, pitch play → PITCH PLAY, stems → NEURAL MIX, instant FX → TOUCH FX. Which button sends which mode is not known

10 Pads
The 8 pads of the current mode (ch 5 / ch 6)

11 The FX UNIT's first knob
The FX wet/dry = that deck's video effect amount

12 **LOAD**
Load the file chosen in the Media Explorer into deck A / B

13 **HIGH**
Filter Morph

14 **MID**
How much the video effects act on the sound

15 **LOW**
Filter Resonance

16 **Headphone CUE**

17 **FILTER**
Filter Cutoff

18 **Channel fader**
That deck's volume and picture strength

19 **Crossfader**
ch 15. Deck 1 on the left side, deck 2 on the right

20 **BROWSE**
Turn, push

Almost the same numbers as the Reloop Buddy (decks ch 1 / ch 2, jog CC 6, touch, PLAY, CUE, SYNC, channel fader, headphone CUE, crossfader ch 15, HOT CUE pads on ch 5 / ch 6). The differences: the jog uses 64-centre values and the tempo is pitch bend. No official MIDI list from Reloop could be found, so the numbers come from a mapping file published by a djay (DJ app) user. The jog tick count and tempo direction are provisional. The pad modes come from the same mapping file. The number of FX ON (the paddle) is not in any document, so it is not mapped (you can check it in the `Learn` tab of setup). Not yet tested on the real hardware.

Controls without a number (GAIN, the FX UNIT's other knobs and buttons, KEY SYNC, IN, OUT, PARAM ◂ ▸, CHANNEL 3 / 4, the MASTER column's knobs and so on) are not used by ReaScratch.

111 · 112 Reloop Touch deck 1 / deck 2

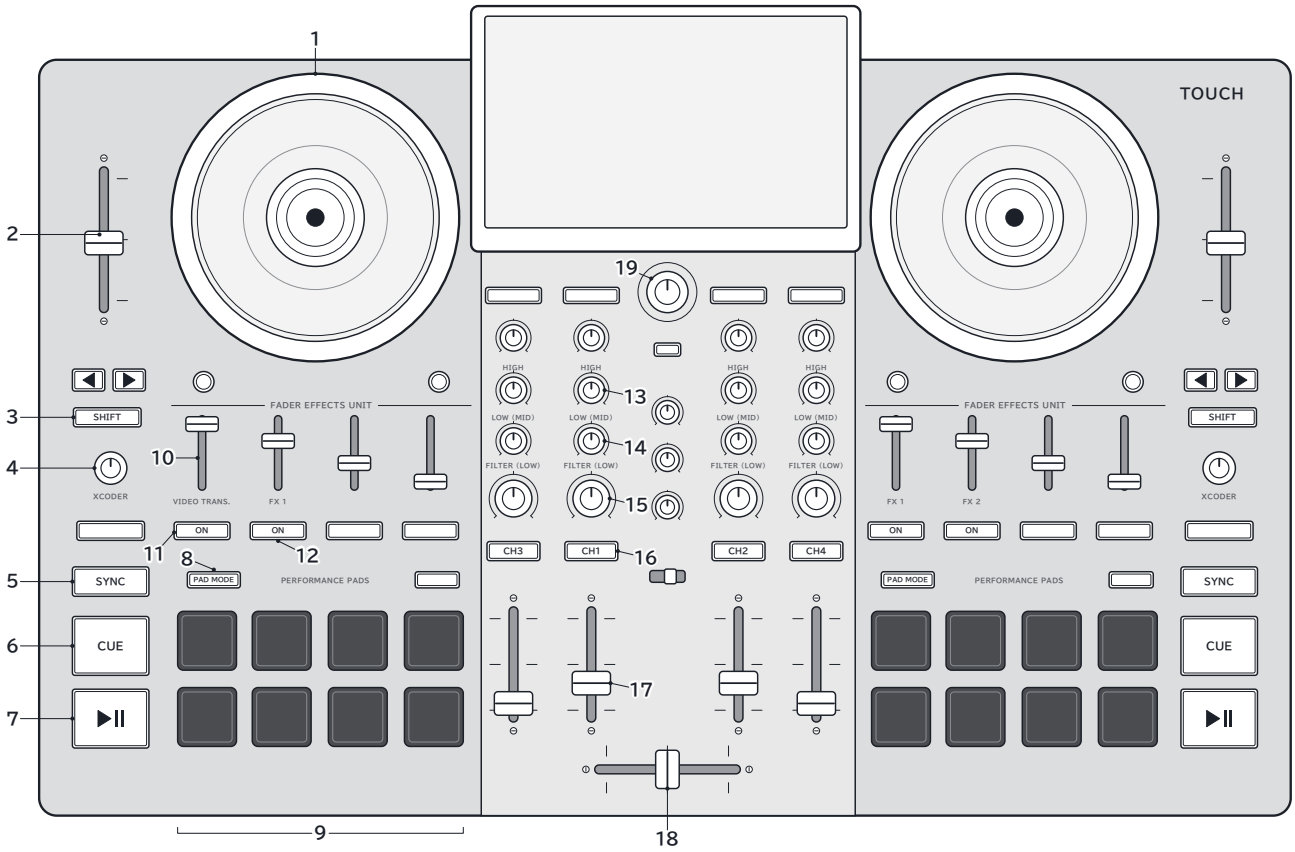


Figure 80 The Reloop Touch's controls (numbered on the left deck, the mixer's CH1 and the centre; the right deck and CH2 are the same, with the TEMPO and the SYNC / CUE / PLAY column mirrored to the outer edge. The right FADER EFFECTS UNIT is printed FX 1, FX 2, FX 3, VIDEO FX, but ReaScratch reads it by position: its first fader and ON from the left are 10 and 11)

Deck

1 Jog
Top scratch (CC 96, 64-centre values) and touch

2 TEMPO
Tempo (pitch bend). The reverse of the Beatmix: the maximum value is on the slow side

3 SHIFT
Hold it with another control for its second function. SHIFT adds 64 to the pad number (clears a hot cue)

4 XCODER
The loop knob (turn, push)

5 SYNC

6 CUE

7 PLAY/PAUSE

8 PAD MODE
Hold it and press pad 1–8 to pick a mode (as on the Buddy: Hot Cue → HOT CUE, Roll → BOUNCE LOOP, Slicer → SLICER, Sampler → FILTER TYPE, Cue Loop → AUTO LOOP, Saved Loop → NEURAL MIX, Beat Jump → TOUCH FX, Key Cue → PITCH PLAY)

9 Pads
The 8 pads of the current mode

FADER EFFECTS UNIT (Buddy-style)

10 **The first fader from the left (DRY/WET; printed VIDEO TRANS.)**
The Buddy's FX knob (the effect amount)

11 **The first ON from the left**
The Buddy's FX ON. SHIFT + FX ON = all effects off while held

12 **The second ON from the left (printed FX 1)**
The Buddy's `FX PARAM` ◂ (the right unit's second ON from the left, printed FX 2, is ▸)

Mixer and centre

13 **HIGH**
Filter Morph

14 **LOW (MID)**
Filter Resonance

15 **FILTER (LOW)**
Filter Cutoff (set the unit's EQ mode to FILTER; in 3-band mode the same knob is LOW)

16 **CH1**
Headphone CUE (note 60)

17 **Channel fader**
That deck's volume and picture strength

18 **Crossfader**
ch 6. Deck 1 on the left side, deck 2 on the right

19 **BROWSE**
Turn and push both work

Almost the same numbers as the Reloop Beatmix 2/4 (decks ch 2 / ch 3, jog CC 96, touch, PLAY, CUE, SYNC, tempo (pitch bend), channel fader, crossfader ch 6). From Reloop's official Traktor mapping (version preserved on the Internet Archive). The jog values and the tempo direction come from the same mapping. The tick count is assumed to be the same as the Beatmix. What 10–12 act on depends on what you hold (as in the Reloop Buddy's note). Not yet tested on the real hardware.

LOAD (ch2 57 / ch3 57) loads the file chosen in the Media Explorer into an empty sample slot. SHIFT + LOAD is the preview (the same as the `P` key). Controls without a number (the touchscreen, PITCH BEND ◂ ▸, MODE, SLIP, DECK 3, the FADER EFFECTS UNIT's second fader from the left and its third and fourth faders and ONs, PARAM, GAIN, CH3, CH4, MASTER, CUE VOL, CUE MIX and so on) are not used by ReaScratch.

113 · 114 Reloop Beatpad deck 1 / deck 2

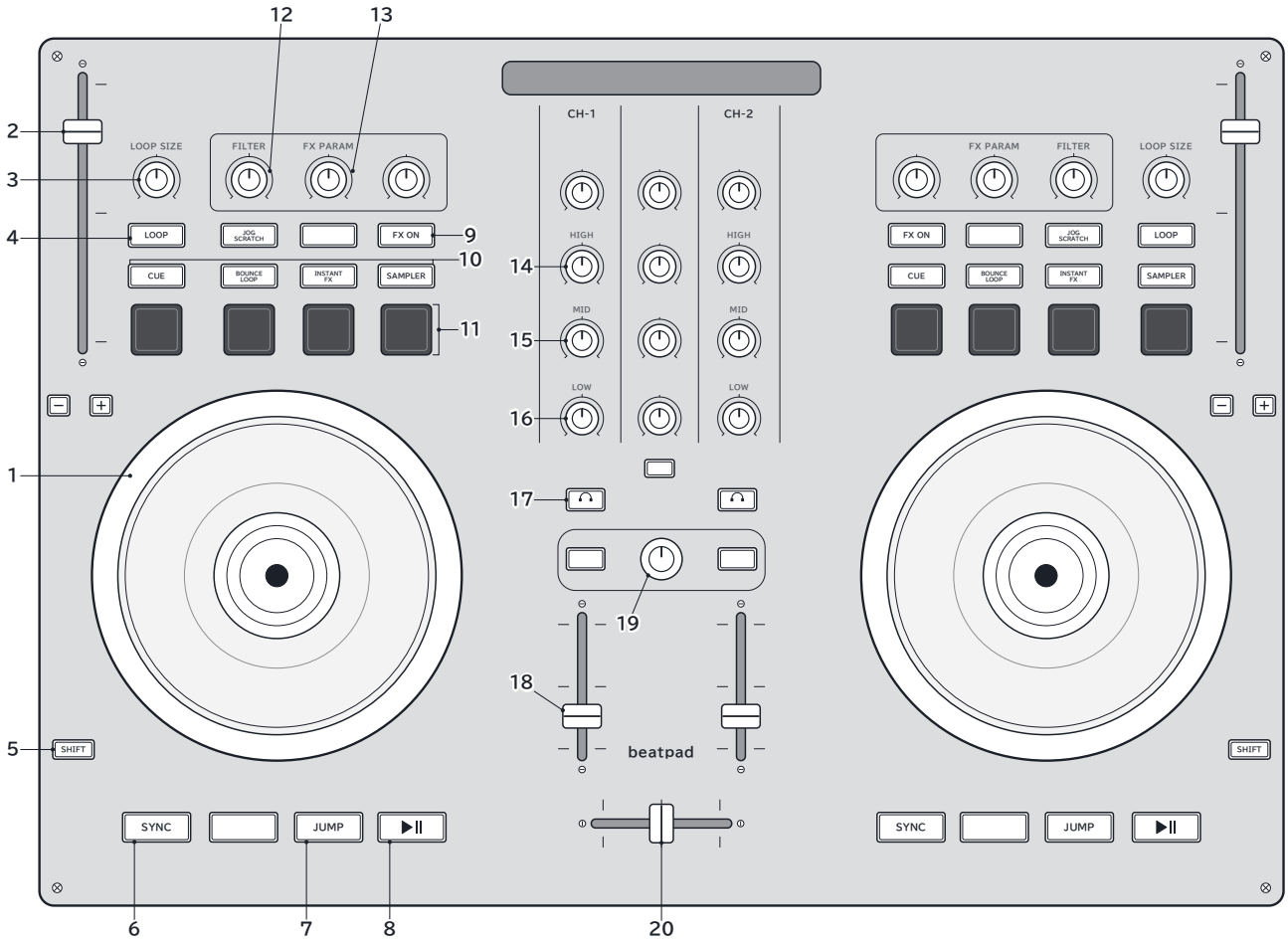


Figure 81 The Reloop Beatpad's controls (numbered on the left deck and the centre; on the right deck the knobs and the upper button row are mirrored, the rest is the same)

Deck

1 **Jog**
Top scratch (touch). Set the controller's jog mode to Vinyl with `JOG SCRATCH`

2 **PITCH**
Tempo fader (pitch bend)

3 **LOOP SIZE**
The loop knob (turn = length)

4 **LOOP**
Loop (press)

5 **SHIFT**
Hold it with another control for its second function

6 **SYNC**

7 **JUMP**
CUE (see the note below)

8 **PLAY**
SHIFT + PLAY plays in reverse while held

9 **FX ON**
Video effect on/off

10 **Pad mode buttons**
`CUE` → HOT CUE, `BOUNCE LOOP` → AUTO LOOP, `INSTANT FX` → TOUCH FX, `SAMPLER` → sample switching (Mixxx's mapping calls the second button LOOP)

11 **Pads**
The 4 pads of the current mode. In `HOT CUE`, SHIFT + pad clears a hot cue

12 **FILTER**
Filter Cutoff

13 **FX PARAM**
The FX knob (effect amount)

Mixer

14 **HIGH**
Filter Morph

15 **MID**
How much the video effects act on the sound

16 **LOW**
Filter Resonance

17 **Headphone button**
Listen to this deck on headphones (SLIP with SHIFT)

18 **Channel fader**
That deck's volume and picture strength

19 **BROWSE**
Turn = how much the video effects act on the sound, push = switch the deck the knob works on, SHIFT + turn = the headphone Cue mix

20 **Crossfader**
(ch 5): Deck 1 on the left side, deck 2 on the right

Note (7): CUE is note 0x5F. In Mixxx's mapping 0x5F is the `JUMP` button (0x5E is `CUE`). In the official MIDI list (v1.2) this number is `CuePlay` and 0x5E is `MainCue` (not checked on the real hardware).

LOAD 1 / 2 (ch1 98 / ch2 98) loads the file chosen in the Media Explorer into an empty sample slot. Hold a pad mode button (10) and press LOAD to preview (the same as the `P` key). Controls without a number (GAIN, JOG SCRATCH (used only to set up), JOG SEEK, FX SELECT, PITCH BEND, SET, REC, MASTER, PHONES, CUE MIX, AUX) are not used by ReaScratch. For the Beatpad, the official MIDI list and Mixxx's mapping disagree in places (such as the jog CC); Mixxx's is used. Tick count from the official list's 600 ticks per turn. Not yet tested on the real hardware.

115 · 116 Reloop Beatpad 2 deck 1 / deck 2

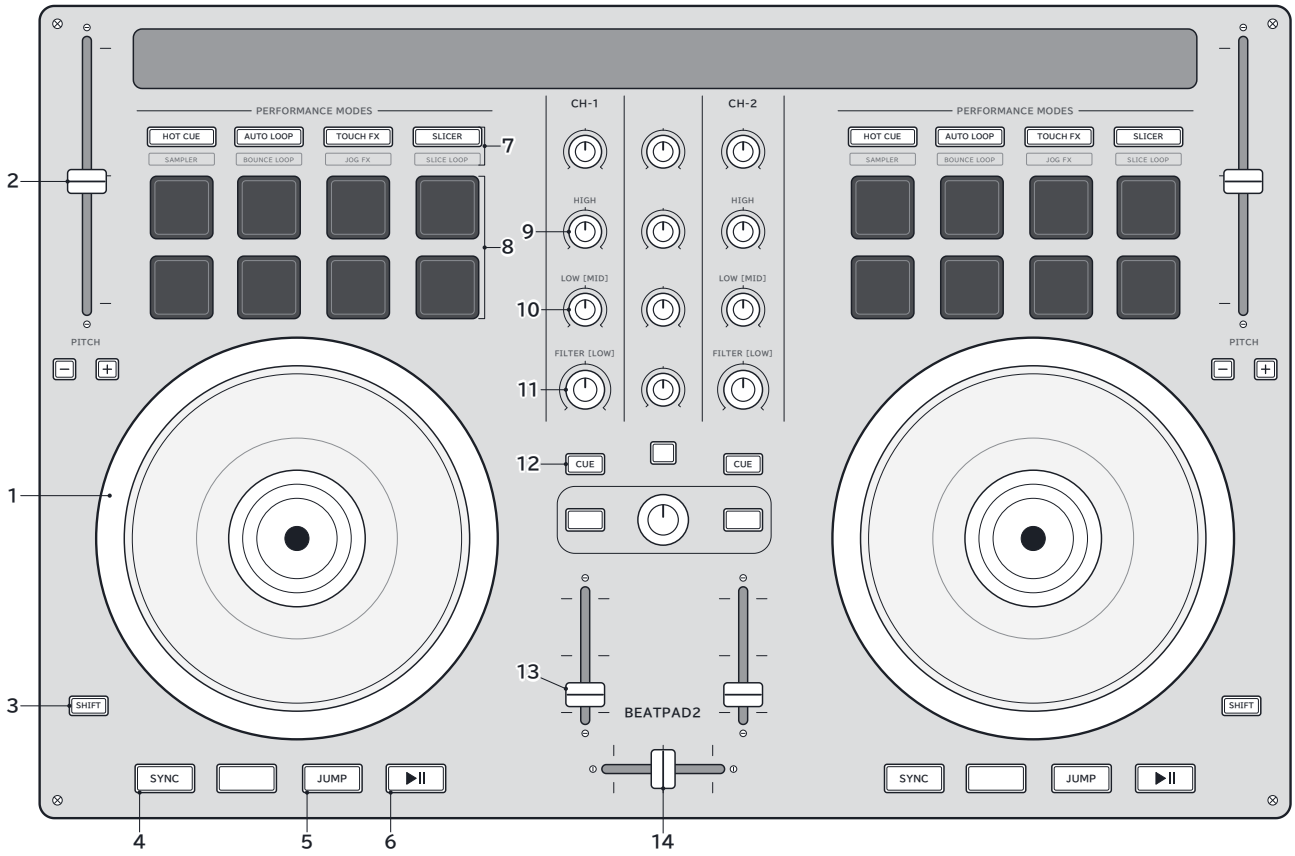


Figure 82 The Reloop Beatpad 2's controls (numbered on the left deck and the centre; the right deck is the same)

Reloop's Beatpad 2 MIDI list covers only the pads, mode buttons and EQ; the rest uses the Beatpad's numbers.

Deck

1 **Jog**
Top scratch (touch). Set the controller's jog mode to Vinyl

2 **PITCH**
Tempo fader (pitch bend)

3 **SHIFT**
Hold it with another control for its second function

4 **SYNC**

5 **JUMP**
CUE (as in the Beatpad's note)

6 **PLAY**

7 **Pad mode buttons**
Each of the 8 modes (printed on the upper and lower lines) arrives as its own number. `HOT CUE` = HOT CUE, `AUTO LOOP` = AUTO LOOP, `TOUCH FX` = TOUCH FX, `SLICER` = SLICER, `SAMPLER` → sample switching, `BOUNCE LOOP` = BOUNCE LOOP, `JOG FX` → NEURAL MIX, `SLICE LOOP` → PITCH PLAY (how the unit picks the lower-line modes is not checked on the real hardware)

8 **Pads**
The 8 pads of the current mode (ch 3 / ch 4). In `HOT CUE`, SHIFT + pad clears a hot cue

9 **HIGH**
Filter Morph

10 **LOW [MID]**
With the controller's EQ mode on FILTER (LOW) = filter Resonance; in 3-band mode (MID) = how much the video effects act on the sound

11 **FILTER [LOW]**
In FILTER mode = filter Cutoff; in 3-band mode (LOW) = filter Resonance

12 **Headphone CUE**
Listen to this deck on headphones

13 **Channel fader**
That deck's volume and picture strength

14 **Crossfader**
(ch 5): Deck 1 on the left side, deck 2 on the right

LOAD 1 / 2 (ch1 98 / ch2 98) loads the file chosen in the Media Explorer into an empty sample slot (not in Reloop's official list; assumed to be the same as the Beatpad). Hold a pad mode button (7) and press LOAD to preview (the same as the `P` key). Controls without a number (GAIN, PITCH − / +, SET, LIBRARY, BROWSE, MASTER, PHONES, CUE MIX, AUX) are not used by ReaScratch. Not yet tested on the real hardware.

117 · 118 Reloop Beatmix 2/4 deck 1 / deck 2

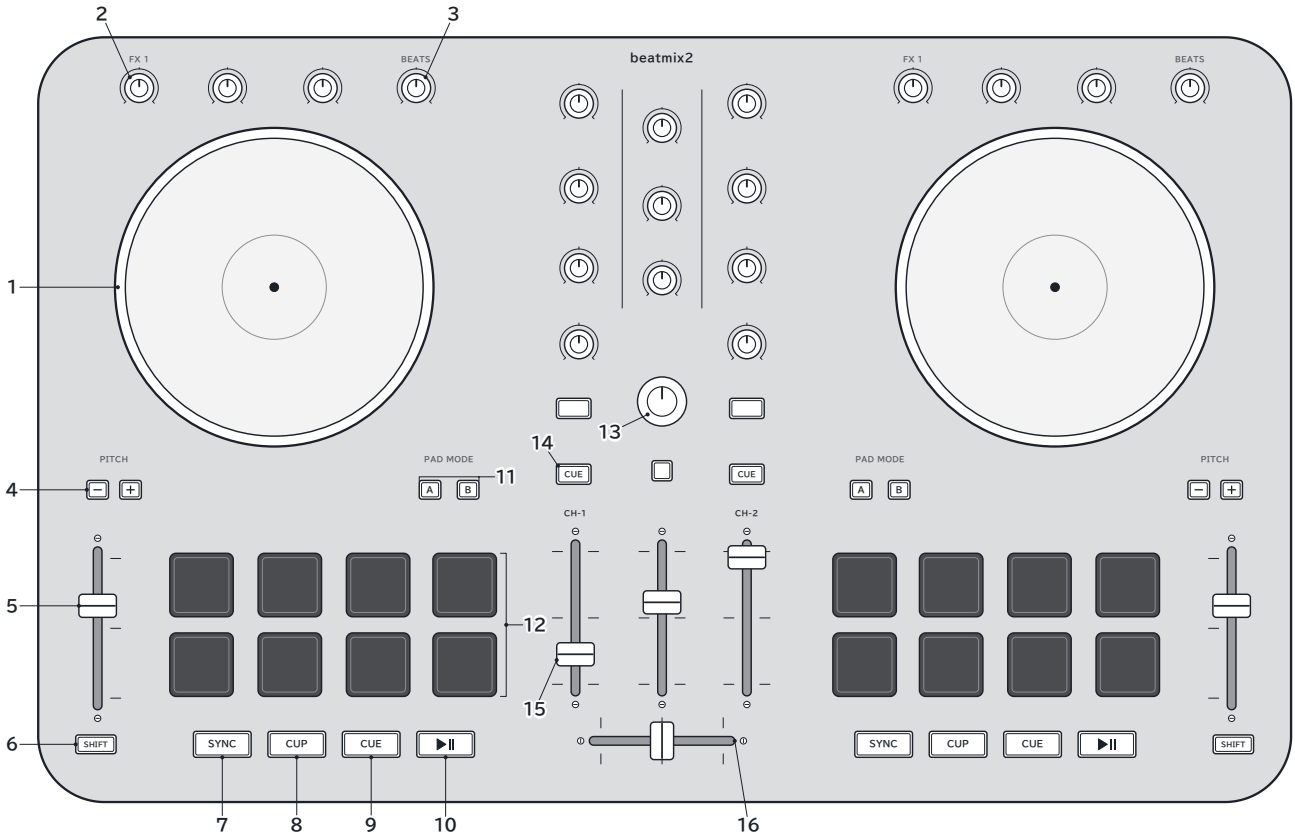


Figure 83 The Reloop Beatmix 2's controls (numbered on the left deck and the centre; the right deck is the same. The Beatmix 4 has a 4-channel mixer and is otherwise the same)

Decks on ch 2 / ch 3. Numbers from Reloop's official MIDI map (version preserved on the Internet Archive).

Deck

1 Jog
Top scratch (CC 96, 800 ticks per turn), touch

2 FX 1
The FX knob (Buddy-style; no ◂ ▸)

3 BEATS
The loop knob (turn, push)

4 PITCH − / +
`SHIFT` + `PITCH −` = FX ON (Buddy-style)

5 Tempo fader
Pitch bend

6 SHIFT
Hold it with another control for its second function

7 SYNC

8 CUP
SHIFT + CUP switches the tempo range

9 CUE

10 PLAY

11 PAD MODE A · B
HOT CUE in the pads' blue mode (A; SHIFT + pad clears) and sample switching in the red mode (B). In the violet mode (both mode buttons together) the top row is hot cues and the bottom row samples

12 Pads
The 8 pads of the current mode

13 **TRAX**
BROWSE (Buddy-style: turn = the headphone Cue mix; while FX ON is on, that effect's sound amount; push = switch the deck the knob works on)

14 **Headphone CUE**
Listen to this deck on headphones

15 **Channel fader**
That deck's volume and picture strength

16 **Crossfader**
(ch 6): Deck 1 on the left side, deck 2 on the right

The notes the unit sends when FX knobs 1–3 leave 0 are not used. LOAD 1 / 2 (ch2 80 / ch3 80) loads the file chosen in the Media Explorer into an empty sample slot. SHIFT + LOAD (ch2 64 / ch3 64) replaces the slot in use. Controls without a number (FX 2, FX 3, GAIN, HIGH, MID, LOW, BACK, the SAMPLER fader, MASTER, PHONES, CUE MIX) are not used by ReaScratch. Not yet tested on the real hardware.

119 · 120 Reloop Terminal Mix 2/4 deck 1 / deck 2

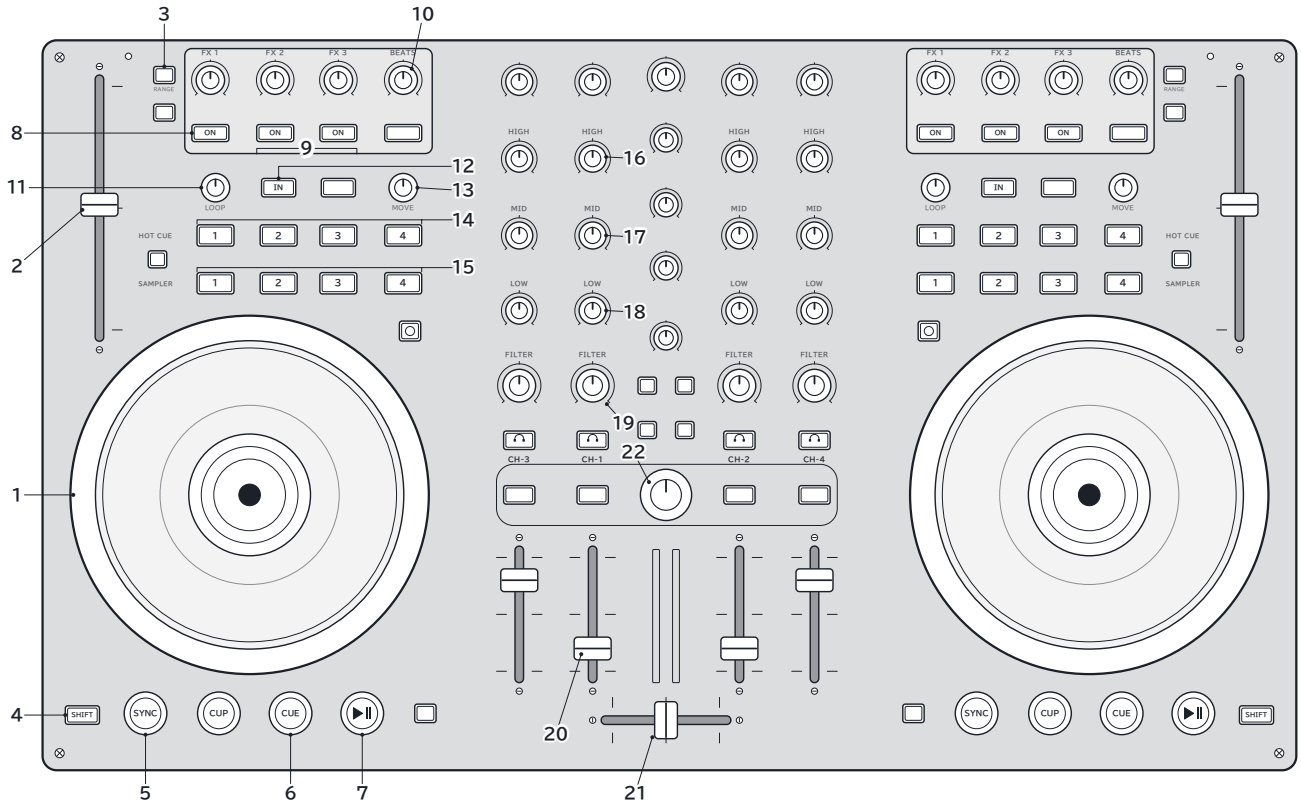


Figure 84 The Reloop Terminal Mix 4's controls (numbered on the left deck and the centre; the right deck is the same. The Terminal Mix 2 has a 2-channel mixer and is otherwise the same)

From Mixxx's mapping. No headphone CUE.

Deck

1 Jog
Top scratch (touch), outer ring = nudge (1/4 speed)

2 Tempo fader
Pitch bend

3 RANGE
Switches the tempo range

4 SHIFT
Hold it with another control for its second function

5 SYNC

6 CUE

7 PLAY

8 FX 1's ON
FX ON (Buddy-style)

9 FX 2 / FX 3's ON
◂ ▸ (Buddy-style)

10 BEATS
The FX knob (Buddy-style; DRY/WET in Mixxx's mapping)

11 LOOP
Turn = loop length, SHIFT + push = RELOOP/EXIT

12 IN
Loop (press)

13 MOVE
Turn to move the loop

14 The HOT CUE row
Hot cues 1–4 (5–8 are the next numbers in Mixxx's mapping; which control on the unit sends them is not checked on the real hardware)

15 The SAMPLER row
Samples 1–4

16 **HIGH**
Filter Morph

17 **MID**
How much the video effects act on the sound

18 **LOW**
Filter Resonance

19 **FILTER**
Filter Cutoff

20 **Channel fader**
That deck's volume and picture strength

21 **Crossfader**
Deck 1 on the left side, deck 2 on the right

22 **TRAX**
BROWSE (Buddy-style: turn = the headphone Cue mix; while FX ON is on, that effect's sound amount)

LOAD (ch1 50 / ch2 50) loads the file chosen in the Media Explorer into an empty sample slot. SHIFT + LOAD is the preview (the same as the `P` key). Controls without a number (KEYLOCK, TAP, OUT, VINYL, DECK, GAIN, the headphone buttons, CRATES, VIEW, BACK, PREP, MASTER, BOOTH, PHONES, CUE MIX, SAMPLER VOL) are not used by ReaScratch. Not yet tested on the real hardware.

121 · 122 Reloop Terminal Mix 8 deck 1 / deck 2

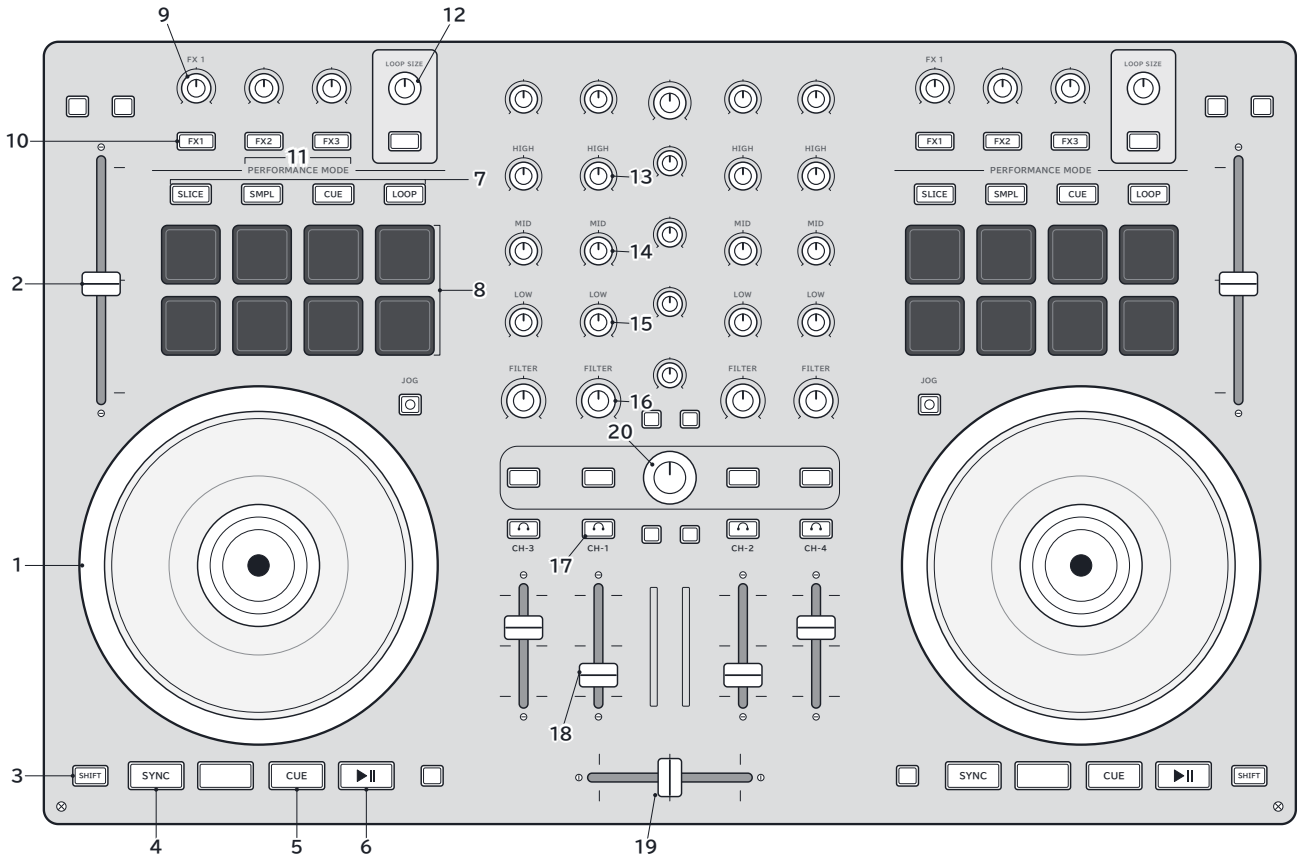


Figure 85 The Reloop Terminal Mix 8's controls (numbered on the left deck and the centre; the right deck is the same)

From Reloop's official MIDI map and Mixxx's mapping.

Deck

1 Jog
With the `JOG` button (VINYL) on, the top scratches; with it off, the outer ring nudges

2 Tempo fader
Pitch bend

3 SHIFT
Hold it with another control for its second function

4 SYNC

5 CUE

6 PLAY

7 Pad mode buttons
`CUE` → HOT CUE, `LOOP` → AUTO LOOP, `SMPL` → sample switching, `SLICE` → SLICER. With SHIFT held (MODE 1–4) → BOUNCE LOOP, TOUCH FX, PITCH PLAY, NEURAL MIX

8 Pads
The 8 pads of the current mode (ch 5 / ch 6). SHIFT + pad clears a hot cue

9 FX 1's knob
The FX knob (Buddy-style)

10 FX1
FX 1's ON = FX ON (Buddy-style)

11 FX2 · FX3
FX 2 / FX 3's ON = ◂ ▸ (Buddy-style)

12 LOOP SIZE (BEATS)
The loop knob (turn = length, push = loop, with SHIFT move by a beat / reloop)

Mixer

13 HIGH
Filter Morph (echo length with SHIFT)

14 MID
How much the video effects act on the sound

15 LOW
Filter Resonance (echo feedback with SHIFT)

16 FILTER
Filter Cutoff (key with SHIFT)

17 Headphone CUE
Listen to this deck on headphones

18 Channel fader
That deck's volume and picture strength

19 Crossfader
Deck 1 on the left side, deck 2 on the right

20 TRAX
BROWSE (Buddy-style)

LOAD (ch1 16 / ch2 16) loads the file chosen in the Media Explorer into an empty sample slot. SHIFT + LOAD (ch1 80 / ch2 80) replaces the slot in use. Hold a pad mode button (7) and press LOAD to preview (the same as the `P` key). Controls without a number (DECK FX ASSIGN, the ON under LOOP SIZE, STUTTER, DECK, GAIN, AREA, VIEW, BACK, PREPARE, MASTER, BOOTH, PHONES, CUE MIX, SAMPLER VOL) are not used by ReaScratch. Not yet tested on the real hardware.

---

0 Custom

Set up by hand using the table below.

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123 FLX4 Emu test (loopMIDI deck 2)

For development (only when connecting `tools/flx4_emu.py`, which is not in the kit; not for normal use).

Manual mapping (Custom)

Set each parameter to match your controller (the setup screen's `Jog` and `Touch & buttons` tabs; what each one means is also in the tables of "3. Jog" (p. 42) and "4. Touch and buttons" (p. 43)).

Parameter	What to set / guide values
MIDI channel	The channel the controller sends on (on DJ controllers often the deck number)
Jog CC number	CC number of the jog / knob (FLX4: 34, APC40 TRACK KNOB 1 / 5: 48 / 52)
Jog CC mode	How the jog values are read (8 modes; listed below the table)
Jog ticks per second at 1x	Ticks per second that correspond to 1x playback. Smaller values move faster. - DDJ-FLX4: `400` - DDJ-FLX2: `256` - Coarse encoders (about 24 clicks per turn): `20 – 40` - The APC40's `TRACK KNOB` (`Absolute knob turned`): `120`
Fine ring CC	CC number for fine adjustment (FLX4 outer ring: 33). 0 turns it off

Parameter	What to set / guide values
Touch note	Note number of the top touch sensor (FLX4: 54). Set to `0` on controllers without touch.
Auto release (ms)	Without touch (`Touch note` = 0), the grace time from when movement stops to when it counts as "hand off". `0` freezes where it stopped (hold); stop it with the `Release note` button.
PLAY note / CUE note / Release note	Note numbers of each button (FLX4: PLAY 11, CUE 12 / APC40: Play 91, Record 93). Release note 0 = off
Encoder smoothing (ms)	Time (ms) to estimate a rotary encoder's turning speed and move smoothly. `0` for jogs; `100`–`150` recommended for encoders.

The 8

Jog CC mode

modes:

- **Relative 64-centre (FLX4)**: DJ controllers from Pioneer and others (64 = still, 65 = +1, 63 = −1)
- **Relative two's complement**: many encoders (such as the Reloop Buddy's jog; 1–63 positive, 127–65 negative)
- **Relative sign bit**: 0–63 positive, 64–127 negative
- **Absolute 7-bit**: ordinary knobs and faders (0–127 sets the position directly)
- **Absolute 14-bit**: high-resolution knobs (a pair of CC n and n+32)
- **Speed (CDJ MIDI)**: CDJ-2000NXS2 MIDI mode. The value is the turning speed (64 = still, 65–127 forward at 0.06–4x, 63–0 backward). Moves only while the top is touched, and the `Fine ring CC` (the side) is also read as speed
- **Platter position (Denon)**: Denon DJ Prime 4 Computer Mode. Receives the platter position (14-bit, 0–16383) as `Jog CC number` (high) and the CC +22 (low), and reads the difference from the previous position as the amount turned
- **Absolute knob turned (APC40)**: uses a knob that sends absolute values 0–127 (the APC40's `TRACK KNOB`) as a jog. The difference from the previous value is read as the amount turned, and when it nears an end (below 20 or above 107) the knob's value is sent back to 64, so you can keep turning in the same direction

The numbers for pad modes, loops, the filter, BROWSE, CDJ buttons and so on are entered on the setup screen's `More` tab (group 13 of the parameter reference).

> **Checking the input** When you move a knob on the controller and the `midi` lamp on the JSFX screen lights up, MIDI is reaching REAPER. If it does not light, check that the controller's input is enabled in `Options > Preferences > Audio > MIDI Inputs`.

Check by touch / make a preset (setup `Learn` tab)

You can check whether your controller matches its preset, or build settings for a controller that has no preset (presets marked ⚠ have not been tested on the real hardware yet).

1. In `Preset`, choose the controller to check (for controllers without a preset, a similar model or `Custom`). Open the `Learn` tab of the setup screen.
2. In `The controller's deck:`, choose which deck of the controller this is (`deck 1` = left, `deck 2` = right) and press `Start`.
3. Follow the on-screen steps one control at a time: PLAY → CUE → SHIFT → touch the top of the jog and rock it back and forth a little → hold the top and turn exactly one full turn clockwise → turn the side without touching the top → move the tempo fader from end to end and stop at the + end → channel fader → crossfader → headphone CUE → pads (1–8 in order in HOT CUE mode) → SHIFT + pad 1 → pads (1–8 in order in SAMPLER mode) → FX ON (switch the button or paddle on once) → FX knob (the FX amount, wet / dry) from end to end → the button that steps the FX back (FX PARAM ◀, BEAT ◀ or a spare FX button) → the button that steps it forward (▶). It advances to the next step when you stop moving the control. Use `Skip` for controls the controller does not have, and `Done` after the last pad if it has fewer than 8. To redo, use `Again` (this step) or `Back` (the step before). While a step is waiting, MIDI does not reach the deck (pressing pads does not set cues).
4. At the end a list appears. ✓ = same as the preset, ✗ = differs (for example `tempo direction: preset Reversed (+ = high), here Normal (+ = low)`), – = skipped steps or steps with nothing to compare.
5. `Apply` puts the measured values into this deck (`Preset` becomes `Custom`; skipped steps keep their values). Enter a name and press `Save` to write it to `Data/ReaScratch_controllers/<controller name>.reascratch-preset` in the REAPER resource folder (saving one deck rewrites only that deck's part of the file; saving both decks in turn puts them together in one file). Saved presets can be chosen from `user presets` to the right of the `Preset` search box.
6. Saving and `user presets` work while the video sync script (`reascratch_video_sync.lua`) is running.
7. The written file contains the measured values, the comparison results and some of the MIDI received. Send it to the author and it can be added as a built-in preset.
8. On controllers that send the jog as speed or platter position, like CDJs and Denon, the ticks per turn are not counted.

Button lights

With many presets, ReaScratch lights the controller's pads and buttons.

- **Pads:** hot cues in the same colour as their waveform lines, the sample in use in green, other loaded samples in dim white. On controllers with pad modes, only the pads of the current mode light up, in colours that change by mode (AUTO LOOP = green with the current loop brighter, TOUCH FX = a colour for each effect, FILTER TYPE = yellow with the chosen one brighter, and so on). The mode buttons light for the current mode too
- **Buttons:** PLAY (while playing), CUE (while stopped), SYNC (while on), headphone CUE (while on)

- **Controllers with colour lights:** Reloop Buddy, Ready, Mixon 8 Pro and Mixon 4; DDJ-800, DDJ-1000, DDJ-FLX10, DDJ-REV5, DDJ-SX3, XDJ-RX3 and XDJ-XZ; Hercules Inpulse 500; Denon DJ Prime 4, SC5000 and SC6000; APC40 mkII (the mkI has three colours: green, red and orange)
- **On/off-only controllers:** controllers that cannot show colours (DDJ-FLX4, DDJ-FLX2, DDJ-400 and others, Numark, the other Hercules and so on) light the pads that have a hot cue
- **Controllers that do not light:** CDJs (no way to light them over MIDI)
- **When REAPER closes:** if the video sync script is running, all lights are turned off as REAPER closes (Windows; how it works is in the paragraph below)
- **NI Traktor Kontrol connected over HID:** the bridge program lights them (see `ReaScratch LED` under "HID controllers" (p. 197) below)

To get the lights working:

1. In REAPER's settings (`Options` > `Preferences` > `Audio` > `MIDI Outputs`), enable the controller as an **output** as well. With reset disabled (so REAPER does not send reset messages), the lights do not go out when REAPER stops.
2. Run `reascratch_setup.lua` again. The `scratch engine` track's MIDI output is set to the output that matches the `Preset` (an output that contains every word of the preset name, for example "APC40" and "mkII"; the maker's name and words in brackets may be missing). If no output matches, it is the first output whose name contains a DJ controller name (Buddy, DDJ, APC40, Mixtrack, Inpulse and so on). While the video sync script runs, changing `Preset` also moves the output to the matching one within about 2 seconds
3. Set `Preset` to that controller

If buttons do not light, or unrelated buttons light, check in the `scratch engine` track's routing window that the MIDI hardware output is the same controller as the `Preset` (sending to a different device lights whatever buttons have matching numbers; to use two controllers at once, see "Two controllers at once" below). The APC40 mkII and Reloop Buddy were tested on real hardware. The others come from official MIDI lists or Mixxx's mappings and have not been tested on the real hardware yet. When ReaScratch starts, it switches the APC40 into Ableton Live mode and clears all previous lights before lighting its own.

When REAPER closes, all lights go off (Windows). REAPER closes its MIDI outputs first when it quits, so `reascratch_lights_off.ps1`, which starts with the video sync script, opens the outputs after REAPER has closed and turns the lights off. If the video sync script is not running, or if you stop it by running it again, the lights stay on.

Two controllers at once (e.g. Reloop Buddy + APC40 mkII)

You can use two controllers at the same time, for example the jog and pads on the Buddy and the effect knobs and faders on the APC40.

1. Enable both as MIDI inputs and outputs in REAPER (`Options` > `Preferences` > `Audio` > `MIDI Inputs` and `MIDI Outputs`)
2. On each deck's setup page, set `Preset` to the first controller (e.g. deck A `Reloop Buddy deck 1`, deck B `deck 2`) and `2nd controller` on the `Jog` tab to the second one (e.g. `APC40 mkII deck 1` / `deck 2`)

3. While the video sync runs, three tracks appear within about 2 seconds, "controller 2 in", "lights 1" and "lights 2", and the routing changes (running `reascratch_setup.lua` again does the same)
    - The `scratch engine` tracks (A and B) take only the first controller as input; the second one's MIDI arrives from "controller 2 in" on MIDI bus 2
    - The lights go to the first controller from "lights 1" and to the second from "lights 2". The input and output devices are found by the presets' names ("Reloop", "Buddy", ...)
4. Set `2nd controller` back to `none` and the three tracks go away; the routing is as before (all MIDI inputs)
5. Controls both have (PLAY, CUE, tempo, LOAD, jog, crossfader, filter, hot cue and sample pads) work from either. The crossfader follows whichever was moved last. The pads light on both
6. A deck has one pad mode. With an APC40 mkII as the second controller, the mode is switched with the Buddy's mode button (the APC40's `DEVICE` buttons turn effects on and off, as described below). The modes in the order of the Buddy's printed buttons: HOT CUE, AUTO LOOP, TOUCH FX, FILTER TYPE (the Buddy's SAMPLER button), SLICER, BOUNCE LOOP, PITCH PLAY (the Buddy's NEURAL MIX button), NEURAL MIX = memory cues and held moves (the Buddy's LOOPER button). The APC40 always has NEURAL MIX's controls on its grid's 4th and 5th rows, so in mode 8 its top two rows' 8 pads pick the video blend (Add, Subtract, Multiply, Overlay, Soft light, Hard light, Luma key, Chroma key; the same pad again = Mix; one blend for both decks). There is no SAMPLER mode: on the Buddy, switch samples with the keys, the mouse or the APC40
7. The second controller keeps its preset's settings (the setup page edits the first one's only)
8. The Reloop Buddy and APC40 mkII together were tested on real hardware

With the APC40 mkII as the second controller, it works as the Buddy's helper:

- The APC40's crossfader becomes the effect boost (far left = none, far right = every effect that is on pushed toward full, picture and sound). Switch decks with the Buddy's crossfader
- `SCENE LAUNCH` 1–5 choose the page of the clip grid (the current page lights up)
    - 1: as before (hot cues, samples, memory cues and so on)
    - Pressing the current page's button again turns that page's effect on or off (2 = no transition and the last one, 3 / 4 = that deck's video effect, 5 = Off and the last type after the mix)
    - 2: video blends and transitions. From the top row: the 9 blends (Mix to Chroma key), no transition, then the 29 transitions. The bottom right button turns the blend-with-transitions combination on or off. The current choices light up brightly
    - 3, 4, 5: the video effect types for deck A, deck B, and after the mix. The top two rows hold Off and the 15 types. On a deck's page, pressing the chosen type again turns it on or off; on the after-the-mix page it switches to Off. The bottom three rows are the video effect LFO (an oscillation synced to the beat): row 3 the shape (Sine to Saw up), row 4 the rate (1/4 beat to 8 beats), row 5 the depth (far left = none, 8 steps). To prevent accidental changes, the LFO settings change only when a pad is pressed with `SHIFT` held (a pad alone only shows a message on screen). While the depth is greater than 0, the lit pads in the depth row blink. Pages 3 and 5 set deck A's (the effect after the mix follows deck A's oscillation), page 4 deck B's
- The 8 `DEVICE` buttons (`◂` to `DETAIL VIEW`): turn the effects of the 8 DEVICE knobs above (the type chosen for each knob) on or off. The knobs keep their amounts. They light up while on

- `CLIP STOP` 1–8: triggers the corresponding effect while held (at the amount that deck remembers for the type; 100% at first). Acts on the selected deck
- `TRACK KNOB` 1 / 5 (the `PAN` page): the sound amount of the video effect chosen on deck A / B (the knob's ring shows the value)
- `TRACK KNOB` 1–8 (pages 2–5): the page's detailed settings (the rings show the values)
  - Page 2: 1 = the transition edge softness, 2 = the key threshold, 3 = the camera moves' blur, 4 = overshoot, 5 = shake
  - Pages 3–5: 1 = the chosen type's amount, 2 = its sound amount, 3 = the LFO depth, 4 / 5 = settings of that type only (Tint's dark and light colours, the Strobe step)
- Track faders 1 / 5: that deck's picture strength (the sound stays)
- Master fader: lowers the overall volume and picture brightness together (all the way down = silent and black). Set the amount after the mix with page 5's knob 1
- The faders take effect only once they reach the current value, so nothing jumps after a change elsewhere
- `RECORD` = REAPER's record, `STOP ALL CLIPS` = stop both decks (as printed)
- `METRONOME` = ReaScratch's metronome on/off (the same as the `N` key; when `Metronome follows` is `Each deck`, both decks switch, each its own). It lights while deck A's metronome is on. This is because REAPER's own metronome cannot be heard while the decks play ("Listening to a metronome while you play" (p. 203))
- `MASTER`: both decks. While you hold it (it lights up), `PLAY`, `CUE` and `SYNC` on the APC40 and the Buddy act on both decks (`PLAY` stops both if either is playing, and plays both if both are stopped). The knobs still act on one deck
- `SHIFT` + `MASTER`: blackout. Press it to instantly cut all sound and video; press again to bring them back (`MASTER` stays lit during the blackout; the master fader stays where it is)
- `SESSION`: saves the REAPER project. If the project has never been saved, it is not saved; a message appears at the bottom of the screen instead (so no save dialog pops up during a set).

HID controllers (NI Traktor Kontrol S4, S3, S2)

The Traktor Kontrol S4 MK3, S4 MK2, S3, S2 MK3, S2 MK2 and S2 MK1 do not send MIDI; they communicate using a different protocol called HID. ReaScratch can only read MIDI, so keep the bridge program in the `HID bridge` folder running while you play. It reads the HID data and turns it into MIDI with the DDJ-FLX4's numbers. No manufacturer driver is needed (the S4 needs its power adapter). The numbers come from Mixxx's mappings, and none of the models has been tested on the real hardware yet. Model differences:

- **S3, S4 MK2, S2 MK2, S2 MK1**: the jog ticks per turn are Mixxx's values (768 on the S3, 1024 on the others). If they do not match, start the program with `--s3-ticks` or `--mk2-ticks` followed by a number.
- **S2 MK2, S2 MK1**: 4 pads. Each press of the `SAMPLES` button switches the pads between hot cues and sample switching
- **S4 MK2**: the 4 pads of the `HOTCUE` row are hot cues; the 4 of the `REMIX` row switch samples
- **S2 MK1**: top touch arrives as pressure, so "touching" is decided from the value stored in the unit (set at the factory). If that cannot be read, the value right after connecting is taken as "not touching" (in that case, do not touch the jogs while connecting)

Windows

1. Install loopMIDI (https://www.tobias-erichsen.de/software/loopmidi.html) and create one port (keep the default name `loopMIDI Port`)
2. Connect the controller over USB and double-click `HID bridge¥ReaScratch-HID-Bridge.exe`. If the black window shows `MIDI out: loopMIDI Port` (followed by a number) and `connected: Traktor Kontrol S4 MK3` (the name of your controller), it is running. If a warning stops it from starting, see “If a warning stops it from starting” below. Keep the window open while you play.
3. In REAPER's "MIDI Inputs" settings, enable `loopMIDI Port`, set the `Preset` of `scratch engine` to `HID bridge (NI S2/S3/S4) deck 1`, and deck B to `deck 2`
4. To light the buttons, create one more port in loopMIDI and name it `ReaScratch LED`. In REAPER's settings, enable `ReaScratch LED` **as an output only**, with the input disabled (loopMIDI sends back whatever it receives, so with the input enabled the lights come back as button presses; `reascratch_setup.lua` warns you if the input is enabled). Then run `reascratch_setup.lua` again and the `scratch engine` output becomes `ReaScratch LED`. At start-up the bridge shows `lights from: ReaScratch LED` (without the port, it says it will not use the button lights and runs without them)

Mac

1. Install Python 3 (https://www.python.org/downloads/)
2. Connect the controller and double-click `HID bridge/run_mac.command` (the first time, it installs the parts it needs into `.venv` in the same folder, which takes a little while. If it will not open, right-click → "Open"). A virtual MIDI port `ReaScratch HID Bridge` appears
3. In REAPER's "MIDI Inputs" settings, enable `ReaScratch HID Bridge`; the presets are the same as on Windows. The bridge also creates the virtual port `ReaScratch LED` for the button lights, so enable it in REAPER **as an output**
4. On a Mac it cannot be used at the same time as Traktor (quit Traktor first). To build a single program that runs without Python, use `build_mac.command`

If a warning stops it from starting

`ReaScratch-HID-Bridge.exe` and the `.command` files are not signed (signing needs a paid certificate). So when you open them for the first time after downloading, Windows or macOS may show a warning and block them. The exe is just `hid_bridge.py` (in the same folder) packed into a single program, and the `.command` is a short script that runs that `hid_bridge.py`.

- **Windows "Windows protected your PC" (blue SmartScreen window)**: press `More info` and a `Run anyway` button appears; press it. It will not appear again
- **To prevent it on Windows in advance**: before extracting the zip, right-click the zip → `Properties` → tick `Unblock` next to `Security: This file came from another computer…` at the bottom → `OK`. Extract it after that and the files inside will not trigger the warning.

- **Your antivirus deleted or quarantined it**: Python programs turned into exes with PyInstaller are sometimes wrongly flagged by antivirus software. Restore it from the quarantine list and add the `HID bridge` folder to the exclusions. If you would rather not use the exe, install Python and run the same thing with `python hid_bridge.py` (needs `pip install hidapi mido python-rtmidi`)
- **Mac "cannot be opened because the developer cannot be verified" / "Apple could not verify … is free of malware"**: right-click the `.command` (or Control-click) → `Open` → `Open` in the confirmation. If that does not work on macOS 15 (Sequoia) or later, double-click once and close the warning, then press `Open Anyway` near the bottom of `System Settings` → `Privacy & Security`
- **To allow everything at once on a Mac**: in Terminal, type `xattr -dr com.apple.quarantine` (with a space at the end), drag the `HID bridge` folder onto the Terminal window and press Enter. The "downloaded" mark is removed from every file in the folder and the warnings stop. The same works for the included `tools/allintra_ffmpeg.command`

Sending a report (for testers)

The HID bridge has not been tried on any of these controllers yet, and it has never been run on a Mac. If you own one, a report file made with the steps below would help a lot.

1. Connect the controller and quit Traktor
2. Double-click `HID bridge/report_mac.command` on a Mac or `HID bridge/report_windows.cmd` on Windows (on a Mac, the first run does the same setup as `run_mac.command`)
3. When it says `connected`, within 15 seconds turn both jogs (touching the top, then the side), press `PLAY` and `CUE` on both decks, and move the tempo faders and the crossfader
4. `hid_bridge_report.txt` appears in the same folder. Attach it to a GitHub issue or the forum thread

The report contains the OS and Python package versions, the self-check result, the MIDI port names and a summary of the data the controller sent. It contains no username or folder path. Nothing is sent to the controller (the lights and motors are left alone). If the controller is not found or cannot be opened, that result is saved too, so please send it anyway.

Other notes

- The left deck is deck A (MIDI channel 1) and the right deck is deck B (channel 2). The S4 MK3's deck C/D switching is not used.
- The lights are PLAY, CUE, SYNC, headphone CUE, SHIFT, the pad mode buttons and the pads (hot cue colours, samples). Colours become the closest one the controller can show. Change the name of the port that receives the lights with `--led-port name`, and leave the lights alone with `--no-lights`
- The tempo fader direction (which end is +) is unknown. If it is reversed, start the program with `--tempo-invert` (on Windows, in a Command Prompt: `ReaScratch-HID-Bridge.exe --tempo-invert`). If the jog is too fast or too slow, change the count per turn with `--s4-ticks` (default 2880) or `--s2-ticks` (default 600)

Two decks with layered video (deck B)

1. Answer Yes when setup asks whether to create a second deck (deck B). Deck B goes on the opposite side of the crossfader (if deck A is `Off`, deck A is set to `Left` and deck B to `Right`). If deck A's preset is "... deck 1" ("(MIDI ch 1)" for CDJs), deck B gets "... deck 2" ("(MIDI ch 2)") of the same model. `Cut-in`, `Fade width` and `Fade curve` start with deck A's values.
2. A video import goes to the deck whose window (the FX window of `scratch engine` or `scratch engine B`) was opened last. If neither window was open, you are asked “Which deck should it go to?” (`Deck A` / `Deck B`). Deck B's video goes on the `video B` track and its audio on the `audio B` track.
3. The video sync script keeps the edit cursor on deck A's position and moves the item on the `video B view` track to deck B's position. That way REAPER's video window shows both decks layered, each at its own frame.
4. For smooth blends, set both decks' `Fade width` to 100%, `Cut-in` to 0 and `Fade curve` to `Equal power` (setup `Crossfader` tab). For scratch cuts, set `Fade width` small (about 2%) and `Fade curve` to `Linear`. By default, it is set in between: `Cut-in` 2%, `Fade width` 30%, and `Fade curve` `Equal power`.
5. Choose the blend or transition with `Video blend` on deck A's setup `Video` tab. With a blend (`Mix` to `Chroma key`), the crossfader at deck A's end shows only A, at deck B's end only B, and in between the chosen blend. With a transition (`Wipe right` and after), the crossfader's position is how far the transition has gone (see “Blends and transitions” for how they look). If both decks' sound is cut, the picture goes black too.
6. With deck B, cutting the sound does not mute the `video` track (the picture is faded within the blend instead).
7. The Video processor makes video processing heavier (about one CPU core on the PC we tested). Blending and effects are calculated at up to 1280×720, and larger videos are scaled down to that size. Small videos (for example, 320×240) are scaled up to fit within a 960×720 frame while maintaining their aspect ratio before processing, so effect lines and noise do not look coarse. When the project's video size is set to automatic, blending and output use the larger of the two decks' video sizes, scaled to fit within the range of 960×720 to 1280×720. Converting the samples to 720p all-intra keeps it light (on import, videos taller than the height chosen in setup, 720 by default, are offered a downscaling conversion even if they are already all-intra).
8. Keep `video B` and `audio B` muted (deck B's picture is shown only through `video B view`). The sync script moves the `video B view` item, so there is no need to move it by hand.

Projector output

1. Set Windows' display to "Extend" and set the projector's resolution to match the projector (for example 1280×800). With "Duplicate", the whole REAPER screen is shown.
2. Move REAPER's video window to the projector screen and make it fullscreen with the action `Video: fullscreen` (search for "fullscreen" in the action list).
3. A 16:9 video on a 16:10 projector gets black bars above and below (the aspect ratio is preserved).
4. If the projector drops out and comes back, the video window returns to the projector still fullscreen and keeps running. While it is disconnected, the video window is not visible.
5. If the video window still stays frozen (this has happened after a monitor was switched off, for example), run `reascratch_video_reopen.lua`. Assign it to a key so you are not caught out during a show.
6. If you close and reopen the video window, REAPER treats it as "not fullscreen" even though it still is. Pressing `Video: fullscreen` then leaves fullscreen, so press it once more.
7. Some computer and projector combinations keep connecting and disconnecting. In that case, put a switcher (such as an ATEM Mini) or an HDMI EDID emulator (a small device that makes the computer believe a screen is always connected) in between.

Listening to a deck on headphones (Cue)

Like the CUE (PFL) on a DJ mixer, you can preview a deck on headphones even while the crossfader has cut it.

To hear the cue apart from the main output, your headphones need an output of their own, other than the main channels 1/2 (3/4, for example): an audio interface with four or more output channels, or a DJ controller with a built-in audio interface. In the cases below you cannot hear them apart. The main output plays as usual in all of them.

- **Only the PC's built-in audio**: there is one pair of outputs (1/2), so the cue and the main output mix on it. REAPER cannot use separate Windows audio devices (the built-in headphone jack and HDMI, for example) at the same time.
- **A unit whose headphone jack plays the same sound as the main output** (such as one with two output channels): you cannot give the headphones an output of their own.
- **A unit whose own mixer sets what goes to the headphones** (such as RME TotalMix): if the headphone output also has the main sound mixed in, they will not be separated. In the unit's mixer, send only the `Cue` track's output to the headphones.

If separate outputs are not available, do not pick the main output in the setup output list (the cue would be added to the main output). Instead, close the list without choosing (`Esc` or a click outside it): the `Cue` track then goes to no output (setup asks again the next time you run it). To hear the metronome in the main output, set `Metronome out` to `Headphones and main` ("Listening to a metronome while you play" (p. 203)).

1. Running setup creates a `Cue` track. Each deck (`scratch engine`, `scratch engine B`) becomes 8 channels and sends its cue sound from channels 5/6 and its metronome from channels 7/8 to the `Cue` track.
2. Cue is switched on/off per deck. On the Reloop Buddy, each deck's headphone button toggles it; on the DDJ-FLX4 and DDJ-FLX2, each deck's headphone `CUE` button (set by their presets). Clicking the `phones` lamp on the JSFX screen or the `H` key also toggles it. The lamp lights while it is on.
3. The headphones receive the pre-fader and pre-crossfader sound (post-effects). Switching fades over 5 ms, so there is no click.
4. `Cue mix` in the `Cue` track's FX (`DJ cue mix (headphones)`) sets the blend between the cue sound and the main sound: 0 = cue only, 1 = main only, 0.5 = an equal mix. When you turn a controller knob (such as BROWSE on the Buddy), the screen shows both shares, such as `headphones: cue 70 % / master 30 %`. `Level` is the headphone volume (dB).
5. The main sound is each deck track's sound after its fader (the master's effects are not applied).
6. The `Cue` track's fader and mute work too. To change the headphone output, choose it under `Hardware output` in the `Cue` track's `ROUTE` button (the chosen output is kept even if you run setup again, unless the current audio device does not have that output; then you are asked again).
7. The note number of the headphone button is `Cue button note` on the setup screen's `More` tab (`Buttons` group) (on the channel of `Mixer MIDI channel`, where 0 = the deck's channel; −1 = not used). 27 for the Buddy, 84 for the DDJ-FLX4 and DDJ-FLX2; other controllers' presets set it too (−1 on controllers without a headphone CUE button).

Listening to a metronome while you play

ReaScratch can play a click in the headphones on a deck's beats. The beats are the deck's own (its BPM and beat grid, or its beat map for a song whose tempo changes), and the first beat of each bar (the thicker line on the waveform) is higher. REAPER's own metronome only sounds while REAPER plays or records; the ReaScratch decks play with REAPER stopped, so you do not hear REAPER's metronome.

- **On/off**: the `N` key, a click on the window in the record's corner, or a MIDI button entered as `METRONOME button` in the `Metronome` group of the setup screen's `More` tab (its channel and note). Most controllers get one from their preset ("Metronome and preview buttons by controller"). With two controllers, the second APC40's `METRONOME` button switches it ("Two controllers at once" (p. 195)). While it is on, the rim of the window in the record's corner glows in the lamp colour, brighter for a moment each time a click sounds on this deck's beats.
- **The sample needs a BPM.** A sample without one gets no clicks (tap `T`, or enter it in setup).
- The settings are in the `Metronome` group of the setup screen's `More` tab. Deck A's settings work for both decks (changing them on deck B's screen changes deck A's).

Setting	Choices	Default
`Metronome`	On/off	Off
`Metronome follows`	`Playing deck` = the beats of the deck that plays (both playing: the one that started first; both stopped: the one that played last) / `SYNC base deck` = the deck the other deck follows with SYNC (without SYNC, as `Playing deck`) / `Each deck` = each deck on its own (its own on/off, its own beats)	Playing deck
`Metronome out`	`Headphones` = the headphones only / `Headphones and main` = the headphones and the main output	Headphones
`Stopped / in hand`	`Keeps time` = it keeps clicking at the deck's BPM while the deck is stopped, touched or searched / `Silent` = no clicks then	Keeps time
`Metronome volume`	−30 to 0 dB	−12 dB

- While the deck plays, the clicks fall exactly on the beats of the read position. Jumping onto a beat with a hot cue or a loop clicks too. With `Keeps time`, after you stop or touch the deck it keeps clicking at the deck's BPM; when it plays again, the clicks follow the beats of the read position again.
- In the headphones, the click sounds at the same volume wherever the `Cue mix` of the `Cue` track is set, and `Headphones and main` does not change its level there.
- Without headphones (no `Cue` track), set `Metronome out` to `Headphones and main` to hear it in the main output. A click in the main output is also in the "Rec Mix" track (the finished sound) of a performance capture, but not in the "Rec A" and "Rec B" tracks of the decks.
- In a project set up before 0.7, the video sync script adds the sends from each deck's channels 7/8 to the `Cue` track when it runs (no need to run setup again).

Metronome and preview buttons by controller

Choosing a preset also sets the metronome on/off button and the preview button. For the metronome, a beat button (SYNC, TAP and so on) pressed with `SHIFT` came first; where none was free, an unused button. For the preview: on controllers with pad mode buttons and `LOAD`, `LOAD` pressed while holding the mode button; on the others, an unused button. Where it says none, or to use another button, enter it on the setup screen's `More` tab (the metronome in the `Metronome` group, the preview in the `Browser` group). In brackets: the channel and note as setup shows them. On controllers marked ⚠ (not tried on hardware) the numbers come from their documents. A button pressed with `SHIFT` may not light on some controllers.

Unit	METRONOME button (deck 1 / deck 2)	Preview (deck 1 / deck 2)
APC40 mkI	REC ARM 3 (ch3 48) / REC ARM 7 (ch7 48)	REC ARM 2 (ch2 48) / REC ARM 6 (ch6 48)
APC40 mkII	REC ARM 3 (ch3 48) / REC ARM 7 (ch7 48)	REC ARM 2 (ch2 48) / REC ARM 6 (ch6 48)
Behringer CMD Micro	BEATLOOP 8 (ch1 19 / ch1 36)	previous playlist (ch1 48) / next playlist (ch1 49)
Denon SC5000/SC6000	MASTER DECK (ch1 21 / ch1 21)	none
Denon Prime 4	GRID EDIT (ch5 27 / ch6 27)	mode button + LOAD
Gemini FirstMix	SCRATCH (ch1 72 / ch1 53)	PREVIEW (ch1 13) / push of BROWSE (ch1 79)
Hercules DJ Control MP3 e2	MASTER TEMPO (ch1 19 / ch1 39)	FOLDER (ch1 44) / FILES (ch1 43)
Hercules DJControl Compact	pad 4 in FX mode (ch1 12) / FX pad 4 (ch1 60)	none
Hercules DJControl Starlight	SHIFT + SYNC (ch5 5 / ch6 5)	none
Hercules DJControl MIX	SHIFT + SYNC (ch5 5 / ch6 5)	none
Hercules DJ Control Instinct	FX 4 (ch1 4 / ch1 30)	browser: previous playlist (ch1 57) / browser: next playlist (ch1 56)
Hercules DJ Control AIR	beat grid adjust (ch1 2 / ch1 24)	FOLDERS (ch1 54) / FILES (ch1 53)
Hercules DJ Console Mk4	K4 (ch1 4 / ch1 24)	FOLDER (ch1 44) / FILES (ch1 43)
Hercules DJ Console 4-Mx	Pitch scale + (ch1 19 / ch1 51)	Nav FOLDER (ch1 63) / Nav FILES (ch1 62)
Hercules DJ 4Set	MOD + (ch1 19 / ch1 51)	FOLDERS (ch1 63) / FILES (ch1 62)
Hercules DJControl Jogvision	SHIFT + SYNC (ch1 53 / ch2 53)	EJECT A (ch1 83) / EJECT B (ch2 83)

Unit	METRONOME button (deck 1 / deck 2)	Preview (deck 1 / deck 2)
Hercules Inpulse 200/300 (MK2)	SHIFT + SYNC (ch5 5 / ch6 5)	mode button + LOAD
Hercules Inpulse 500	SHIFT + SYNC (ch5 5 / ch6 5)	mode button + LOAD
ION Discover DJ	SCRATCH (ch1 72) / SCRATCH (both decks) (ch1 72)	LIBRARY (ch1 79) / none
ION Discover DJ Pro	KEYLOCK (ch1 81 / ch1 82)	SCRATCH (ch1 72 / ch1 80)
Korg KAOSS DJ	TAP (ch7 11) / TAP (both decks) (ch7 11)	SCRATCH MODE (ch8 22 / ch9 22)
HID bridge (NI S2/S3/S4)	none	none
Numark Party Mix	SCRATCH (ch1 7 / ch2 7)	mode button + LOAD
Numark Party Mix II/Live	none	mode button + LOAD
Numark DJ2GO2	none	none
Numark DJ2GO2 Touch	none	none
Numark Mixtrack	KEYLOCK (ch1 81 / ch1 82)	SCRATCH (ch1 72 / ch1 80)
Numark Mixtrack Pro	KEYLOCK (ch1 81 / ch1 82)	SCRATCH (ch1 72 / ch1 80)
Numark Mixtrack 2	TAP (ch1 92 / ch1 96)	SCRATCH (ch1 72 / ch1 80)
Numark Mixtrack (Pro) 3	TAP (ch2 10 / ch3 10)	PFL (ch1 14 / ch1 15)
Numark Mixtrack Platinum	TAP (ch9 4 / ch10 4)	mode button + LOAD
Numark Mixtrack Pro/Platinum FX	TAP (ch9 9 / ch10 9)	mode button + LOAD
DDJ-FLX2	SHIFT + CUE (ch1 72 / ch2 72)	none
DDJ-FLX4	SHIFT + IN (ch1 76 / ch2 76)	mode button + LOAD
DDJ-FLX6	SHIFT + MASTER (ch1 96 / ch2 96)	mode button + LOAD
DDJ-FLX10	SHIFT + TEMPO RESET (ch1 96 / ch2 96)	mode button + LOAD
DDJ-REV1	SHIFT + SYNC (ch1 92 / ch2 92)	mode button + LOAD
DDJ-REV5	SHIFT + SYNC (ch1 26 / ch2 26)	mode button + LOAD
DDJ-200	SHIFT + CUE (ch1 72 / ch2 72)	none
DDJ-RB	SHIFT + SYNC (ch1 92 / ch2 92)	none

Unit	METRONOME button (deck 1 / deck 2)	Preview (deck 1 / deck 2)
DDJ-400	SHIFT + LOOP IN (ch1 76 / ch2 76)	mode button + LOAD
DDJ-800	SHIFT + BEAT SYNC (ch1 92 / ch2 92)	mode button + LOAD
DDJ-1000	SHIFT + MASTER TEMPO (ch1 96 / ch2 96)	mode button + LOAD
DDJ-SB	SHIFT + SYNC (ch1 92 / ch2 92)	none
DDJ-SB2	SHIFT + SYNC (ch1 92 / ch2 92)	none
DDJ-SB3 / SX	SHIFT + SYNC (ch1 92 / ch2 92)	mode button + LOAD
DDJ-SX3	SHIFT + SYNC (ch1 92 / ch2 92)	mode button + LOAD
CDJ-2000NXS2	MASTER (ch1 30 / ch2 30)	none
CDJ-3000/3000X	MASTER (ch1 21 / ch2 21)	none
XDJ-RX3 / XZ	MASTER (ch1 30 / ch2 30)	mode button + LOAD
Reloop Buddy	SHIFT + MODE (ch13 17 / ch14 17)	mode button + LOAD
Reloop Ready	SHIFT + KEY LOCK (ch1 41 / ch2 41)	mode button + LOAD
Reloop Mixage	SHIFT + push of the loop length knob (ch1 95 / ch1 97)	SHIFT + LOAD (ch1 76 / ch1 90)
Reloop Mixon 4	SHIFT + SYNC (ch5 78 / ch6 78)	mode button + LOAD
Reloop Mixon 8 Pro	SHIFT + SYNC (guessed) (ch1 3 / ch2 3)	mode button + LOAD
Reloop Touch	SHIFT + SYNC (ch2 97 / ch3 97)	SHIFT + LOAD (ch2 121 / ch3 121)
Reloop Beatpad	SHIFT + SYNC (ch1 29 / ch2 29)	mode button + LOAD
Reloop Beatpad 2	SHIFT + SYNC (ch1 29 / ch2 29)	mode button + LOAD
Reloop Beatmix 2/4	SHIFT + SYNC (ch2 97 / ch3 97)	none
Reloop Terminal Mix 2/4	SHIFT + TAP (ch1 76 / ch2 76)	SHIFT + LOAD (ch1 120 / ch2 120)
Reloop Terminal Mix 8	SHIFT + SYNC (ch1 66 / ch2 66)	mode button + LOAD

Scratching from envelopes (patterns)

Write the scratching hand (the record position) and the crossfader into REAPER envelopes, and when the deck plays that spot it scratches exactly that way. Place the included patterns and rearrange them as automation items: copy, move, stretch and edit points.

How it works

- The envelopes are written on two lanes of the deck's `scratch engine` track
  - **Envelope scratch**: the record position. The value is "how many beats after the grab point" (0 = the grab point, 0.5 = half a beat later). The very bottom (−0.25) means "hand off". The length of a beat is set by that deck's sample BPM
  - **Envelope fader**: the crossfader. 1 = open, 0 = closed. It is multiplied with the crossfader's own volume, and the picture fades with it
- The picture follows the crossfader opening and closing (including the envelope) directly (up to about 16 switches a second are shown). Very short switches that would end between two redraws are still shown for the next two redraws.
- The lanes' timeline positions match the time in that deck's sample (sample 1 from 0 s, sample 2 from 1 hour, and so on, in the same places as the videos). When the deck plays through a pattern's position, it grabs the record there, and when the pattern ends it lets go (with `slip` on, it returns to the normal play position after letting go)
- REAPER's own playback is not used (the video sync script reads the envelopes and passes them to the JSFX, so keep the video sync running). It works only while the deck is playing and you are not holding it with the jog or the mouse
- If you grab by hand, your hand wins. If you let go in the middle of a pattern, the pattern continues from that position

Placing a pattern

1. Move the deck's read position to where you want the pattern (for deck A, the edit cursor position)
2. Run `reascratch_pattern.lua` and choose the deck (with two decks) and the pattern
3. It is placed as an automation item on the beat nearest the read position ("Snap" below changes this). From there you can drag it, copy it, stretch it by its edges, and add or move points like any automation item (making the lane taller makes editing easier).
4. Patterns are grouped as "Basic", "Chirp", "Tear", "Flare", "Transform", "Stutter" (stutter = repeating the first 16th, 8th or 32nd; freeze and hold = stopping the sound on the spot) and more. Patterns that move the crossfader also place an item at the same position on `Envelope fader` (select both together when you move them)
5. Choosing a pad (1–8) under "Put on a PATTERN pad" in the menu puts the next pattern you pick on that pad (it is only assigned, not placed; the PATTERN mode plays it)

6. "Again: name" at the top of the menu places the pattern you placed last. Patterns you saved yourself are marked with "★"
7. "Snap" at the bottom of the menu lets you choose where it lands: nearest beat (default), next beat (never before the head), nearest half beat, or next bar. Choosing one reopens the menu, and the choice is kept for next time. The beats are counted from the BPM and beat position where the deck is now (in a song whose tempo changes, beats far from the read position can be off)
8. If you place a pattern over another, REAPER plays the two added together (to keep only one, delete or move the item underneath)
9. To place the same pattern in many places, use the automation item pool (Pool); fixing one fixes them all
10. **Swing**: when placing, the times of the pattern's points are shifted to the track's swing (the off-beat 16ths move later). Which value is used, typical values and the steps are in "Fitting the swing" (p. 209)

Showing only the lanes you use

A deck's `scratch engine` track has about 90 parameters that can have envelopes (the ReaScratch lanes and the performance lanes, `DJ performance lanes`). REAPER's envelope list shows all of them, so `reascratch_lanes.lua` lets you show only the ones you use, by group.

1. Run `reascratch_lanes.lua` from the action list. A menu opens at the mouse cursor
2. Choose a group to show. Choose it again to hide it. A group that is shown has a check mark
   - **Hand-made scratch**: `Envelope scratch` and `Envelope fader`. Use this group to make a scratch by hand with points
   - **Recorded effects**: the video FX, filter and pad lanes written when you record a performance (laid out in the sample's time)
   - **Performance: play, sample, tempo** / **Performance: crossfader, volume, picture** / **Performance: loops, cues** / **Performance: effects**: the performance lanes (laid out on REAPER's timeline; see "Recording and playing a performance on REAPER's timeline" (p. 212))
3. The menu opens again after each choice. To finish, click outside it
4. "Show only the lanes with points" shows the lanes that have points or automation items and hides the rest
5. "Hide all ReaScratch lanes (points kept)" hides them all. Hidden lanes keep their points and still play
6. Both decks change the same way. One run is one undo step (Ctrl+Z)

Saving a rearranged pattern

1. On the `Envelope scratch` lane, select the item to save (the `Envelope fader` item at the same position is saved with it)
2. Run `reascratch_pattern_save.lua` and enter a name and group
3. It is saved to `Data/ReaScratch_patterns_user.lua` in the REAPER resource folder and appears in the pattern menu from then on (the same name in the same group is overwritten)

Saving removes the swing and keeps a straight version (see "Fitting the swing" (p. 209)).

All `.lua` files in the `Data` folder whose names start with `ReaScratch_patterns_` are loaded into the menu (the format is the same as `reascratch_patterns.lua`).

Fitting the swing

Swing is a rhythm where, of the two 16ths in each 8th, the off-beat 16th is moved later. ReaScratch can make a deck's playback swing, and can place patterns to fit a track's swing.

The two settings, and which one is used

- **The deck's `Swing (16th, %)`** (setup `Beat` tab; per deck; 50–75, default 50 = straight): makes the deck's playback itself swing. Patterns placed on the deck and patterns played from the PATTERN pads use this value too
- **REAPER's grid swing** (REAPER's "Snap/Grid" settings): used only when placing patterns (no effect on the deck's playback or the PATTERN pads; see “What it does not affect” below)
- When placing a pattern, the deck's value is used if it is above 50. At 50, REAPER's grid swing is used (when swing is on, its amount is not 0 and the grid is an 8th or finer). Otherwise the pattern is placed straight
- A REAPER grid of 1/16 gives a 16th swing, 1/8 an 8th swing (the off-beat 8th moves later), and 1/32 a 32nd swing. The deck's `Swing` is for 16ths only
- There is no key or MIDI assignment. Change it in setup or in REAPER's parameter list (it can also be written as an envelope)

Which one to use

- **The track already swings**: leave the deck at 50 and set REAPER's grid swing to fit the track. Raising the deck's `Swing` would add more swing on top of the track's own
- **You want a straight track to swing**: raise the deck's `Swing`. Patterns you place and patterns played from the PATTERN pads get the same swing

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Typical values

The swing value is the share of an 8th's length taken by its first 16th. A REAPER swing amount of X% corresponds to a swing of 50% + X ÷ 4.

Swing	Deck `Swing`	REAPER swing amount	Feel
50%	50	0%	Straight
54%	54	16%	Very light swing
58%	58	32%	Light swing (measured by the author on a test track)

Swing	Deck `Swing`	REAPER swing amount	Feel
62%	62	48%	Clear swing
66.7%	67 (nearly the same)	67%	Triplet swing (shuffle)
75%	75	100%	Dotted 8th + 16th

A negative REAPER swing amount moves the off-beat 16th earlier (the deck's `Swing` cannot go below 50).

What it affects

- The deck's normal-speed playback (while it runs on with your hand off; also with the tempo changed, and while a reverse or half-speed pad is held): within each 8th, the first 16th is read at 0.5 ÷ swing times the speed and the second at 0.5 ÷ (1 − swing) (about 0.86x and 1.19x at 58%). Grains are used, so the pitch does not change
- The background playback of SLIP works the same way
- The picture shows the position with the swing taken out, so it moves on evenly instead of jerking every 16th
- The starts of the 8ths (beats and half beats) do not move, so beat-quantized jumps, loops and SYNC stay on the beat
- The times of the points of patterns placed with `reascratch_pattern.lua` (counted from that deck's beat grid)
- Patterns played from the PATTERN pads (when the deck's `Swing` is above 50): the times of their points are shifted as when placing. The swing is counted from the sample's beat grid (with `Q` on, from the beat the pad waited for; with `Q` off, even when it starts between beats, from the beat position where you pressed, so the 16ths change speed at the same places as the deck's playback). The `Swing` value at the start is used; changing it while the pattern plays takes effect the next time you press a pad. The picture shows the position with the swing taken out, as in normal-speed playback

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What it does not affect

- A sample with a BPM of 0 (with no beats known, it plays straight)
- While you hold the jog or the mouse, while searching, and the spin after letting go
- REAPER's grid swing has no effect on the deck's playback or the PATTERN pads (the JSFX cannot read REAPER's grid settings). If you fit a swinging track with REAPER's grid swing and leave the deck's `Swing` at 50, the pad patterns play straight
- Patterns already placed: the swing is in the times they were placed at, so changing the value does not change them until you place them again

Example (placing patterns on a swinging track)

1. Set the BPM and beat position of the deck's sample ("9. Beat-quantized jumps" (p. 48); the `T` key and `Shift` + `T`)
2. Leave the deck's `Swing` at 50. In REAPER's "Snap/Grid" settings, set the grid to 1/16, turn swing on and enter an amount (start around 32% from the table above)
3. Move the read position to where you want the pattern and place it with `reascratch_pattern.lua`
4. Play and compare with the track's 16ths by ear. If it does not fit, delete the placed item, change the amount and place it again
5. After rearranging, save it with `reascratch_pattern_save.lua` ("Saving a rearranged pattern" (p. 208))

Saving, and never swinging twice

Saving takes out the swing in use at that moment (the deck's value or REAPER's grid swing) and keeps a straight version. Placing puts the swing in use at that moment back in, so it never swings twice. Save with the same swing you placed with (if you change the value before saving, the wrong amount is taken out and the shape is saved distorted).

Recording a performance into envelopes

The scratching you do while REAPER is recording (record moves made by grabbing with the jog, mouse or keyboard) and the crossfader are written as automation items to the `Envelope scratch` and `Envelope fader` lanes when you stop recording. When the deck plays that section again, the scratch is reproduced as recorded.

1. Keep the video sync script running (the video sync does the writing)
2. Press REAPER's record button, play the deck and perform (to keep the sound too, prepare a track to record on; writing works even with tracks that record nothing).
3. When you stop recording, an `Envelope scratch` item is created for each stretch where your hand was on the record (named `Rec A-1`, `Rec A-2`, ...; `Rec B-1`, ... for deck B). If you moved the crossfader during it, an `Envelope fader` item is created at the same position too. Stretches where you only moved the crossfader without touching the record become `Envelope fader` items on their own. A stretch where the crossfader stayed closed for more than 2 beats without touching the record is taken as the deck being out of the mix, not a cut, and is not written
4. To play it back, set the crossfader to the middle and play the deck.
5. Only movement while the deck is playing is written (scratching a stopped deck is not written, because envelopes only act while the deck is playing)
6. The crossfader is written as "how far it closed compared with the middle position". Channel volume faders are not written.
7. The items can be moved, copied and have their points edited like other patterns, and saved as patterns with `reascratch_pattern_save.lua`

8. If you play the same spot twice (with slip off you passed the same place again from where you let go, you went back with a hot cue, and so on), the item written later is on top. Only performances that pass each section of the sample once are reproduced exactly as recorded (with slip on, letting go returns to the normal position, so you get exactly that).
9. The video effect (this deck's type and amount, and on deck A the effect after the mix), the filter (cutoff, resonance, morph) and the TOUCH FX pads are also written to their own lanes (`Envelope vfx type`, `Envelope vfx amount`, `Envelope filter cutoff`, `Envelope filter resonance`, `Envelope filter morph`, `Envelope pad FX`, `Envelope master type`, `Envelope master amount`; one item per stretch where an effect or the filter was on). When played back, the video effect uses whichever of the recording and what you play live has the larger amount (with its type), a TOUCH FX pad is on if either has it on, and the filter recording takes effect only while the live cutoff knob is at its centre (off)
10. Hot cues, loops, tempo, play/stop, sample switching and the filter type are not written (they are recorded on the lanes of the next section, "Recording and playing a performance on REAPER's timeline" (p. 212))
11. If the position moves further than the lane's range (2.2 beats) while you hold the record, the excess is clipped (if you started the video sync by hand, the REAPER console shows how many points were clipped when you stop recording)
12. If you use your hand over a section where an envelope (pattern) is already playing, the `Envelope scratch` items there are cut out for the length of the hand item, which replaces them (REAPER plays overlapping items added together, and when you recorded, the hand took priority over the pattern). `Envelope fader` items are not cut
13. Each hand item ends with a 30 ms let-go mark (the bottom of the lane). When played back, the deck slips back there just as when you let go
14. Nothing is written after recording stops
15. For recordings you do not want written (such as keeping the sound of a performance made from envelope patterns), run `reascratch_record_lanes.lua` to turn it off (run it again to turn it on. The setting is saved, and if you assign it to a toolbar button, the button shows on/off)

Recording and playing a performance on REAPER's timeline

While REAPER records, the deck operations are recorded on the "performance lanes" on REAPER's timeline. With "play the performance" on, REAPER's play button plays that performance again. It is played at the times you operated, including while a deck is stopped and inside loops (the lanes of the previous section follow the deck's sample time, so they cannot hold play / stop or loops).

What is recorded:

- Play / pause, the play position (with scratches, brakes, spinbacks, reverse and patterns), sample switching, tempo, key, PITCH PLAY
- Loops (on / off, range, roll), the CUE position, setting and clearing hot cues 1-8, adding and deleting memory cues
- The crossfader (position, the `Z` key's cut, assignment), the channel volume, mute, video opacity, the blend mode with its key threshold and softness (deck A)

- The filter (type, cutoff, resonance, Morph), the video FX type and amount (this deck's, and on deck A the master's), FX knobs 1-8 and Strobe
- Echo / bitcrusher / vibrato, the sound of the video FX, TOUCH FX (rows 1 and 2) and their amounts, the vocoder, the pattern fader
- The second APC40: the effects while CLIP STOP is held, the video FX LFO (shape, rate, depth), the picture mute, FX boost, master level and blackout

Not recorded: loading (LOAD) and what is in the slots, the setup settings, editing the BPM and the grid, the crossfader's curve, Cut-in and Fade width. SYNC is recorded but not used when playing (the recorded tempo is played).

To record:

1. Have the video sync script running
2. Press REAPER's record button, play, and stop recording
3. Far along the project (the "performance zone" from 10 hours = 36000 s on) a region `Perf 1` appears (the next recording `Perf 2` ...). On each deck's engine track, the `DJ performance lanes` lanes get one automation item per lane (`Perf A-1` , deck B `Perf B-1` ...)
   - The FX window of `DJ performance lanes` shows each lane's current value in a table (a lane without a value shows a grey "-"). The number of columns follows the window's width. The values cannot be changed in the window; edit them as envelopes ("Showing only the lanes you use" (p. 208))
   - A recording started inside the performance zone overwrites the place where it was recorded

To play it:

1. If REAPER stops playing at the end of the project, playback stops at once in the performance zone. Uncheck `Stop/repeat playback at end of project` in Options > Preferences > Audio > Playback (setup and turning "play the performance" on tell you when it is on)
2. Press `PERF PLAY` (`reascratch_perf_play.lua`) on the toolbar to turn "play the performance" on (the button lights; press again for off)
3. Put the edit cursor at the start of the region to play (`Perf 1` ...) and press REAPER's play button. Started from the middle, the decks are first set to the state at that point (loops, hot cues, effects ...)
4. When REAPER stops, the decks that played the lanes stop too
5. While "play the performance" is on, the video sync does not stop REAPER's playback and does not move the edit cursor to the deck. In the performance zone the video window shows the decks (at the edit cursor while stopped). When stopped outside the zone, the cursor follows the deck as before
6. An operation by hand while it plays wins until that lane's value changes again. Grabbing the jog or the record and letting go returns playback to the recorded position (as slip does)
7. When REAPER stops, the `Z` key's cut, the pattern fader, the TOUCH FX pads and CLIP STOP go back to none. Other values (effect amounts ...) stay as they were at the stop
8. To play one deck only, run `reascratch_perf_play_decks.lua` from the action list and choose "Deck A only" or "Deck B only" (both by default). The other deck follows only your hands
9. Recording while it plays records it again (operations added by hand are included)

10. While "play the performance" is on, the deck-time lanes of the previous section (`Envelope scratch` ...) are not used. Recording writes to both kinds of lanes
11. The lane items are REAPER automation items, so they can be moved, copied and edited point by point

Capturing a performance

A performance recorded in REAPER is gathered per deck as "the scratched audio" and "the video that goes with it" (raw material for audio-visual edits). It also produces a "finished mix", the sound that came out of the speakers and the picture shown in the video window, exactly as they were.

1. Run `reascratch_take.lua` to turn it on (it creates the tracks "Rec A" and "Rec B" for each deck, and "Rec Mix" for recording the finished sound. Run it again to turn it off and remove the tracks. In projects made with older versions, the tracks with Japanese names are renamed to these automatically when you run `reascratch_setup.lua` or `reascratch_take.lua`).
2. With the video sync script running, press REAPER's record button and perform
3. When you stop recording (choose to save if asked), a new project opens in another tab (the project you are performing in stays in front). It is saved in REAPER's `Data/ReaScratch_takes/`
   - "A video" track ("A 映像" when the language is Japanese): items cut and pasted from the original video, following the frames the deck showed. If you converted to all-intra on import and the original video is next to the converted one, the original (original quality) is used
   - "A audio" track ("A 音"): the deck's sound (before the crossfader and the channel volume fader)
   - "mix audio" track ("完成 音"): both decks mixed through the crossfader and the channel volume faders, then through a copy of the master's effects (limiters and so on; meters and visualizers are left out)
4. If FFmpeg is available, these videos are also made in the same folder (in the background; when done, `done` is written at the end of the `.log` file of the same name. Needs FFmpeg 5.1 or later; checked with 5.1, 6.1, 7.1 and 9.0; with 4.4 it warns you and skips video creation. To choose the FFmpeg binary to use, put its full path in REAPER's ExtState `ReaScratch` / `ffmpeg`)
   - `<name>_A.mp4` (`_B.mp4` for deck B): each deck's video with not a single frame out of place
   - `<name>_mix.mp4`: the finished mix. The crossfader (including the behaviour that keeps a switch on screen for 2 frames) and the blend (Mix, Add, Subtract, Multiply, Overlay, Soft light, Hard light) are drawn as on screen, with the "完成 音" (mix audio) attached. Luma key, Chroma key and the transitions (`Wipe right` and after) are drawn as Mix, and video effects are not drawn (the number of frames not drawn is written on the first line of the `.log`, separately for the keys, the transitions (camera moves included) and the video effects; the keys and transitions count only the frames with the crossfader between the ends)
5. When REAPER renders this project, frames can be off by 1–2 (that is how REAPER's rendering behaves). When the frames must be exact, use the `.mp4` files made by ffmpeg
6. The picture is aligned to when the sound was heard (the deck's sound comes about half a grain length later than the position shown in the picture, so the picture is shifted by that much)
7. The per-deck videos do not contain the crossfader or video effects (they are the deck's own sound and picture)

8. The finished-mix video is redrawn from the compositing values (crossfader, blend, effects) recorded by the video sync and the JSFX during recording, reproducing the video window's picture. REAPER's rendering is not used
9. While recording, the video sync moves the video items to the recording position (during recording REAPER shows the picture at the recording position, not the edit cursor; they go back when you stop). REAPER draws video ahead and buffers it, so by default the crossfader and video effects appear late in the video window (about 1 s by default while playing or recording, 0.2–0.5 s even when only the decks are playing). Right-click the video window → Performance tweaks (experimental) → Video cache → **All media and video FX are not cached**. The delay drops to about 0.07 s (setup also tells you if this is not set. It takes effect immediately; no need to restart REAPER). With this setting, while a heavy video effect such as VHS is on, REAPER's screen updates slow down, and other scripts (such as visualizers) can move slowly. If that bothers you, set `Video cache` in the same menu to 0.1 s instead (a delay of about 0.1–0.2 s remains)
10. While capturing, the deck tracks have at least 4 channels (8 if you are using Cue. Channels 3/4 carry the sound before the crossfader. Only channels 1/2 are sent to the master). Turning it off restores them.

Heavy video (all-intra conversion)

Ordinary MP4 (H.264) is compressed with reference to neighbouring frames, so playing backwards or seeking sharply makes the preview stutter. Converting every frame into a standalone image (keyframe) makes scratching smooth.

Method A: convert with the included script (recommended)

1. Run `reascratch_reencode_mjpeg.lua` (this is also what happens if you answer Yes in the import dialog). Videos that are already all-intra are skipped (unless they are taller than the height below; then they are converted to scale them down).
   - If ffmpeg is on the PC, it is used automatically for the conversion (frames match the original one for one). Without ffmpeg, REAPER's own features are used (nothing extra to install, but cases have been seen where the picture is 1–2 frames off from the original or black frames get in). The REAPER console shows which was used.
   - Where FFmpeg is searched for: the full path in REAPER's ExtState `ReaScratch` / `ffmpeg`, the same folder as this script, REAPER's `UserPlugins` folder, `PATH`, `C:¥Program Files¥FFmpeg` (and its `bin`) and `C:¥ffmpeg¥bin` on Windows, and Homebrew on macOS (`/opt/homebrew/bin`, `/usr/local/bin`). REAPER's screen freezes during an ffmpeg conversion (it comes back when done).
2. `<name>_mjpeg.avi` is written in the same folder as the original, and the video on the track is swapped automatically (the original file is kept).
   - The height is reduced to the height chosen in setup (720, 540 or 360; default 720). When converting on import, you can choose a different height for that video only after answering Yes. Videos already smaller are not reduced (or enlarged). Files get bigger (as a guide, per minute at 30 fps: around 100 MB at 720p, 70–80 MB at 540p, 40–50 MB at 360p).
   - The cost of reading one frame (a rough guide from the PC we tested), taking 1080p as 1: about 1/2 at 720p, 1/3 at 540p, 1/6 at 360p.
   - To change the resolution or quality, open the script in a text editor and replace the right-hand side of the `local MAX_HEIGHT = …` line at the top with a number (0 keeps the original; heights other than 720, 540 and 360 also go here), or change `JPEG_QUALITY` (a number in `MAX_HEIGHT` takes priority over the height chosen in setup or on import).
   - When converting with REAPER's own features, containers other than mp4 / mov / avi (MKV and so on) cannot report their resolution automatically, so a window asking for width, height and frame rate appears (it does not appear when converting with FFmpeg).

Method B: use the FFmpeg scripts (external tool)

If FFmpeg is installed on the PC, the included scripts can convert to all-intra H.264 (smaller files than MJPEG).

- **Windows**: drag and drop the video file onto `tools/allintra_ffmpeg.cmd`.

- **macOS**: run `chmod +x tools/allintra_ffmpeg.command` once in Terminal, then double-click the file, drag and drop the video into the window that opens, and press Enter.
- Import the resulting `<name>_allintra.mp4` with `reascratch_import_video.lua` (it is all-intra, so no conversion is offered, unless it is taller than the height chosen in setup; then you are asked whether to convert it smaller). The scripts scale videos down to a height of 720 (to change it, edit `MAXH` in the script).

Why MJPEG, not HAP

HAP (a GPU-oriented format used in TouchDesigner and elsewhere) also stores every frame on its own, but this kit does not use it.

- REAPER cannot write HAP. Reading it also needs FFmpeg installed separately (this kit puts working without extra installs first)
- With FFmpeg 9 installed in REAPER, HAP playback once showed red and blue swapped (2026-10-02). When tested again with the same REAPER and FFmpeg, the issue could not be reproduced; the cause is unknown
- For showing in REAPER's video window, MJPEG runs light enough

If you already have HAP videos, the import check reports them as all-intra (they play if REAPER has FFmpeg).

Testing on a low-spec PC (load test)

`reascratch_load_test.lua` checks whether ReaScratch runs comfortably on this PC.

1. With the audio settings you normally use (ASIO and so on, and the same buffer size as for the show), open a project with imported videos, open the video window and keep the video sync running (without it, the result says no video load was included). In a project with deck B, import into deck B too (the test does not start while a deck is empty).
2. Run `reascratch_load_test.lua` from the action list (after running setup, searching for "`reascratch`" finds it; if not, load it with `New action...` > `Load ReaScript...`).
3. You are asked “Start the load test (about 40 s)?”: “Yes: with sound (turn the volume down first)”, “No: measure muted”, “Cancel: do not start”.
4. It takes about 40 seconds (the first 10 seconds with things as they are, then each deck scratches automatically for 30 seconds and reloads a sample once along the way). Do not touch the mouse or keyboard meanwhile. To stop early, run it again.
5. When it finishes, the results are shown and saved to `ReaScratch_load_test_<date>_<time>.txt` in the REAPER resource folder.

Reading the results:

- **Dropouts**: the number of times audio processing fell behind, counted by ReaScratch itself (REAPER's own counter is shown too, but it has been found to miss some even with ASIO, so go by ReaScratch's count). Even one during the automatic scratching means the sound may drop out during a show as it is. Make the video height smaller in setup (540 or 360) or the audio buffer larger, and test again. With WaveOut or DirectSound, dropouts caused by tracks or plug-ins other than ReaScratch may not be counted.
- **JSFX load**: the share of each processing time slot that ReaScratch used. If the "heaviest moments" of all decks add up to more than 70%, there is little headroom left when other work (such as video) piles on.
- **Screen updates**: the longest time the screen drawing and the video following stalled. Above 0.25 s, the picture may look like it hitches.

Troubleshooting

Symptom	Likely cause	What to do
No sound	Monitoring settings / no file loaded	- Check that record arm (red circle) and input monitoring (speaker) are ON for the `scratch engine` track. - Check that the JSFX window's waveform view shows a waveform (if it says "no sample", run the import). - If `Untouched transport` is `Stop`, there is no sound while you are not touching the jog.
The sound stops 0.15 s after the knob stops	Touch mode setting	- With a touch sensor: set `Touch note` to the right note number (54 on the FLX4). - Without a touch sensor: this is normal. To keep sounding while stopped, set `Auto release` to `0` (stop it with the `Release note` button).
Pressing the jog does not scratch (touch does not work)	The controller's jog mode	On controllers with a VINYL (scratch) mode switch, check that it is ON (when OFF, some controllers do not send the top touch, and the jog acts as a pitch nudge). If the JSFX's `touch` lamp lights when you press the top, the touch is arriving
Silence after reaching the end	End-of-sample setting	With `At end of sample` set to `Stop (silent)`, reaching the end with your hand off gives silence (if you are touching the jog, the last grain keeps sounding so you can pull back). To keep playing from the start, change it to `Loop to start`.
The video window froze after a projector/monitor was switched off/on or unplugged	REAPER's video window stopped drawing	Run `reascratch_video_reopen.lua` to reopen the video window (see "Projector output" (p. 201))
The picture does not follow	The sync script is not running	- Check that `reascratch_video_sync.lua` is running (it toggles, so running it again stops it; even with auto-sync chosen in setup, running it once by hand stops it). - Check that the video window is open with `View` > `Video`.
Opening `ReaScratch-HID-Bridge.exe` or a `.command` shows a warning and it will not start	Windows / macOS block unsigned programs	See "If a warning stops it from starting" under "HID controllers (NI Traktor Kontrol S4, S3, S2)" (p. 197)
The crossfader and video effects show up late in the picture	REAPER draws video ahead and buffers it	Right-click the video window → `Performance tweaks (experimental)` → `Video cache` → `All media and video FX are not cached`. It takes effect immediately (setup also tells you if it is not set)
Video seeking is jerky	Keyframes are far apart	The video is still compressed H.264 or similar. Convert it to MJPEG as described in "Heavy video (all-intra conversion)" (p. 216).

Symptom	Likely cause	What to do
The sound drops out / crackles	Too much load on the PC	Measure as described in "Testing on a low-spec PC (load test)" (p. 218). A larger audio buffer, a smaller video height in setup, or closing other background software helps.
The imported audio goes silent partway through	An old version of the video sync script	A warning appears on import. An old version of the video sync script stopped playback while REAPER was rendering. Update the kit to the latest version and import the video again.
Playback stops partway through the track / the waveform is cut off	The track is too long	If the waveform view shows "only the first ○:○○ of ○:○○ loaded" in red, the track exceeds the length that can be loaded. Cut the video shorter and import it again.
Turning the Reloop Buddy does nothing (Windows)	The MIDI driver is not working	With the standard Windows driver, the Buddy's MIDI part (jog and buttons) does not start ("Code 10" in Device Manager). Install the "ASIO Driver" from the Reloop product page and reconnect the Buddy (how to check: below the table). No driver is needed on a Mac.
With two decks, deck B's picture is not layered / breaks up when stopped / video effects do not work on B	The project's video settings	This does not happen in projects made from the kit's template (`File` > `Project templates` > `ReaScratch`). In the current project, run `reascratch_setup.lua` again (it moves the `video B view` track to the place that suits the project's video settings; restarting the video sync does the same). If that does not help, close REAPER, open the project file (.rpp) in a text editor, delete the one line starting with `VIDEO_CONFIG`, save, and open it again (the cause is below the table).
The JSFX cannot be found	Files in the wrong place	Check that `Effects/ReaScratch/reascratch.jsfx` is inside the `Effects` folder of the REAPER resource folder (the folder opened by `Options` > `Show REAPER resource path in explorer/finder`).

Turning the Reloop Buddy does nothing: Once `Reloop Buddy In` appears in REAPER's `Options` > `Preferences` > `Audio` > `MIDI Inputs`, it is ready.

With two decks, deck B's picture is not layered (cause): Depending on REAPER's video settings (`Video` in `Project settings`), the picture of an upper track covers the lower one. Setup and the video sync put `video B view` in the place that suits them, but when the tracks have been reordered by hand, for example, the `video B view` track's picture can be shown as is, hiding the processing that layers the two pictures (the Video processor on the `video` track).

FAQ (how it works and other software)

This section summarizes questions asked since the release. Answers about other software cover what could be checked in public documents and source code in October 2026.

Why not just use DJ software such as rekordbox?

ReaScratch was made because these two could not be found:

- Scratching without a pitch change
- Editing scratches as automation inside a DAW ("Scratching from envelopes (patterns)" (p. 207))

Hot cues, memory cues, BPM, beat positions and key can be loaded from an XML exported from rekordbox ("Loading cues from rekordbox" (p. 26)).

Is there DJ software that scratches without changing the pitch?

None was found. Most DJ software turns key lock (Master Tempo in rekordbox) off only while you scratch. Mixxx, for example, turns key lock off while scratching and below 0.1x or above 1.9x speed, and switches to a read where the speed is the pitch (`enginebuffer.cpp` in its source code). For other software, too, there are many reports that key lock turns off by itself while scratching, or that the sound breaks up if it stays on.

Can the Bitwig–rekordbox integration edit scratches?

No. The integration added in Bitwig Studio 6.1.4 (October 2026) brings in the rekordbox library and its analysis (BPM, key, hot cues, beatgrid, phrases). It does not bring in scratches, jog movements or recorded performances.

How does it scratch without changing the pitch?

It plays short slices of the sound, called grains (50 ms by default), always at normal speed. The jog only moves the place the grains are taken from. So the pitch stays the same when you move slowly, turn backwards or stop. When you move slowly, the grains are placed at a slowly advancing position; the sound itself is not stretched.

The place a grain is taken from falls between samples, so it is read with interpolation. Normally this is linear (a straight line between the two neighbouring samples). With `Pitch mode` above 0 or a changed key, it is 6-point Lagrange interpolation (a curve through the 6 samples around the position, which keeps more of the highs).

REAPER's own playback (item playback and time stretching) is not used. The sample is loaded into the JSFX's memory and the JSFX plays it itself. From REAPER, ReaScratch uses the tempo and beat information, and envelopes to record and edit what you play.

Don't the grains sound muddy when they are simply lined up?

They would. Three things keep it clean:

- **Hann window**: each grain fades in and out smoothly, which removes clicks at the edges
- **WSOLA** (`WSOLA search`, ±10 ms by default): before placing the next grain, it is shifted slightly back and forth and starts where its waveform best matches the continuation of the previous grain. This reduces the muddiness of waves cancelling each other where the grains overlap (a flanger-like sound)
- **Jitter** (`Freeze jitter`, 8 ms by default): when the record is held still, the same grain repeats and makes a periodic buzz. The place the grain is taken from is moved slightly at random to break it up. Above `Jitter off above` (0.15x by default) the jitter is gone

How is this different from key lock (time stretching) in DJ software?

The idea of cutting the sound into small pieces and putting them back together is similar. SoundTouch, one of Mixxx's key lock engines, is described in its official documentation as a WSOLA-like method. The difference is the movement it is made for. Key lock is made to sound clean when playback moves forward at a nearly constant speed. ReaScratch prioritizes keeping the pitch while you go back and forth between stopping, turning backwards and speeding up suddenly. In exchange, at a constant speed more of the grainy character remains than with key lock. The key lock methods of rekordbox, Serato and Traktor are not public, so they were not compared.

Does a record-like movement, where the pitch follows the speed, need a separate engine?

No. The same grains are used; only the speed at which they are read changes. The `pitch — tape` bar under the record (`Pitch mode`) sets how far the pitch follows the speed: 0 = grains with no pitch change, 1 = tape (like a record; silent when held still), and at 0.5 double speed sounds half an octave higher. `Key` (±12 semitones, without changing the tempo) and the PITCH PLAY pads also multiply the speed at which the grains are read.

At first, the in-between values mixed two sounds, the grains and a tape read. They sounded like two copies and cancelled out in places, so it was rebuilt as the single method used now.

Does it use a time-stretching library such as Rubber Band?

No. Rubber Band, Signalsmith Stretch and Bungee were compared before starting. Rubber Band does not support backward playback (a negative ratio), so it was left out.

If Rubber Band does not play backwards, how does Mixxx play in reverse?

In Mixxx's source code, Rubber Band only gets the absolute value of the speed. In reverse, Mixxx reads the track from the end and passes the samples in that order, so Rubber Band itself only stretches forward

(`enginebufferscalerubberband.cpp`, `readaheadmanager.cpp`, `cachingreader.cpp`). The moment the direction changes is joined with a crossfade. Key lock is off while scratching, so this is not used then.

Things that depend on your setup

What changes with your PC, your equipment or REAPER's settings. Where it says “not tried”, the author has not tested it.

Sound

- **Hearing the cue apart on headphones** needs an output other than the main one (“Listening to a deck on headphones (Cue)” (p. 202)).
- **The video's audio**: only the first audio track of a video is imported. In a video with several (another language, a commentary), it may not be the one you want. 5.1 and other layouts are mixed down to stereo.
- **Sample rate**: the imported audio keeps the video's sample rate; when it differs from your audio interface's, it is converted as it plays. The length limit also depends on it (about 11.5 minutes for a 96 kHz video; see “Step 1: Import a video” (p. 19)).
- **The deck's track**: a deck sounds while the `scratch engine` track's record arm and input monitoring are on. Turning either off, or muting the track in REAPER (`M`), stops not only the sound but also the loading of samples and the deck itself. To make it quieter, lower the track's fader instead.
- **Audio buffer**: a larger buffer means fewer dropouts but a later response to the jog. The video does not allow for the audio output's delay, so the larger the buffer, the later the sound is heard after the picture (how much has not been measured). A look-ahead limiter or the like on the master delays it the same way.
- **The headphone (Cue) output**: the chosen output is kept in the project by its number. If you open the project on another PC or with an audio device with fewer outputs and that output is not there, run setup again to choose anew. If the number exists but is now another jack, choose again under the `Cue` track's `ROUTE`.
- **If REAPER crashes**: with WaveOut as the audio system, a forced quit while playing once left the last sound looping on the Windows side. Playing another sound (a silent file, for example) on Windows' default output stopped it.

Video

- **REAPER's video decoder order**: REAPER uses the decoders in the order of `Video decoder priority` under `Options` > `Preferences` > `Media` > `Video`. With the default order (`wmf vlc ffmpeg …`), the all-intra AVI (MJPEG) files were also read by WMF, and seeking was about 3 times slower than with FFmpeg. WMF sometimes showed an mp4's picture 2 frames off. If REAPER can use its FFmpeg decoder, putting `ffmpeg:avi` first reads only AVI files with FFmpeg (for example `ffmpeg:avi wmf vlc ffmpeg qt:mov,qt,m4v,mp4`). Turning on `Show decoder information` on the same page shows in the video window which decoder is used.
- **The video cache setting**: with `All media and video FX are not cached`, which setup recommends, other scripts running in REAPER (visualizers and the like) may slow down while a heavy video effect such as VHS is on.

- **Frame rates and large files**: a 60 fps video is heavier than 30 fps and its converted file larger (not tried). Videos with a variable frame rate (from a phone, for example) and converted files over 2 GB have not been tried either. If it is heavy, cut the video shorter or choose a smaller conversion height in setup.
- **Screen size**: the JSFX screen grows only in whole multiples with its window. The kit's window layout was made on a 1920×1080 screen. On a smaller screen or with display scaling (125%, 150% …), windows may fall off the screen (not tried). Move them and save the window set again.
- **External screens on a laptop**: on laptops whose external screen ports are wired to an NVIDIA (or other) GPU, the GPU driver's processing grew long enough to cause dropouts. Setting `reaper.exe` to "High-performance NVIDIA processor" with power management "Prefer maximum performance" in the NVIDIA Control Panel fixed it (at the cost of more heat and power). Which port goes to which GPU differs between models.
- **Capturing with OBS**: OBS's "Window Capture" could not capture a floating JSFX window. Use "Display Capture" and crop.

Controllers and keyboard

- **Other devices and programs**: a deck listens to all MIDI inputs. If another MIDI device (a keyboard, a foot switch …), ReaLearn, a REAPER control surface or an action shortcut uses the same channel and number, the deck moves too. Disable the inputs a deck does not need in REAPER's MIDI inputs. On Windows, REAPER may not be able to use a controller that rekordbox, Serato, Traktor or another DJ program is using (not tried). Quit that program first.
- **Two of the same unit, or plugged in again**: the lights go to the first output found by the preset's name, so with two of the same unit they may go to the other one. When the name or number of a MIDI output changes (another USB port, for example), run setup again.
- **The light-off helper (Windows)**: to turn the lights off after REAPER closes, the video sync starts PowerShell in a hidden window. On a work or school PC, or with some antivirus software, it may be blocked and the lights stay on (not tried; REAPER itself is not affected). The helper sends 0 for every note and CC on all channels of the light outputs. If those outputs go to another program through loopMIDI or the like, that program's volume (CC7) and so on go to 0 too.
- **Keyboard layouts**: the keys are laid out for a Japanese keyboard. For `Shift` + `7` on an English keyboard, see the keys under "Step 3: Scratch" (p. 20). German, French and other layouts, an input method (IME) that is on, and F1–F8 on a Mac laptop (which may need the Fn key) have not been tried.
- **The HID bridge (Mac)**: if asked for Input Monitoring permission, allow it (not tried).

REAPER and the PC

- **REAPER version**: tested with REAPER 7.80 (the template was saved with 7.82). With older versions the JSFX may not load, the length limit may be shorter, or a menu in this manual may be missing. Systems other than Windows and macOS (Linux) have not been tried.
- **REAPER-wide settings that setup changes**: with the SWS extension installed, setup turns off `Limit scrub rate to 1.0x` on the `Scrub/Jog` page of `Preferences`. This applies to all projects (tick it again to undo). When toolbar 3 or a window set is replaced, the old content is kept in `reaper-menu.ini.before_reascratch`

and `reaper-screensets.ini.before_reascratch` in REAPER's resource folder. To go back, close REAPER and replace the file with its copy, taking `.before_reascratch` off the name (later changes to menus or window sets are undone too).

- **Updating the kit**: after copying a new kit over the old one, restart REAPER. A running video sync and open JSFX keep the old version until then.
- **Moving the kit or REAPER's resource folder**: the autostart lines (`__startup.lua`) and the toolbar buttons remember where the kit is. If that changes (for example, a portable REAPER on a USB stick gets another drive letter), run setup again (when the autostart cannot start the sync, REAPER's console says so).
- **Starting the video sync again and again**: starting it again from an action several times once brought up REAPER's "ReaScript task control" window, which kept coming back after closing. Restart REAPER then.
- **Power and memory**: run a laptop on its AC adapter with a high-performance power setting. On battery or power saving it is heavier and drops out more easily (the author's PC runs REAPER on its fast cores, so other PCs may be heavier). Long samples can use up to about 1 GB of memory per deck (not measured).

Where files go

- **Imported sound and captures**: imported sound (WAV) is kept in `Data/ReaScratch_samples` in REAPER's resource folder, captures in `Data/ReaScratch_takes`. While a capture makes its video, images of every frame are also made there for a while. Check the free space on the drive that holds the resource folder (usually C:).
- **Opening a project on another PC**: the JSFX remembers samples by their filenames in `ReaScratch_samples`. If you move only the .rpp to another PC or reinstall REAPER, the samples show `ERR`. Move the files in `Data/ReaScratch_samples` with it.
- **Where the all-intra video goes**: into the original video's folder (about 100 MB per minute at 720p). It fails in a read-only place. In a folder synced by OneDrive or the like, uploading the large file may go on during the show. Copy the video to a working folder before converting.
- **Folder names and usernames**: Japanese characters and spaces in filenames have been tried. A PC with a Japanese Windows username has not.

Tested environment

- **OS / host**: Windows 11 / REAPER 7.80 (AVI/MJPEG writing and playback checked without FFmpeg)
- **Test hardware**: Akai APC40 mkII (real hardware), Reloop Buddy (MIDI mapping checked on real hardware; Windows needs the driver), Pioneer DDJ-FLX4 (tested on real hardware only early in development, not with current features (channel faders, headphone CUE button, tempo fader direction and so on); treated as untested like the other models until it is tested again), AlphaTheta DDJ-FLX2 (settings based on the official MIDI message list; not yet tested on the real hardware)
- **Other controllers**: not tested on the real hardware, including the NI Traktor Kontrol models connected over HID (marked as untested in the "Preset list" (p. 65)).
- **macOS**: an early version (2026-10-03) was checked on macOS (REAPER 7.47) by a test user. Features added since (deck B, video blending, the HID bridge and so on) have not been checked on macOS. Linux has not been tried. If something goes wrong, please let us know.

License & credits

ReaScratch is free software under GPL-3.0-or-later (`LICENSE`). Some parts are code ported from other licenses: the DJ filter (Saike FM Filter 2, MIT License, `Effects/ReaScratch/LICENSE_Saike_filters.txt`), the VHS video (zhuker/ntsc, Apache License 2.0, `Effects/ReaScratch/LICENSE_ntsc_zhuker.txt`), the key detection (libKeyFinder, GPL-3.0-or-later) and the loudness measurement for auto gain (libebur128, Jan Kokemüller, MIT License, https://github.com/jiixyj/libebur128; its K-weighting filter constants and gating are written in Lua as libebur128 does them). The licenses of the parts in the HID bridge's exe are in `HID bridge/THIRD_PARTY_LICENSES.txt` . The references and thanks are in `CREDITS.md` .

Version history

The main changes in each version, newest first.

0.7.3 (2026-10-21)

- Stopping a deck with the echo on now lets the echo tail ring out (before, stopping cut the tail too). Turning the echo down to 0 also lets the tail finish before cutting out
- The crossfader on the screen now works with the mouse (click the track to jump there, drag to follow, double-click to center)
- With Quantize on, PLAY on a stopped deck waits for the other deck's next bar before it starts
- Fixed PLAY starting slightly early (by 10 to 25 ms)
- With loop on, scratching on the platter or the waveform now wraps across the loop's edges
- The deck now responds faster to the first movement after holding the platter, and sound processing is slightly lighter
- New chapter in the manual: "FAQ (how it works and other software)"
- CREDITS: added OtoMadSampler, and the TIME-1 link now points to its own repository

0.7.2 (2026-10-17)

- New script `reascratch_lanes.lua` and the `LANES` button on toolbar 3: show or hide the envelope lanes by group (hand-made scratch, recorded effects, performance lanes)
- The FX window of the performance lanes (`DJ performance lanes`) now shows each lane's value in a table (the number of columns follows the window's width). The parameter names are shorter
- Updated screenshots in the manual to reflect the current screen with LANES

0.7.1 (2026-10-17)

- New section in the manual, "Things that depend on your setup" (p. 224) (sound, video, controllers and keyboard, REAPER and the PC, where files go), and what you need to hear the headphone cue apart from the main output
- Setup: keeps what was on toolbar 3 or in a window set before replacing it, says so in REAPER's console when the autostart cannot start the video sync (the kit moved, for example), and asks for the Cue output again when the current audio device does not have it
- Fixes in the manual (the names of the MIDI settings pages and more)

0.7 (2026-10-16)

- More controllers: presets grew from 62 to 122 (sorted A–Z by maker; those not yet tested on the hardware are marked ⚠), four more models for the HID bridge including the Traktor Kontrol S3, and diagrams of 66 controllers in the manual
- Eight pad modes (auto loop, TOUCH FX, slicer, BOUNCE LOOP and more), loops, CUE LOOP, eight DJ filter types, and both decks from a single APC40 mkII or mkI
- Button lights (in the hot cue colours, turned off when REAPER closes), two controllers at once (Reloop Buddy + APC40 mkII), and the setup `Learn` tab (make your own preset) and `More` tab
- 29 video transitions (wipes, irises, camera moves, Brush wipe, Shatter and more), driven directly by the crossfader position
- Video effects expanded to 15 (VHS, Disc, Line art, Halftone and Zoom blur added), an LFO that moves the amount with the beat, a matching sound effect for each, and effects recorded into envelopes when you record a performance
- `LOAD` button: put a file chosen in the Media Explorer into a deck while you play (audio-only files show a spinning speaker; heavy videos are converted to all-intra in the background). The LOAD button now works on 13 more units (Hercules Inpulse, Reloop Terminal Mix and others; before, it did nothing on them even though they have one). Loading by dragging a file from the Media Explorer or the Explorer onto the window
- Previewing a Media Explorer file (the `P` key, or a button on most controllers: the sound in the headphones only, the picture only in the deck's record, never in the video window)
- A metronome (a click on a deck's beats, the first beat of each bar higher; the `N` key, the window in the record's corner, a MIDI button; most presets set the button; the second APC40's `METRONOME` now turns it on/off instead of REAPER's own metronome)
- Importing rekordbox XML (hot cues, memory cues, BPM and beat grid, key), loading Traktor and VirtualDJ track data, writing out track data (an XML for rekordbox or ReaScratch track data), beat maps for songs whose tempo changes partway, and key detection with KEY SYNC
- Tempo and BPM: BPM-based tempo ranges (±5 to ±60 BPM, default ±20 BPM), typing the BPM on the screen, sync with REAPER's own tempo, BPM detection for short loops, and beat, bar and loop lines on the waveform
- Performance lanes (what you do on the decks while REAPER records is recorded on REAPER's timeline and plays back from REAPER's play button in `PERF PLAY` performance playback mode)
- Auto gain per sample (the loudness is measured on import and LOAD and brought to a target; setup's `Samples` tab), and swing for patterns played from the PATTERN pads
- Brake length in beats, PAUSE quantize, and a fix for the Reloop Buddy tempo fader (not working, and running the wrong way)
- Fixed: an empty deck showing the other deck's picture (the black picture of an empty slot did not loop), and the preview's picture flickering on an empty deck B
- Projects saved with 0.6 or earlier are no longer converted from the old save format (choose the preset and other settings again in setup). The old names of the capture tracks (before 2026-10-08: "撮る A / B / 完成") are no longer converted either

0.6 (2026-10-07)

- Cue: listen to a deck on headphones (Cue mix, and the output device is chosen from a list and remembered)
- More controllers: presets grew from 6 to 62 (DDJ-FLX2 and other DDJ models, CDJ-2000NXS2, CDJ-3000, and models from Reloop, Numark, Hercules and Denon DJ)
- The setup `Preset` menu is now three levels: maker → model → deck
- HID bridge for HID controllers (NI Traktor Kontrol S4 MK3 and S2 MK3)
- Channel faders and headphone CUE on the DDJ-FLX4
- Fine adjustment, crossfader, volume, Cue and video effects from the keyboard
- Projector output (handles screens being plugged in and out)
- Running setup again keeps each deck's settings, saved data and envelopes
- Fixed: quantized hot cue landings drifting further each time, and wrong choices on the setup screen

0.5 (2026-10-06)

- Recording a performance into envelopes
- Capturing a performance (each deck's sound and video, and a finished video that matches the screen; older FFmpeg versions supported)
- The crossfader while stopped, and video effects on still-image decks, now show in the video
- ANSI-style block-letter logo, a new top row (`sync`, `Q`, `memory`) and `help`
- A warning in setup when REAPER's video cache is long enough to delay the crossfader in the video
- Fixed: Trails getting darker; with the Reloop Buddy, the crossfader video lagging, the jog drifting and hot cues landing off the beat

0.4 (2026-10-05)

- Optimised for the Reloop Buddy (deck 2 preset, pad modes, 10-bit tempo, volume, FX)
- Eight hot cues, and PLAY quantize
- Swing (16th-note shuffle)
- Scratching from envelopes (place the included patterns, rearrange and save them, fit them to the swing)
- A crossfader picture in the bottom strip, and even the briefest crossfader cut always shows in the video
- Shorter help text in the bottom strip, with long text scrolling
- Fixed: a stopped deck B flashing black and jittering

0.3 (2026-10-04)

- A second deck (deck B), with the crossfader layering the two videos (nine blend modes)
- Ten video effects (on each deck's picture and on the mixed picture)
- Tempo (change the speed without changing the pitch) and SYNC

- Sample switching (up to 8; each sample keeps its own position and cues)
- Memory cues (up to 32), and hot cue buttons above the frame
- Beat-quantized jumps, with BPM detected on import
- Brake, start-up and spinback
- Left-handed layout (`Layout`), and a list of keyboard shortcuts (`keys` button or `?` key)
- Large videos scaled down on import (height 720, 540 or 360), samples loaded a little at a time, and a load test script that counts dropouts

0.2 (2026-10-03)

- Crossfader cuts (adjustable cut-in gap and width; the video also disappears while the sound is cut)
- A record platter you turn with the mouse
- Four hot cues (a short press jumps; hold to return on release), and a CDJ-style CUE button
- SLIP mode
- Search by clicking or dragging the whole-track waveform, and a redesigned waveform
- Tape-style sound where the pitch changes (the `tape` bar), with the record coasting on after you let go
- Maximum song length raised to about 23 minutes, and FFmpeg preferred for all-intra conversion
- Fixed: the Space key being taken by REAPER itself, and imported audio going silent partway when rendered

0.1 (2026-10-02)

- First release
- Pitch-locked scratching (grains always play at normal speed), with the video showing the frame at the playback position
- Video import (writes the audio to a WAV) and conversion to all-intra (MJPEG)
- Presets: DDJ-FLX4 (deck 1 and deck 2), APC40 mkII (two single-deck presets), Reloop Buddy (deck 1) and Custom
- DOS-style screen (two-row waveform of the whole track and a zoomed view, knobs, lamps, setup screen)
- Use without a controller (drag the zoomed waveform to scratch; play/stop, CUE and freeze from the keyboard)
- An option to start the video sync automatically when REAPER starts